component maintenance manual

Endura Beverage Maker

component maintenance manual
(with illustrated parts list)

25-33-21

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REVISION NUMBER 05 DATED 07 OCTOBER 2020

HIGHLIGHTS

This Component Maintenance Manual (CMM) with Illustrated Parts List (IPL) has been reviewed for technical changes resulting from B/E Service Bulletins, B/E Service Information letters, configuration changes and part improvements. Any existing Temporary Revisions (TRs) against this CMM have been incorporated with this revision.

This CMM and IPL have been revised. Please replace the old copies of this manual with this Revision Number 05 dated 07 Oct 2020. A description of the changes follows.

SECTION AND PAGES	CHANGE DESCRIPTION
LEP-1 thru LEP-3	Revised to reflect current revision.
1002	Revised Table 1001 to delete “Test Brew Cup” to reflect current revision.
1005	Revised step 3.E.(7) to update the relief valve pressure value.
1009	Revised step 3.J.(1) to update the Brew Test procedure.
1015	Revised Table 1002 to delete fault isolation procedure for checking brew cup water temperature.
3030	Revised step 2.L.(10) to correct the figure reference from Figure 9003 to Figure 9008.
4001	Revised Table 4001 to delete “Sodium Bicarbonate (Baking Soda)”.
7002	Revised Table 7001 to update specification for Thread Lubricant.
7003	Revised step 2.A.(3) to correct the figure reference from Figure 9003 to Figure 9008.
7004	Revised step 2.B.(4) to correct the figure reference from Figure 9003 to Figure 9008 and updated step 2.B.(4) Note from sealant to thread lubricant.
7005	Revised Figure 7003 and Figure 7003 title.
7006	Revised Figure 7005 and Figure 7005 title.
7007	Revised Figure 7006 and Figure 7006 title.
7008	Revised Figure 7007 and Figure 7007 title.
7014	Revised to add step 2.K.(2) for the installation of PC board insulator and sequence rearranged.
7023	Revised Figure 7022 to add new items. Revised step 2.M.(3) to correct the figure reference from Figure 7020 to Figure 7022. Revised step 2.M.(4) to remove the figure reference. Revised step 2.M.(5), 2.M.(5)(a) thru 2.M.(5)(c) to correct the item numbers and updated tea tube funnel installation procedures. Revised step 2.M.(5)(a) to delete Note. Revised step 2.M.(5)(d) Note to update the o-ring item number.

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SECTION AND PAGES	CHANGE DESCRIPTION
7036 thru 7038	Revised to add step 2.M.(13)(a) for the installation of controller assembly and sequence rearranged.
7037	Revised Figure 7034 to add new item call-out and updated the relay retainer item number in step 2.M.(13)(e).
9003	Revised Table 9001 to delete “Sodium Bicarbonate (Baking Soda)”. Updated specification for thread lubricant and corrected the nomenclature of “Silicone Rubber Adhesive” to “Red Silicone Rubber Adhesive”.
10000	Revised Illustrated Parts List to reflect current revision.

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Endura Beverage Maker

component maintenance manual
(with illustrated parts list)

This manual includes data for the equipment that follows:

Unit	Part No
Endura Beverage Maker	4510-31UF-00
	4510-31LF-00
	4510-32UF-00
	4510-32LF-00
	4510-33UF-00
	4510-33LF-00
	4510-34UF-00
	4510-34LF-00
	4510-35UF-00
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	4510-36LF-00
	4510-37UF-00

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CMM OCT 07/2020

523-0832004-501F73
1st Edition, 22 Dec 2004
5th Revision, 07 Oct 2020

Unit	Part No
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	4510-38UF-00
	4510-38LF-00
	4510-41UF-00
	4510-41LF-00
	4510-42UF-00
	4510-42LF-00
	4510-43UF-00
	4510-43LF-00
	4510-44UF-00
	4510-44LF-00
	4510-45UF-00
	4510-45LF-00
	4510-46UF-00
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	4510-36UG-00
	4510-36LG-00
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	4510-41UG-00
	4510-41LG-00
	4510-42UG-00

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Unit	Part No
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	4510-43UG-00
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	4510-45UG-00
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	4510-48UG-00
	4510-48LG-00

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REV NO.	ISSUE DATE	DATE INSERTED	BY	REV NO.	ISSUE DATE	DATE INSERTED	BY
1	14 May 2013						
2	08 Sep 2014						
3	18 Mar 2016						
4	06 Aug 2018						
5	07 Oct 2020						

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All Temporary Revisions will be incorporated into the Component Maintenance Manual (CMM).

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SERVICE BULLETIN LIST

All Service Bulletins will be incorporated into the Component Maintenance Manual (CMM).

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3520-0534-25-001 / Rev 003	16 Aug 2016	06 Aug 2018	New Screws for Brewhead Assembly, PN 3520-0534-01
3520-0594-25-001 / Rev 004	06 Nov 2017	06 Aug 2018	Beverage Maker/coffee Maker Vent Valve Ball Replacement
3520-0594-25-002 / Rev 001	19 Oct 2016	06 Aug 2018	Beverage Maker/coffee Maker Vent Valve Ball Replacement
3520-0594-25-003 / Rev 003	18 Jul 2017	06 Aug 2018	Improved Durability Vent Valve Components
3530-0121-25-01 / Rev 04	18 May 2018	No Effect	Printed Circuit Board Assembly Replacement
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4510-25-13 / Rev 001	25 Apr 2016	No Effect	Beverage Maker Disable Platen Heater Function
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SIL-4510-2500-101	04 Apr 2017	No Effect	Beverage Maker Installation & Operating Instructions
3520-0579-SIL-001	23 Apr 2014	No Effect	Improved IR Sensor PN 3520-0579-01
3520-0579-SIL-002	30 Mar 2017	No Effect	Improved IR Sensor PN 3520-0579-01 with new LED PN LP-50098
3530-0261-SIL-001 / Rev 001	14 Jan 2015	No Effect	Platen Heater PN 3530-0261 Product Improvement.
3575-0650-SIL-001 / Rev 003	15 Jan 2015	No Effect	Clarification of Secondary Retention Device on Units Shown in Effectivity Table
4510-SIL-01	23 Dec 2004	No Effect	Optional Replacement of Microcontroller PN 3530-0270-01 from Rev. A01 to Rev. B00 for all 4510 Series mfg before 1/1/2005.
4510-SIL-02	06 Nov 2006	14 May 2013	Replacement of OT Thermostat PN 3530-0268-01 and 3530-0280-01 to 352-0983-3 for all 4510 Series mfg before 10/27/2006.
4510-SIL-03	02 Oct 2009	14 May 2013	SIL to explain the various platen assy (PN 3520-0561-0X) and washer configurations on the Endura Beverage Makers.
4510-SIL-05 / Rev 001	26 Jan 2012	No Effect	Current Revision Level of Microcontroller PN 3530-0270-01
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INTRODUCTION

1. General

A. This manual has been prepared to fulfill the requirements of ATA iSpec 2200 for Component Maintenance Manuals.

B. All procedures established in this manual have been performed and verified acceptable by trained personnel at B/E Aerospace, Inc. A competent shop mechanic who is unfamiliar with these units should, with a minimum amount of effort and tools, be able to isolate the problems of a nonfunctional unit and restore it to service.

C. Procedures are provided for the full range of maintenance and include all the work functions that could or may be done to the unit throughout its entire service life.

D. A competent shop mechanic should use this manual first to test and isolate a faulty component. The mechanic should then go directly to the applicable manual subject to effect a repair in the specific problem area without disturbing other areas.

2. Abbreviations

The following abbreviations may be used in this manual:

ABBREVIATION	MEANING
AR	As Required
ASSY	Assembly
°C	Celsius or Centigrade
CAGE	Commercial and Government Entity
CM	Centimeter
DIA	Diameter
DIP	Dual In-line Package
EFF	Effectivity
ESD	Electro Static Devices
°F	Fahrenheit
FIG	Figure
FH	Flat Head (screw)
FL	Flat
GA	Gauge
ID	Inner Diameter
IN	Inch
IN-LBS	Inch-Pounds
IPL	Illustrated Parts List
KPAG	Kilo Pascals Gauge
LBS	Pound
LK	Lock

ABBREVIATION	MEANING
LKG	Locking
MM	Millimeter
NA	Not Applicable
NHA	Next Higher Assembly
NO	Number
OD	Outer Diameter
PN	Part Number
PH	Pan Head (screw)
PSIG	Pounds per Square Inch Gauge
QTY	Quantity
REQ	Required
RF	Reference
RTD	Resistive Temperature Detector
SB	Service Bulletin
SIL	Service Information Letter
SUPLD	Supplied
SUPSD	Superseded
SUPSDS	Supersedes
TTL	Total
V	Vendor
W/	With

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1. General

A. Beverage Maker PNs 4510-3 and 4510-4 series are manufactured by B/E Aerospace, Inc., Commercial Aircraft Products Group.

B. The Beverage Maker (refer to Figure 1) is designed for use in aircraft galleys. The Beverage Maker is easily removed from the permanently installed rail assembly after a locking pin is released.

2. Description

A. The Beverage Maker is of modular design, utilizing cast aluminum-magnesium-titanium construction for maintainability and reliability. The front of the casting serves as a framework for the housing platen, electrical controls, brew assembly, and water faucet. The rear of the casting serves as a framework for the hot water tank, power module, plumbing and controls, and provides close dimensional control for accurate connection to the aircraft electrical system, water supply, and drain.

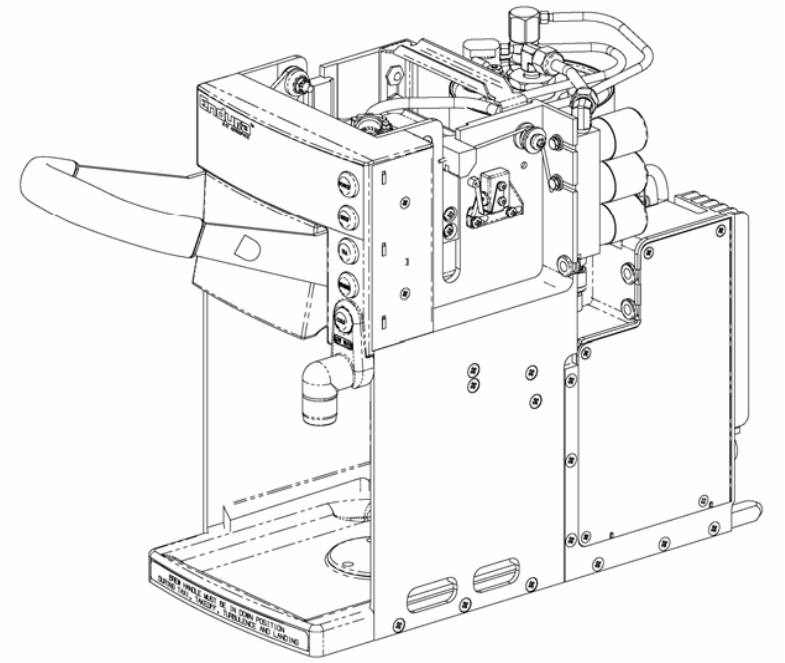

Figure 1
Beverage Maker

B. The Beverage Maker heats water for a full server of coffee. Optional controls are available to heat a server full of hot water for tea or other uses. Optional controls are available to provide hot or cold water from a faucet on the front of the Beverage Maker. The faucet aerator decreases likelihood of splashing.

C. The water is heated in a single tank by three 900 watt immersion type heaters mounted in the bottom of the tank.

D. An optional 15 watt WARMER, mounted in the bottom of the platen will maintain liquid temperature between 165 °F (74 °C) to 195 °F (91 °C) for approximately 30 minutes.

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E. An insulated server is available. It should not be used when the WARMER is on.

F. The brew chamber uses the standard size 4.25 in (10.79 cm) x 5.25 in (13.33 cm) coffee bag (pillow pack). The brew chamber is sealed during the brew cycle to retain grounds in the event that the pillow pack is torn or broken.

G. The server and brew cup are restrained under all flight conditions while the brew handle is in the down position. The server is also covered by the server retainer when the brew handle is down.

H. The Beverage Maker provides a vent valve to allow automatic filling or draining of the Beverage Maker when interfaced with an aircraft water system.

3. Operation (refer to Figure 1)

NOTE: The server must be in place and the brew handle in the down position before the BREW button or TEA button can be pressed to start a brew cycle.

A. Three-phase 115 VAC, 400 Hz is required at the power connection for operation of the Beverage Maker.

B. When the Beverage Maker is placed on an energized rail, power is supplied to the Controller board and the electrical system.

C. If the water level in the tank is low, the LOW WATER indicator light will come on until the tank is filled.

D. Pressing the POWER button starts the preheat cycle so water is heated for brewing.

E. When the POWER button is pressed, with the brew handle in the down position and the server in place in the server cavity, brewing may be started by pressing the BREW button or the optional TEA button.

F. When the temperature in the water tank reaches the set point [approximately 196 °F (91 °C)], the brew solenoid valve opens and lets water flow. The water flow will continue until the level sensors detect the liquid in the server or the brew handle is lifted. Completion of an electrical circuit across the level sensor will signal the Controller board to close the brew solenoid valve.

G. A time delay causes the BREW button light to stay on for 10-20 seconds after the brew solenoid valve closes. This time lets most hot liquid drain into the server.

H. If the brew cycle is interrupted, it may be reset by pressing the POWER button OFF and ON again.

CAUTION: DO NOT USE THE WARMER WHEN USING AN INSULATED SERVER. THE WARMER CAN MAKE THE OUTER SURFACE OF THE INSULATED SERVER VERY HOT.

J. Power may be applied to the optional WARMER by pressing the POWER button and the WARMER button to the ON positions. With the server in the server cavity, the WARMER keeps coffee temperature between 165 °F (74 °C) to 195 °F (91 °C) for approximately 30 minutes.

K. In the event of failures disabling the electronic safety features, control circuits prevent Beverage Maker operation when internal tank temperature is below 32 °F (0 °C) or above 235 °F (113 °C).

L. Additional safety features provided in the Beverage Maker design include:

(1) If the water tank external temperature exceeds 225 °F (107 °C), power to the heaters will automatically be removed by three resettable over temperature thermostats.

(2) A pressure relief valve is set at the factory to open in the event that the water tank internal pressure exceeds 80 to 85 psig (5.51 to 5.86 barg).

(3) A low water sensor prevents operation of tank heaters and the LOW WATER indicator light comes ON any time the tank water level drops below the sensor mounted in the tank.

(4) A coffee level sensor provides accurate control of liquid level in the server and stops a brew cycle with a full server.

(5) Over temperature detection trips the safety latch if the tank temperature goes above 225 °F (107 °C). The POWER button light goes into double-blink mode while the Beverage Maker is connected to a power supply until the temperature goes down and/or the required circuits are repaired, and the safety latch is reset.

(6) Phase loss detection detects the loss of one of the phases in the circuit. The POWER button light goes into triple-blink mode while the Beverage Maker is connected to a circuit where one or more phases are closed or disconnected. Once all phases are restored, cycling the POWER button OFF and ON will eliminate the blinking POWER light.

4. Operating Instructions

NOTE: The Beverage Maker is designed to consistently brew coffee or tea of uniform extraction. Control of the coffee blend and water quality are beyond the control of this equipment and remain the responsibility of the user.

A. Brew Coffee

(1) To prepare Beverage Maker to brew coffee:

(a) Press the POWER button. This starts operation of the tank heaters.

(b) Remove brew cup from the brew chamber above the server cavity.

(c) Put a pillow pack of coffee, with its seam down, on top of the brew cup's metal base plate.

NOTE: Make sure all of the pillow pack is in the brew cup. Do not compress or fold the pillow pack. This could cause the pack to break open during the brew cycle.

(d) Put brew cup into brew chamber.

(e) Put an empty server into server cavity.

(f) Fully lower the brew handle to lock brew cup and server in the Beverage Maker.

(2) To brew a server of coffee (after doing the steps above):

CAUTION: DO NOT USE THE WARMER WHEN USING AN INSULATED SERVER. THE WARMER CAN MAKE THE OUTER SURFACE OF THE INSULATED SERVER VERY HOT.

(a) Press the WARMER button (where option is installed). This warmer or platen heater is used to keep coffee hot in the server.

(b) Press the BREW button. Brewing will start when the water comes to brew temperature. From a cold start, this will take approximately three minutes. The Beverage Maker will then brew a server of coffee in approximately three minutes.

(c) When the brew cycle is complete (shown by the BREW button light going OFF) lift the brew handle.

(d) Remove the brew cup and discard the used pillow pack. If there are any coffee grounds in the brew cup, clean the brew cup with clean water.

(e) Put a new pillow pack in the brew cup and put the brew cup in the brew chamber.

(f) Fully lower the brew handle to lock brew cup and server in the Beverage Maker.

B. Brew Tea

(1) To prepare Beverage Maker to brew tea:

(a) Press the POWER button.

(b) Put tea bag(s) in empty server.

(c) Put server into server cavity.

(d) Put an empty brew cup into brew chamber.

(e) Fully lower the brew handle to lock brew cup and server in the Beverage Maker.

(f) Press TEA button.

(g) When TEA button light goes OFF, the tea is ready.

(h) Lift the brew handle to access the server.

5. Leading Particulars

A. Refer to Table 1 for leading particulars of the Beverage Maker.

Table 1
Leading Particulars

OPERATING CHARACTERISTICS			
Preheat Time	3 minutes 30 seconds (maximum)		
Brew Cycle Time (50 oz ±3 oz)	2:30 to 3:35 minutes		
Tea Cycle Time (50 oz ±3 oz)	2:30 to 3:20 minutes		
Unit Recovery Time	3 minutes 45 seconds (maximum)		
Initial or Subsequent Cycle, Server Temperature	175 °F (79 °C) to 195 °F (91 °C)		
ELECTRICAL DATA			
Operating Voltage	115 VAC, 400 Hz, 3-phase		
Wattage (Phase A, B, C)	2760 Watts Total (Phase A - 960 Watts, Phase B - 900 Watts and Phase C - 900 Watts)		
OVERALL DIMENSIONS			
Length	Width	Height (with rail)	Weight (Dry)
16.92 in (430 mm)	6.31 in (160 mm)	12.03 in (306 mm)	17.5 lbs (7.9 kg)

NOTE: Overall dimensions vary with options installed. For example, an extended platen and handle affect length.

NOTE: The (dry) weight shown is an upper limit. Some installed options may weigh less.

Table 2
Part Numbers and Product Distinctions

PN	Drain	Optional Features	Brew Cup	Faucet
4510-31UF-00	Sump	Hot Water from Faucet	Upper Plate	Standard
4510-31LF-00	Sump	Hot Water from Faucet	Lower Plate	Standard
4510-32UF-00	Sump	Hot Water from Faucet, Tea Brew	Upper Plate	Standard
4510-32LF-00	Sump	Hot Water from Faucet, Tea Brew	Lower Plate	Standard
4510-33UF-00	Sump	Hot or Cold Water from Faucet	Upper Plate	Standard
4510-33LF-00	Sump	Hot or Cold Water from Faucet	Lower Plate	Standard
4510-34UF-00	Sump	Hot Water from Faucet, Platen Heater	Upper Plate	Standard
4510-34LF-00	Sump	Hot Water from Faucet, Platen Heater	Lower Plate	Standard
4510-35UF-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Standard

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Table 2 (Continued)
Part Numbers and Product Distinctions

PN	Drain	Optional Features	Brew Cup	Faucet
4510-35LF-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Standard
4510-36UF-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-36LF-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-37UF-00	Sump	Cold Water from Faucet, Tea Brew	Upper Plate	Standard
4510-37LF-00	Sump	Cold Water from Faucet, Tea Brew	Lower Plate	Standard
4510-38UF-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-38LF-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-41UF-00	Rear Connection	Hot Water from Faucet	Upper Plate	Standard
4510-41LF-00	Rear Connection	Hot Water from Faucet	Lower Plate	Standard
4510-42UF-00	Rear Connection	Hot Water from Faucet, Tea Brew	Upper Plate	Standard
4510-42LF-00	Rear Connection	Hot Water from Faucet, Tea Brew	Lower Plate	Standard
4510-43UF-00	Rear Connection	Hot or Cold Water from Faucet	Upper Plate	Standard
4510-43LF-00	Rear Connection	Hot or Cold Water from Faucet	Lower Plate	Standard
4510-44UF-00	Rear Connection	Hot Water from Faucet, Platen Heater	Upper Plate	Standard
4510-44LF-00	Rear Connection	Hot Water from Faucet, Platen Heater	Lower Plate	Standard
4510-45UF-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Standard
4510-45LF-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Standard
4510-46UF-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-46LF-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-47UF-00	Rear Connection	Cold Water from Faucet, Tea Brew	Upper Plate	Standard
4510-47LF-00	Rear Connection	Cold Water from Faucet, Tea Brew	Lower Plate	Standard
4510-48UF-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Standard
4510-48LF-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Standard
4510-31UG-00	Sump	Hot Water from Faucet	Upper Plate	Swivel
4510-31LG-00	Sump	Hot Water from Faucet	Lower Plate	Swivel
4510-32UG-00	Sump	Hot Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-32LG-00	Sump	Hot Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-33UG-00	Sump	Hot or Cold Water from Faucet	Upper Plate	Swivel
4510-33LG-00	Sump	Hot or Cold Water from Faucet	Lower Plate	Swivel
4510-34UG-00	Sump	Hot Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-34LG-00	Sump	Hot Water from Faucet, Platen Heater	Lower Plate	Swivel
4510-35UG-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-35LG-00	Sump	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Swivel

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Table 2 (Continued)
Part Numbers and Product Distinctions

PN	Drain	Optional Features	Brew Cup	Faucet
4510-36UG-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-36LG-00	Sump	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel
4510-37UG-00	Sump	Cold Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-37LG-00	Sump	Cold Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-38UG-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-38LG-00	Sump	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel
4510-41UG-00	Rear Connection	Hot Water from Faucet	Upper Plate	Swivel
4510-41LG-00	Rear Connection	Hot Water from Faucet	Lower Plate	Swivel
4510-42UG-00	Rear Connection	Hot Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-42LG-00	Rear Connection	Hot Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-43UG-00	Rear Connection	Hot or Cold Water from Faucet	Upper Plate	Swivel
4510-43LG-00	Rear Connection	Hot or Cold Water from Faucet	Lower Plate	Swivel
4510-44UG-00	Rear Connection	Hot Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-44LG-00	Rear Connection	Hot Water from Faucet, Platen Heater	Lower Plate	Swivel
4510-45UG-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Upper Plate	Swivel
4510-45LG-00	Rear Connection	Hot or Cold Water from Faucet, Platen Heater	Lower Plate	Swivel
4510-46UG-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-46LG-00	Rear Connection	Cold Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel
4510-47UG-00	Rear Connection	Cold Water from Faucet, Tea Brew	Upper Plate	Swivel
4510-47LG-00	Rear Connection	Cold Water from Faucet, Tea Brew	Lower Plate	Swivel
4510-48UG-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Upper Plate	Swivel
4510-48LG-00	Rear Connection	Hot Water from Faucet, Tea Brew, Platen Heater	Lower Plate	Swivel

6. Airworthiness Limitations

A. The Airworthiness Limitation Section is FAA approved and specifies maintenance required under 14 CFR 43.16 and 91.403, unless an alternative program has been FAA approved.

B. The Beverage Maker PNs 4510-3 and 4510-4 series and its components have no life limited parts, no mandatory replacement times for parts, and no defined structural inspection intervals.

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1. General

WARNING: THE BEVERAGE MAKER AND WATER IN THE TANK MAY BE VERY HOT.

WARNING: REMOVE THE BEVERAGE MAKER FROM POWER AND WATER SUPPLY BEFORE DISASSEMBLY FOR TESTING PURPOSES.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE DISASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORKSTATION IN A STATIC ELECTRICITY FREE ENVIRONMENT.

A. A Beverage Maker should have the full Functional Test at every shop visit.

NOTE: After replacing components, a functional test should be performed to make sure the Beverage Maker operates correctly. Refer to ILLUSTRATED PARTS LIST for item descriptions and locations.

NOTE: The following tests are performed with a non-insulated server. Use of an insulated server requires that the WARMER not be used. An insulated server may affect water temperature tests.

B. Completely read and thoroughly understand the entire TESTING AND FAULT ISOLATION section. Become thoroughly familiar with the test sequences and their interaction before performing any tests.

C. Perform a visual inspection to detect obvious problems or defects. Refer to INSPECTION/CHECK, DISASSEMBLY, and ASSEMBLY.

2. Test Equipment

A. Refer to Table 1001 for test equipment. See SPECIAL TOOLS, FIXTURES, EQUIPMENT AND CONSUMABLES for more information. Verification of the equivalence for substitutions for the following items is the responsibility of the repair facility.

NOTE: Equivalent substitutes may be used for listed items.

Table 1001
Test Equipment and Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
1/8 inch NPT plug	AN933-1D or MIT-022P	Commercially Available or FasTest, Inc. 1646 Terrace Drive Roseville, MN 55113-1317
Electrical connector cable between Beverage Maker and power supply	998552	B/E Aerospace Inc. (V63367)
High Low Temperature RTD wiring harness with 4-position connector	997112	B/E Aerospace Inc. (V63367)
Megohmmeter	None	Commercially Available
Milliohmmeter	Model 1880	QuadTech Inc (V0PK96)
Ohmmeter	None	Commercially Available
Power Supply, including: 3-phase 115 ±3 VAC, 400 Hz, 10 amps per phase; 3 ammeters (0-25 amp); and 3 voltmeters (0-250 volt) (refer to Figure 1001)	None	Commercially Available
Server	MISC-3201	B/E Aerospace Inc. (V63367)
Stopwatch	None	Commercially Available
DELETED		
Thermometer, 50-250 °F (10-121 °C)	None	Commercially Available
Water Supply, including: pressure regulator (0-150 psig, 0-10.34 barg) and flow meter (0-0.5 GPM, 0-1.9 LPM) (refer to Figure 1001)	None	Commercially Available

B. Set-up for Functional Tests (refer to Table 1001 and Figure 1001)

(1) Provide a power supply that meets the requirements described in Table 1001.

(2) Provide a water supply that meets the requirements described in Table 1001.

(3) Provide a Server and Beverage Maker for testing.

3. Functional Tests (refer to Figure 1001)

NOTE: When testing the Beverage Maker, each test should be performed in the sequence listed below.

NOTE: If any of the test fails, refer to Table 1002, Fault Isolation.

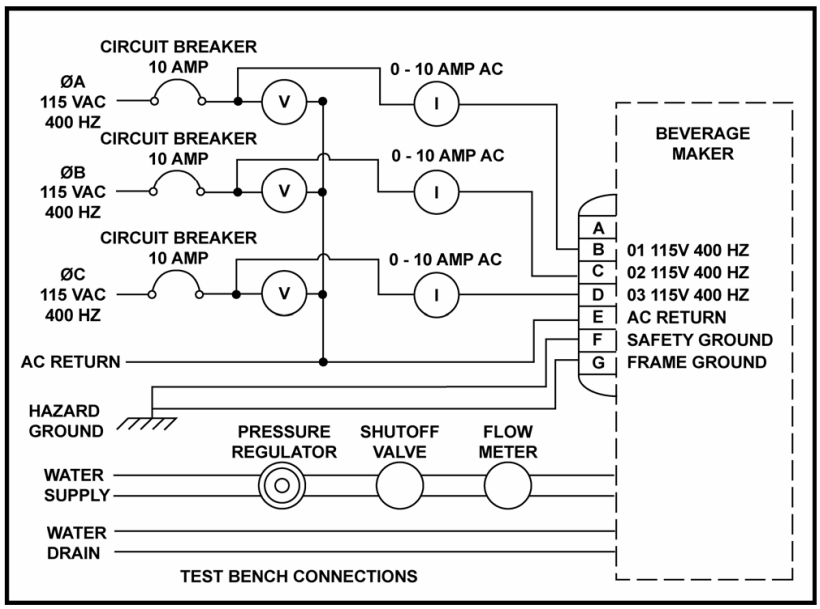

Figure 1001
Functional Test Set-up

A. Dielectric Test

NOTE: Do a dielectric test only after a major rewiring has been done. A dielectric test is not needed every time the unit comes into the shop for repair.

NOTE: Some components of the Beverage Maker have had separate dielectric tests performed by the vendor, as noted on their stock code sheets or specification control drawings and some components are not designed to withstand such a test. These components are

Heaters HR1, 2, 3

Power Control Board 3530-0121

Warmer-HR 4

(1) Remove P1 connector from J1 connector on circuit board.

(2) Connect one lead of the Hi Pot tester to the PP1 (B, C, D, and E) and the other lead to the PP1 (pins F & G). Increase the voltage of the Hi Pot test set in increments of 250 to 500 volts per second until 1500 volts are applied across test connection. Maintain the voltage at the 1500 volt level for 60 seconds. Current flow shall not exceed a maximum of 2.0 milliamperes during the test period.

(3) Connect megohmmeter between the of PP1, pins B, C, D, and E and the Beverage Maker housing. The megohmmeter indication must be greater than 200 megohms.

B. Resistance Check

(1) Connect a resistance bridge (refer to Table 1001) between PP1-G and unanodized brew shelf on the Beverage Maker. The measured resistance shall not be greater than 2.5 milliohms.

(2) If applicable, connect a resistance bridge between PP1-G and metal disc located in the center of platen. The measured resistance shall not be greater than 3.2 milliohms.

(3) Connect a resistance bridge between PP1-F and faucet bar. The measured resistance shall not be greater than 3.0 milliohms.

C. Water Supply Temperature Test

(1) If water supply is connected to the Beverage Maker, disconnect water supply from the Beverage Maker.

(2) Measure the temperature of the water from the water supply using a thermometer. The temperature of the water should be 62 °F (17 °C) to 72 °F (22 °C).

NOTE: Temperatures outside this range will affect correct Beverage Maker operation. Results of other tests may not meet specifications.

D. Ambient Temperature Test

(1) Using a thermometer, measure the temperature in the immediately adjacent area of the test environment. The temperature should be between 70 °F (21 °C) and 85 °F (29 °C).

NOTE: Temperatures outside this range will affect correct Beverage Maker operation. Results of other tests may not meet specifications.

E. Tank Pressure Test

NOTE: The following test is for some safety features, correct operation of plumbing system, and for detection of water leaks. If any of the test fails, refer to Table 1002, Fault Isolation.

(1) Replace relief valve with a 1/8 inch NPT plug prior to testing.

(2) Connect the Beverage Maker to water supply.

(3) Let tank fill with water. Open faucet to verify the tank is full.

(4) Adjust water pressure to 130 psig (8.96 barg). Hold for a minimum of 5 minutes. Inspect tank for leaks. No leaks are allowed.

(5) Release pressure then remove plug.

(6) Put relief valve back into the tank using thread seal tape.

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(7) Gradually increase the water pressure to the tank. The relief valve should remain closed at pressures below 75 psi (5.17 barg). Continue to increase pressure ensuring the relief valve is fully open prior to or at 100 psi (6.89 barg).

NOTE: The relief valve should be examined and have a pressure test for correct operation every six months or 1500 flight hours, whichever comes first.

(8) Disconnect water supply and drain the Beverage Maker. Make sure vent valve operates correctly.

NOTE: The vent valve lets the tank fill and drain, when required. If the tank fills (without continuous overflow) and drains, the vent valve is operating. A small amount of water may come out of the vent valve drain line as the tank completes filling. This is normal operation.

F. POWER Button and LOW WATER Indicator Light Tests

NOTE: The following test is for some safety features and correct operation of water level systems. If any of the test fails, refer to Table 1002, Fault Isolation.

(1) Connect the power supply to the Beverage Maker.

NOTE: The small amount of current required for operation of the control circuits will not usually show on ammeters. Zero on the ammeters, described in these tests, includes the small amount of current required by the control circuits after heaters go off.

(2) If connected, disconnect the water supply from the Beverage Maker.

NOTE: The Beverage Maker tank must be empty to continue this test. If the tank is not empty, drain the tank before continuing.

(3) Press the POWER button.

(4) The POWER button light should come on, the LOW WATER indicator light should come on, and the heaters should not come on (as indicated by the ammeters).

(5) Press the POWER button. Make sure the POWER button light goes off.

(6) Connect a water supply to the Beverage Maker, adjust water pressure to 24 to 29 psig (1.65 to 2.0 barg).

(7) Make sure the Beverage Maker tank fills (as shown by the flow meter) and does not leak.

(8) At the end of fill, press the POWER button. Make sure the POWER button light comes on. Make sure the LOW WATER indicator light is off, and then press the POWER button so the POWER button light goes off.

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G. RTD Circuit Test

WARNING: REMOVE THE BEVERAGE MAKER FROM POWER AND WATER SUPPLY BEFORE DISASSEMBLY FOR TESTING PURPOSES.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE DISASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORKSTATION IN A STATIC ELECTRICITY FREE ENVIRONMENT.

NOTE: The following test is for correct operation of brew temperature control systems. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

(1) With the Beverage Maker disconnected from the power supply and water supply, remove the cover from the power module assy and disconnect the RTD assy plug from the J4 connector on the controller board assy.

(2) Toggle the switch on the High Low Temperature RTD wiring harness (refer to Table 1001) to the high side (RTD harness simulates 235 °F (113 °C) water in the tank) and connect the High Low Temperature RTD wiring harness to the controller board assy.

(3) Connect the Beverage Maker to the power supply.

(4) Press the POWER button. This should trip the safety latch. If the safety latch is tripped, there will be no current draw to the heaters and the POWER button light should go into the double-blink mode.

(5) Disconnect the power supply from the Beverage Maker.

(6) Toggle the switch on the High Low Temperature RTD wiring harness (refer to Table 1001) to the low side (RTD harness simulates 205 °F (96 °C) water in the tank) and connect the High Low Temperature RTD wiring harness to the controller board assy.

(7) Connect the power supply to the Beverage Maker.

(8) Reset the safety latch by pressing the manual switch on the back of the power module assy under the grommet (refer to Figure 1002).

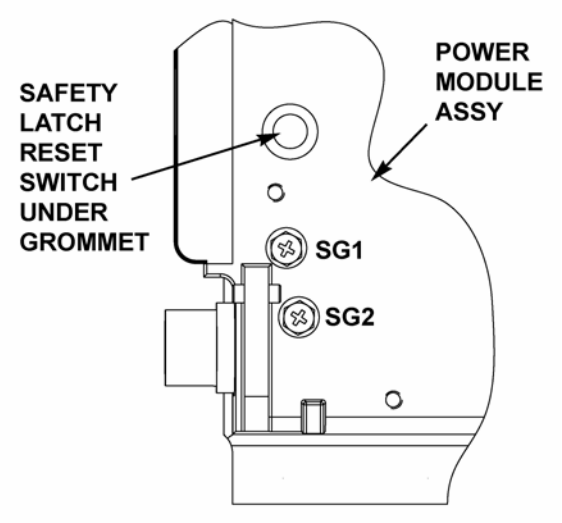

Figure 1002
Safety Latch Location

(9) Press the POWER button. When there is no current drawn by the heaters, the safety latch should not trip and the POWER button light should not go into double-blink mode.

(10) Disconnect the power supply from the Beverage Maker.

(11) Disconnect the High Low Temperature RTD wiring harness from the controller board assy.

(12) Connect the RTD assy to the controller board assy.

(13) Put the cover on the power module assy.

(14) Connect the power supply to the Beverage Maker.

(15) Press the POWER button. This should not trip the safety latch. Make sure the POWER button light does not go into double-blink mode, and then press the POWER button again to turn off the Beverage Maker.

H. Tank Heater and Preheat Test

NOTE: The following test does a check of safety features and correct operation of heater control systems. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

NOTE: Preheat time is the elapsed time from heaters on to heaters off after the tank is first filled with water.

(1) With the Beverage Maker connected to the power supply and water supply and the tank filled with water, be prepared to time preheating before pressing the POWER button.

NOTE: The LOW WATER indicator light should not come on during this test.

(2) Press the POWER button and start the stopwatch at the same time. Turn ON WARMER (if applicable).

(3) Make sure the amperes for the phases are as follows:

Phase A	8.1 +0.6/-0.9 amperes
Phase B	7.8 +0.4/-0.7 amperes
Phase C	7.8 +0.4/-0.7 amperes

(4) Stop the stopwatch when the current on Phase B goes to zero. The elapsed time should be 3 minutes, 30 seconds maximum, when starting with a cold tank.

NOTE: The small amount of current required for operation of the control circuits will not usually show on ammeters. Zero on the ammeters, described in these tests, includes the small amount of current required by the control circuits after the heaters go off.

NOTE: A small amount of water may come out of the pressure relief valve drain line as the tank completes preheating. This is normal operation.

J. BREW Test

NOTE: The following test puts the Beverage Maker through several brew cycles to do a check of safety features and correct operation. If the Beverage Maker fails any of these tests, refer to Table 1002 for recommendations.

(1) Start this test by making sure the tank is filled with water, the water supply pressure is set to 24 to 29 psig (1.66 to 2.0 barg), the Beverage Maker is connected to the water supply and power supply, and the POWER button light is on. The WARMER (where option is installed) must be on before starting this test. Put an empty server in the Beverage Maker and lower the brew handle.

(2) Press the BREW button, observe the flow meter and when flow starts, start the stopwatch.

NOTE: Brew does not start until the water in the tank is heated. After the start of flow, there will be an interruption of flow for approximately 10 seconds. This lets water make coffee pillow pack wet during a brew cycle. This is normal operation and the time is included in the brew cycle time.

(3) Stop the stopwatch when the flow meter goes to zero flow a second time. Elapsed time between when flow starts and stops should be 2 min. 30 sec. to 3 min. 35 sec.

NOTE: The time between when the flow meter shows zero flow until the BREW button light goes off may be 10 to 20 seconds. This is a time that is not adjustable by a repair facility. This time between when the flow goes to zero and when the BREW button light goes off is a safety feature that lets most hot liquid have time to drain from the brew cup before the server is removed from the Beverage Maker.

(4) Remove, empty, and install server in the Beverage Maker.

(5) With the BREW button light off, press the BREW button to start a second brew.

(6) Lift brew handle after the normal 10 second flow interruption. Lifting the brew handle should interrupt the brew cycle, as shown by the flow going to zero.

(7) Lower the brew handle. This should let the brew cycle continue.

(8) When the BREW button light goes off, press the BREW button. Another brew cycle should not start.

(9) Lift the brew handle and measure the temperature of liquid in the server. It should be between 175 °F (79 °C) and 195 °F (91 °C). The server should be full.

(10) Put the full server in the Beverage Maker and lower the brew handle.

(11) Press the BREW button. The BREW button light should come on for only a short time then go off again. The full server should be detected by the backup liquid level sensor for the server.

(12) Lift the brew handle, remove, empty, and install server in the Beverage Maker. Lower the brew handle.

K. Install a server and a brew cup, then lower the brew handle. Pull on the server to make sure the server retainer is working correctly.

L. Optional Feature Tests

NOTE: The following tests are for optional features and correct operation. If the Beverage Maker fails any of these tests, refer to the FAULT ISOLATION Section (Paragraph 5) for recommendations. Start these tests by making sure the tank is filled with water, the Beverage Maker is connected to the water supply and power supply, and the POWER button light is on.

(1) TEA Test (if option is installed)

(a) The WARMER (where option is installed) must be on before starting this test. Put an empty server in the Beverage Maker and lower the brew handle. The water supply pressure is set to 24 to 29 psig (1.66 to 2.0 barg).

(b) Press the TEA button, observe the flow meter and when flow starts, start the stopwatch.

NOTE: TEA brew does not start until the water in the tank is heated.

(c) Stop the stopwatch when the flow meter goes to zero flow. Elapsed time between when flow starts and stops should be 2 min. 30 sec to 3 min. 20 sec.

(d) Lift the brew handle and measure the temperature of liquid in the server. It should be between 175 °F (79 °C) and 195 °F (91 °C). The server should be full.

(e) Remove, empty, and install server in the Beverage Maker. Lower the brew handle.

(2) Platen Heater (WARMER) Test (if option is installed)

(a) Press the WARMER button when POWER button light is on. Make sure the WARMER light comes on and make sure the platen heater becomes warm.

NOTE: The amperes required for the platen heater is so small it may not show on the ammeter for Phase A.

(3) COLD Water from Faucet Test (if option is installed)

(a) Press the COLD button for cold water. Make sure cold water comes out of the faucet.

(4) HOT Water from Faucet Test (if option is installed)

(a) Press HOT button for hot water. Make sure hot water comes out of the faucet. Continue to press button so hot water comes out of the faucet until heaters come on (as shown by ammeters).

4. Bench Test

A. RTD Assy Test

NOTE: Do this check of the RTD only if necessary for isolation of a fault in temperature control of the water heating system.

(1) Disconnect RTD assy from the controller board assy.

(2) Check the resistance across the disconnected wires with a standard ohmmeter. Resistance should be 1.1 kOhms ±5% at room temperature (75 °F (23 °C) nominal).

(3) If RTD assy is out of tolerance, replace RTD assy.

B. Pressure Relief Valve Calibration (PN 62-0223-1 Only) (refer to Figure 1003)

NOTE: Calibration of the pressure relief valve is necessary if it is disassembled.

(1) Assemble the valve.

(2) Attach the valve to an adjustable pressure-regulated water supply.

(3) Loosen the lock nut (140).

(4) Turn the adjusting cap (145) on the valve body (150) so that water begins to come out of the fitting (180) when water pressure is at 80 psig. Water must flow freely out of the fitting at 100 psig.

(5) When the adjusting cap (145) is set so that the valve opens at 80 psig and is fully open at 100 psig, turn the lock nut (140) on the valve body (150) so that it keeps the adjusting cap (145) from turning.

(6) After the lock nut (140) is tight, do a check of the valve to make sure it begins to open at 80 psig and becomes fully open at 100 psig.

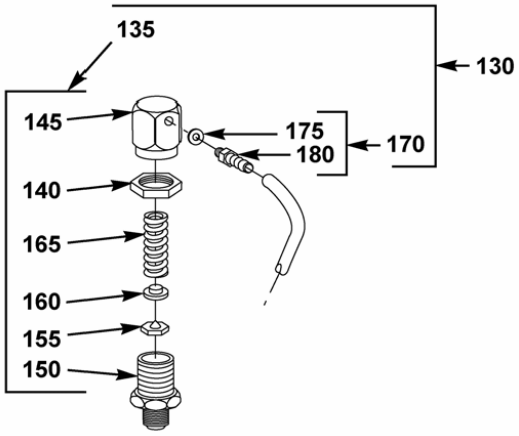

Figure 1003
Valve Assembly

5. Fault Isolation

A. Table 1002 lists the probable causes for faults seen during use or testing.

Table 1002
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Brew handle hard to operate.	Chassis assy bent or broken.	Attempt to repair chassis assy before replacing.
	Hardware and/or fasteners bent or broken.	Replace hardware and/or fasteners.
	Brew handle binding (metal-to-metal contact).	
	Brew handle bent or broken.	Replace brew handle.
	Handle bushing(s) is deformed or broken.	Replace handle bushing(s).
	Brew head assy loose or binding.	
	Brew head casting bent or broken.	Attempt to repair brew head before replacing.
	Brew head guides worn or broken.	Replace brew head guides.
	Return spring weak or broken.	Replace spring. (Check bushing for abnormal wear.)
	Server retainer assy binding (metal-to-metal contact).	
	Server retainer assy deformed or broken.	Attempt to repair server retainer assy before replacing.
	Server retainer spring(s) weak or broken.	Replace spring(s). (Check spring bushing(s) for abnormal wear.)

Table 1002 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
POWER button does not turn the Beverage Maker on or off. (Make sure unit is connected to main power source.) or POWER button does not illuminate when required. or POWER button illuminates when not required. or WARMER button does not turn the platen heater on or off. (Make sure platen heater is good.)	PC board switch assy failure. Faulty wiring and/or connector(s). Controller board assy failure.	Replace PC board switch assy. Replace wiring and/or connector(s). Replace fuses. Replace controller board assy.
POWER button displays double-blink mode (overtemperature indication – safety latch is tripped).	Faulty wiring and/or connectors. RTD assy failure (fails OPEN simulating overtemperature condition in tank). Excessive steam head in tank assy. Low water sensor assy failure (fails SHORT simulating a full tank condition). Pressure relief valve failure (fails CLOSED allowing tank pressure to rise). Solid state relay failure (fails CLOSED).	Replace wiring and/or connector(s). Replace RTD assy. Replace low water sensor assy. Attempt to clean pressure relief valve before replacing. Replace controller board assy.

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Table 1002 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
POWER button displays triple-blink mode (phase loss indication).	Faulty wiring and/or connectors.	Replace wiring and/or connector(s).
	Thermostat buss bar failure (fails OPEN).	Repair/replace thermostat buss bar.
	Overtemperature thermostat(s) failure (tripped or failed OPEN).	Reset/replace over-temperature thermostat(s).
	Tank heater assy (s) failure.	Replace tank heater assy (s).
Beverage Maker will not accomplish a BREW or TEA cycle.	LLS backup probes failure (SHORTED simulating full server condition).	Replace infrared sensor assy. (Check for “wicking” and towel dry as needed. Check for water intrusion into cable assy.)
or		
Beverage Maker will not dispense HOT water.	Brew head assy interlock failure (fails OPEN simulating brew handle up condition).	Repair/replace brew head assy.
	Solenoid valve assy failure.	Attempt to clean manifold assy before replacing.
	Solenoid valve assy plunger assy stuck.	Replace plunger assy.
	Manifold assy clogged.	Attempt to clean manifold assy before replacing.
Beverage Maker will not accomplish a BREW or TEA cycle.	Tank assy sensors.	
or		
Beverage Maker will not dispense HOT water.	Low water sensor assy failure (fails OPEN simulating false tank empty condition).	Attempt to clean low water sensor assy before replacing.
or		
POWER button displays double-blink mode (overtemperature indication).	RTD assy failure (fails OPEN simulating overtemperature condition in tank or fails SHORT simulating frozen condition in tank).	Replace RTD assy.

Table 1002 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Beverage Maker will not dispense HOT or COLD water.	Main utilities (water and power) are not connected.	Connect to main utilities.
	Water inlet clogged or damage.	Attempt to clean before replacing.
	Faucet assy clogged or damaged.	Attempt to clean before replacing.
	Solenoid valve assy failure (failed CLOSED).	Repair/replace solenoid valve assy.
	Solenoid valve assy plunger assy stuck.	Replace plunger assy.
	Manifold clogged.	Attempt to clean manifold assy before replacing.
DELETED		
Tank heaters will not heat	Thermostat buss bar failure (fails OPEN).	Repair/replace thermostat buss bar.
	Overtemperature thermostat(s) failure (tripped or failed OPEN).	Reset/replace over-temperature thermostat(s).
	Faulty wiring and/or connectors.	Replace wiring and/or connector(s).
	Low water sensor assy failure.	Attempt to clean low water sensor assy before replacing.
	RTD assy failure (fails OPEN simulating overtemperature condition in tank or fails SHORT simulating frozen condition in tank).	Replace RTD assy.
	Controller board assy failure.	Replace controller board assy.
Tank heater(s) will not turn off.	Tank assy sensors.	
	Low water sensor assy failure (failed SHORT through inner sleeve).	Replace low water sensor assy.
	RTD assy failure (reporting false low temperatures).	Replace RTD assy.
	Controller board assy failure.	Replace controller board assy.

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Table 1002 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
Faucet leaks.	Solenoid valve assy failure (fails OPEN). Solenoid valve assy plunger assy does not seal against internal valve seat.	Repair/replace solenoid valve assy. Replace plunger assy.
Server does not completely fill.	Supply water pressure too low. Plumbing (tubing) is clogged. LLS backup probes failure (SHORTED simulating false full server condition).	Adjust supply water pressure to 24 to 29 psig. Attempt to clean before replacing. Replace infrared sensor assy. (Check for “wicking” and towel dry as needed. Check for water intrusion into cable assy.)
Server overflows.	Faulty wiring and/or connectors. LLS malfunction/failure. (If TP6 on controller board assy is 5 VDC, then photo transistor SHORTED.) If voltage across R43 on controller board assy does not drop after BREW cycle initiation plus 35 seconds, then infrared LED is OPEN.) LLS backup probes failure (failed OPEN - will not detect an overflowing server condition.) Controller board assy failure.	Replace wiring and/or connector(s). Replace infrared sensor assy. Replace controller board assy.
Water tank will not drain correctly.	Vent valve failure (failed CLOSED). Plumbing (tubing) is clogged. Water inlet clogged or damage.	Repair/replace vent valve assy. Attempt to clean before replacing. Attempt to clean before replacing.

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Table 1002 (Continued)
Fault Isolation

FAULT	PROBABLE CAUSE	CORRECTIVE ACTION
BREW or TEA cycle too short or too long.	Flow washer failure (causing excessive or restricted water flow).	Replace flow washer.
	Supply water pressure too low.	Adjust supply water pressure to 24 to 29 psig.
	Plumbing (tubing) is clogged. (check manifold, brew head, tank, tubing, and water inlet).	Attempt to clean before replacing.

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1. General

A. Schematic and wiring diagrams are shown with associated parts and assemblies within this manual.

B. For general reference, see the Beverage Maker plumbing schematic (refer to Figure 2001) and the wiring diagram (refer to Figure 2002).

NOTE: Plumbing schematic (Figure 2001) is shown with all options. Three (3) solenoids (maximum) may be installed.

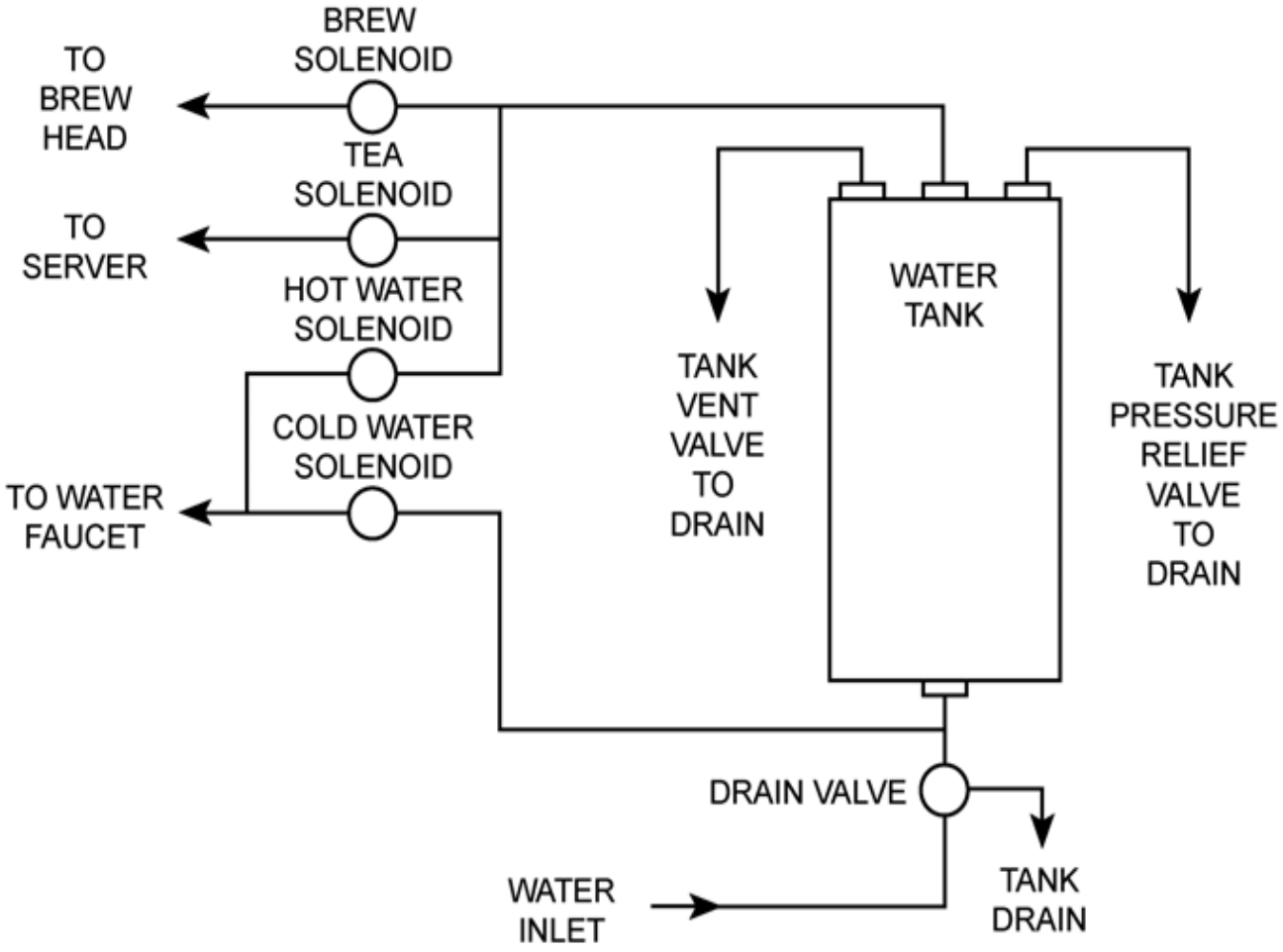

Figure 2001
Plumbing Schematic

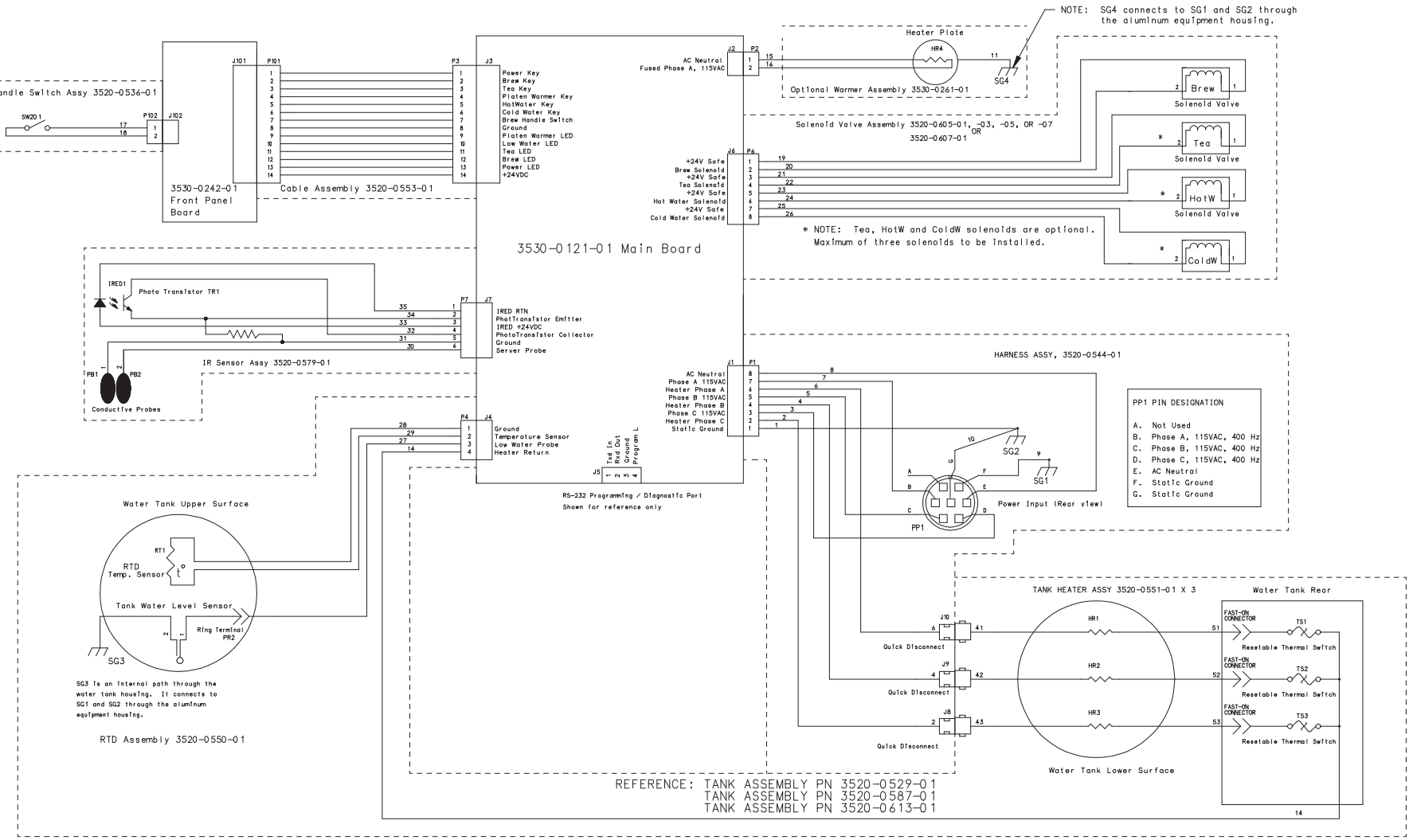

Figure 2002
Wiring Diagram

DISASSEMBLY

1. General

WARNING: THE BEVERAGE MAKER AND WATER IN THE TANK MAY BE VERY HOT.

WARNING: REMOVE THE BEVERAGE MAKER FROM POWER AND WATER SUPPLY BEFORE DISASSEMBLY.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE DISASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORKSTATION IN A STATIC ELECTRICITY-FREE ENVIRONMENT.

CAUTION: MAINTENANCE ENGINEER MUST WEAR A CONDUCTIVE WRIST STRAP TO PROTECT THE PC BOARD ASSEMBLY AGAINST ESD.

A. Refer to the TESTING AND FAULT ISOLATION and INSPECTION/CHECK Sections prior to disassembly to establish the condition of the faulty component or most probable cause of its malfunction. This will determine the extent of disassembly required without completely tearing down and rebuilding the Beverage Maker or faulty component.

B. Refer to the ILLUSTRATED PARTS LIST (IPL) Section for disassembly and assembly relationships. The number parentheses after the noun name of a part refers to the item number in the IPL Figure.

C. Provide a clean space and working surface before beginning any disassembly operation.

NOTE: Removal of some screws may require use of an adjustable-torque screwdriver due to the use of thread sealant.

NOTE: Beverage residue and mineral deposits can be cooked into part connections making them hard to remove. The cleaning of some parts can help in their removal, especially parts exposed to coffee residue.

D. Before removing electronic components, note the polarity and position of the component.

2. Procedures

A. Beverage Maker (refer to IPL Figures 1, 1A, 1B, or 1C)

(1) Remove brew cup assembly (20A or 20B) from the Beverage Maker subassembly (10A thru 10H).

(a) With the brew handle assembly (325, IPL Figures 2, 2A, 2B, and 2C) in the raised position, remove brew cup assembly (20A or 20B) from the Beverage Maker subassembly (10A thru 10H).

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Table 3001
Brew Cup Assembly PN Reference

UNIT PN	BREW CUP ASSEMBLY PN	DESCRIPTION
4510-XXUX-XX*	3520-0540-01	Brew cup assembly (upper)
4510-XXLX-XX*	3520-0540-03	Brew cup assembly (lower)

*X represents other numbers or letters in the Unit Part Number

(b) Remove two nuts (35), two screws (40), and spacer plate (30) from brew cup (25A or 25B) (refer to Figure 3001).

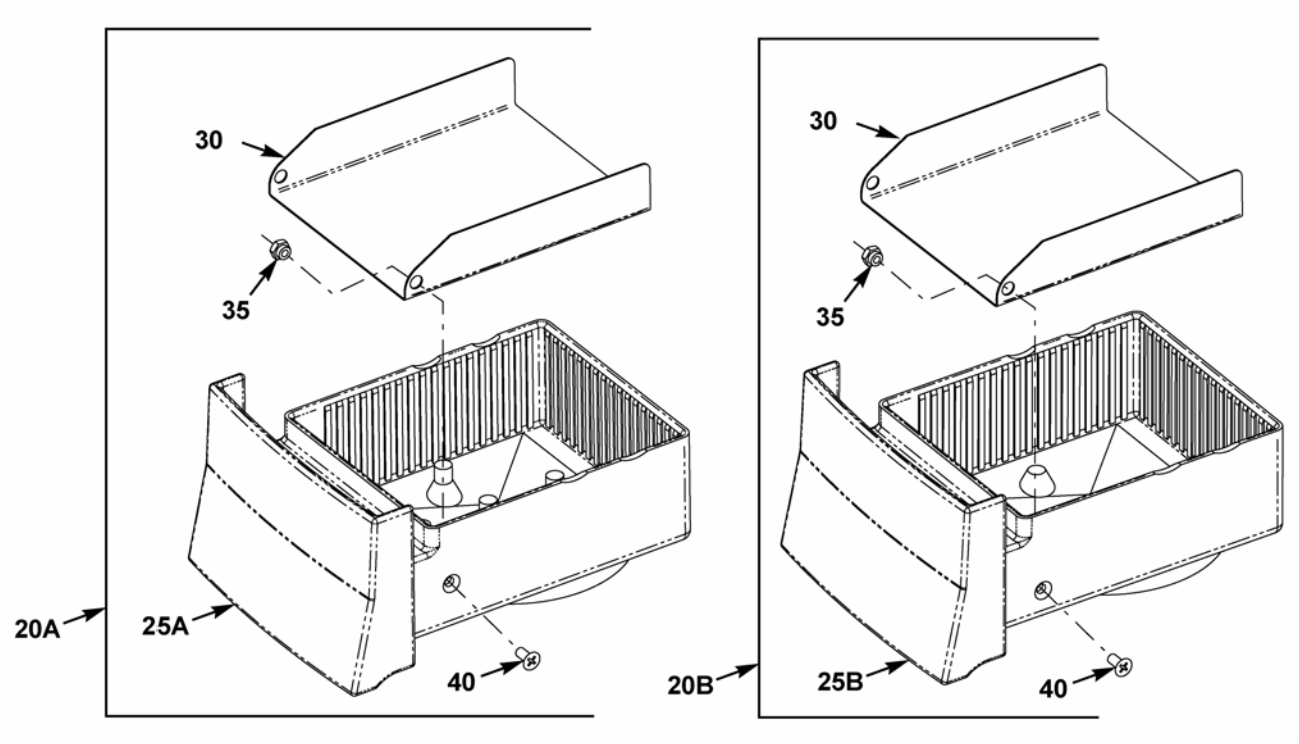

Figure 3001
Brew Cup Disassembly

(2) Refer to the DISASSEMBLY Section, Paragraph 2.B, for disassembly of the Beverage Maker subassembly (10A thru 10H).

B. Beverage Maker Subassembly (refer to IPL Figures 2, 2A, 2B, 2C and Figures 3002 thru 3014).

CAUTION: DRAIN WATER FROM TANK ASSEMBLY (15) BEFORE DISASSEMBLY.

NOTE: Remove nylon wire tie downs as required to release wiring during the disassembly process.

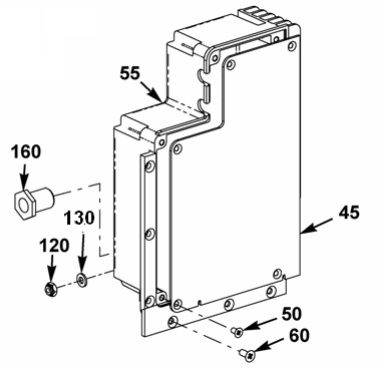

Figure 3002
Removal of Power Module Cover

(1) Remove power module housing cover (45) from power module assembly (55).

(a) Remove five screws (50) and power module housing cover (45) from the power module assembly (55).

NOTE: This gives access to wiring harness assembly (65) connections to the PC board assembly (75).

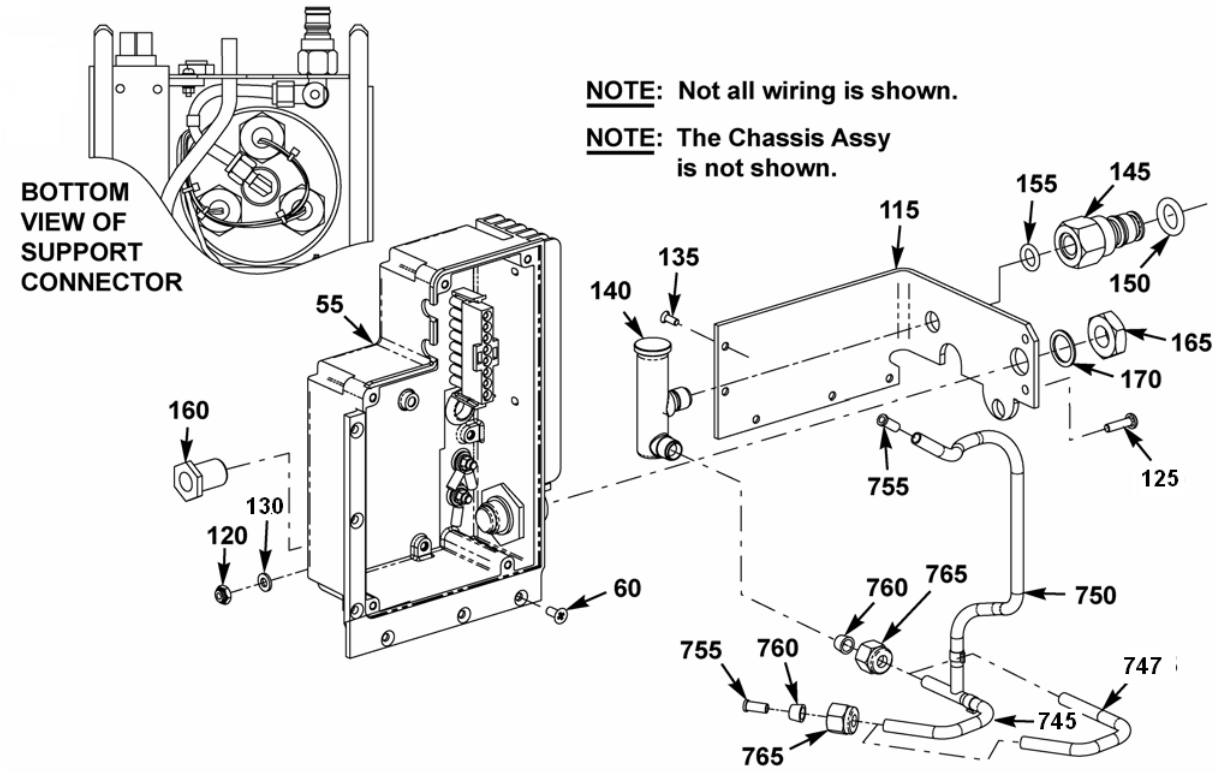

Figure 3003
Removal of Support Connector

(2) Remove connector support (115) from chassis assembly (10) (refer to Figure 3003).

(a) Disconnect water supply tube assembly (750, 747, or 745) from drain valve assembly (140) and tank assembly (15).

1 Remove tubing ferrule (760), fitting nut (765), and tube insert (755) from tubing (745) only if replacement or cleaning is required.

(b) Remove screw (125), flat washer (130), and lock nut (120) from support connector (115) and power module assembly (55).

(c) Remove hexagon nut (165), lock washer (170) and alignment pin guide (160) from connector support (115) and power module assembly (55).

(d) Remove two screws (135) from the connector support (115) and the chassis assembly (10).

(e) Remove three screws (135) from the connector support (115) and the left side rail guide (175).

CAUTION: AFTER REMOVING ALL FASTENERS FROM THE CONNECTOR SUPPORT, CAREFULLY SLIDE THE SUPPORT CONNECTOROFF THE REAR DRAIN TUBE (IF OPTION INSTALLED) TO AVOID DAMAGE TO THE DRAIN TUBE.

(f) Remove the water shutoff plug assembly (145), o-ring (155), and drain valve assembly (140) from connector support (115). Refer to DISASSEMBLY, Paragraph 2.F for disassembly of the drain valve assembly (140).

(g) Remove o-ring (150) from water shutoff plug assembly (145).

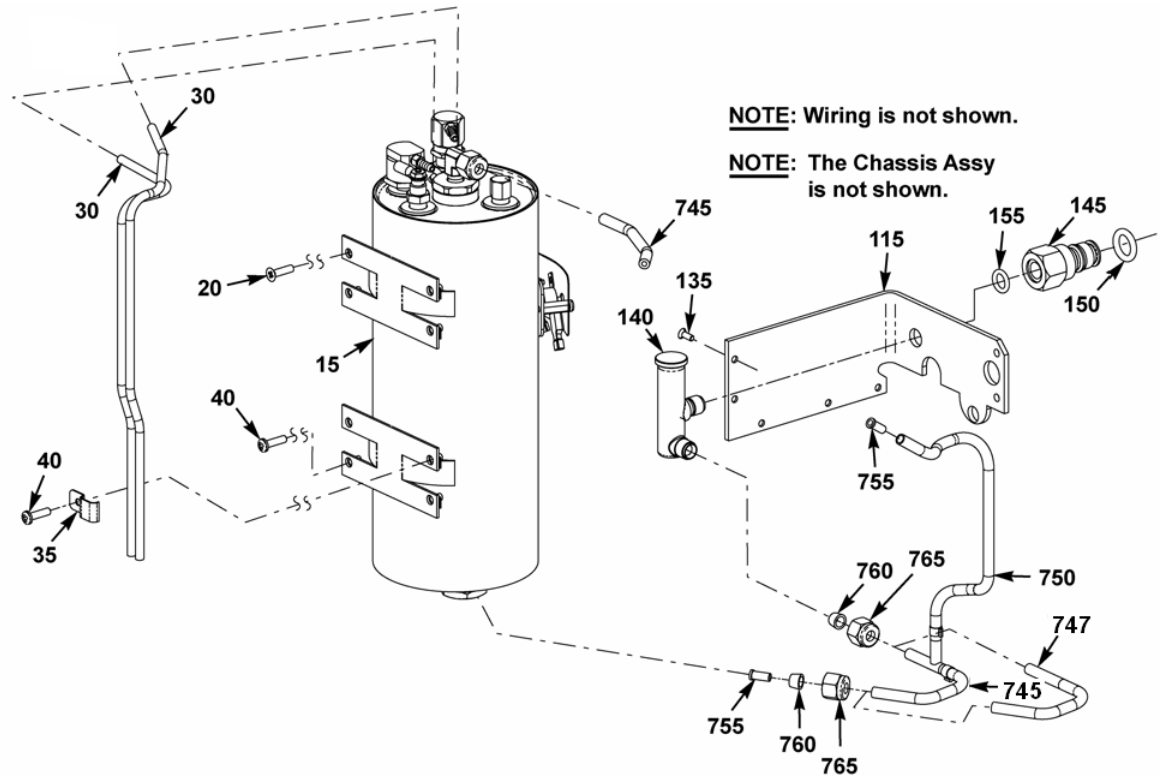

Figure 3004
Removal of Tank Assembly

(3) Remove the tank assembly (15) from the chassis assembly (10) (refer to Figure 3004).

NOTE: The tank assembly (1, IPL Figure 3) includes the welded tank assembly (10, IPL Figure 3), tank heaters (15, IPL Figure 3), over temperature thermostats (30, IPL Figure 3), filter assembly (65, IPL Figure 3), vent valve assembly (100, IPL Figure 3), low water sensor assembly (120, IPL Figure 3), RTD assembly (125, IPL Figure 3), and the pressure relief valve (135, IPL Figure 3). Refer to the DISASSEMBLY Section, Paragraph 2.C for disassembly of the tank assembly (15).

CAUTION: IDENTIFY HEATER TERMINALS AND WIRE HARNESS ASSEMBLY TERMINALS THAT GO TOGETHER. THE WIRING TERMINALS MUST BE TURNED AND PULLED TO BE REMOVED FROM EACH OTHER. THEY ARE NOT EASY TO DISCONNECT BY HAND, SO TOOLS MAY BE REQUIRED. DO NOT USE TOO MUCH FORCE TO HOLD OR REMOVE THE PLASTIC TERMINALS FROM EACH OTHER.

(a) Disconnect terminals of heaters (15, IPL Figure 3) from the wire harness terminals.

(b) Disconnect tubing (745) from elbow fitting (60, IPL Figure 3) of water filter on top of tank assembly (15).

(c) Inside the server cavity, remove screw (40) from tube retainer clip (35) on the drain tubes (30) that come from the top of the tank assembly (15).

(d) Remove the tubes (30) from barbed fittings (105 and 170, IPL Figure 3) on the top of the tank assembly (15) only if replacement or cleaning is required.

(e) If this has not been done, remove five screws (50) and power module housing cover (45) from the power module assembly (55).

NOTE: This gives access to the RTD assembly (125, IPL Figure 3) wiring plug connection to the PC board assembly (75).

(f) Disconnect plug of RTD assembly (125, IPL Figure 3) from PC board assembly (75) in the power module assembly (55).

NOTE: THE ID PLACARD MUST BE KEPT WITH THE BEVERAGE MAKER WHEN THE SCREWS HOLDING THE PLACARD AND TANK ASSEMBLY (15) ON THE CHASSIS ASSEMBLY (10) ARE REMOVED. THIS ID PLACARD IS NOT AVAILABLE AS A SPARE PART.

(g) Remove four flat head screws (20), two pan head screws (45, IPL Figures 1, 1A, 1B, or 1C) and two pan head screws (25 and 40) from the chassis assembly (10) and tank assembly (15). Removal of all eight screws frees the tank assembly (15) from the chassis assembly (10). Refer to the DISASSEMBLY Section, Paragraph 2.C for disassembly of the tank assembly (15).

NOTE: Removal of two screws (45, IPL Figures 1, 1A, 1B, or 1C) low inside the chassis assembly (10) server cavity frees the ID placard from the Beverage Maker (1, IPL Figures 1, 1A, 1B or 1C).

NOTE: The brew head assembly (300) may need to be removed to get access to screws (20) on the upper part of the chassis assembly (10) and the tank assembly (15). Refer to the DISASSEMBLY Section, Paragraph 2.B.(12) for brew head assembly (300) removal.

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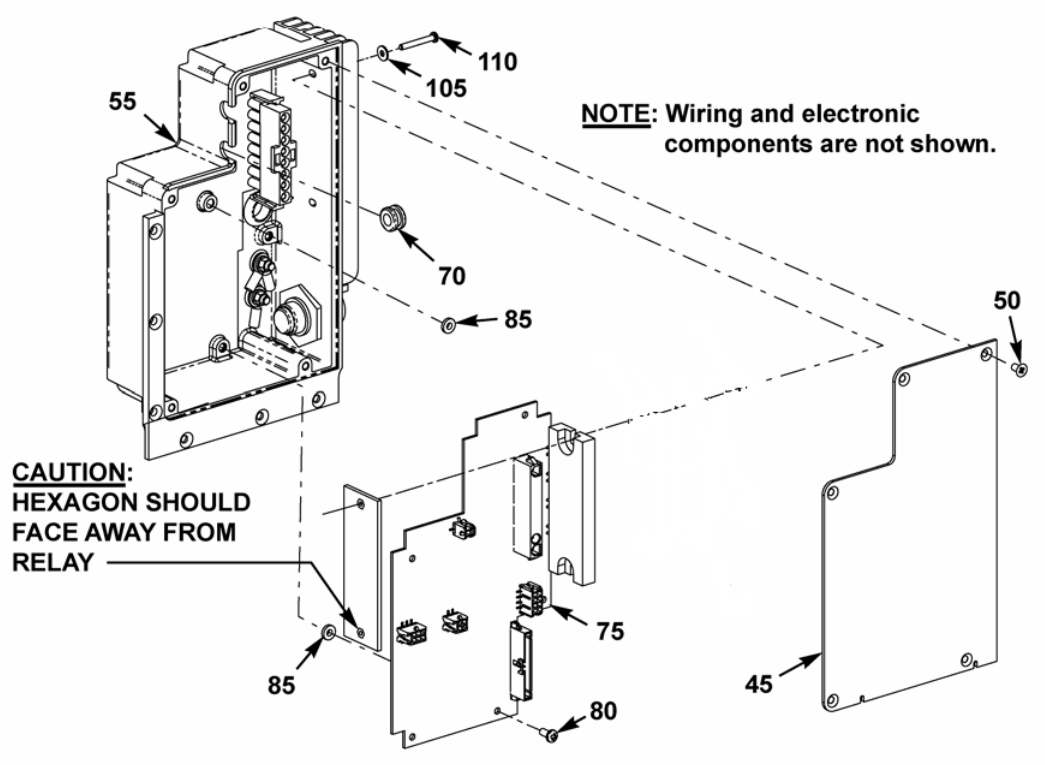

Figure 3005
Removal of Controller Board Assembly

(4) Remove the power module assembly (55) from the chassis assembly (10) (refer to Figure 3003 and 3005).

NOTE: Remove the connector support (115) from the power module assembly (55) and chassis assembly (10) before removing the power module assembly (55) from chassis assembly (10). Refer to the DISASSEMBLY, Paragraph 2.B.(2) to remove the connector support (115) from the chassis assembly (10).

(a) Remove three screws (60) from the power module assembly (55) and right side rail guide (175).

(b) Remove three screws (60) from the power module assembly (55) and chassis assembly (10).

CAUTION: WHEN TOUCHING THE PC BOARD ASSEMBLY (75) OR ANY OF ITS INDIVIDUAL PARTS, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ESD.

(c) Disconnect all wiring plugs from sockets in the power module assembly (55). Refer to the DISASSEMBLY Section, Paragraph 2.D for disassembly of the power module assembly (55).

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(d) Remove all wiring plugs from the PC board assembly (75).

NOTE: Identify all wires and points of connection before disassembly to help in ASSEMBLY.

(e) Remove wiring and grommets (70) from holes in power module assembly (55).

(f) Remove five screws (80) and six shoulder washers (85) from the PC board assembly (75) and the power module assembly (55) (refer to Figure 3004).

NOTE: The six shoulder washers (85) are on six bosses in the power module assembly (55).

(g) Remove two screws (110) and two flat washers (105) from the power module assembly (55) and the relay retainer (100).

CAUTION: DO NOT TOUCH THE RELAY OR PRY THE PC BOARD ASSEMBLY (75) FROM THE POWER MODULE HOUSING ASSEMBLY (10, IPL FIGURE 4).

(h) Carefully remove relay and controller board assembly (75), and relay retainer (100) from the power module assembly (55). Isolate the controller board assembly (75) from electrostatic discharge and exposure to water. Refer to DISASSEMBLY, Paragraph 2.E for disassembly of the controller board assembly (75).

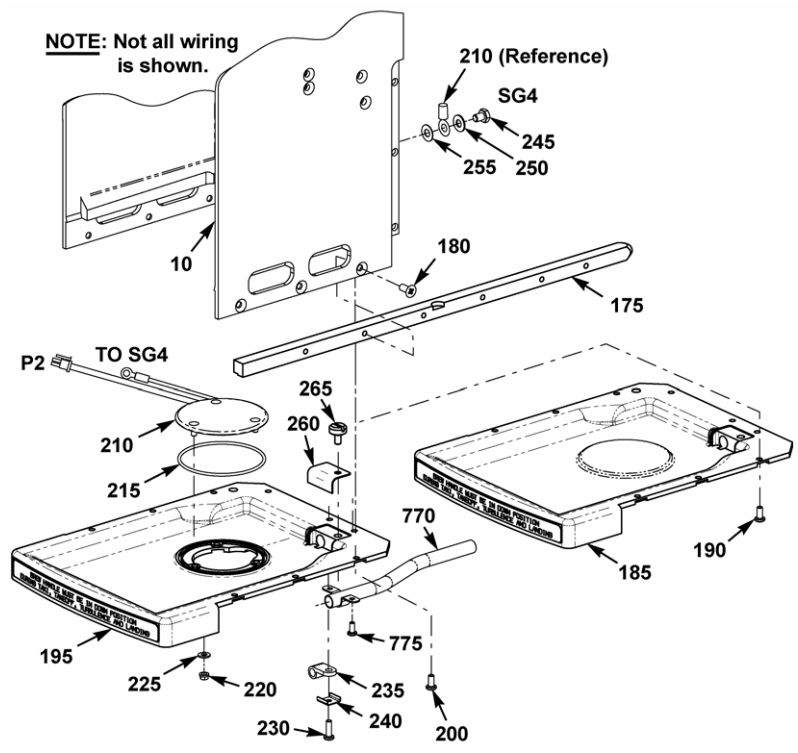

Figure 3006
Removal of Platen Assembly

(5) Remove the platen assembly (185 or 195) from the chassis assembly (10) (refer to Figure 3006).

Table 3002
Platen Assembly Part Number Reference

UNIT PN	PLATEN ASSEMBLY PN
4510-X1XX-XX*	3520-0561-05 (no warmer)
4510-X2XX-XX*	3520-0561-05 (no warmer)
4510-X3XX-XX*	3520-0561-05 (no warmer)
4510-X4XX-XX*	3520-0561-07 (used with warmer)
4510-X5XX-XX*	3520-0561-07 (used with warmer)
4510-X6XX-XX*	3520-0561-07 (used with warmer)
4510-X7XX-XX*	3520-0561-05 (no warmer)
4510-X8XX-XX*	3520-0561-07 (used with warmer)

*X represents other numbers or letters in the Unit Part Number

(a) Remove lock nut (220), flat washer (225), and o-ring (215) from platen heater (210) and platen assembly (195), where this option is installed.

(b) Remove the platen heater (210) plug from its connection on the controller board (75). The power module assembly (55) may need to be opened to get access to the controller board (75).

NOTE: Remove the guide rails (175) and screws (180) from the chassis assembly (10) and the support connector(115) to get access to the screws that hold the platen assembly (185 or 195) on the chassis assembly (10).

(c) Remove screw (230 and 775, or just 775) and the drain tube (770) from the platen assembly (185, 195), where this option is installed.

Table 3003
Platen Drain Part Number Reference

UNIT PN	PLATEN DRAIN
4510-3XXX-XX*	Sump drain, no drain tube required
4510-4XXX-XX*	Rear drain, drain tube (770) required

*X represents other numbers or letters in the Unit Part Number

(d) Remove screws (190) and platen assembly (185, 195) from the chassis assembly (10).

(e) Remove screw (230), washer (240), and wire clamp (235) from wires for the platen heater (210) and the platen assembly (195), where this option is installed.

(f) Remove the drain screen fastener (265) and drain screen (260) from the platen assembly (185 or 195).

(g) Remove screw (775) and drain tube (770) from platen assembly (185 or 195) and chassis assembly (10). There may be adhesive sealant on the drain tube where it meets the platen assembly (185 or 195).

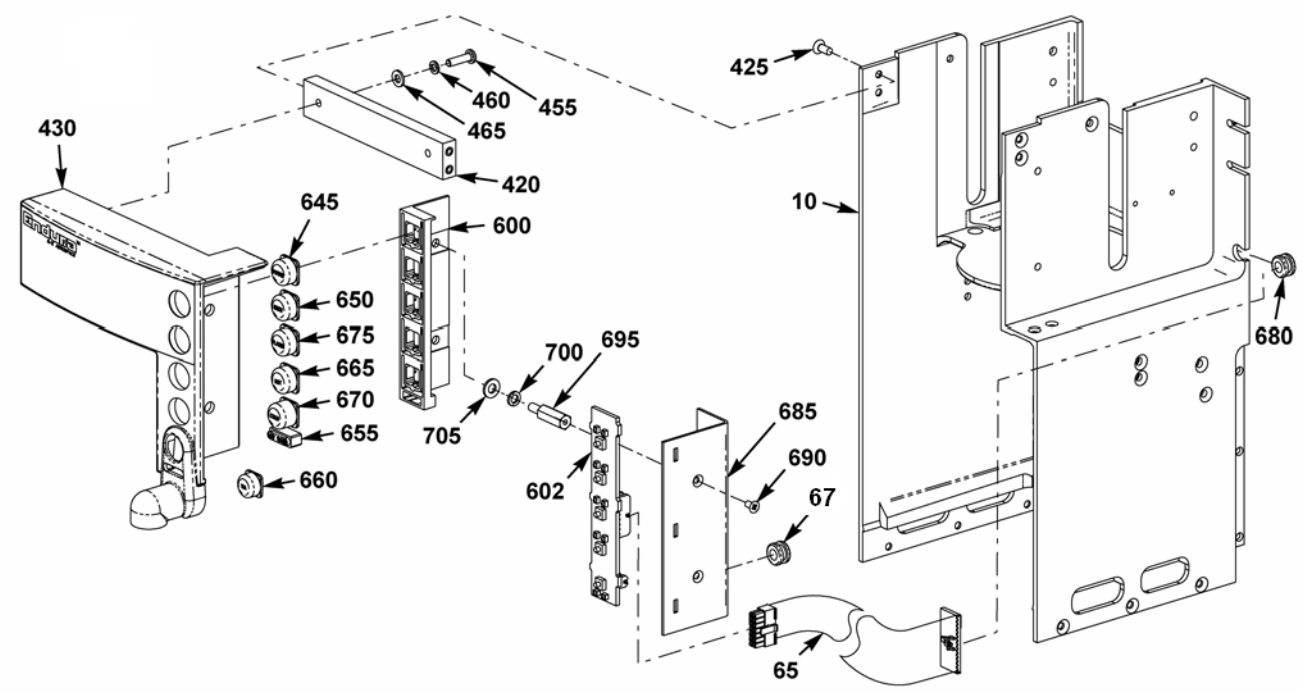

Figure 3007
Removal of Front Panel Assy

(6) Remove the front panel assembly (430, 435, 440, 445, or 450), brewhead assembly (300), and the brew handle (325) from the chassis assembly (10) (refer to Figure 3007).

(a) Remove two screws (690) from the switch cover (685).

CAUTION: WHEN TOUCHING THE SWITCH PC BOARD, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.

(b) Carefully remove switch PC board (605, 610, 615, 620, 625, 630, 635, or 640) tabs from slots in the button frame (600) and the switch cover (685).

(c) Remove the wire harness assembly (65) plug and switch assembly (585) plug from the switch PC board (605, 610, 615, 620, 625, 630, 635, or 640). Isolate the switch PC board (605, 610, 615, 620, 625, 630, 635, or 640) from electrostatic discharge and exposure to water.

(d) Remove two standoffs (695), two lock washers (700), and two flat washers (705) from the button frame (600), front panel assembly (430, 435, 440, 445, or 450), and chassis assembly (10).

(e) Remove the harness assembly (65), wires from switch assembly (585), and grommet (67) from the switch cover (685).

(f) Remove the grommet (67) from the harness assembly (65) and wires from switch assembly (585).

(g) Remove harness assembly (65) and grommet (67) from slot in chassis assembly (10) near the manifold assembly (710, 715, 720, or 725).

(h) Remove two screws (455), two lock washers (460), and two flat washers (465) from the front panel assembly (430, 435, 440, 445, or 450) and chassis brace assembly (420).

(j) Remove the front panel assembly (430, 435, 440, 445, or 450) from the brew handle (325), and faucet assembly (270) or swivel faucet assembly (275).

(k) Remove the buttons (645, 650, 660, 665, 670, or 675) and indicator (655), where options are installed, from the button frame (600), only if necessary for cleaning or replacement.

NOTE: The buttons are press fit into the button frame (600) and should only be removed if they need to be replaced or cleaned.

(l) Remove four screws (425) from the chassis assembly (10) and chassis brace assembly (420).

NOTE: The switch cover (685) must be removed to get access to screws (425) that attach the chassis brace assembly (420) and screws (280) that attach the faucet assembly (270) or the swivel faucet assembly (275) to the chassis assembly (10).

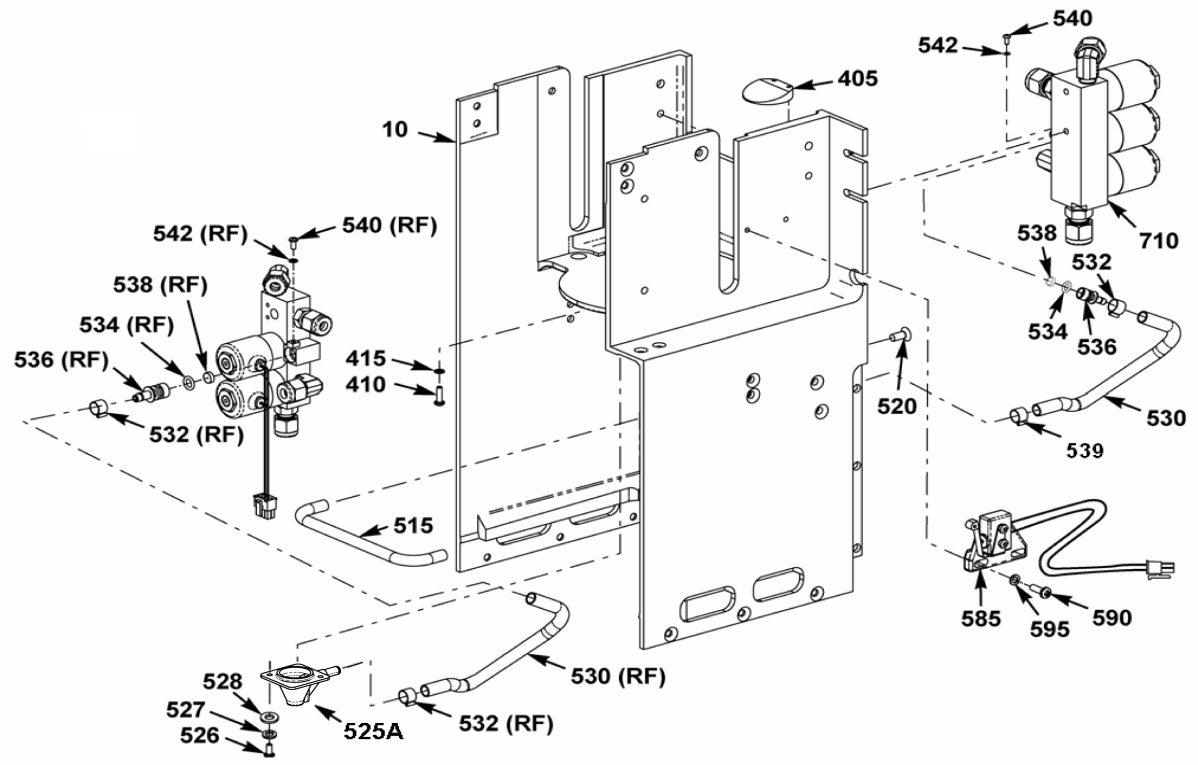

Figure 3008
Removal of Switch Assembly and Other Assembly

(7) Remove the switch assembly (585) from the chassis assembly (10) (refer to Figure 3008).

CAUTION: WHEN TOUCHING THE SWITCH ASSEMBLY (585), USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.

(a) The switch cover (685) may need to be removed to get access to the switch PC board (605, 610, 615, 620, 625, 630, 635, or 640).

CAUTION: WHEN TOUCHING THE SWITCH PC BOARD, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.

(b) Disconnect the switch assembly (585) plug from the switch PC board (605, 610, 615, 620, 625, 630, 635, or 640).

(c) Remove two screws (590), two lock washers (595), and the switch assembly (585) from the side of the chassis assembly (10).

(8) Remove two screws (520) and handle (515) from the chassis assembly (10) (refer to Figure 3008).

(9) Remove the tea tube funnel (525A) from the chassis assembly (10) (refer to Figure 3008).

(a) Remove two screws (526), two lock washers (527), and two flat washers (528) from the tea tube funnel (525A) and chassis assembly (10).

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(b) Remove screw (540), washer (542), from the manifold assembly (710 or 725).

(c) Remove flow washer (538), o-ring (534), regulator fitting (536), tube clamp (532 and 539) and tube (530 or 530A) from the tea tube funnel (525A) and manifold assembly (710 or 725).

NOTE: Removal of this tube clamp (532, or 539) requires destruction of this clamp that is usually installed with a crimping tool.

(10) Remove two screws (410), two lock washers (415), and cup retainer (405) from the chassis assembly (10) (refer to Figure 3008).

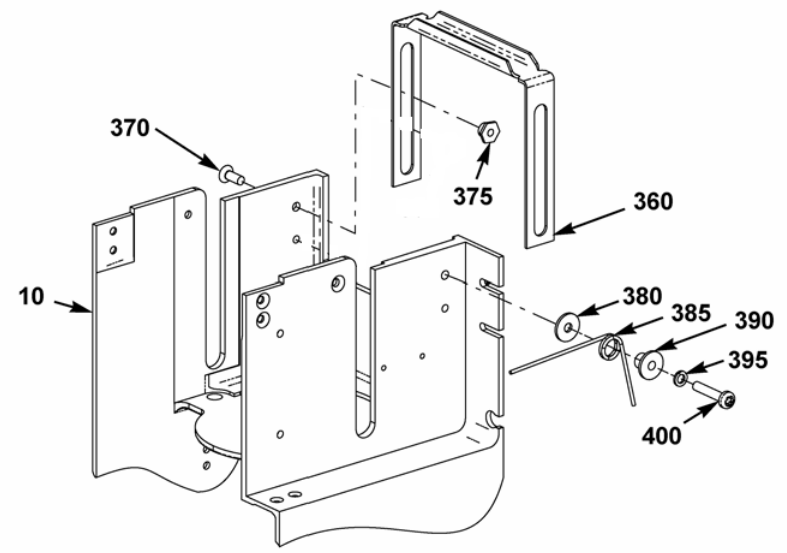

Figure 3009
Removal of Carrying Handle

(11) Remove the carrying handle (360) from the chassis assembly (10) (refer to Figure 3009).

(a) Remove the tension spring (385) from slot in right side guide of brewhead assembly (300).

(b) Remove screw (400), lock washer (395), spring retainer (390), tension spring (385), metallic washer (380), and handle bushing (375) from right side of the chassis assembly (10).

(c) Remove screw (370) and handle bushing (375) from left side of chassis assembly (10).

(d) On units manufactured after 2006/01/20, remove two screws (370) and two handle bushings (375) from the left side of the chassis assy (10) holding the carrying handle (360).

(e) On units manufactured after 2006/01/20, remove a screw (370) and a handle bushing (375) from the right side of the chassis assy (10) holding the carrying handle (360).

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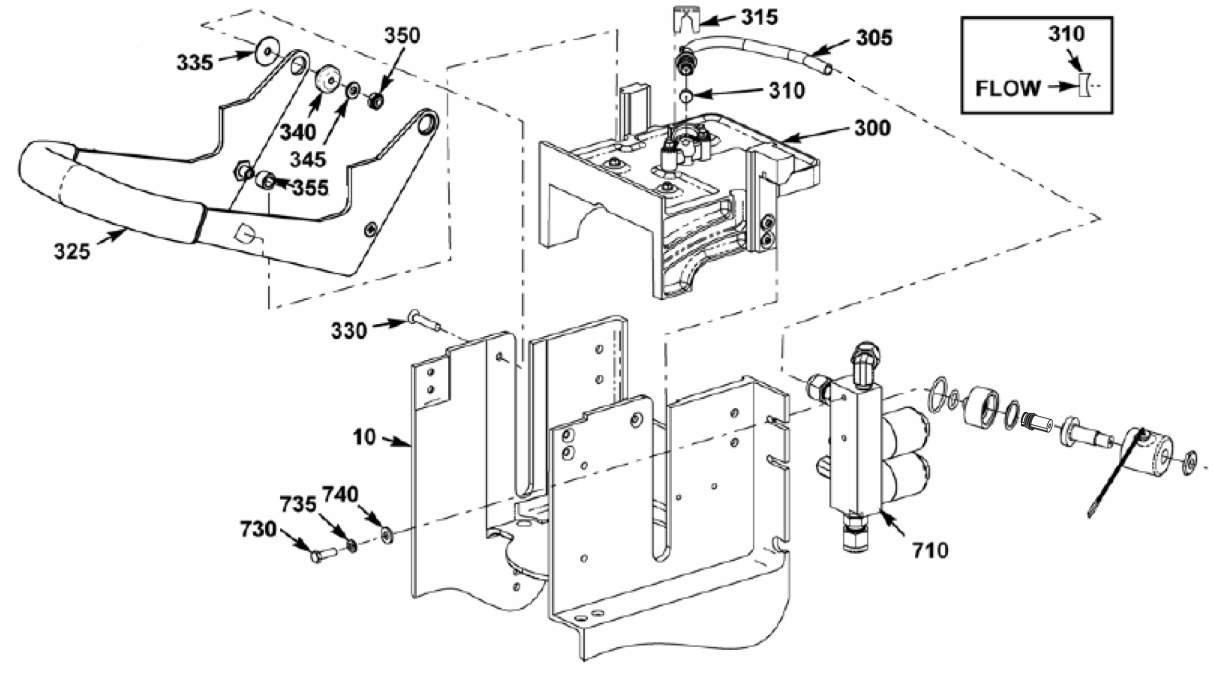

Figure 3010
Removal of Brewhead Assembly

(12) Remove brew handle (325) and brewhead assembly (300) from the chassis assembly (10) and manifold assembly (710, 715, 720, or 725) (refer to Figure 3010).

NOTE: Remove the front panel assembly (430, 435, 440, 445, or 450) and chassis brace assembly (425) from the chassis assembly (10) (refer to DISASSEMBLY, Paragraph 2.B.(6)) before removing the whole brewhead assembly (300) and brew handle (325) from the chassis assembly (10).

(a) Remove two screws (330), two handle washers (335), brew handle (325), two threaded bushings (340), two flat washers (345) and two lock nuts (350) from the chassis assembly (10).

(b) Remove the clip (315) and flow washer (310) from the brewhead assembly (300) and brew tube assembly (305).

(c) Remove tube fittings from the brew tube assembly (305) and the manifold assembly (710, 715, 720, or 725).

(d) Remove the brewhead assembly (300) and brew handle (325) from the chassis assembly (10).

(e) Remove the brew handle (325) from the brewhead assembly (300) then remove two bushings (355) from the brew handle.

(f) Refer to DISASSEMBLY, Paragraph 2.J for disassembly of brewhead assembly (300).

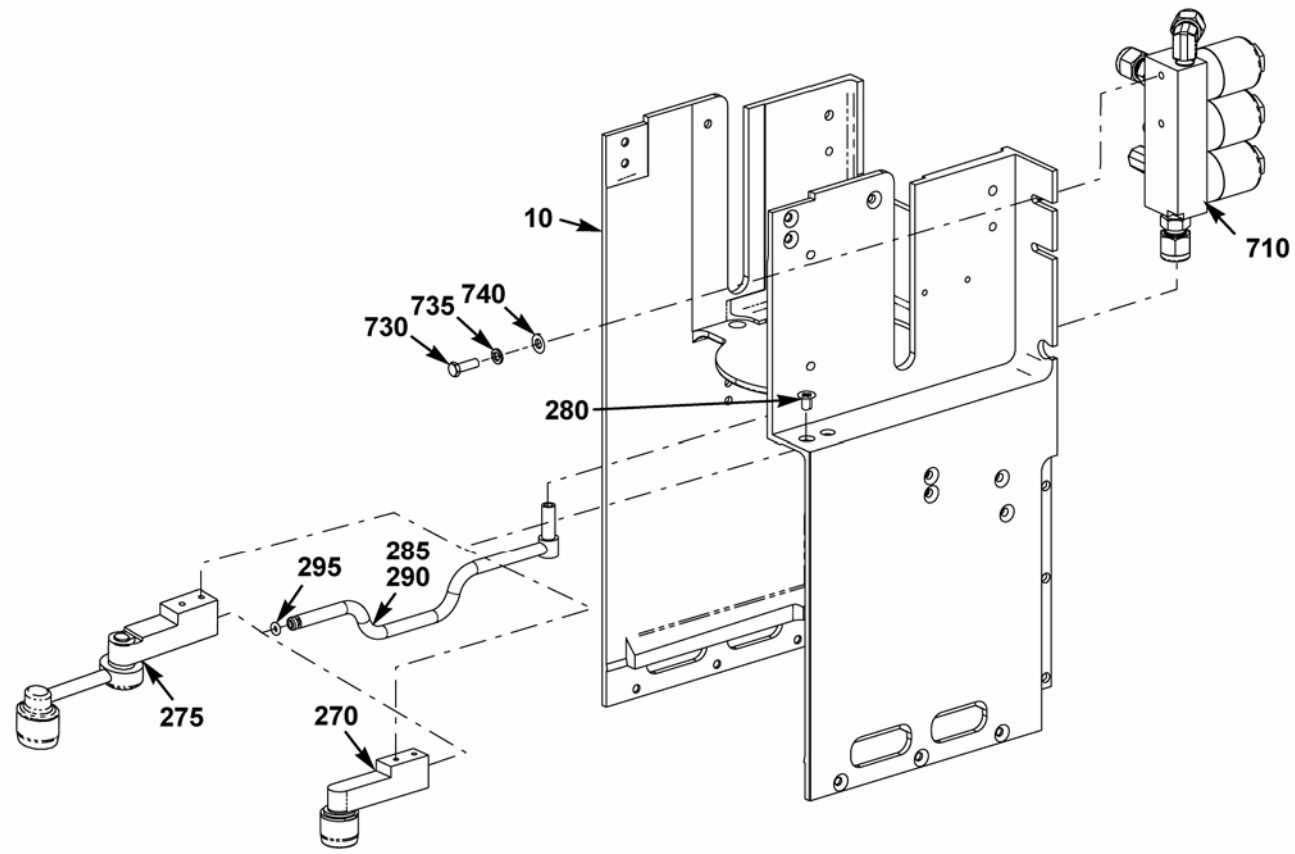

Figure 3011
Removal of Faucet Assembly or Swivel Faucet Assembly

(13) Remove the faucet assembly (270) or swivel faucet assembly (275) from the chassis assembly (10) (refer to Figure 3011).

Table 3004
Faucet or Swivel Faucet Assy Part Number Reference

UNIT PN	FAUCET OR SWIVEL FAUCET ASSEMBLY PN	DESCRIPTION
4510-XXXF-XX*	3520-0525-01	Standard Faucet Assy
4510-XXXG-XX*	3520-0559-01	Swivel Faucet Assy

*X represents other numbers or letters in the Unit Part Number

NOTE: The switch cover (685) must be removed to get access to screws (425) that attach faucet assembly (270) or swivel faucet assembly (275) to the chassis assembly (10).

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(a) Remove two screws (280) from the faucet assembly (270) or swivel faucet assembly (275) and the chassis assembly (10).

(b) Remove the faucet assembly (270) or swivel faucet assembly (275) from the faucet tube assembly (285 or 290).

(c) Remove o-ring (295) from faucet tube assembly (285 or 290).

(d) Remove the faucet tube assembly (285 or 290) and fittings from the manifold assembly (710, 715, 720, or 725). Refer to DISASSEMBLY, Paragraph 2.G for disassembly of the faucet assembly (270) or DISASSEMBLY, Paragraph 2.H for disassembly of the swivel faucet assembly (275).

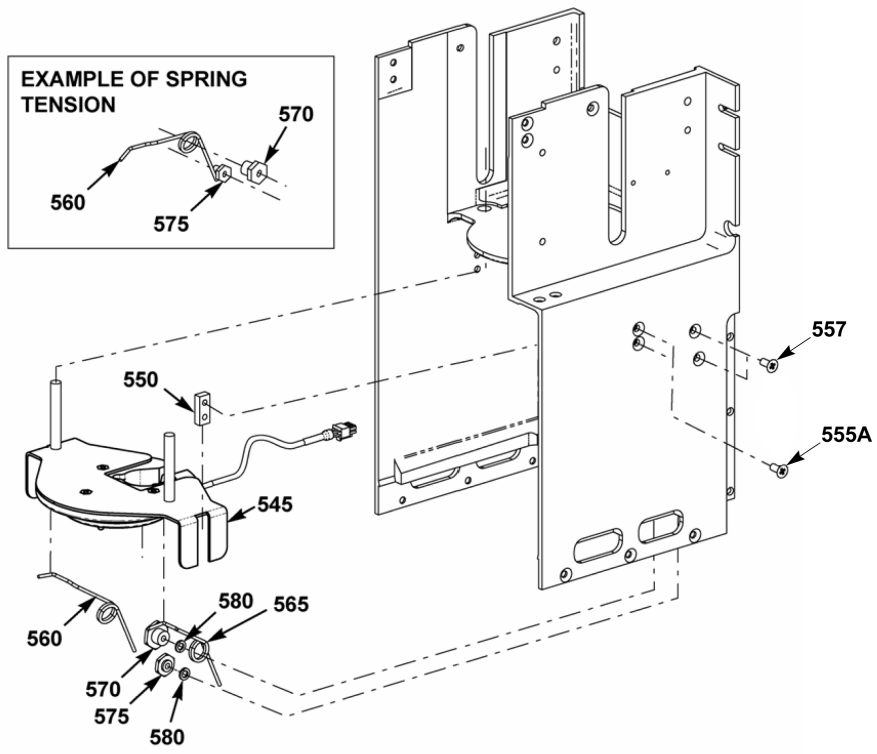

Figure 3012
Removal of Server Retainer Assembly

(14) Remove the server retainer assembly (545) from the chassis assembly (10) (refer to Figure 3012).

NOTE: The LLS potted assembly (15, IPL Figure 10) can be removed without removal of the whole server retainer assembly (545) from the chassis assembly (10). Refer to DISASSEMBLY, Paragraph 2.K for disassembly of the server retainer assembly (545).

NOTE: The server retainer assy (545) is replaced with the new server retainer assy (545A) per Service Bulletin 4510-25-03.

(a) Open the power module assembly (55) to get access to the controller board assembly (75) then remove the electrical connector of the server retainer assembly (545) from the controller board assembly (75).

CAUTION: THE TWO SERVER RETAINER SPRINGS ARE UNDER TENSION SO THAT THEY HOLD UP THE SERVER RETAINER ASSEMBLY CORRECTLY. THEIR BOTTOM LEGS ARE HELD BY THE SPRING GUIDE BUSHINGS.

(b) Remove two spring guide bushings (575) (holding the bottom leg of the server retainer springs (560, 565), two lock washers (580), and two screws (555A) from the chassis assembly (10).

(c) Remove two screws (555A), two lock washers (580), two server retainer springs (560, 565), and two spring bushings (570) from the chassis assembly (10).

(d) Remove four screws (557) and two LLS guide assembly (550) from the chassis assembly (10).

(e) Remove the server retainer assembly (545) from the chassis assembly (10). Refer to DISASSEMBLY, Paragraph 2.K for disassembly of the server retainer assembly (545).

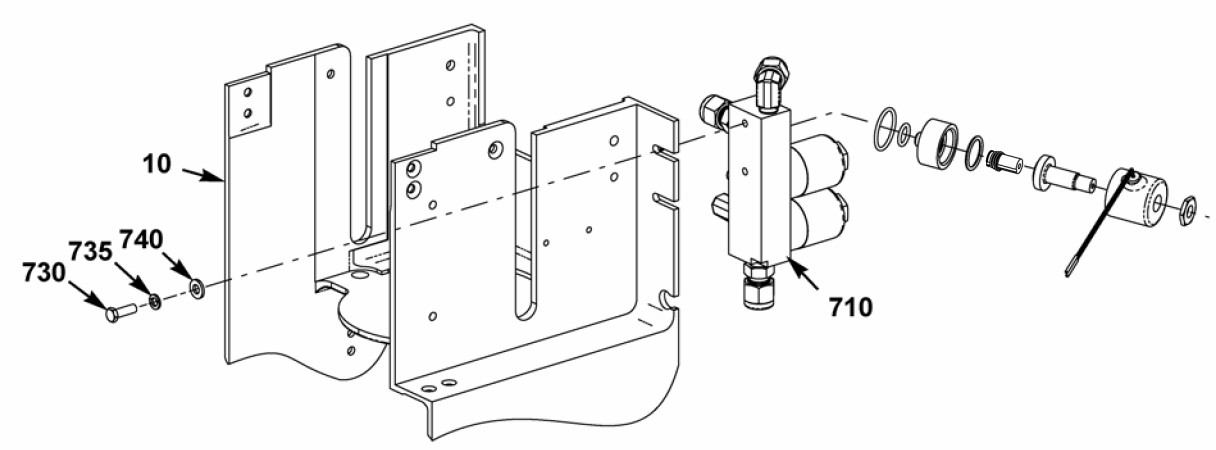

Figure 3013
Removal of Manifold Assembly

(15) Remove manifold assembly (710, 715, 720, or 725) from chassis assembly (10) (refer to Figure 3013).

(a) Remove all plumbing connections from the manifold assembly (710, 715, 720, or 725).

(b) Remove wiring connector for manifold assembly (710, 715, 720, or 725) to power module assembly (55). The power module cover (45) may need to be removed to get access to controller board (75).

(c) Remove two screws (730), two lock washers (735), and two flat washers (740) from the manifold assembly (710, 715, 720, or 725) and chassis assembly (10).

(d) Refer to DISASSEMBLY, Paragraph 2.L for disassembly of manifold and solenoid valve assembly.

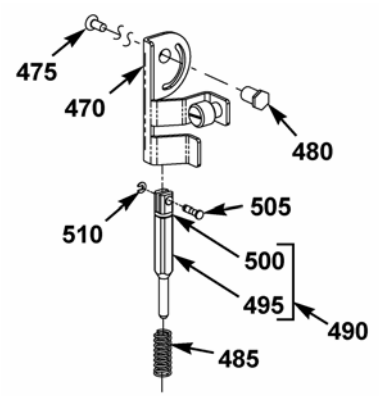

Figure 3014
Removal of Lockdown Lever

(16) Remove the lockdown lever from the chassis assembly (10) (refer to Figure 3014).

(a) Remove screw (475) and pin pivot assembly (480) from the locating pin lever (470) and chassis assembly (10).

(b) Remove pin locating lever (470), locating pin assembly (490), and helical spring (485) from the chassis assembly (10).

1 Remove o-ring (500) from machined pin (495).

(c) Remove retaining ring (510) from pin (505) that holds locating pin assembly (490) on locating pin lever (470).

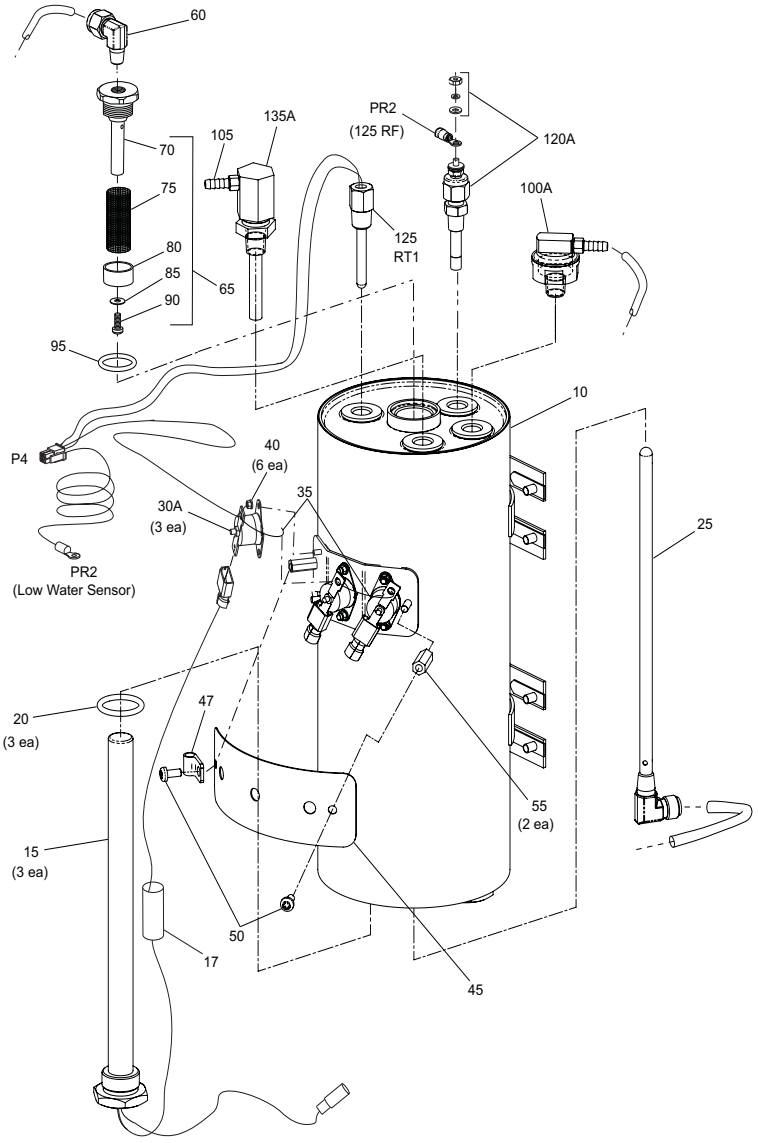

Figure 3015
Disassembly of Tank Assembly

C. Tank assembly (refer to Figure 3015)

(1) Refer to the DISASSEMBLY Section, Paragraph 2.B.(3) for removal of tank assembly (15, IPL Figure 2) from chassis assembly (10, IPL Figure 2).

(2) Remove screws (50), clamp (47), thermal switch cover (45), and hex spacers (55) from welded tank assembly (10).

CAUTION: WHEN HEATING SOLDER, TURN TANK SO HOT SOLDER FALLS AWAY FROM THE TANK (10) AND OVERTEMPERATURE THERMOSTATS (30).

(3) Heat solder and remove two wire busses (35) and an RTD assembly (125) wire from thermostat (30) terminals.

CAUTION: CAREFULLY REMOVE HEATER ASSEMBLY WIRE TERMINAL FROM TERMINAL OF THE OVERTEMPERATURE THERMOSTAT (30) SO THAT THE TERMINAL DOES NOT MOVE ON THE PLASTIC HOUSING OF THE THERMOSTAT. TERMINAL MOVEMENT CAN CRACK THE THERMOSTAT HOUSING AND RESULT IN FAILURE OF THE THERMOSTAT.

(4) Remove three quick connect wiring terminals of heater assembly (15) from terminals of over temperature thermostats (30) on the welded tank assembly (10).

(5) Remove six lock nuts (40) and three thermostats (30) from the welded tank assembly (10).

(6) Remove hex nut, lock washer, flat washer and RTD assembly (125) wire terminal from low water sensor (120).

(7) Remove low water sensor (120) from welded tank assembly (10).

(8) Remove tie downs from wiring for RTD assembly (125).

(9) Remove RTD assembly (125) from welded tank assembly (10).

(10) Remove elbow (60) from filter assembly (65).

(11) Remove filter assembly (65) and o-ring (95) from welded tank assembly (10).

(a) Disassemble filter assembly (65).

1 Remove screw (90), lock washer (85), retainer (80), filter screen (75) from the water jet (70).

(12) Remove vent valve assembly (100) from the welded tank assembly (10) (refer to Figure 3016).

(a) Unscrew vent valve body (102) from adapter (101).

(b) Remove o-ring (104B), ball (104A), liner (103), and washer (104).

(c) Remove barbed fitting (105) from the vent valve assembly (100).

1 Disassemble the barbed fitting by removing the gasket (110) from the fitting (115).

(13) Remove pressure relief valve assembly (130) from welded tank assembly (10).

(14) Remove tube assembly (25) from welded tank assembly (10).

NOTE: Identify tank heater assembly (15) and where they are installed in the bottom of the welded tank assembly (10).

(15) Remove three tank heater assembly (15) and three o-rings (20) from welded tank assembly (10).

NOTE: Only if required for replacement, remove connector (15B), connector pin (15C) and terminal (15A) from heater assy tank (15D).

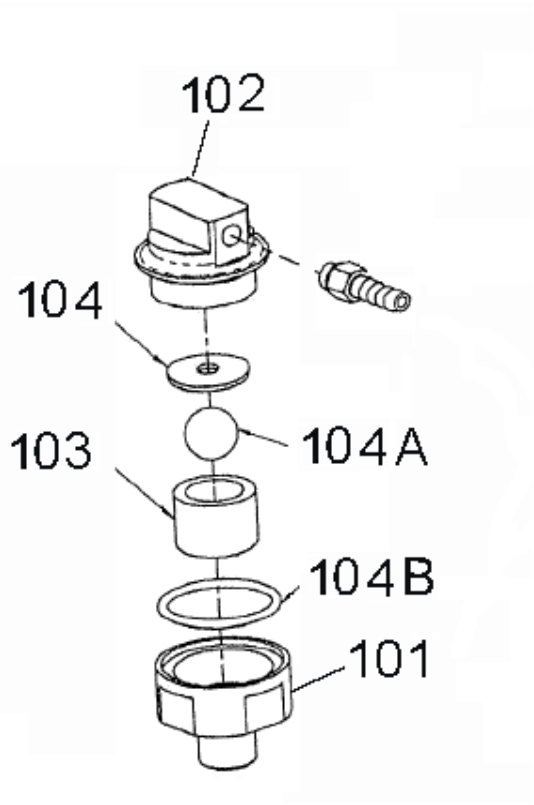

Figure 3016
Disassembly of Vent Valve Assembly

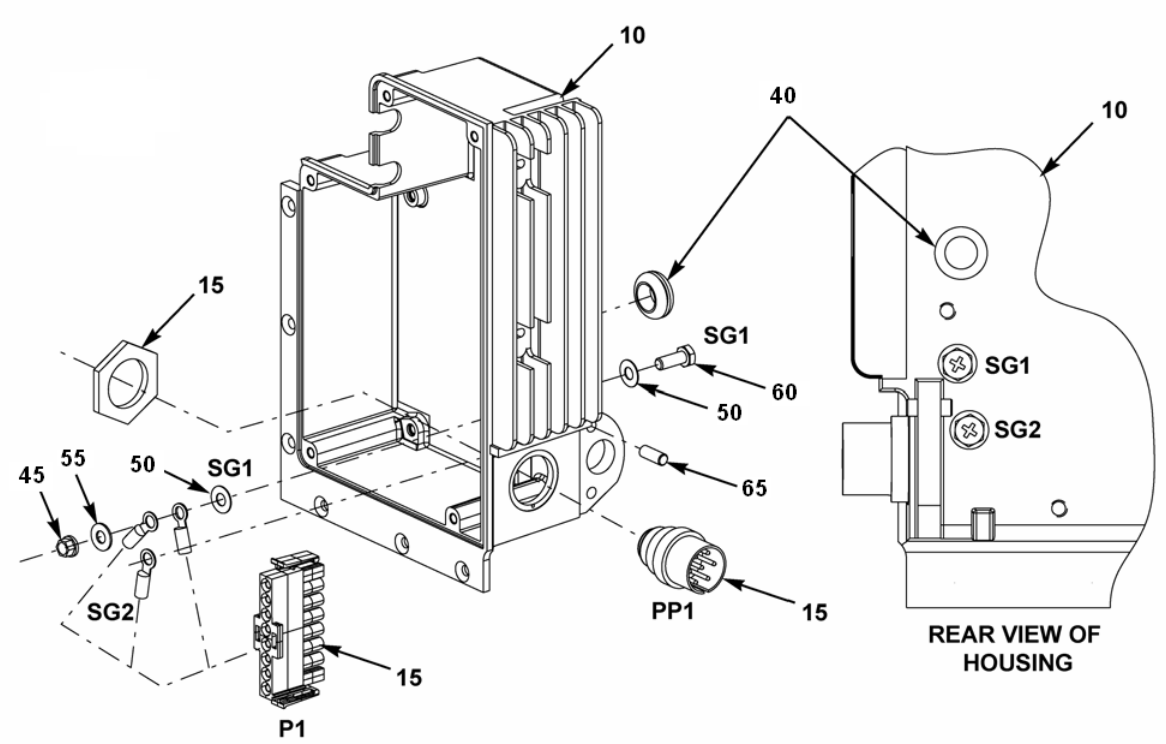

Figure 3017
Disassembly of Power Module Assembly

D. Power Module Assembly (refer to Figure 3017)

(1) Refer to DISASSEMBLY, Paragraph 2.B.(4) for removal of power module assembly (55) from chassis assembly (10).

(2) Remove SG1 and SG2 ground stack ups from the power module housing.

(a) Remove two lock nuts (45), flat washers (50), wire terminals for harness assembly (15), two flat washers (55), and two hex screws (60) from the power module housing assembly (10).

(b) Remove pin (65) from power module housing assembly (10).

CAUTION: THE P1 AND PP1 CONNECTORS MUST BE REMOVED FROM THE HARNESS ASSEMBLY TO REMOVE THE HARNESS ASSEMBLY THROUGH THE GROMMET AND HOLES IN THE POWER MODULE HOUSING ASSEMBLY (refer to Figure 3018).

NOTE: This pin helps align the support connector (115, IPL Figures 2, 2A, 2B, 2C) with the power module assembly (55, IPL Figures 2, 2A, 2B, 2C).

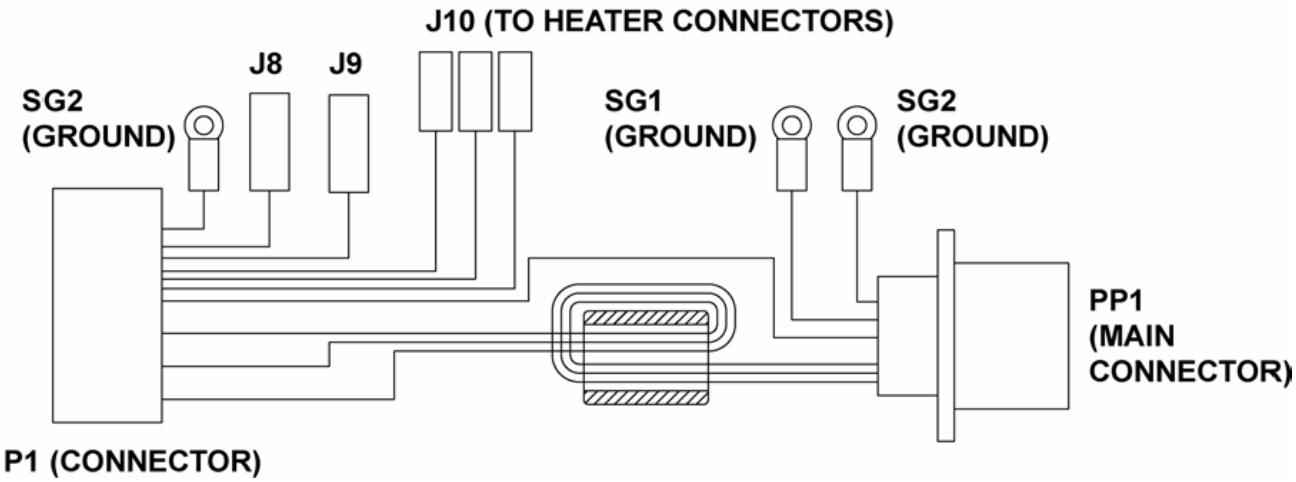

Figure 3018
Diagram of Harness Assembly in Power Module Assembly

(3) Remove the PP1 connector (20) from power module housing assembly (10).

(a) Remove the hex nut from the PP1 connector.

CAUTION: CAREFULLY REMOVE CONNECTOR FROM THE POWER MODULE HOUSING ASSEMBLY TO AVOID DAMAGE TO THE PP1 GUIDE PIN.

CAUTION: CAREFULLY REMOVE TERMINAL PINS (25 OR 30) FROM THE CONNECTOR TO FULLY REMOVE THE CONNECTOR FROM THE POWER MODULE HOUSING.

(b) Only if required for replacement, remove o-ring (35) and pins (25 or 30) from the connector (20).

(4) Remove harness assembly (15) from the power module housing assembly (10).

(5) Remove the grommet (40) from the back of the power module housing assembly (10).

E. Controller Board Assembly (refer to CMM 25-33-32).

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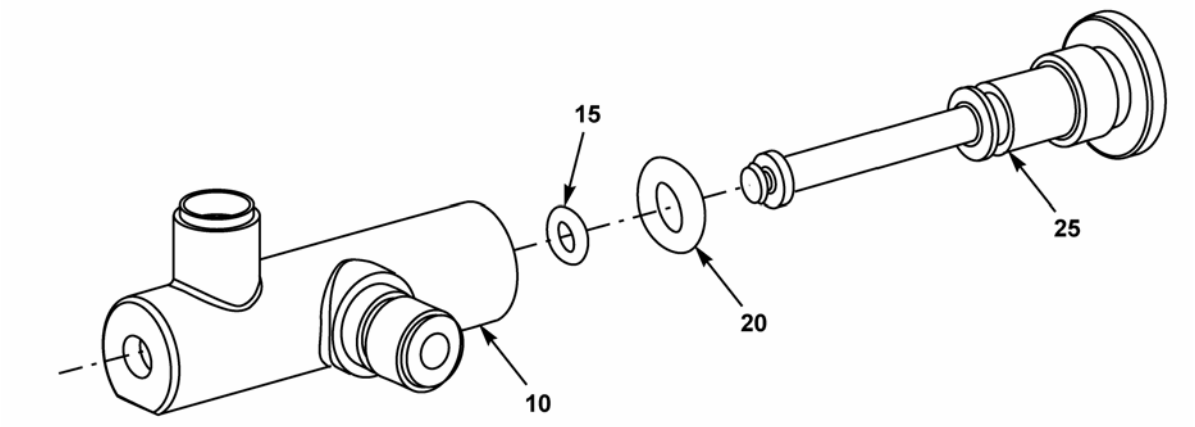


Figure 3019
Disassembly of Drain Valve Assembly

F. Drain Valve Assembly (refer to Figure 3019)

(1) Refer to DISASSEMBLY, Paragraphs 2.B.(2) (f) and (g) for removal of drain valve assembly (140, IPL Figures 2, 2A, 2B, 2C) from support connector (115, IPL Figures 2, 2A, 2B, 2C).

(2) Remove stem assembly (25) from body (10).

(a) Remove upper o-ring (20) and lower o-ring (15) from stem assembly (25).

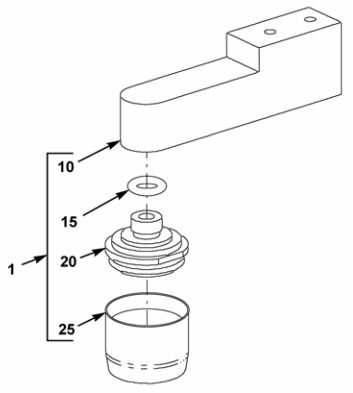


Figure 3020
Disassembly of Faucet Assembly

G. Faucet Assembly (refer to Figure 3020)

(1) Refer to DISASSEMBLY, Paragraph 2.B.(13) for removal of faucet assembly (270, IPL Figures 2, 2A) from chassis assembly (10, IPL Figures 2, 2A), if option is installed.

(2) Remove aerator (25), aerator adapter (20) from faucet bar (10).

(3) Remove o-ring (15) from aerator adapter (20).

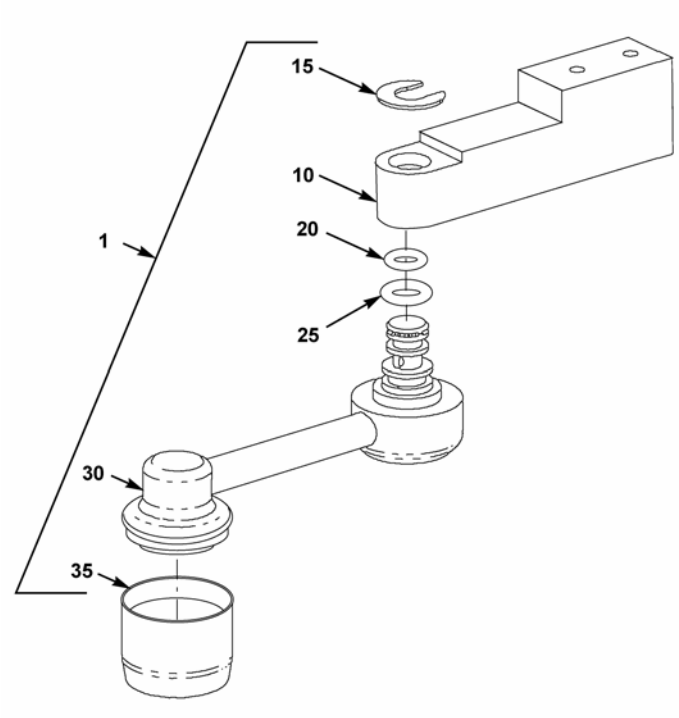


Figure 3021
Disassembly of Swivel Faucet Assembly

H. Swivel Faucet Assembly (refer to Figure 3021)

(1) Refer to DISASSEMBLY, Paragraph 2.B.(13) for removal of swivel faucet assembly (275, IPL Figures 2B, 2C) from chassis assembly (10, IPL Figures 2B, 2C), if option is installed.

(2) Remove aerator (35) from welded faucet assembly (30).

(3) Remove retaining ring (15) from welded faucet assembly (30).

(4) Remove welded faucet assembly (30) from swivel faucet bar (10).

(5) Remove upper o-ring (20) and lower o-ring (25) from welded faucet assembly (30).

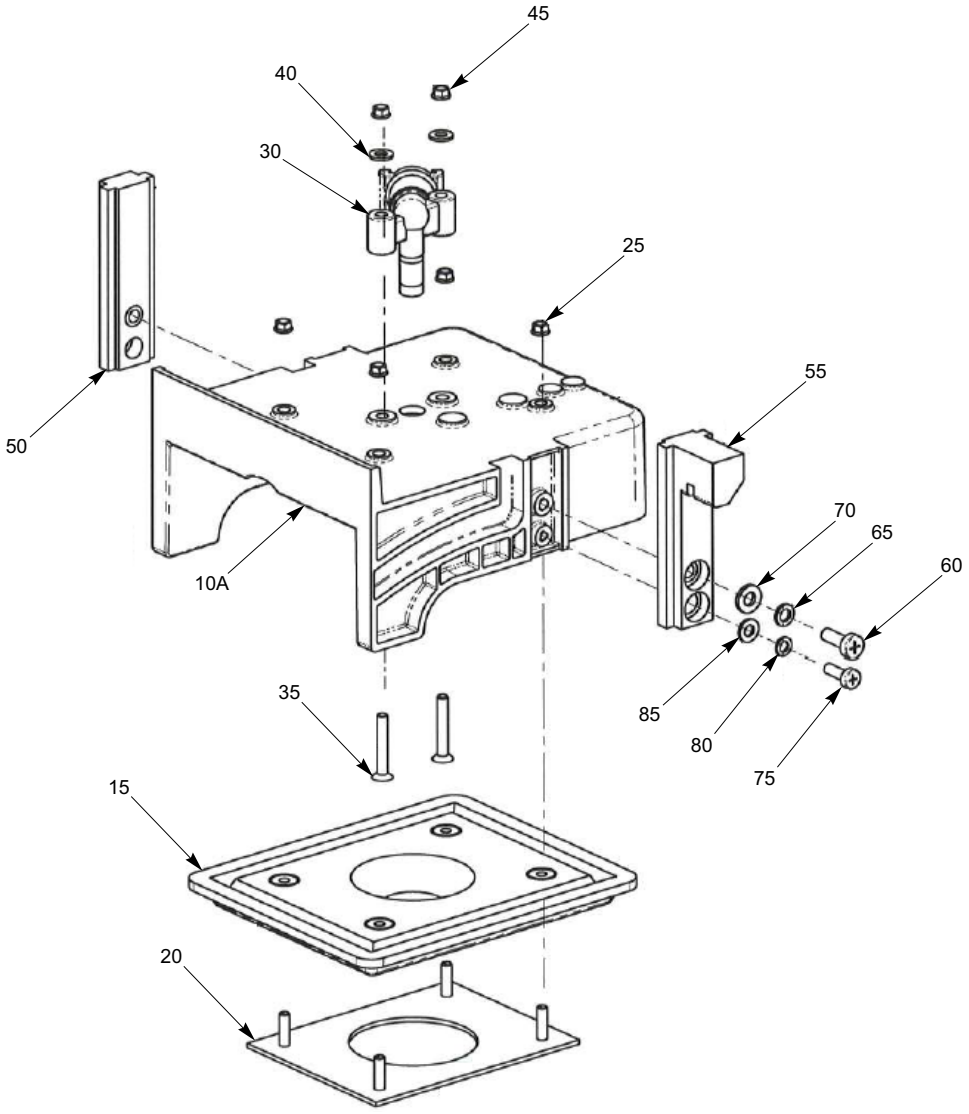


Figure 3022
Disassembly of Brewhead Assembly

J. Brewhead Assembly (refer to Figure 3022)

(1) Refer to DISASSEMBLY, Paragraph 2.B.(12) for removal of brewhead assembly (300, IPL Figures 2, 2A, 2B, 2C) from chassis assembly (10, IPL Figures 2, 2A, 2B, 2C).

(2) Remove four lock nuts (25), gasket retainer (20), and gasket (15) from the brew head (10A).

NOTE: Coffee residue can cause parts to stick together in the brewhead assembly.

(3) Remove two screws (35), two lock nuts (45), two flat washers (40), and brew fitting (30) from the brew head (10A).

(4) Remove four screws (60 and 75), four lock washers (65 and 80), four flat washers (70 and 85), and two guides (50 and 55) from the brew head.

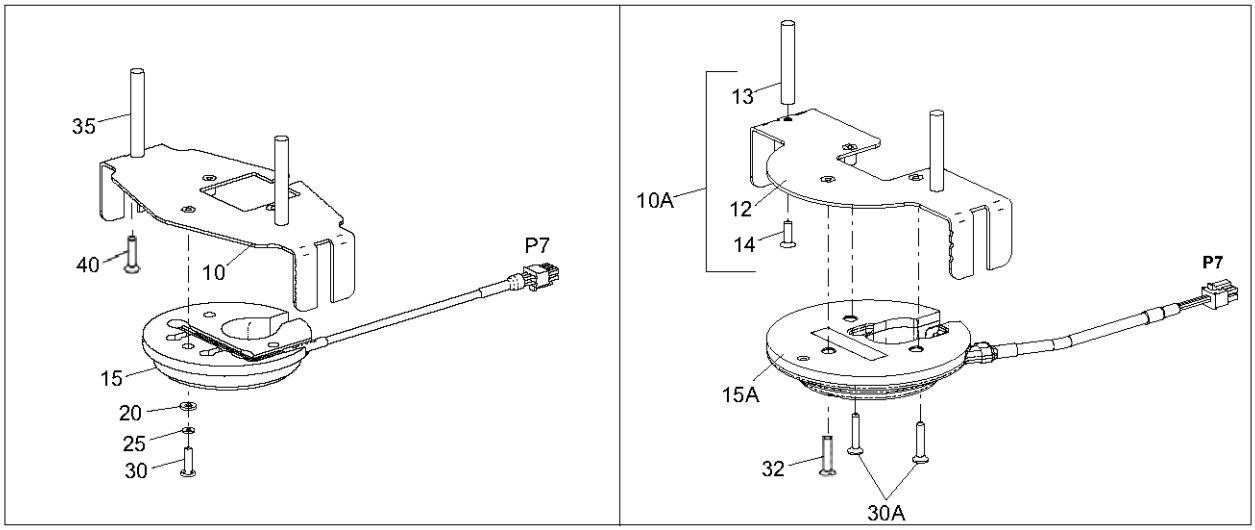


Figure 3023
Disassembly of Server Retainer Assembly

K. Server Retainer Assembly (refer to Figure 3023)

(1) Refer to DISASSEMBLY, Paragraph 2.B.(14) for removal of server retainer assembly (545, IPL Figures 2, 2A, 2B, 2C) from chassis assembly (10, IPL Figures 2, 2A, 2B, 2C).

(2) For PN 3520-0569-01, Remove three screws (30), three lock washers (25), and three flat washers (20) from the liquid level sensor (LLS) assembly (15) and the retainer assembly (10).

(3) For PN 3520-0580-01, Remove two screws (30A), screw (32), from the liquid level sensor (LLS) assembly (15A) and the retainer assembly (10A).

(4) For PN 3520-0569-01, If necessary for repair, remove two screws (40), and two LLS retaining rods (35) from the retainer assembly (10).

(5) For PN 3520-0580-01, If necessary for repair, remove two screws (14), and two LLS retaining rods (13) from the retainer plate assembly (12).

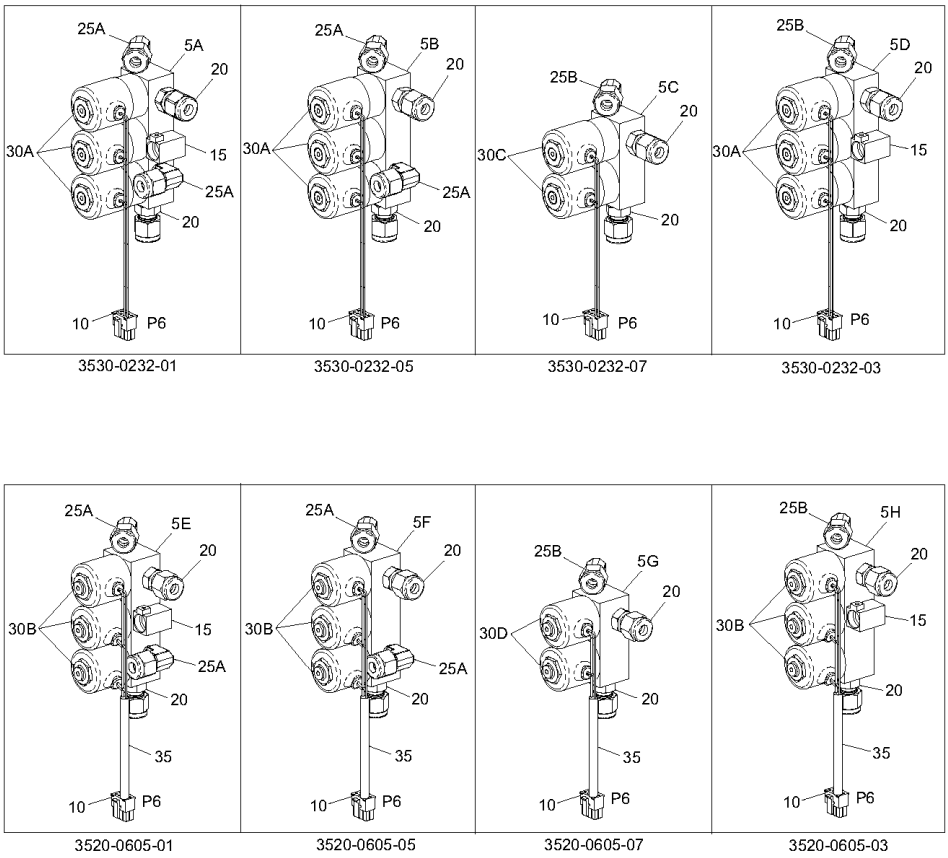


Figure 3024
Disassembly of Manifold Assembly

L. Manifold and Solenoid Valve Assembly (refer to Figures 3024 and 3025)

(1) Refer to DISASSEMBLY, Paragraph 2.K.(15) for removal of manifold assembly (710, 715, 720, or 725, IPL Figures 2, 2A, 2B, 2C) from chassis assembly (10, IPL Figures 2, 2A, 2B, 2C).

(2) Remove regulator fitting (15) from manifold body (5A, 5D, 5E or 5H).

(3) Remove straight fittings (20) from manifold body (5).

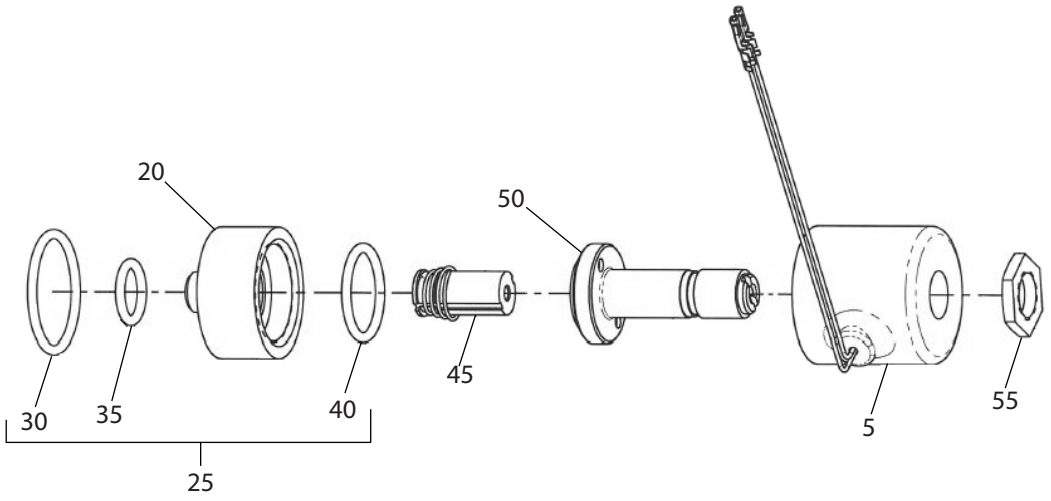
(4) Remove elbow fittings (25A or 25B) from manifold body (5).

NOTE: After removing fittings from manifold body, clean thread seal tape from all threads.

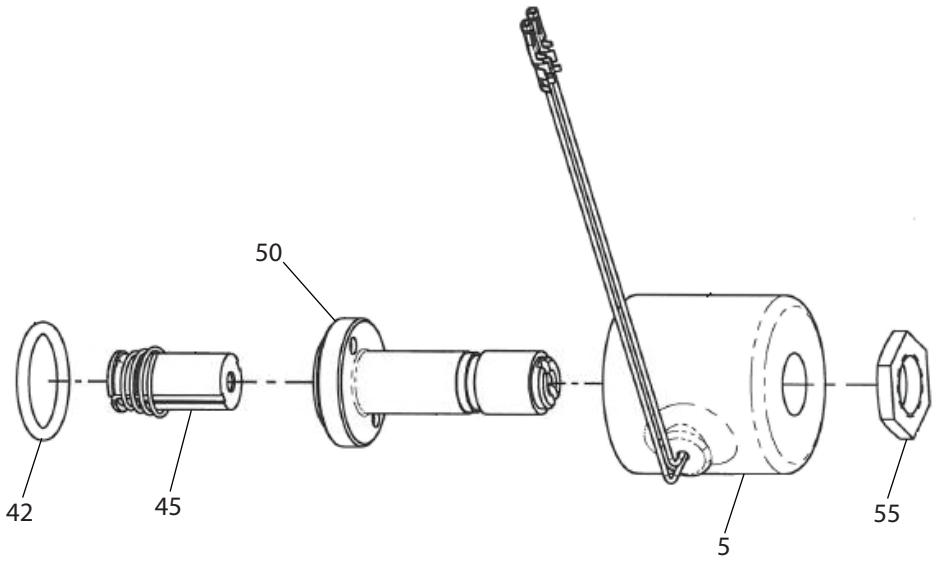
CAUTION: CAREFULLY REMOVE TERMINALS FROM THE P6 CONNECTOR, OR THE CONNECTOR MAY BE DAMAGED, RESULTING IN DAMAGE TO THE CONTROLLER BOARD.

(5) Only if necessary for replacement, remove P6 connector (10) from wires for solenoid valve assy (30) using a Molex extractor tool part number 11-03-0043 or equivalent.

(6) Remove o-rings (30 and 35, IPL Figure 12) for solenoid valve assembly (30) from manifold body (5D).



3530-0279-01



3530-0279-23

Figure 3025
Solenoid Valve Assembly

(7) For PN 3530-0279-01, Remove two o-rings (30 and 35) from the valve body (20) or manifold body (refer to Figure 3024).

(8) Remove nut (55) from solenoid guide (50).

(9) Remove coil assembly (5) from solenoid guide (50).

(10) Remove solenoid guide (50) from valve body (20). Special tools are recommended for disassembly (refer to Table 9001 and Figure 9008).

(11) Remove plunger (45) from inside the solenoid guide (50).

(12) Remove o-ring (40 or 42) from the valve body (20).

(13) Only if necessary for repair, remove crimp terminals (15) from P6 connector using a Molex (V27264) extractor tool part number 11-03-0043 or equivalent.

(14) Only if necessary for repair, remove crimp terminals (15) from the wiring to the solenoid coil (10).

NOTE: In the event that crimp terminals (15) are removed from wiring, the terminals cannot be reused.

M. If CLEANING or INSPECTION/CHECK will not be performed immediately, store all parts in a dry, dust-free container.

CLEANING

1. General

WARNING: DO NOT CONNECT THE BEVERAGE MAKER TO POWER AND WATER SUPPLY DURING CLEANING.

WARNING: WEAR EYE PROTECTION AND WATERPROOF PROTECTIVE GLOVES WHILE WORKING WITH CHEMICALS.

CAUTION: DO NOT LET MOISTURE TOUCH ELECTRICAL PARTS (FOR EXAMPLE, SOLENOID COILS OR PLATEN HEATER), WIRING, OR PC BOARD ASSEMBLY.

CAUTION: DO NOT USE SOLUTIONS OR CONCENTRATIONS OTHER THAN THOSE RECOMMENDED BY B/E AEROSPACE.

CAUTION: USE ONLY THE RECOMMENDED CLEANER CONCENTRATION IN NORMAL ROOM TEMPERATURE SOLUTION WITH CLEAN WATER.

A. Most interior and exterior surfaces of the Beverage Maker can be cleaned with cloth towels that have been moistened with clean water.

NOTE: To avoid scratching surfaces, do not use paper towels.

B. Cleaning materials for the Beverage Maker are listed in Table 4001.

NOTE: Equivalent substitutes may be used for listed items.

Table 4001
Cleaning Materials

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
A food-grade aqueous acidic cleaner that removes calcium, lime, and rust deposits (scale) without damage to unit form, fit, or function	None	Commercially Available
Brush, Wire, Soft-bristled	None	Commercially Available
Clean Water	None	Commercially Available
Cleaner, Mild Liquid Detergent (Dish Soap)	None	Commercially Available
Cloth, Cleaning, Low-Lint, Type 1	A-A-59323	Commercially Available
Cotton Swabs	None	Commercially Available
Isopropyl Alcohol	TT-I-735	Commercially Available
Nitrogen Gas	A-A-59503	Commercially Available
DELETED		

2. Procedure

A. The Beverage Maker assembly is part of a galley configuration subject to heat and pressure variations. Therefore, the following cleanliness requirements must be complied with:

(1) The work area for cleaning, assembling, and testing must be clean and free of hydrocarbon residue, oil, or other petroleum base lubricants.

(2) The pressurized gas or air supply used to dry parts after cleaning must be dry, clean, and free of hydrocarbon contamination and excessive moisture.

(3) New or used parts must be cleaned in accordance with Paragraphs 2.B. through 2.F. of this section before assembling into the Beverage Maker Assembly.

B. Clean the parts (except those parts mentioned in the preceding CAUTION) in a warm mild liquid detergent solution bath.

WARNING: WEAR EYE PROTECTION AND WATER PROOF PROTECTIVE GLOVES WHILE WORKING WITH CHEMICALS.

(1) Clean electrical parts with isopropyl alcohol (refer to Table 4001).

(2) Prepare a warm mild liquid detergent solution with clean water (refer to Table 4001) no hotter than 140 °F (60 °C). A mixture of approximately 13 ounces of cleaner concentrate in one gallon of clean water, or 100 ml of cleaner concentrate in one liter of clean water (refer to Table 4001).

(3) Fully submerge and agitate each part (except those parts mentioned in the preceding CAUTION) in the mild liquid detergent solution. Let the cleaner flush all internal and external surfaces. Scrub the chassis assembly and the top of the brew head assembly with a soft brush.

(4) Mineral deposits may be removed by soaking affected parts in a solution of calcium, lime, and rust remover (refer to Table 4001). Follow manufacturer's precautions for any cleaner. Rinse with clean water (refer to Table 4001).

(5) Thoroughly rinse the parts in clean water (refer to Table 4001) at room temperature. Allow the parts to dry thoroughly at room temperature.

CAUTION: DO NOT ALLOW DAMPNESS OR MOISTURE TO ENTER THE POWER MODULE ASSEMBLY, SOLENOID COILS, SWITCH ASSEMBLIES, OR ANY ELECTRICAL CONNECTIONS.

C. Clean the surfaces of the remaining parts that should not be submerged (refer to the preceding CAUTION and Table 4001).

(1) Clean stained surfaces of the Beverage Maker with a soft, cotton cloth or swab dipped in the prepared mild liquid detergent solution.

(2) Brew Cup

(a) Clean all surfaces inside brew cup with cloth moistened with water and the prepared mild liquid detergent solution.

(b) Wipe off any film with a clean, damp cloth.

(c) Lift flap inside brew cup and clean all surfaces under flap with cloth moistened with water and the prepared mild liquid detergent solution.

(3) Clean tube assemblies by running a flexible wire brush through the interior. Rinse with clear water.

(4) When the tank assembly is disassembled, clean thread seal tape from fittings.

(5) Clean server retainer assembly with a clean damp cloth.

(6) Clean the server and brew cup cavities of the chassis assembly with a soft cotton cloth or swab dipped in the prepared mild liquid detergent solution.

(7) Rinse the parts with a soft cotton cloth dipped in clean water (refer to Table 4001) at room temperature.

D. If coffee residue is on a part, remove the residue using a small cotton swab and isopropyl alcohol (refer to Table 4001). Rinse the parts with a soft cotton cloth dipped in room temperature clean water (refer to Table 4001).

E. Cleaning and De-Calcification of the Tank assembly

(1) Perform cleaning and decalcification of unit per CMM as frequently as is required to maintain reliable and correct operation of unit.

(2) If aircraft water supply is not chemically-pure-water or water containing dissolved minerals over zero parts per million will result in mineral buildup on internal parts of the units plumbing system.

(3) Periodic test of aircraft water supply is recommended per international health and safety regulations.

(4) Frequency of cleaning and decalcification of unit is directly affected by aircraft water supply quality, and water supply quality is subject to the economic and health and safety mandates of the authorities responsible for the operation of the aircraft where unit is installed.

(5) Individual parts or unit damaged by use of high-mineral-content or biologically-and chemically-contaminated water supplies are not covered by manufactures warranty.

F. Allow the parts to dry thoroughly at room temperature, or dry each part using nitrogen gas (refer to Table 4001) or commercially bottled, dry, clean, compressed air.

G. If INSPECTION/CHECK will not be performed immediately, store all parts in a dry, dust-free container.

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INSPECTION/CHECK

1. General

WARNING: DO NOT CONNECT THE BEVERAGE MAKER TO POWER AND WATER SUPPLY DURING INSPECTION/CHECK.

A. Preliminary cleaning before initial inspection will help locate visible defects.

B. Visually inspect working parts under strong light and magnification for cracks. Inspect structure for distortion, cracking, and excessive wear that could impair the function of equipment.

NOTE: The term, excessive wear, shall be defined as obvious deformation or deterioration of a part that might render the Beverage Maker inoperative. If doubt exists regarding the serviceability of a part, refer to the TESTING AND FAULT ISOLATION, REPAIR, and FITS AND CLEARANCES Sections of this book, as well as the guidelines listed below.

NOTE: Prepare a clean space and working surface before beginning any INSPECTION/CHECK Section operation.

NOTE: If repair of individual parts are necessary, refer to the REPAIR Section.

2. Procedures

A. Inspect the Beverage Maker for signs of damage, breaks, overheating, cracks, or deformation. Use a strong light and some means of magnification.

(1) Visually inspect all disassembled threaded parts for stripped or crossed threads.

B. Electrical

(1) Inspect electrical harnesses, wiring, and connection points for damage, deteriorated insulation, and loose wires.

(2) Inspect wire routing and nylon wire tie downs to make sure no wiring is loose or exposed to chafing against other parts or heat sources.

(3) Examine the Beverage Maker power connector PP1 for loose, bent, or broken pins, deformed shell, and evidence of arcing or flashover at pin insert insulation.

(4) Examine wiring harnesses and terminals for cracked insulation and loose terminals.

(5) Examine the pc board assembly for deformation, overheating, water intrusion, loose, bent or broken components and connections, and evidence of arcing or flashover.

(6) Switches should not be broken or excessively worn. Make sure text on the switch buttons can be read and is complete. When pressed, switches should compress smoothly with a slight resistance until circuit connection is complete. The button light should come on when the button is pressed and go off when the button is pressed again.

C. Plumbing

(1) Examine tubing for leakage, flattening, or kinking, and cracked, split, loose, or worn tubing and fittings.

(2) Examine all plumbing for flattening, kinking, cracking, splitting, loose fittings, or ineffective sealing.

(3) Examine tank assembly for leakage, overheating, deterioration, or cracks.

(4) Examine solenoid valve assembly (Figure 5001) on manifold assembly for leakage, overheating, deterioration, flattening, cracking, splitting, loose fittings, or ineffective sealing. Special tools are recommended for disassembly (refer to Table 9001).

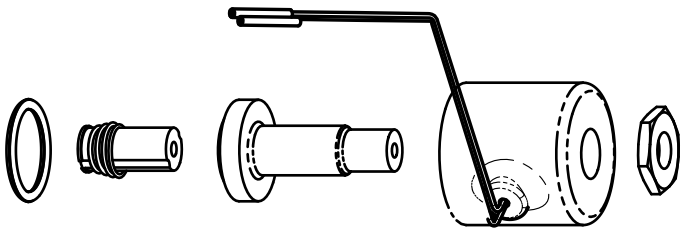

Figure 5001
Solenoid Valve Assembly

D. Vent Valve Assembly (refer to Figure 5002)

(1) Disassemble vent valve assembly and inspect parts for wear/defects (refer to DISASSEMBLY section). Replace worn or defective parts. Clean body (102), adapter (101), and internal components (refer to CLEANING section). Reassemble vent valve assembly (refer to ASSEMBLY section).

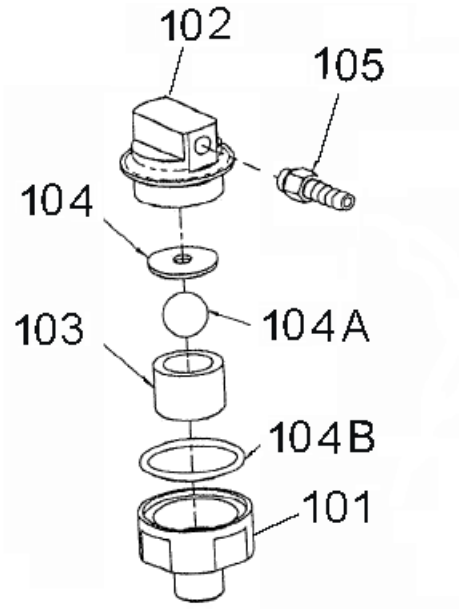

Figure 5002
Vent Valve Assembly

E. Mechanical

(1) Examine the brew handle locking action for damage or excessive wear. Operation should be smooth and easy. Examine bushings on the brew handle pins. Repeated raising and lowering of the brew head assembly contributes to wear.

(2) Examine brew head assembly action for damage or excessive wear. Lubricant is necessary on the sides of the upright channels in the chassis assembly for the brew head assembly guides. Operation should be smooth and easy.

(3) Put a server (refer to Table 9001) in the Beverage Maker and fully lower the brew handle. The server retainer assembly should keep the server in place. Examine the server retainer assembly for damage or wear. Operation should be smooth and easy.

(4) Installation and removal of server and brew cup in chassis assembly cavities should be smooth and easy.

(5) The locking pin mechanism for the Beverage Maker on a rail assembly should operate smooth and easy without binding in either the raised or lowered position.

(6) When lowered, the locking pin should keep the Beverage Maker on the rail assembly. When raised, the locking pin should let the Beverage Maker be removed from the rail assembly.

(7) Installation and removal of the Beverage Maker on a rail assembly should be smooth and easy.

(8) Examine alignment of the platen assembly, chassis assembly, power module assembly, rail guides, electrical connector, and water supply connection. The bottom of the Beverage Maker should rest squarely on a flat surface enabling an easy fit into galley installation locations.

(9) Examine the switch assembly for the brew head assembly that is attached to the right side of the chassis assembly. If parts have been bent or fasteners have become loose, the switch assembly may need to be aligned correctly (refer to the FITS AND CLEARANCES section for more information).

(10) Instruction placards on the front of the Beverage Maker should be complete and easy to read.

3. Recommended Periodic Maintenance

NOTE: It is highly recommended that the relief valve be examined and tested for correct operation every six months or every 1500 flight hours, whichever comes first.

NOTE: Beverage maker should be removed from the aircraft for regular maintenance at a minimum of once a year. In areas where water has high mineral content more frequent maintenance may be required. Water quality and airline experience dictate frequency of maintenance.

NOTE: Failure to perform regular maintenance as described may limit warranty claims.

A. Daily (on aircraft): Fully clean the Beverage Maker infrared sensor assembly, brew cup cavity, and switch panel.

B. Weekly: Clean and inspect for obvious damage. Replace worn components.

C. Every 1500 flight hours or six months (whichever comes first).

(1) Completely flush the entire Beverage Maker water system to remove mineral deposits (refer to the CLEANING Section for cleaning procedures and materials).

(2) Clean or replace vent valve.

(3) Inspect and clean relief valve. If necessary, replace relief valve seat.

(4) Test relief valve (refer to TESTING AND FAULT ISOLATION for calibration instructions).

(5) Tighten all external screws.

(6) Check brew handle operation, springs, and bushings. Replace worn parts and lubricate new parts at installation.

(7) Test the Beverage Maker as described in the TESTING AND FAULT ISOLATION Section. Repair or replace parts so the Beverage Maker meets test requirements.

D. Every 3000 flight hours or 12 months (whichever comes first).

(1) Perform 1500 hour maintenance procedures as outlined in Paragraph 3.C.

(2) Examine brew head flow washer, o-rings, fitting, and brew head gasket. Replace parts that are worn or damaged.

(3) Where TEA option is installed, examine flow washer for tea tube funnel. Replace o-ring and flow washer if worn.

(4) Examine solenoid valve assembly. Replace o-ring if worn.

E. Repeat cycle of maintenance procedures as outlined in Paragraphs 2.C and 2.D.

F. Create and maintain maintenance and repair records for all Beverage Makers by part number and serial number.

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REPAIR

1. General

CAUTION: B/E AEROSPACE RECOMMENDS THE FOLLOWING PROCEDURES BE UTILIZED BY A QUALIFIED, APPROVED, AND CERTIFIED REPAIR FACILITY. THE FOLLOWING INFORMATION IS SUPPLIED AT THE REQUEST OF OPERATING CUSTOMERS. IT IS RECOMMENDED THAT EQUIPMENT REQUIRING THIS LEVEL OF REPAIR BE RETURNED TO THE FACTORY. B/E AEROSPACE ACCEPTS NO LIABILITY FOR THE PERFORMANCE OF THE EQUIPMENT WHEN REPAIRS ARE DONE BY THE CUSTOMER USING THIS INFORMATION.

A. Replace all parts worn or damaged beyond repair. Refer to IPL Figures 1 thru 12 and corresponding Illustrated Parts Lists beginning on page 10001 of this manual, as a guide to identification and location of replaceable components.

NOTE: B/E Aerospace does not recommend the repair of broken metal parts. The integrity of the chassis assembly, power module assembly, rail guides and tank are not assured if these items are broken then welded. These items should be replaced in their entirety.

B. Refer to the TESTING AND FAULT ISOLATION and INSPECTION/CHECK Sections to assist in identification of parts requiring replacement or adjustment.

C. Anchor nuts are permanently attached to various assemblies in the Beverage Maker subassembly. In the event that they become damaged, they may be replaced by drilling out attaching rivets and riveting on new anchor nuts.

D. Some holes in castings use helical coil inserts. In the event that they become damaged, they may be removed and replaced.

E. In the event that tubing is worn or damaged, the tubing should be replaced and not repaired.

2. Electrical

A. Replace any worn or defective parts.

B. Repair or replace any metal parts that have nicks or gouges.

C. Do not repair wiring. Any damaged wiring should be removed and replaced.

D. Refer to the DISASSEMBLY and ASSEMBLY Sections for removal and replacement of electrical components.

E. Controller Board Assembly (refer to CMM 25-33-32).

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ASSEMBLY

1. General

WARNING: DO NOT CONNECT THE BEVERAGE MAKER TO POWER AND WATER SUPPLY DURING ASSEMBLY.

CAUTION: PROTECT ALL COMPONENTS AS THEY ARE ASSEMBLED AND OBSERVE THE NORMAL PRECAUTIONS REGARDING THE CORRECT USE OF TOOLS.

CAUTION: HANDLE THE BEVERAGE MAKER ONLY AT A STATIC SAFE WORK STATION IN A STATIC ELECTRICITY FREE ENVIRONMENT.

CAUTION: MAINTENANCE ENGINEER MUST WEAR A CONDUCTIVE WRIST STRAP TO PROTECT THE PC BOARD ASSEMBLY AGAINST ESD.

A. Refer to the TESTING AND FAULT ISOLATION and INSPECTION/CHECK Sections prior to assembly. This will determine the extent of assembly required for testing procedures after rebuilding the Beverage Maker or removing a faulty component.

B. Refer to the ILLUSTRATED PARTS LIST (IPL) Section for disassembly and assembly relationships. The number in parentheses after the noun name of a part refers to the item number in the IPL Figure.

C. Provide a clean space and working surface before beginning any assembly operation.

NOTE: Installation of some parts may require use of an adjustable-torque wrench due to torque requirements.

NOTE: Beverage residue and mineral deposits can be cooked into part connections making them hard to remove. The cleaning of some parts can help in their installation, especially parts exposed to coffee residue.

D. Before installing electronic components, make sure polarity and position of the component will let the circuit operate correctly.

E. Refer to the figures in the ILLUSTRATED PARTS LIST Section and assemble the Beverage Maker in accordance with the following procedures. Refer to Table 7001 for required consumable items.

CAUTION: DO NOT USE THREAD LOCK ADHESIVE (REFER TO TABLE 7001) ON OR NEAR PLASTIC PARTS. LET PARTS INSTALLED INTO AN ASSEMBLY USING THREAD LOCK ADHESIVE CURE FOR 24 HOURS BEFORE ATTACHING THE ASSEMBLY TO ANY PARTS MADE OF PLASTIC. MAKE SURE TO FULLY CLEAN HANDS AFTER USING THREAD LOCK ADHESIVE SO NO ADHESIVE IS PUT ON OTHER PARTS OF THE UNIT. SOLVENT IN THE ADHESIVE CAN DAMAGE PLASTIC PARTS.

NOTE: Refer to the FITS AND CLEARANCES Section for adjustment of essential assemblies during assembly.

NOTE: Equivalents may be used for consumable materials.

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Table 7001
Assembly Materials and Equipment

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Alodine Chemical Coating	Alodine 1201 CM-50017	Henkel Corp, (V71410) B/E Aerospace (V63367)
Anti-Seize Lubricant, Food Grade	White Paste Lubricant CM-50022	Loctite Corp. (V05972) B/E Aerospace (V63367)
Black Adhesive Sealant	732 Black RTV-103 AD-052	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)
Epoxy Adhesive	E-Cast F-28 Resin with F-14 Hardener AD-50029	United Resin Corp. (V08LA7) B/E Aerospace (V63367)
Heat Sink Compound	10-8109 CM-50019	Waldom Inc. (V72653) B/E Aerospace, Inc. (V63367)
Petroleum Jelly	None CM-50015	Commercially Available B/E Aerospace, Inc. (V63367)
Red Silicone Rubber Adhesive	736 Red AD-50005	Dow Corning Corp. (V71984) B/E Aerospace, Inc. (V63367)
Sleeving, Non-Fray	TU-50039	B/E Aerospace, Inc. (V63367)
Solder	Kester No. 245, No Clean, Low Residue SD-50003	Kester Solder (Div of Litton Systems, Inc.) (V75297) B/E Aerospace Inc. (V63367)
Teflon Tape - 1/4 in	MS-STR-4 TT-7454	Swagelok Companies (V02570) B/E Aerospace, Inc. (V63367)
Thread Lock Adhesive	242 Blue AD-300	Loctite Corp. (V05972) B/E Aerospace, Inc. (V63367)
Thread Lubricant	Never Seez NSWT-14 CM-50016	Bostik Inc. (V1Y0Z7) B/E Aerospace, Inc. (V63367)
Translucent Silicone Rubber Adhesive/Sealant	732 Clear RTV RTV 108 AD-50002	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)

2. Procedure

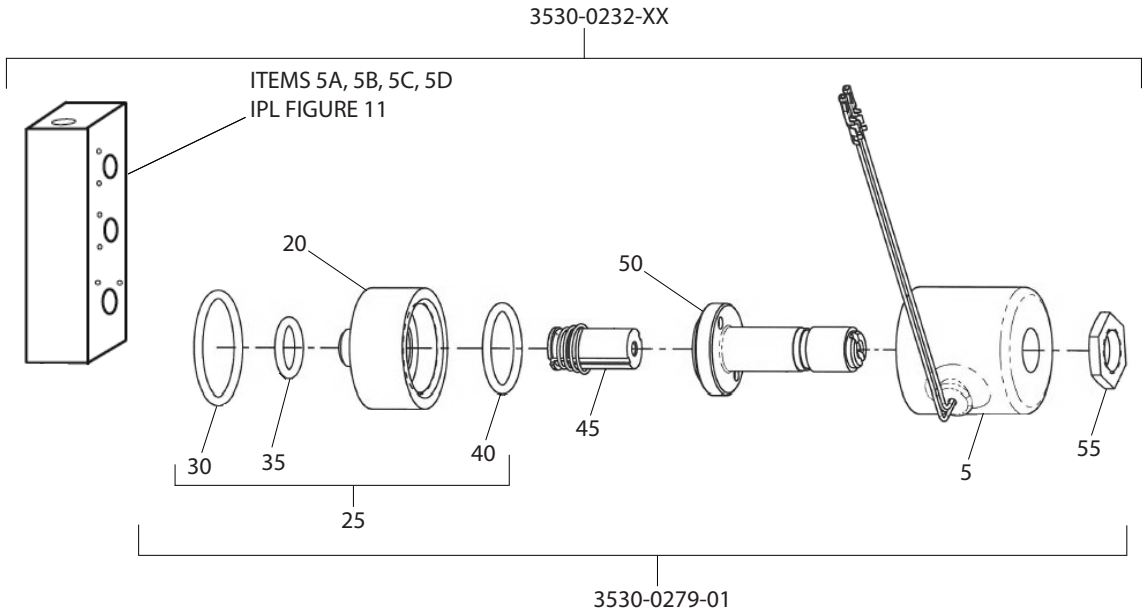

Figure 7001
Solenoid Valve Assembly (PN 3530-0279-01) / Manifold Assembly (PN 3530-0232-XX)

A. Solenoid Valve Assembly (PN 3530-0279-01) / Manifold Assembly (PN 3530-0232-XX) (refer to Figure 7001)

(1) Only if necessary for repair, put crimp terminals (15) on the wiring to the solenoid coil (5), using a Molex terminal crimp tool part number 63819-0000 or equivalent.

(2) Put plunger (45) in the solenoid guide (50), so that the seal end on the base of the plunger faces the open end of the solenoid guide.

(3) Install o-ring (40) in valve body (20). Put solenoid guide (50) on valve body (20). Special tools are recommended for assembly (refer to Table 9001 and Figure 9008).

(4) Install o-rings (30, 35) into the valve body (20). Install valve body (20) into manifold block (5, IPL Figure 11). Torque the solenoid guide (50) to 20 in-lbs to 30 in-lbs. When the torque on solenoid guide (50) is obtained, the valve body (20) will rotate until, torque listed above is obtained.

(5) Put coil assy (5) on solenoid guide (50).

(6) Put nut (55) on solenoid guide (50). Apply a torque of 18 to 22 in-lbs to the nut (55).

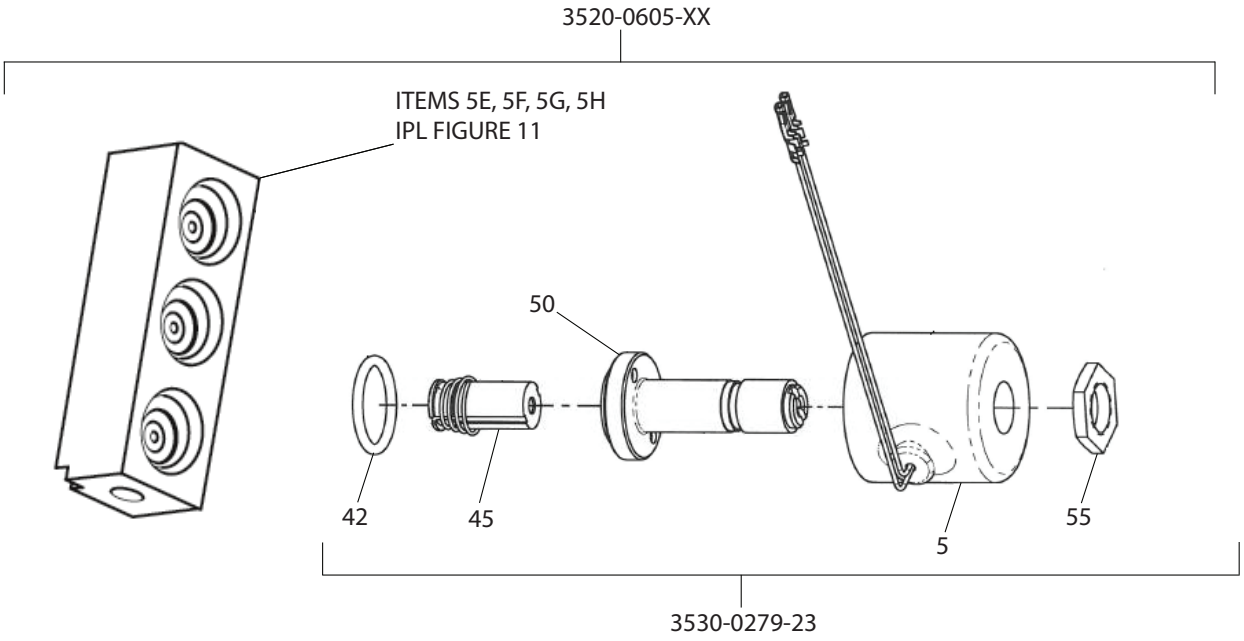

Figure 7002
Solenoid Valve Assembly (PN 3530-0279-03) / Manifold Assembly (PN 3530-0605-XX)

B. Solenoid Valve Assembly (PN 3530-0279-23) / Manifold Assembly (PN 3530-0605-XX) (refer to Figure 7002)

(1) Only if necessary for repair, put crimp terminals (15) on the wiring to the solenoid coil (5), using a Molex terminal crimp tool part number 63819-0000 or equivalent.

(2) Put plunger (45) in the solenoid guide (50), so that the seal end on the base of the plunger faces the open end of the solenoid guide.

(3) Put the o-ring (42) in the solenoid block (5, IPL Figure 11).

(4) Put solenoid guide (50) on solenoid manifold (5, IPL Figure 11). Special tools are recommended for assembly (refer to Table 9001 and Figure 9008). Torque the solenoid guide (50) into solenoid manifold (5, IPL Figure 11) to 20 in-lbs to 30 in-lbs. A minimum amount of thread lubricant should be applied to fill the screw recesses and screw cavity to allow for a smooth finish. Any excess thread lubricant should wiped clean.

NOTE: The thread lubricant should be applied only to clean dry surface.

(5) Put coil assy (5) on solenoid guide (50).

(6) Put nut (55) on solenoid guide (50). Apply a torque of 18 to 22 in-lbs to the nut (55).

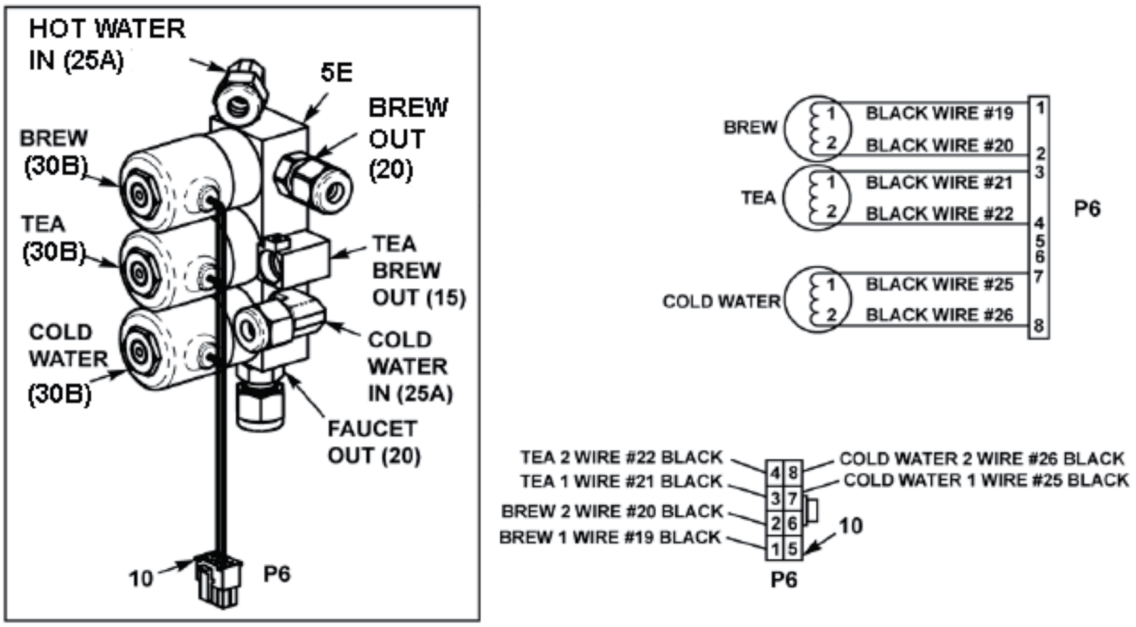

Figure 7003
Manifold Assembly (PN 3520-0605-01)

C. Manifold Assembly (refer to Figures 7003 through 7007)

(1) Put thread seal tape on regulator fitting (15) and then put regulator fitting on manifold body (5A, 5D, 5E or 5H).

(2) Put thread seal tape on straight fittings (20) and then put straight fittings on manifold body (5).

(3) Put thread seal tape on elbow fittings (25A or 25B) and then put elbow fittings on manifold body (5).

CAUTION: CORRECTLY CRIMP TERMINALS ON WIRE FOR SOLENOID COIL USING A MOLEX TERMINAL CRIMP TOOL PART NUMBER 63819-0000 OR EQUIVALENT.

(4) Where necessary, install crimp terminals (15, IPL Figure 11) on solenoid coil (10, IPL Figure 11) wires.

CAUTION: INSTALL CRIMP TERMINALS CORRECTLY IN THE P6 CONNECTOR, OR THE CONNECTOR MAY BE DAMAGED, RESULTING IN DAMAGE TO THE CONTROLLER BOARD.

(5) Put P6 connector (10) in solenoid valve assembly (30A or 30B) (refer to Figure 7004).

(6) Refer to ASSEMBLY, Paragraph 2.M.(2) for installation of manifold assembly (545, IPL Figures 2, 2A, 2B, 2C) on chassis assembly (10, IPL Figures 2, 2A, 2B, 2C).

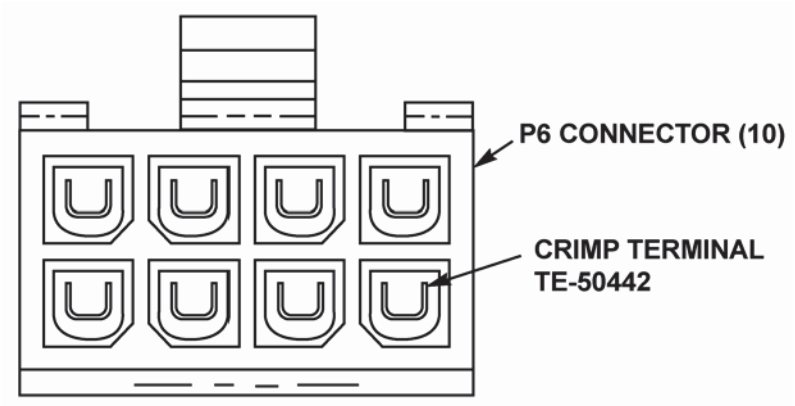

Figure 7004
Correct Crimp Terminal Alignment in P6 Connector (Terminal Side View of Connector)

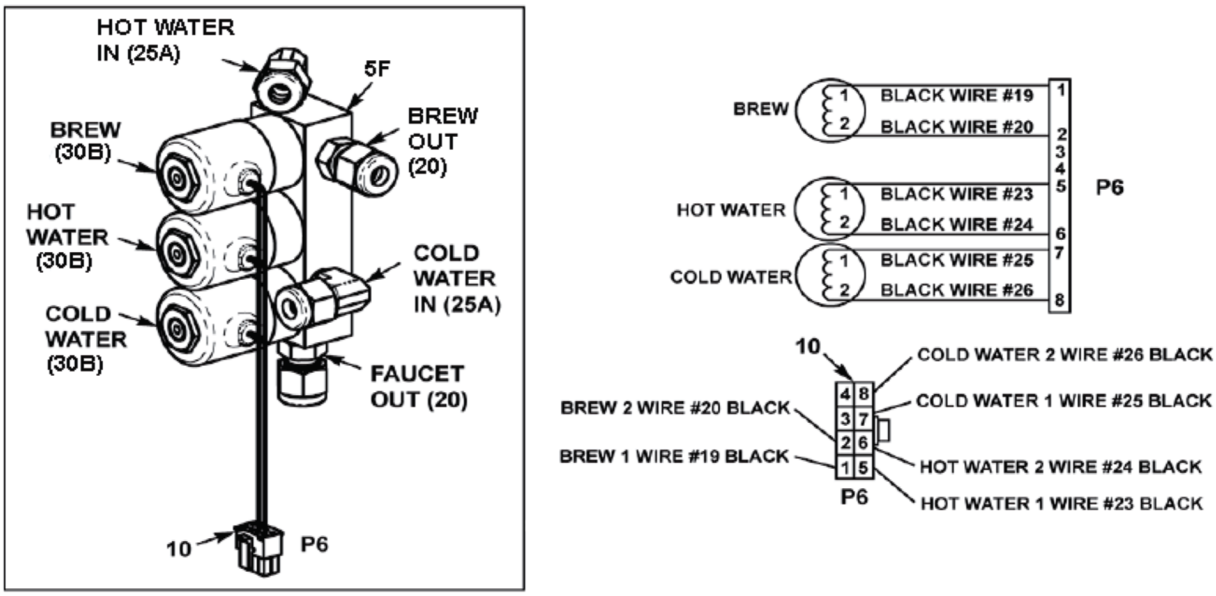

Figure 7005
Manifold Assembly (PN 3520-0605-05)

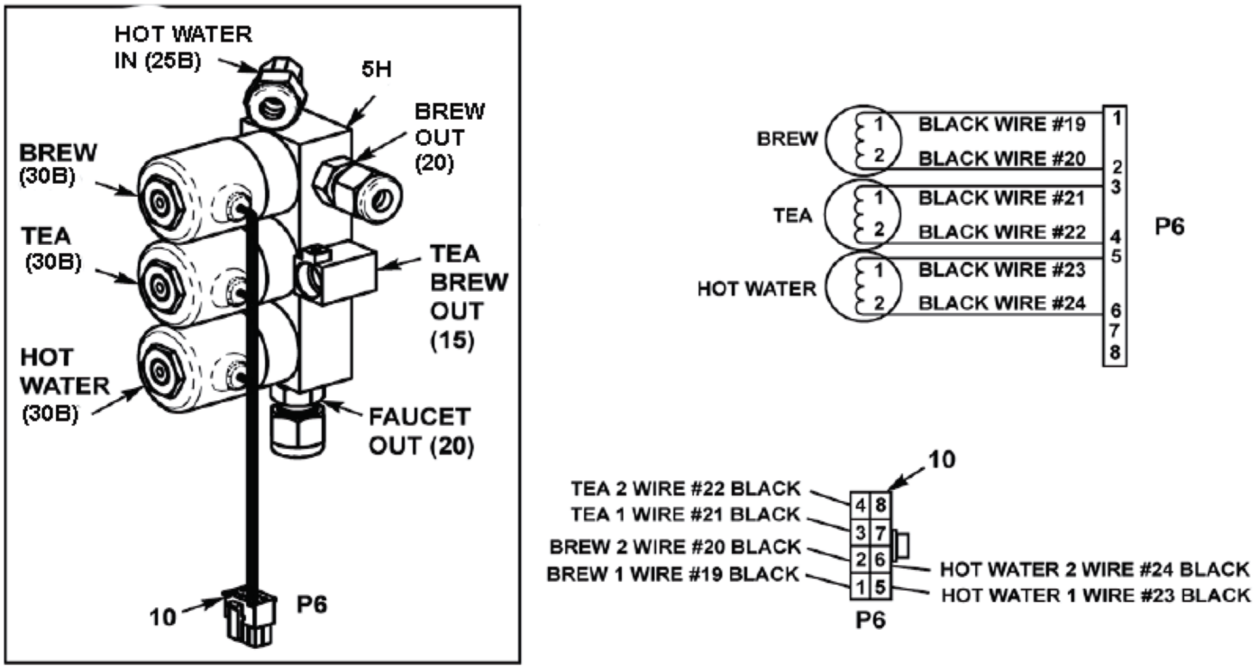

Figure 7006
Manifold Assembly (PN 3520-0605-03)

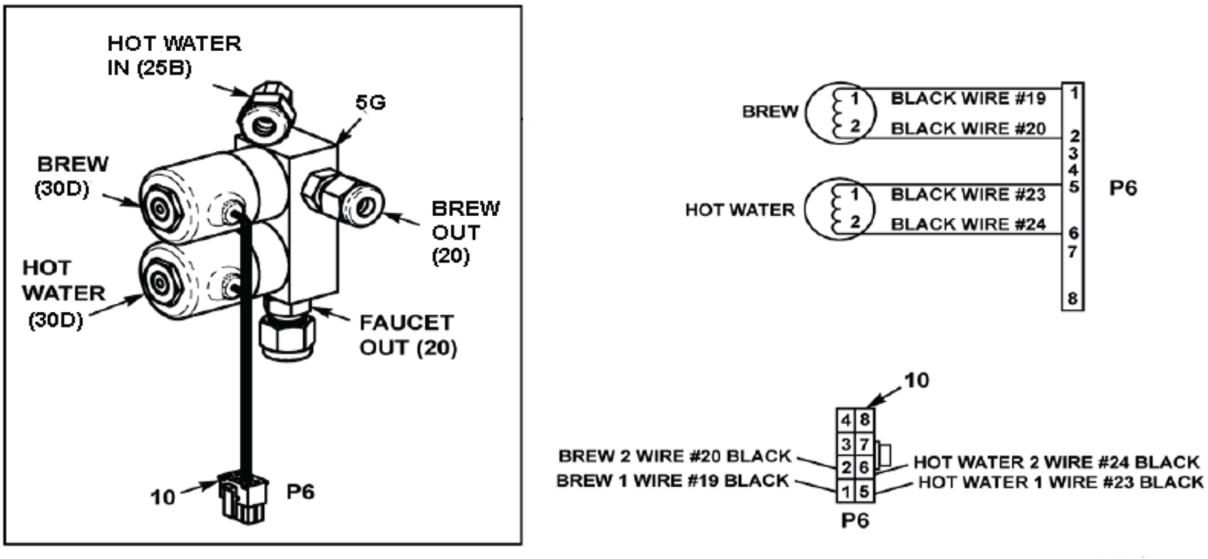

Figure 7007
Manifold Assembly (PN 3520-0605-07)

D. Server Retainer Assembly (refer to Figure 7008)

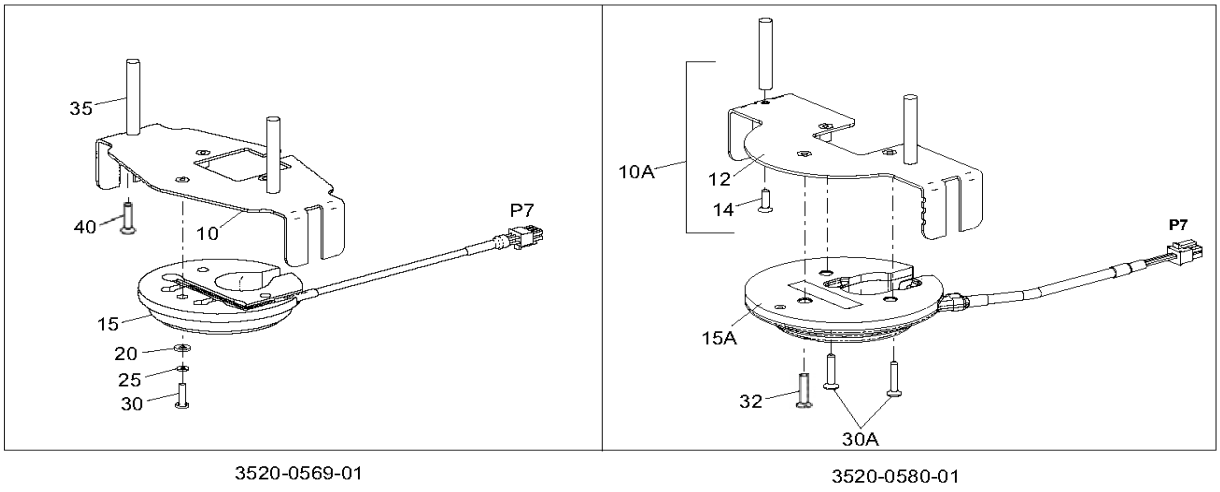

Figure 7008
Assembly of Server Retainer Assembly

NOTE: Thread Lock Adhesive Part Number AD-50029 on the retainer assembly requires 24 hours of curing time before the Liquid Level Sensor (LLS) assembly may be installed on the retainer assembly.

CAUTION: A NYLON PATCH ON THE SCREWS (30A) WILL CAUSE RESISTANCE TO TURNING OF THE SCREWS (30A). DO NOT OVERTIGHTEN THE SCREWS (30A) ON THE RETAINER PLATE AND ROD ASSY (10A) AND INFRARED SENSOR ASSY (15A) OR DAMAGE TO THE INFRARED SENSOR ASSY (15A) CAN RESULT. IN THE EVENT THAT THE SCREWS (30A) ARE OVER-TIGHTENED, STRESS FRACTURES MAY DEVELOP IN THE INFRARED SENSOR ASSY (15A).

(1) For PN 3520-0569-01, Put two LLS retaining rods (35) on the retainer assembly (10) using two screws (40). Secure screws (40) with epoxy adhesive AD-50029 and tighten to a minimum of 15 in-lbs (1.70 N-M).

(2) For PN 3520-0580-01, Put two LLS retaining rods (13) on the retainer assembly (12) using two screws (14). Secure screws (14) with epoxy adhesive AD-50029 and tighten to a minimum of 15 in-lbs (1.70 N-M).

(3) For PN 3520-0569-01, Put liquid level sensor (LLS) assembly (15) on the retainer assembly (10) using three flat washers (20), three lock washers (25), and three screws (30).

(4) For PN 3520-0580-01, Put liquid level sensor (LLS) assembly (15A) on the retainer assembly (10A) using screw (32), and two screws (30A). Apply translucent silicone rubber sealant AD-50002 to threads before installing. Tighten screws (30A) to 5.0 in-lb (0.57 N-m). Tighten screw (32) to 1.0 in-lb (0.11 N-m). Fill screw cross recesses with sealant AD-50002.

NOTE: A minimum amount of sealant should be applied to fill the screw recesses and screw cavity to allow for a smooth finish. Any excess sealant should wiped clean.The sealant should be applied only to clean dry surface.

(5) Refer to ASSEMBLY, Paragraph 2.M.(7) for installation of server retainer assembly (545, IPL Figures 2, 2A, 2B, 2C) on chassis assembly (10, IPL Figures 2, 2A, 2B, 2C).

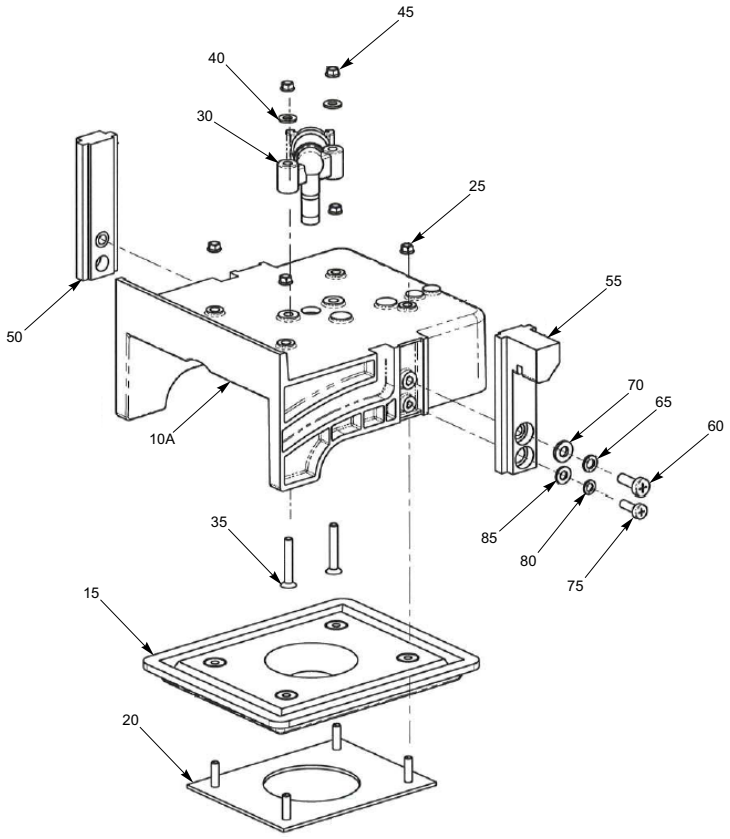

Figure 7009
Assembly of Brewhead Assembly

E. Brew Head Assembly (refer to Figure 7009)

(1) Install the gasket (15) on the brew head (10A) using the gasket retainer (20) and four lock nuts (25).

(2) Install the brew fitting (30) in the brew head using two screws (35), two flat washers (40), and two lock nuts (45).

(3) Install the right sight guide (55) on the brew head (10A) using two screws (60 and 75) by applying torque of 10 to 12 in-lbs, two lock washers (65 and 80), and two flat washers (70 and 85).

(4) Install the left side guide (50) on the brew head (10A) using two screws (60 and 75) by applying torque of 10 to 12 in-lbs, two lock washers (65 and 80), and two flat washers (70 and 85).

(5) Refer to ASSEMBLY, Paragraph 2.M.(9) for installation of brewhead assembly (300, IPL Figures 2, 2A, 2B, 2C) on chassis assembly (10, IPL Figures 2, 2A, 2B, 2C).

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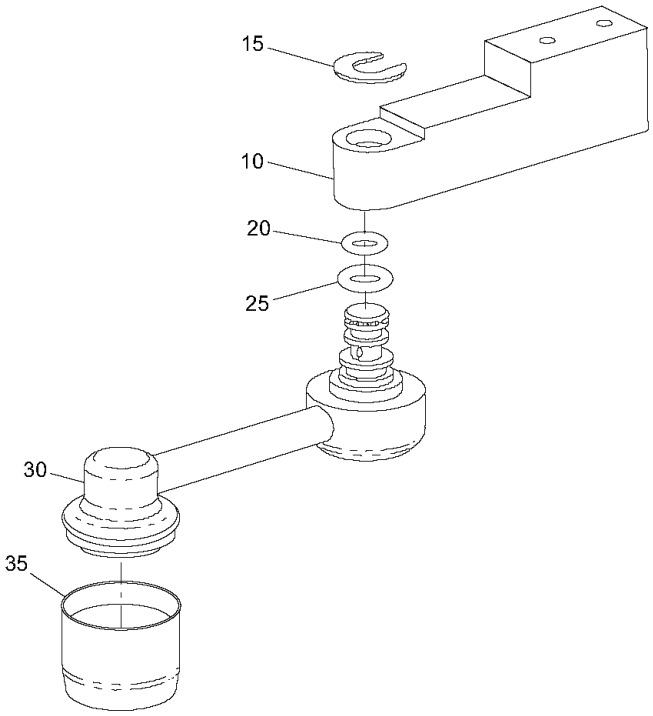

Figure 7010
Assembly of Swivel Faucet Assembly

F. Swivel Faucet Assembly (refer to Figure 7010)

(1) Put upper o-ring (20) and lower o-ring (25) on welded faucet assembly (30).

(2) Put welded faucet assembly (30) in swivel faucet bar (10).

(3) Put retaining ring (15) on welded faucet assembly (30).

(4) Put aerator (35) on welded faucet assembly (30).

(5) Refer to ASSEMBLY, Paragraph 2.M.(8) for installation of swivel faucet assembly (275, IPL Figures 2B, 2C) on chassis assembly (10, IPL Figures 2, 2A, 2B, 2C), if option is installed.

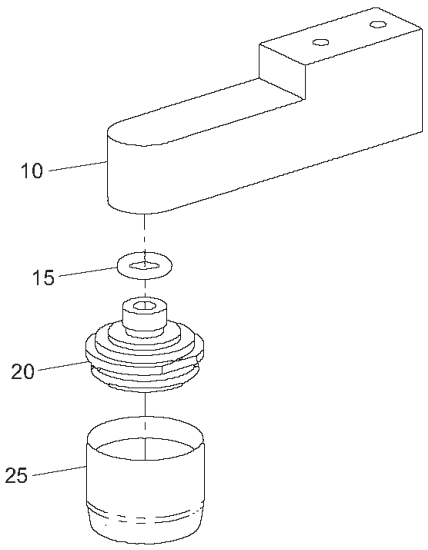

Figure 7011
Assembly of Faucet Assembly

G. Faucet Assembly (refer to Figure 7011)

(1) Install o-ring (15) on aerator adapter (20).

(2) Install aerator (25) and aerator adapter (20) on faucet bar (10).

(3) Refer to the ASSEMBLY Section, Paragraph 2.M.(8) for installation of faucet assembly (270, IPL Figures 2, 2A) on chassis assembly (10, IPL Figure 2).

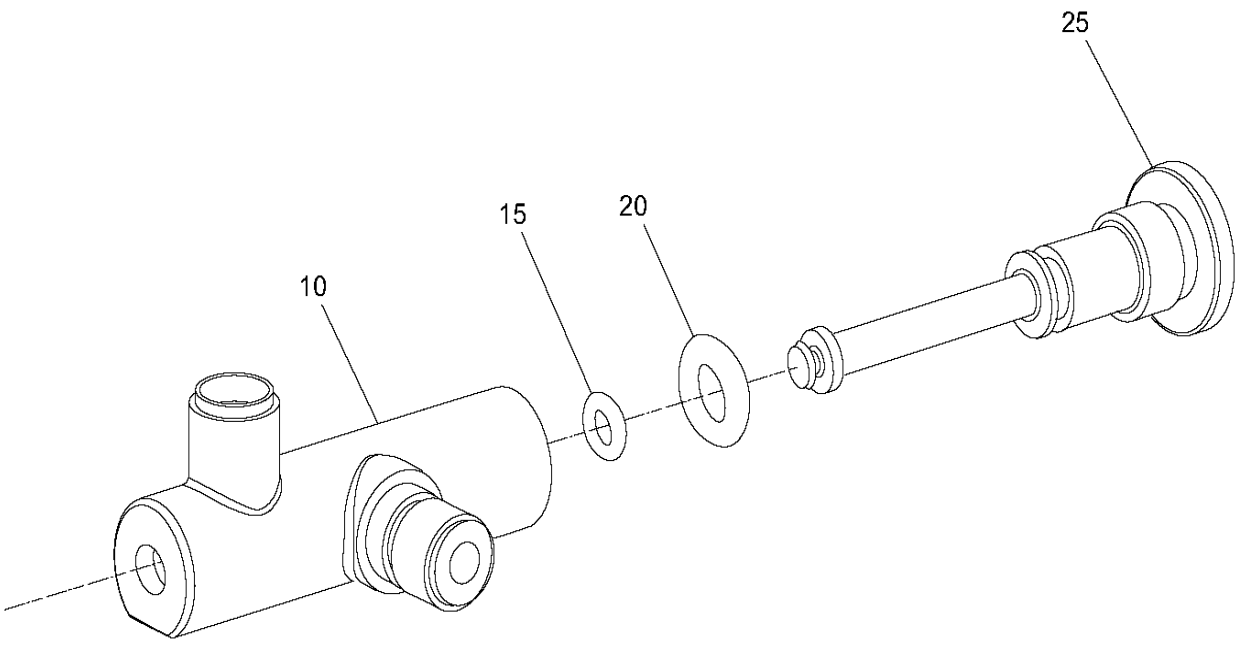

Figure 7012
Assembly of Drain Valve Assembly

H. Drain Valve Assembly (refer to Figure 7012)

(1) Put upper o-ring (20) and lower o-ring (15) on stem assembly (25).

(2) Put stem assembly (25) in body (10).

(3) Refer to ASSEMBLY, Paragraphs 2.M.(15)(b) for installation of drain valve assembly (140, IPL Figures 2, 2A, 2B, 2C) on support connector (115, IPL Figures 2, 2A, 2B, 2C).

J. Controller Board (refer to CMM 25-33-32)

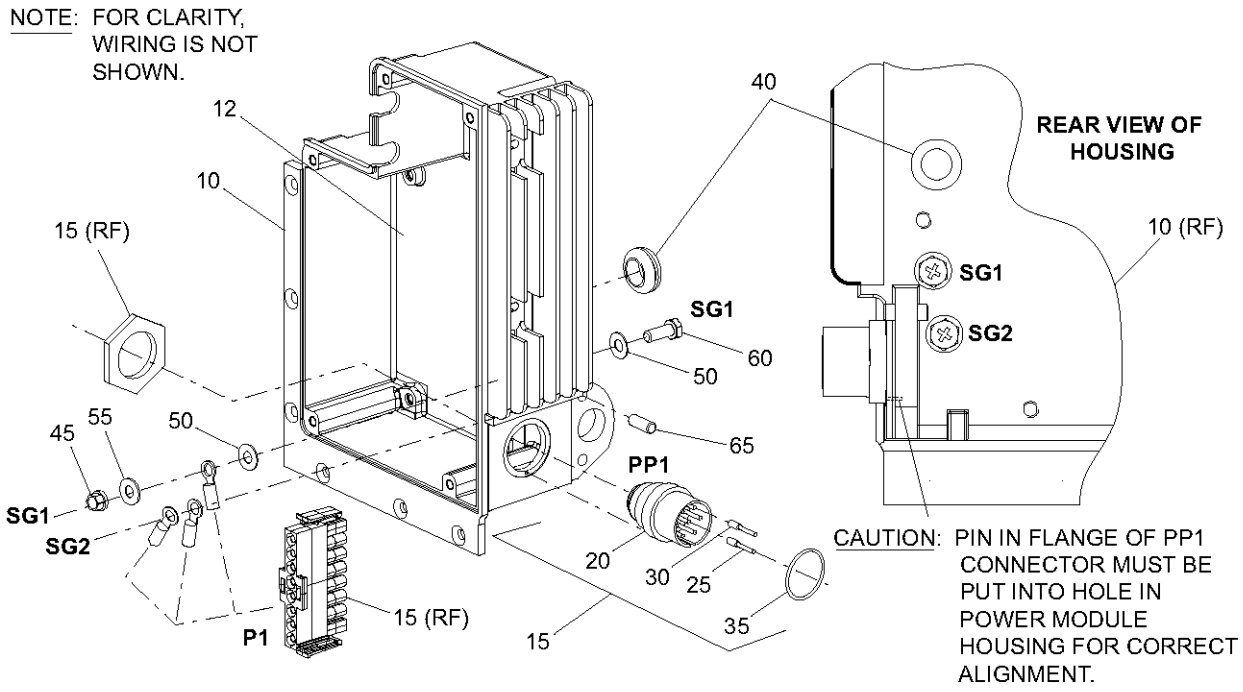

Figure 7013
Assembly of Power Module Assembly

K. Power Module Assembly (refer to Figure 7013)

(1) Put the grommet (40) in hole in back of the power module housing assembly (10).

NOTE: This grommet (40) covers the safety latch switch on the back of the controller board assembly.

CAUTION: THE PP1 AND P1 CONNECTORS MUST BE INSTALLED ON THE HARNESS ASSEMBLY AFTER THE HARNESS ASSEMBLY IS PUT THROUGH THE GROMMET AND HOLE IN THE POWER MODULE HOUSING ASSEMBLY (REFER TO FIGURE 7013).

(2) Install PC board insulator (12) on the power module housing assembly (10) using red silicone rubber adhesive (refer to Table 7001).

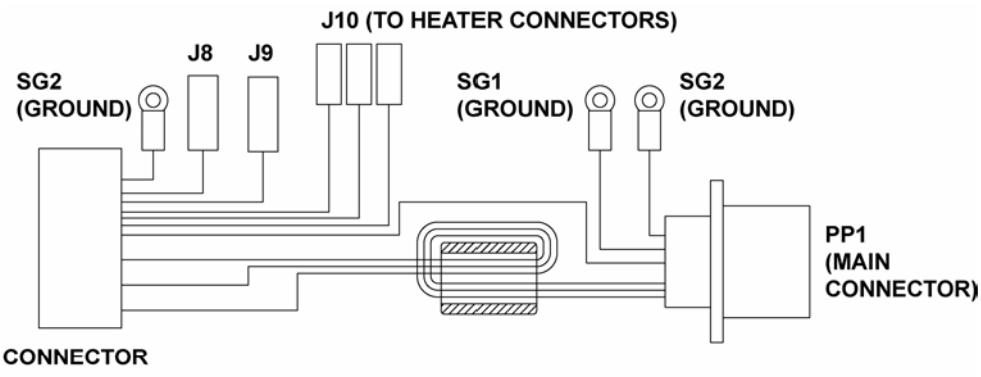

Figure 7014
Harness assembly for Power Module Assembly

NOTE: The hex nut for the PP1 connector must be put on the harness assembly and the PP1 connector must be put into the power module housing assembly before the PP1 connector is attached to the harness assembly (refer to Figure 7014).

(3) Install pin (65) in power module housing assembly (10).

NOTE: This pin helps align the support connector (115, IPL Figures 2, 2A, 2B, 2C) with the power module assembly (55, IPL Figures 2, 2A, 2B, 2C).

(4) Install the SG1 and SG2 ground stack-ups for the harness assembly (15) (refer to Figure 7009) in the power module housing assembly (10).

(a) Install two lock nuts (45), four flat washers (55), wire terminals for harness assembly (15), two flat washers (50), and two hex screws (60) on the power module housing assembly (10).

(5) Refer to ASSEMBLY, Paragraph 2.M.(13) for installation of power module assembly.

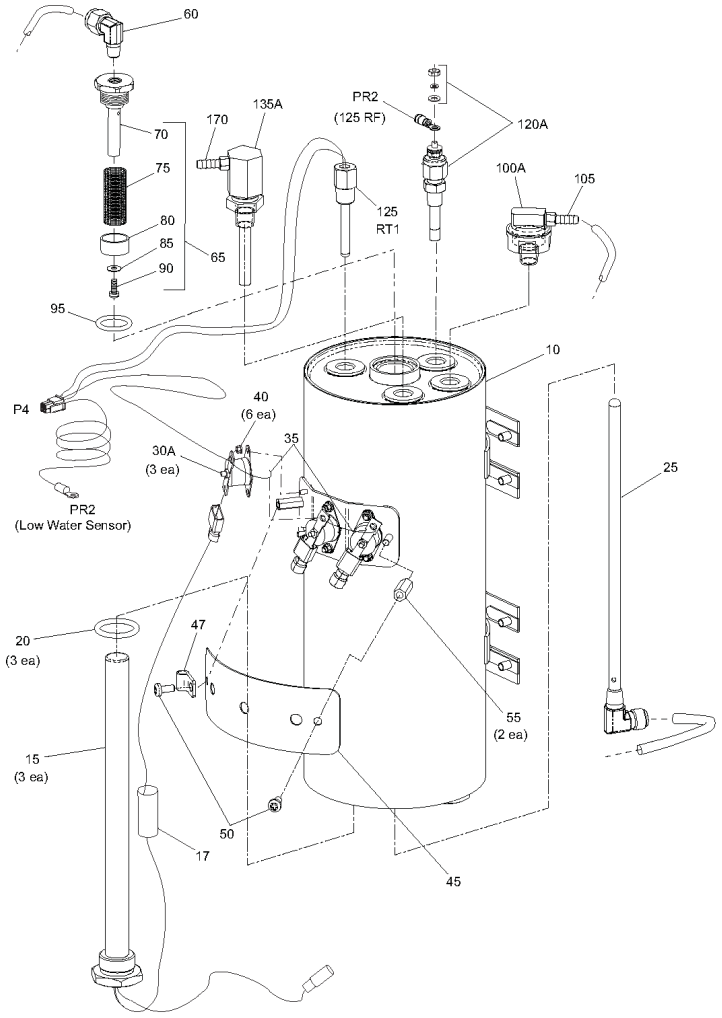

Figure 7015
Assembly of tank Assembly

L. Tank Assembly (refer to Figures 7015, 7016, and 7017)

(1) Install three tank heater assembly (15) and three o-rings (20) in welded tank assembly (10).

(a) Put o-ring (20) on each tank heater assembly (15).

(b) Turn tank heater assembly (15) into threaded hole in welded tank assembly (10) to finger tightness, then use socket (refer to Figure 9001) and torque wrench to turn heater assembly to 10 ft/lbs with a tolerance of -1/+0 ft/lbs torque.

NOTE: Apply Thread Lubricant (refer to Table 7001) to the threads of heater assembly and install tank heater assembly (15) correctly in the bottom of the welded tank assembly (10) (refer to Figure 7015).

NOTE: Move wiring for tank heater assembly (15) and connect it with wire tie downs to keep it from touching the welded tank assembly (10).

NOTE: Only if removed for replacement, put connector (15B), connector pin (15C) and terminal (15A) on heater assy tank (15D).

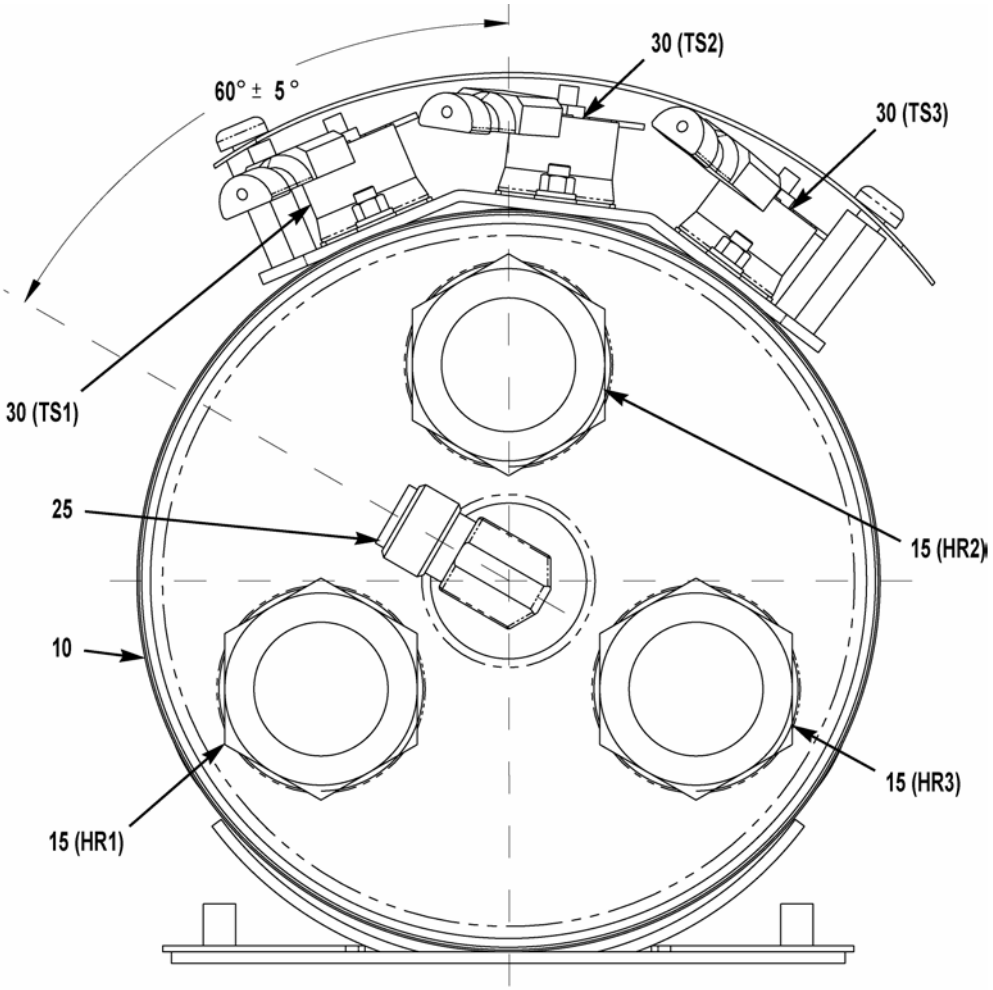

Figure 7016
Bottom View of Tank Assembly

(2) Install tube assembly (25) in welded tank assembly (10) thread seal tape (refer to Table 7001). Align elbow fitting (60) (refer to Figure 7007).

(3) Install pressure relief valve assembly (130) in top of welded tank assembly (10) using thread seal tape (refer to Table 7001 and Figure 7018).

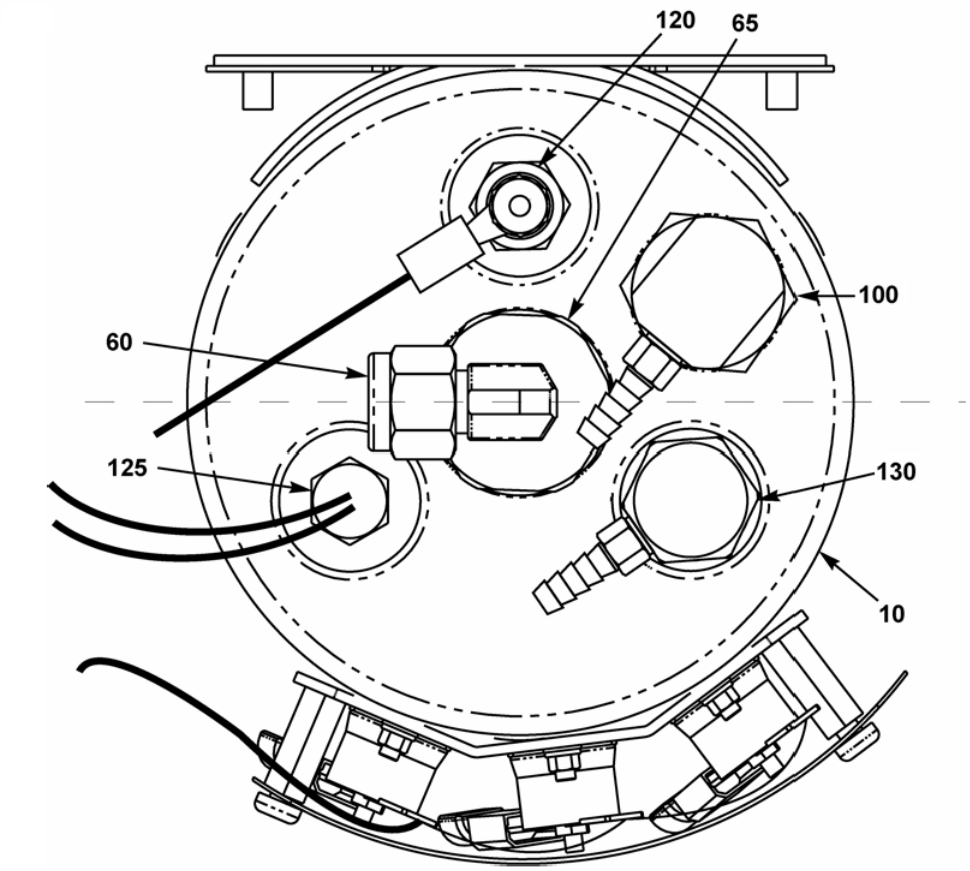

Figure 7017
Top View of Tank Assembly

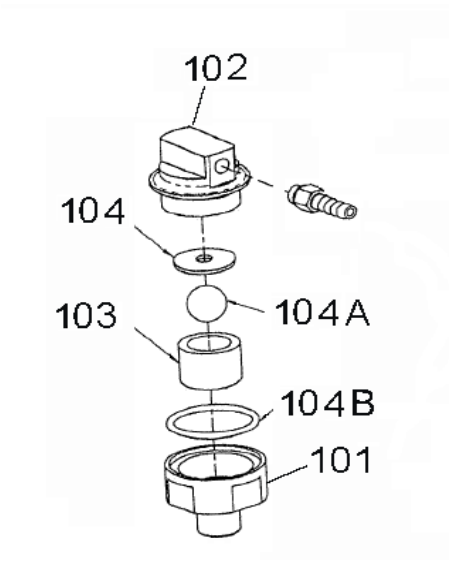

Figure 7018
Assembly of Vent Valve Assembly

(4) Install vent valve assembly (100) in top of welded tank assembly (10) using thread seal tape (refer to Figure 7018).

(a) Install the gasket (110) on the fitting (115) to assemble the barbed fitting (105).

(b) Install the barbed fitting (105) on the vent valve assembly (100).

(c) Fully insert washer (104) into body (102).

(d) Insert liner (103) into body (102).

(e) Insert ball (104A) into liner (103).

(f) Install o-ring (104B) onto adapter (101).

(g) Tighten body (102) onto adapter (101).

(5) Assemble the filter assembly (65) before installing it in the welded tank assembly (10).

(a) Install the filter screen (75) and retainer (80) on the water jet (70) using a screw (90) and lock washer (85).

(6) Install an elbow fitting (60) on the filter assembly (65) using thread seal tape.

(7) Install the filter assembly (65) and o-ring (95) in the top of the welded tank assembly (10).

(a) Turn the filter assembly (65) to finger tightness in the threaded hole in center of the top of the welded tank assembly (10) then use socket (refer to Figure 9001) with torque wrench to turn filter assembly to 9 to 10 ft/lbs torque.

(8) Install the RTD assembly (125) in the top of the welded tank assembly (10) using thread seal tape.

(9) Install the low water sensor (120) in the welded tank assembly (10) using thread seal tape.

(10) Install three thermostats (30) on the welded tank assembly (10) using six lock nuts (40).

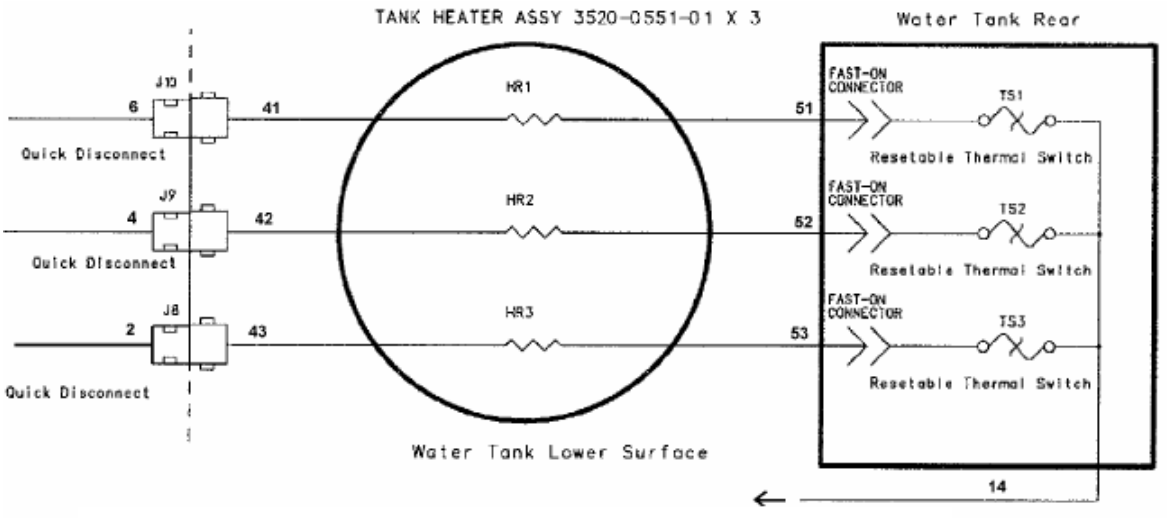

Figure 7019
Schematic of Tank Heater Assembly Wiring

CAUTION: WHEN HEATING SOLDER, TURN TANK SO HOT SOLDER FALLS AWAY FROM THE TANK AND THERMOSTATS.

(11) Install two wire busses (35) and one wire from RTD assembly (125) on the upper terminals of the over temperature thermostats (30) using solder.

(12) Attach RTD assembly (125) wire that has ring terminal on it to the low water sensor (120) using hex nut.

(13) Attach plug of RTD assembly (125) to controller board (75, IPL Figures 2, 2A, 2B, 2C). Power module assembly (55, IPL Figures 2, 2A, 2B, 2C) and controller board (75, IPL Figures 2, 2A, 2B, 2C) may need to be installed before attaching wiring plug from RTD assembly (125).

(14) Install two hex spacers (55) on the welded tank assembly (10).

(15) Install a thermal switch cover (45) on the two hex spacers (55) using two screws (50).

(16) Refer to ASSEMBLY, Paragraph 2.M.(14) for installation of tank assembly (15) (refer to Figure 7034) on chassis assembly (10).

M. Beverage Maker Subassembly (refer to IPL Figures 2, 2A, 2B, 2C, and Figures 7020, 7021, 7022, 7023, 7024, 7025, 7026, 7027, 7028, 7029, 7030, 7031, 7032, 7033, 7034, 7035, 7036, 7037, and 7038).

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(1) Install the lockdown lever on the chassis assembly (10) (refer to Figure 7020).

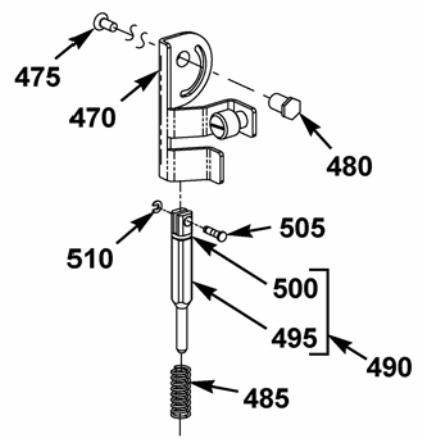

Figure 7020
Assembly of the Lockdown Lever

(a) Install locating pin assembly (490) by applying petroleum jelly (refer to Table 7001) on locating pin lever (470) using retaining ring (510) and pin (505).

1 Install the o-ring (500) by applying thread lubricant (refer to Table 7001) on the machined pin (495) that make up the locating pin assembly (490).

(b) Coat the locating pin assembly (490) and helical spring (485) with petroleum jelly and put them into the hole in the boss in the left corner inside the chassis assembly (10) server cavity.

NOTE: The platen assembly (185 or 195) will have to be installed on the chassis assembly (10) to hold the helical spring (485) in the chassis assembly.

(c) Install the locating pin lever (470) on the chassis assembly (10) using a screw (475) and pin pivot assembly (480).

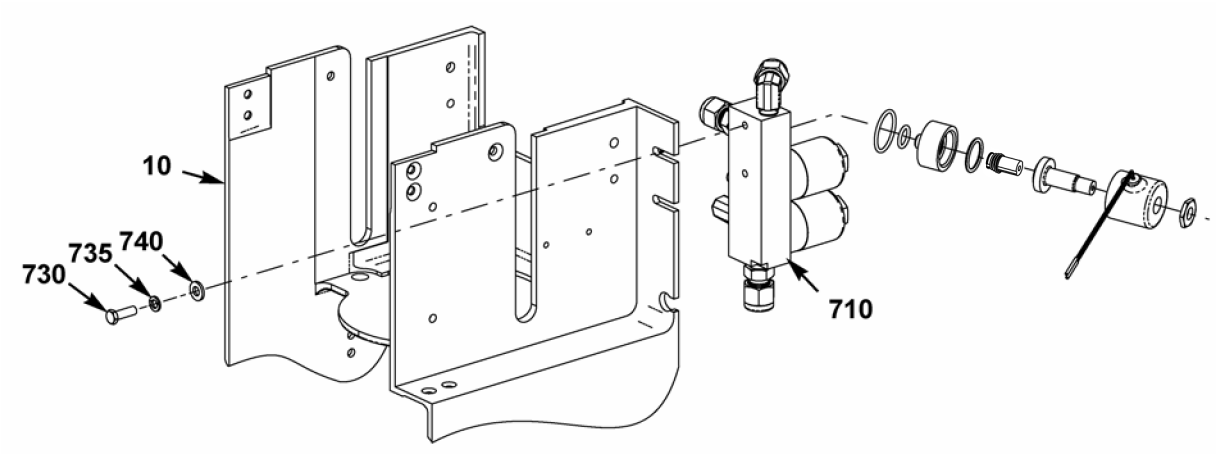

Figure 7021
Installation of Manifold Assembly

(2) Install manifold assembly (710, 715, 720, or 725) on the chassis assembly (10) (refer to Figure 7021).

(a) Install the manifold assembly (710, 715, 720, or 725) on chassis assembly (10) using two screws (730), two lock washers (735), and two flat washers (740).

(b) Attach wiring connections for the manifold assembly (710, 715, 720, or 725) in the power module assembly (55). Install the power module assembly (55) on the chassis assembly (10) before attaching wiring.

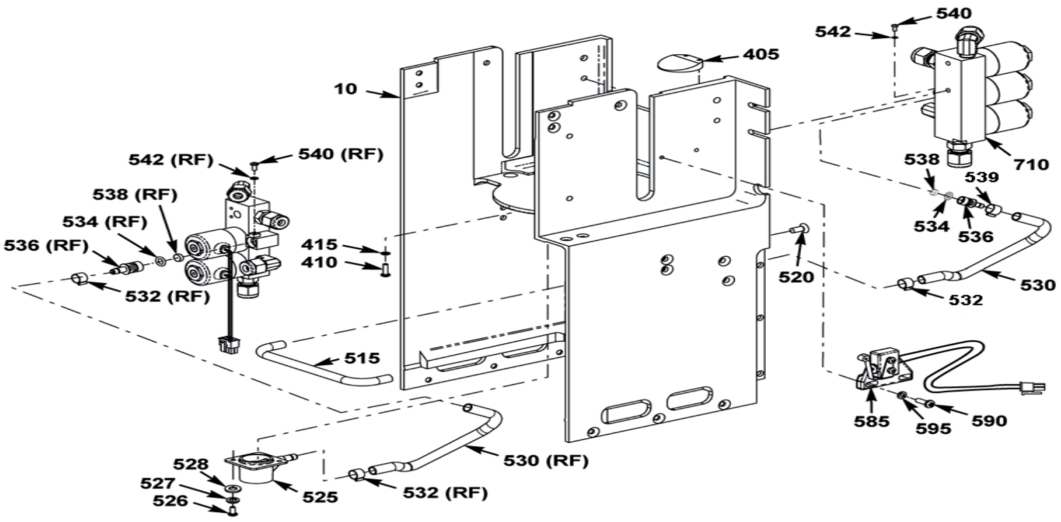

Figure 7022
Installation of Tea Tube Funnel

(3) Install the cup retainer (405) on the chassis assembly (10) using two lock washers (415) and two screws (410) (refer to Figure 7022).

(4) Install handle (515) on chassis assembly (10) using two screws (520).

(5) Install the tea tube funnel (525) on the chassis assembly (10), where this option installed.

NOTE: Apply black adhesive sealant (refer to Table 7001) on the perimeter of the tea tube funnel (525) prior to installation on the chasis assembly (10).

(a) Install a tube (530) on the tea tube funnel (525) using a tube clamp (532).

NOTE: DELETED

(b) Install tea tube funnel (525) on chassis assembly (10) using two screws (526), two lock washers (527), and two flat washers (528).

(c) Attach tea tube (530) to regulator cartridge (536) using a tube clamp (539).

(d) Put o-ring (534) on regulator cartridge (536).

NOTE: Apply a thin layer of thread lubricant (refer to Table 7001) on the outer surface of o-ring (534). Remove excessive lubricant, if any.

(e) Put flow washer (538) in regulator fitting (536). Put the concave side of flow washer (538) so it is facing out of the regulator fitting (536).

(f) Attach regulator cartridge (536) to the correct elbow fitting of the manifold assembly (710A or 725A) using a screw (540) and lock washer (542).

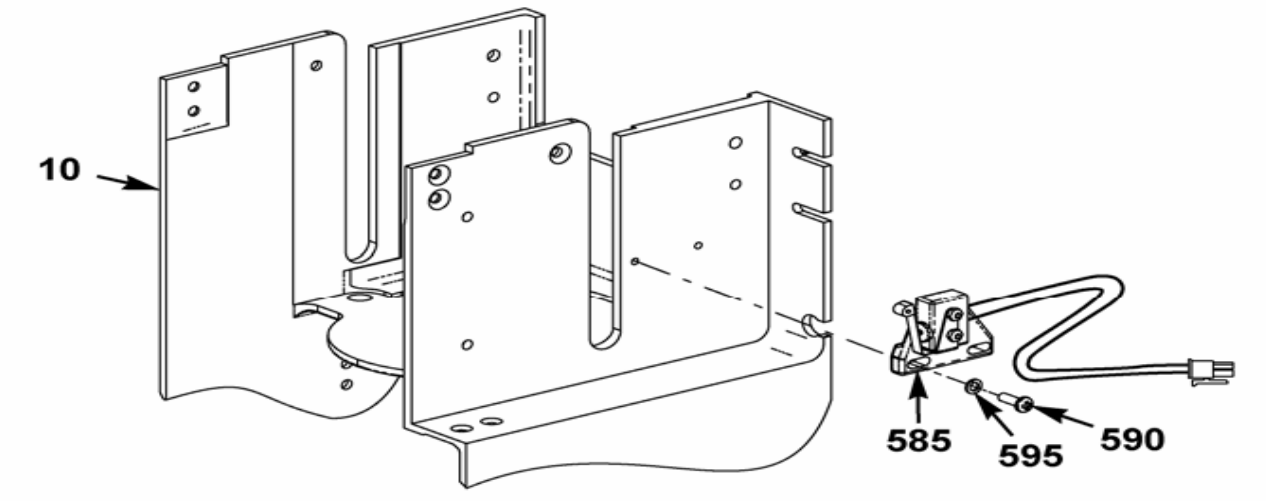

Figure 7023
Installation of Switch Assembly

(6) Attach the switch assembly to the chassis assembly (10) (refer to Figure 7023).

(a) Put the harness assembly (65) and wire for the switch assembly (585), into the grommet (67), and then put the grommet into the slot in the chassis assembly (10) next to the manifold assembly (710, 715, 720, or 725).

(b) Attach the switch assembly (585) to the right side of the chassis assembly (10) using two lock washers (595) and two screws (590).

NOTE: The screws holding the switch assembly on the chassis assembly should be tightened only after the switch actuator is correctly aligned with the guide on the brewhead assembly (refer to FITS AND CLEARANCES).

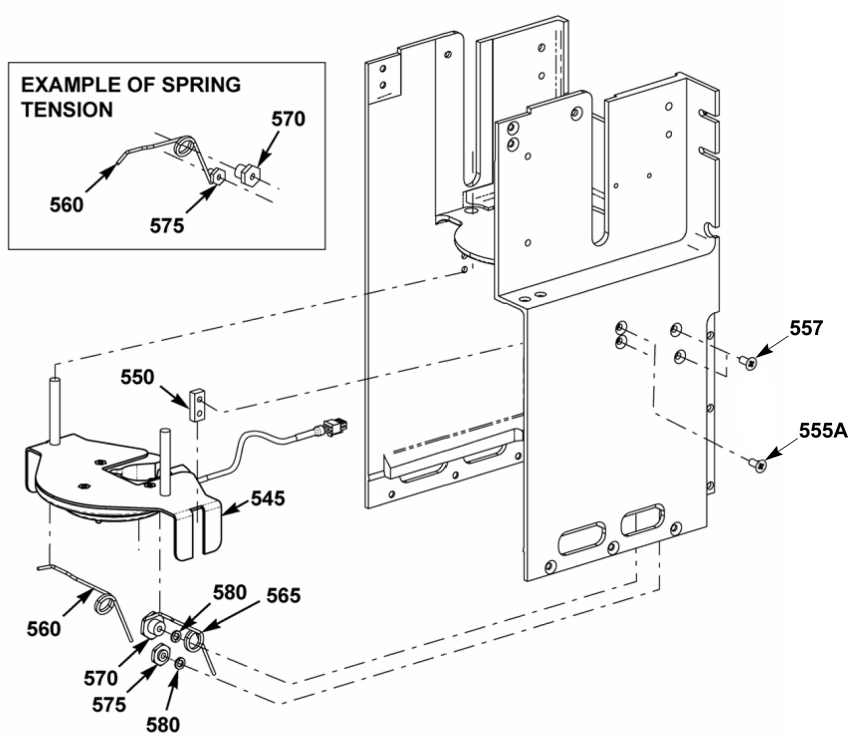

Figure 7024
Installation of Server Retainer Assembly

(7) Install the server retainer assembly (545) on the chassis assembly (10) (refer to Figure 7024). Refer to ASSEMBLY, Paragraph 2.D for assembly of the server retainer assembly (545).

(a) Put the posts for the server retainer assembly (545) into holes in the chassis assembly (10).

(b) Install two LLS guide assembly (550) on the chassis assembly (10) using four screws (557).

(c) Install two spring guide bushings (575) on the chassis assembly (10) using two lock washers (580) and two screws (555A).

CAUTION: THE TWO SERVER RETAINER SPRINGS MUST BE PUT UNDER TENSION DURING INSTALLATION, SO THAT THEY HOLD UP THE SERVER RETAINER ASSEMBLY CORRECTLY.

(d) Install two server retainer springs (560, 565), on the chassis assembly (10) using two spring bushings (570), two lock washers (580) and two screws (555A). Put the bottom legs of the server retainer springs (560, 565) in front of the spring guide bushings (575) (refer to Figure 7023).

NOTE: Apply petroleum jelly (refer to Table 7001) on server retainer springs (560, 565) prior to installation.

(e) Install two spring guide bushings (575) on the chassis assembly (10) using two lock washers (580), and two screws (555A).

(f) Connect the plug for the server retainer assembly (545) to the controller board (75). The cover (45) of the power module assembly (55) may need to be removed to get access to the controller board assembly (75).

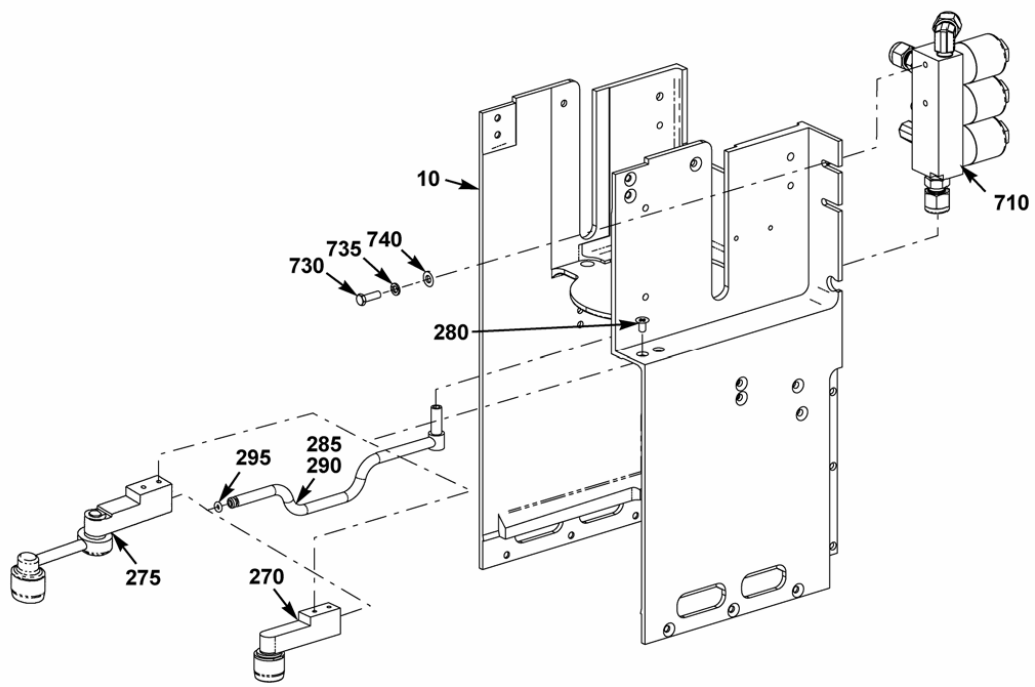

Figure 7025
Installation of Faucet Assembly or Swivel Faucet Assembly

(8) Install the faucet assembly (270) or swivel faucet assembly (275) on the chassis assembly (10) (refer to Figure 7025).

Table 7002
Faucet or Swivel Faucet Assy Part Number Reference

UNIT PN	FAUCET OR SWIVEL FAUCET ASSEMBLY PN	DESCRIPTION
4510-XXXF-XX*	3520-0525-01	Standard Faucet Assy
4510-XXXG-XX*	3520-0559-01	Swivel Faucet Assy

*X represents other numbers or letters in the Unit Part Number

(a) Refer to ASSEMBLY, Paragraph 2.F for assembly of swivel faucet assembly (275) or ASSEMBLY, Paragraph 2.G for assembly of faucet assembly (270).

(b) Install o-ring (295) on faucet tube assembly (285 or 290).

NOTE: Apply a thin layer of thread lubricant (refer to Table 7001) on the outer surface of o-ring (295). Remove excessive lubricant, if any.

(c) Install the faucet tube assembly (285 or 290) on swivel faucet assembly (275) or faucet assembly (270).

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(d) Install faucet tube assembly (285 or 290) on manifold assembly (710, 715, 720, or 725).

(e) Attach the swivel faucet assembly (275) or faucet assembly (270) on chassis assembly (10) using two screws (280).

NOTE: The switch cover (685) must be removed to get access to screws (425) that attach faucet assembly (270) or swivel faucet assembly (275) to the chassis assembly (10).

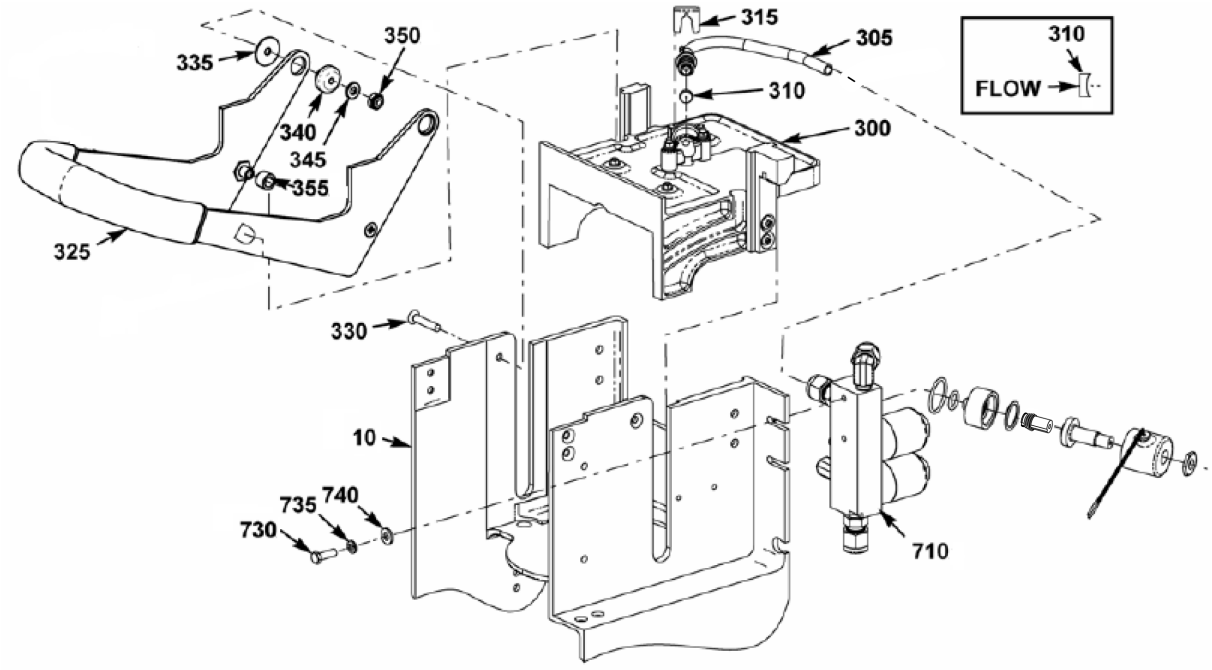

Figure 7026
Installation of Brewhead Assembly

(9) Attach the brewhead assembly (300) to the manifold assembly (710, 715, 720, or 725) and chassis assembly (10) and brew handle (325) to the chassis assembly (10) (refer to Figure 7026).

(a) Refer to ASSEMBLY, Paragraph 2.E for assembly of brewhead assembly (300).

CAUTION: CORRECTLY INSTALL FLOW WASHER INTO BREWHEAD FITTING WITH FLAT SIDE OF THE FLOW WASHER FACING THE DIRECTION FROM WHICH THE WATER FLOWS (I.E, CONCAVE SIDE SHOULD FACE INWARDS INTO BREW HEAD FITTING). IN THE EVENT THAT THE FLOW WASHER IS NOT INSTALLED CORRECTLY, FLOW RATE AND BEVERAGE QUALITY WILL BE AFFECTED.

(b) Install flow washer (310) into brewhead fitting, which is located on top of brewhead assembly (300), with concave side facing inward. Install brew tube assembly (305) into brewhead fitting and secure with clip (315).

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(c) Remove nut and ferrule from compression fitting on manifold assembly (710) and slide onto brew tube assembly.

(d) Put two bushings (355) on brew handle (325).

(e) Coat with petroleum jelly the sides of guides on brewhead assembly (300), and channel in brewhead assembly for the two brew handle bushings (355).

(f) Put brew handle (325) on brewhead assembly (300).

(g) Put brew handle (325) and brewhead assembly (300) on chassis assembly (10).

NOTE: The guides on the brewhead assembly (300) are put into the channels in the chassis assembly (10) that are coated with petroleum jelly.

NOTE: Screws (20) holding the tank assembly (15) on the chassis assembly (10) will be covered when the brewhead assembly (300) is put on the chassis assembly. The tank assembly (15) should be attached to the chassis assembly (10) before the brewhead assembly (300) is attached.

(h) Attach the brew handle (325) to the chassis assembly (10) using two screws (330), two handle washers (335), brew handle (325), two threaded bushings (340), two flat washers (345) and two lock nuts (350).

(j) Attach brew tube assembly (305) to the manifold assembly (710, 715, 720, or 725).

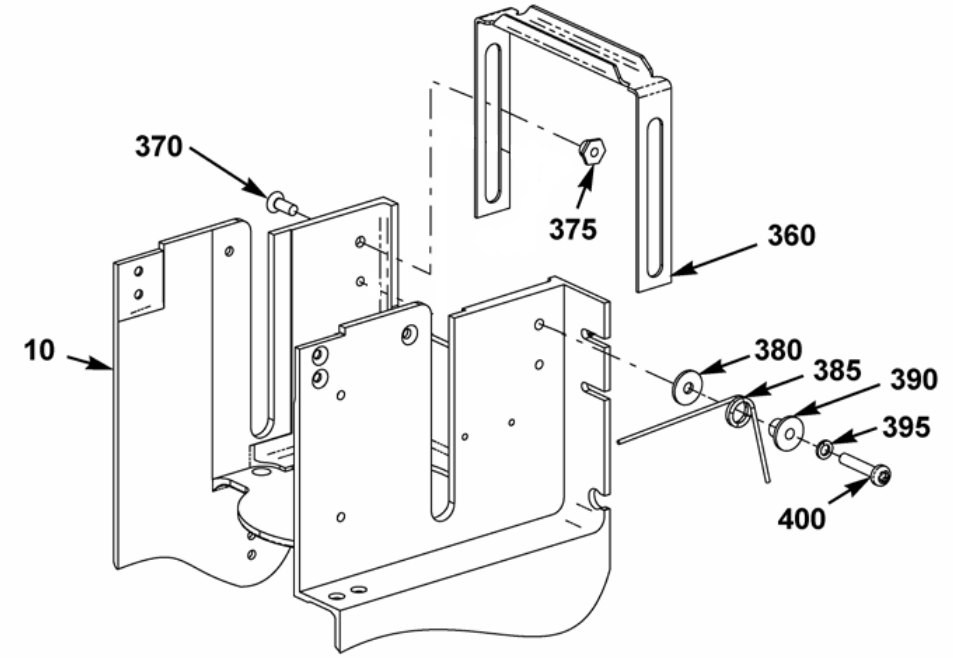

Figure 7027
Installation of Carrying Handle

(10) Attach the carrying handle (360) to the chassis assembly (10) (refer to Figure 7027).

(a) Attach the carrying handle (360) to the left side of the chassis assembly (10) using a screw (370) and handle bushing (375).

(b) Attach the carrying handle (360) to the right side of the chassis assembly (10) using a screw (400), lock washer (395), spring retainer (390), tension spring (385), metallic washer (380), and handle bushing (375).

NOTE: The end of the tension spring must be placed in the channel in the guide on the right side of the brewhead assembly.

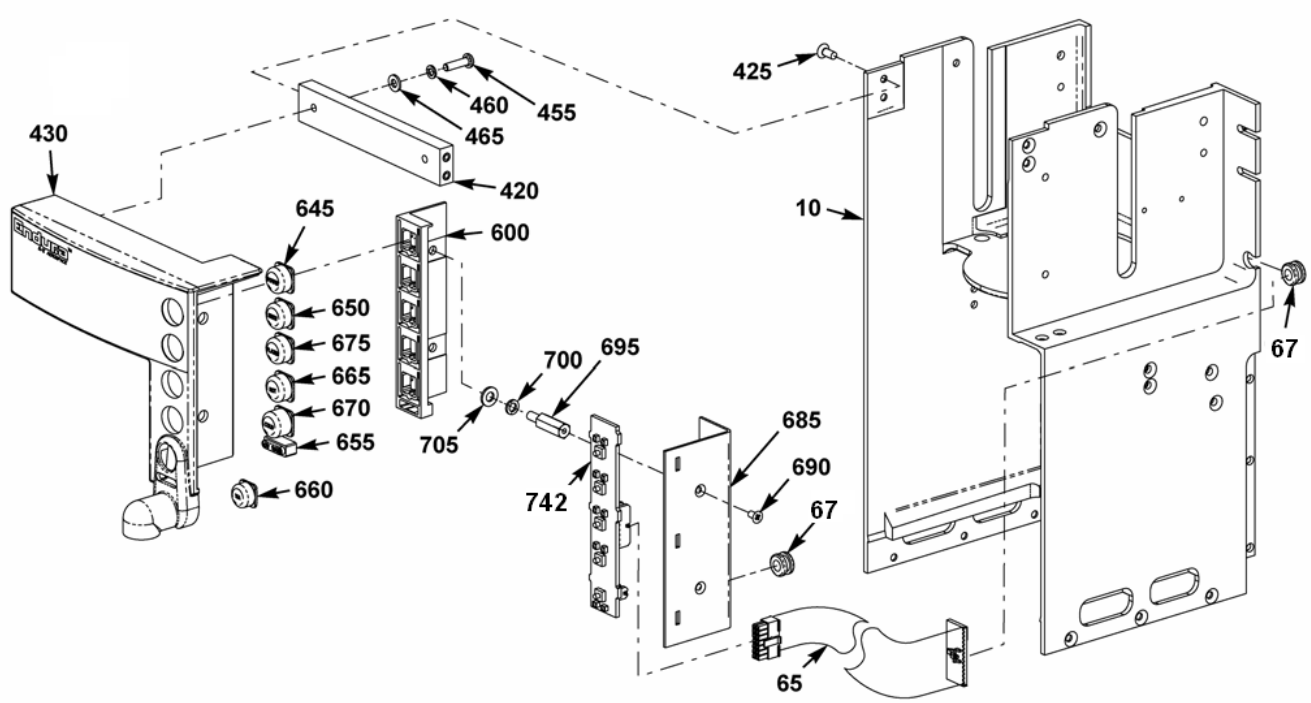

Figure 7028
Installation of Front Panel Assembly

(11) Attach the front panel assembly (430, 435, 440, 445, or 450) and switches to the chassis assembly (10) (refer to Figure 7028).

Table 7003
Front Panel Assembly Part Numbers

UNIT PN	FRONT PANEL ASSEMBLY PN
4510-X1XX-XX*	3520-0560-05
4510-X2XX-XX*	3520-0560-03
4510-X3XX-XX*	3520-0560-09
4510-X4XX-XX*	3520-0560-03
4510-X5XX-XX*	3520-0560-07
4510-X6XX-XX*	3520-0560-01
4510-X7XX-XX*	3520-0560-03
4510-X8XX-XX*	3520-0560-01

*X represents other numbers or letters in the Unit Part Number

(a) Attach the chassis brace assembly (420) to the chassis assembly (10) using four screws (425).

(b) Put the buttons (645, 650, 660, 665, 670, or 675), where options are installed, in the button frame (600) (refer to Figure 7028).

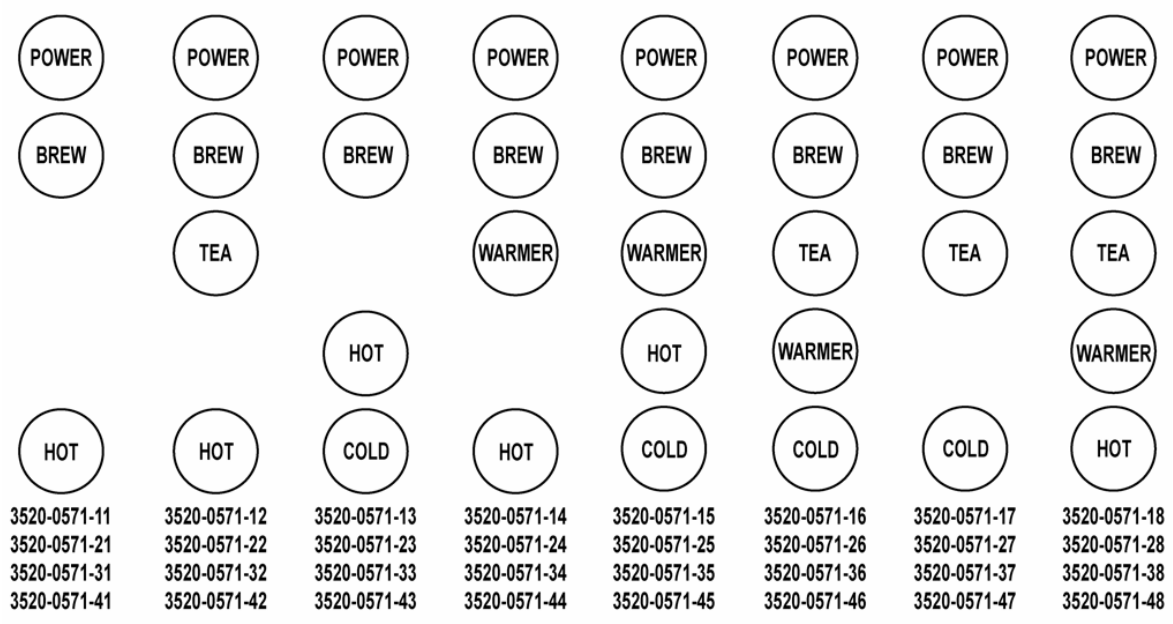

Figure 7029
Installation of Buttons in Button Frame

(c) Put the indicator (655) in the front panel assembly (430, 435, 440, 445, or 450) (refer to Figures 7029 and 7030).

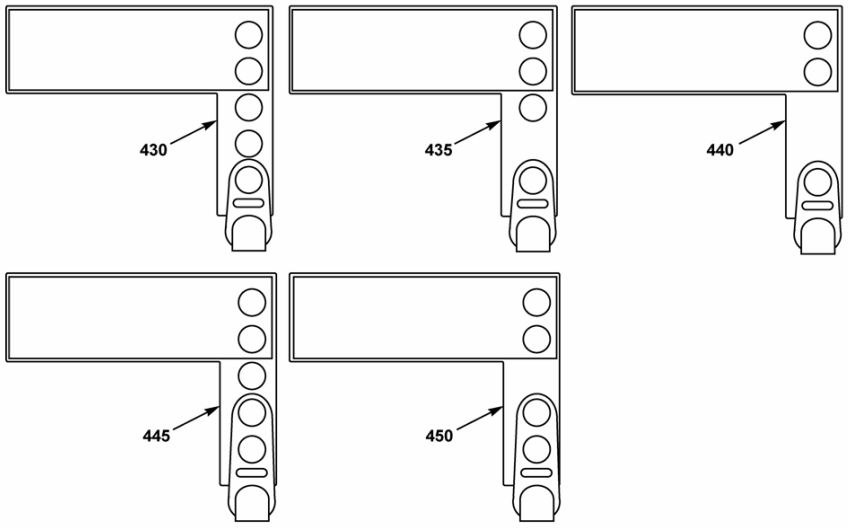

Figure 7030
Identification of Front Panel Assembly

(d) Put the front panel assembly (430, 435, 440, 445, or 450) over the brew handle (325), and faucet assembly (270) or swivel faucet assembly (275).

(e) Attach the button frame (600) and front panel assembly (430, 435, 440, 445, or 450) to the chassis assembly (10) using two flat washers (705), two lock washers (700), and two standoffs (695).

(f) Attach the front panel assembly (430, 435, 440, 445, or 450) to the chassis brace assembly (420) using two flat washers (465), two lock washers (460), and two screws (455).

(g) Put harness assembly (65) into grommet (67).

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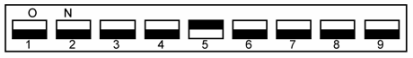
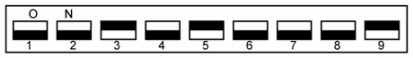
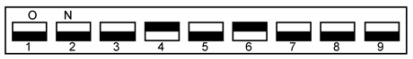
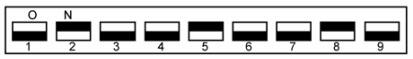
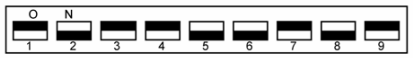
SWITCH ASSEMBLY PN	BEVERAGE MAKER OPTION (FOR REFERENCE ONLY)	DIP SWITCH CONFIGURATION
3520-0556-01	POWER, BREW, HOT WATER	
3520-0556-02	POWER, BREW, TEA, HOT WATER	
3520-0556-03	POWER, BREW, COLD WATER, HOT WATER	
3520-0556-04	POWER, BREW, WARMER, HOT WATER	
3520-0556-05	POWER, BREW, WARMER, COLD WATER, HOT WATER	
3520-0556-06	POWER, BREW, TEA, WARMER, COLD WATER	
3520-0556-07	POWER, BREW, TEA, COLD WATER	
3520-0556-08	POWER, BREW, TEA, WARMER, HOT WATER	

NOTE: Black = Switch position. For example, if bottom half of box is filled, then switch is in down or "off" position.

Figure 7031
Position of DIP Switches

CAUTION: WHEN TOUCHING THE SWITCH PC BOARD USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE. USE ONLY A NON-CONDUCTIVE TOOL TO MOVE THE DIP SWITCHES.

(h) Use a non-conductive tool to put DIP switches on unconfigured switch PC board in correct configuration positions (605, 610, 615, 620, 625, 630, 635, or 640) for options on Beverage Maker (refer to Figure 7031).

(j) Attach plugs on harness assembly (65) to switch PC board (605, 610, 615, 620, 625, 630, 635, or 640).

(k) Attach switch PC board (605, 610, 615, 620, 625, 630, 635, or 640) and switch assembly PC board (642) to button frame (600). Tabs on the switch PC board (605, 610, 615, 620, 625, 630, 635, or 640) go into slots in the button frame (600).

(l) Put harness assembly (65) and grommet (67) into slot on switch cover (685).

(m) Push wires under the switch cover (685) then put tabs on the switch PC board (605, 610, 615, 620, 625, 630, 635, or 640) in slots in the switch cover.

(n) Attach switch cover (685) to two standoffs (695) using two screws (690).

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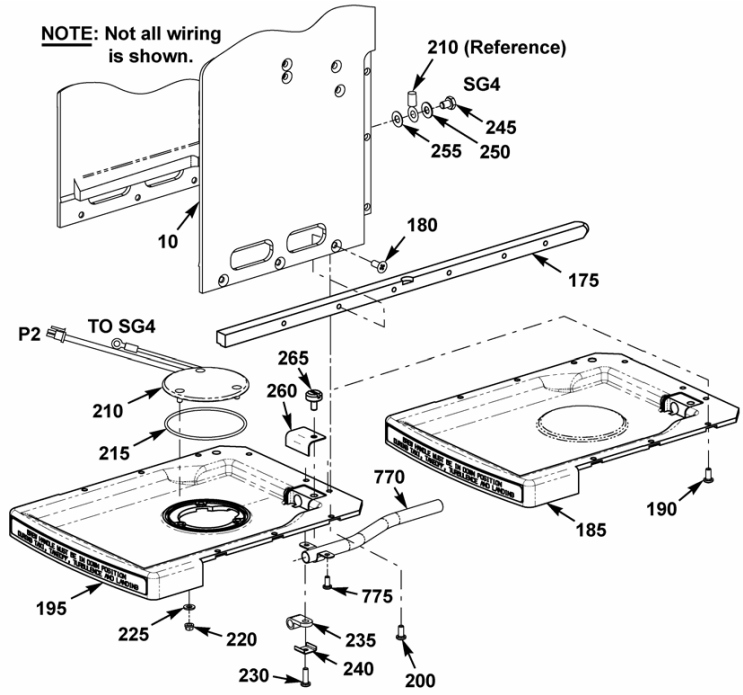

Figure 7032
Installation of Platen Assembly

(12) Attach platen assembly (185, 195) to chassis assembly (10) (refer to Figure 7032).

Table 7004
Platen Assembly Part Number Reference

UNIT PN	PLATEN ASSEMBLY PN
4510-X1XX-XX*	3520-0561-05 (no warmer)
4510-X2XX-XX*	3520-0561-05 (no warmer)
4510-X3XX-XX*	3520-0561-05 (no warmer)
4510-X4XX-XX*	3520-0561-07 (used with warmer)
4510-X5XX-XX*	3520-0561-07 (used with warmer)
4510-X6XX-XX*	3520-0561-07 (used with warmer)
4510-X7XX-XX*	3520-0561-05 (no warmer)
4510-X8XX-XX*	3520-0561-07 (used with warmer)

*X represents other numbers or letters in the Unit Part Number

(a) Where this option is installed, attach drain tube (770) to platen assembly (185 or 195) using black adhesive sealant and a screw (775).

NOTE: The platen assembly (185 or 195) may need to be installed on the chassis assembly (10) to attach the screw (775) for the drain tube (770).

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Table 7005
Platen Drain Part Number Reference

UNIT PN	PLATEN DRAIN
4510-3XXX-XX*	Sump drain, no drain tube required
4510-4XXX-XX*	Rear drain, drain tube (770) required

*X represents other numbers or letters in the Unit Part Number

(b) Attach drain screen (260) to the platen assembly (185 or 195) using drain screen fastener (265).

(c) Put o-ring (215) in slot on platen assembly (195) where platen heater (210) option is installed.

NOTE: Apply a thin layer of thread lubricant (refer to Table 7001) on the outer surface of o-ring (215). Remove excessive lubricant, if any.

(d) Where this option is installed, attach platen heater (210) to platen assembly (195) using screws (200), three flat washers (225) and three lock nuts (220).

(e) Attach platen assembly (185, 195) to the chassis assembly (10) using screws (190).

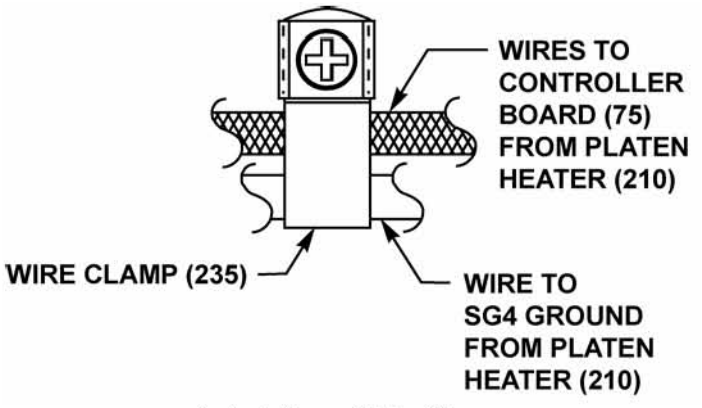

Figure 7033
Installation of Wire Clamp

CAUTION: MAKE SURE WIRES IN WIRE CLAMP ARE SIDE BY SIDE AND NOT ON TOP OF EACH OTHER (REFER TO FIGURE 7033).

(f) Where this option is installed, attach platen heater (210) wiring to chassis assembly (10) and platen assembly (195) using wire clamp (235), washer (240), and screw (230 or 232) (refer to Figure 7032).

(g) Where this option is installed, attach platen heater (210) ground wire to chassis assembly (10) using a flat washer (255), wire terminal, flat washer (250), and hex screw (245).

NOTE: Apply alodine (refer to Table 7001) on flat washer (255) prior to installation.

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(h) Attach the platen heater (210) plug to its connection on the controller board (75). The power module assembly (55) may need to be installed for correct routing of the wiring to the controller board (75).

NOTE: The guide rails (175) and screws (180) to the chassis assembly (10) and the support connector(115) will have to be installed after the platen assembly (185 or 195) is installed on the chassis assembly (10).

(13) Install the power module assembly (55) on the chassis assembly (10) (refer to IPL Figure 2).

CAUTION: WHEN TOUCHING THE POWER CONTROLLER BOARD OR ANY OF ITS INDIVIDUAL PARTS, USE STANDARD SAFETY GROUNDING METHODS TO PREVENT ELECTROSTATIC DAMAGE.

(a) Before installing controller assembly (75), apply a thin film of heat sink compound CM-50019 to the K2 relay heat sink (refer to CMM 25-33-32).

(b) Install controller board assembly (75) in power module assembly (55) using five screws (80) and six shoulder washers (85). One shoulder washer (85) needs to centered on the back of the controller board, where there will be no screw hole, and no interference with board circuitry (refer to ASSEMBLY, Paragraph 2.J for assembly of controller board assembly (75)).

(c) Put two screws (110) and two flat washers (105) into the power module assembly (55).

CAUTION: PUT THE SIDE OF THE RELAY RETAINER (100) THAT HAS VISIBLE THREADS IN THE HOLES NEAR THE RELAY, AND THE SIDE THAT SHOWS THE HEXAGON OF THE THREADED INSERT SHOULD FACE AWAY FROM THE RELAY. DAMAGE TO THE RELAY CAN RESULT FROM INSTALLATION WITH THE RELAY RETAINER (100) HEXAGONS OF THE FASTENERS FACING TO THE RELAY.

(d) Before inserting the screws (110) place a small amount of thread lock AD-300 on the threaded holes in the relay retainer (100).

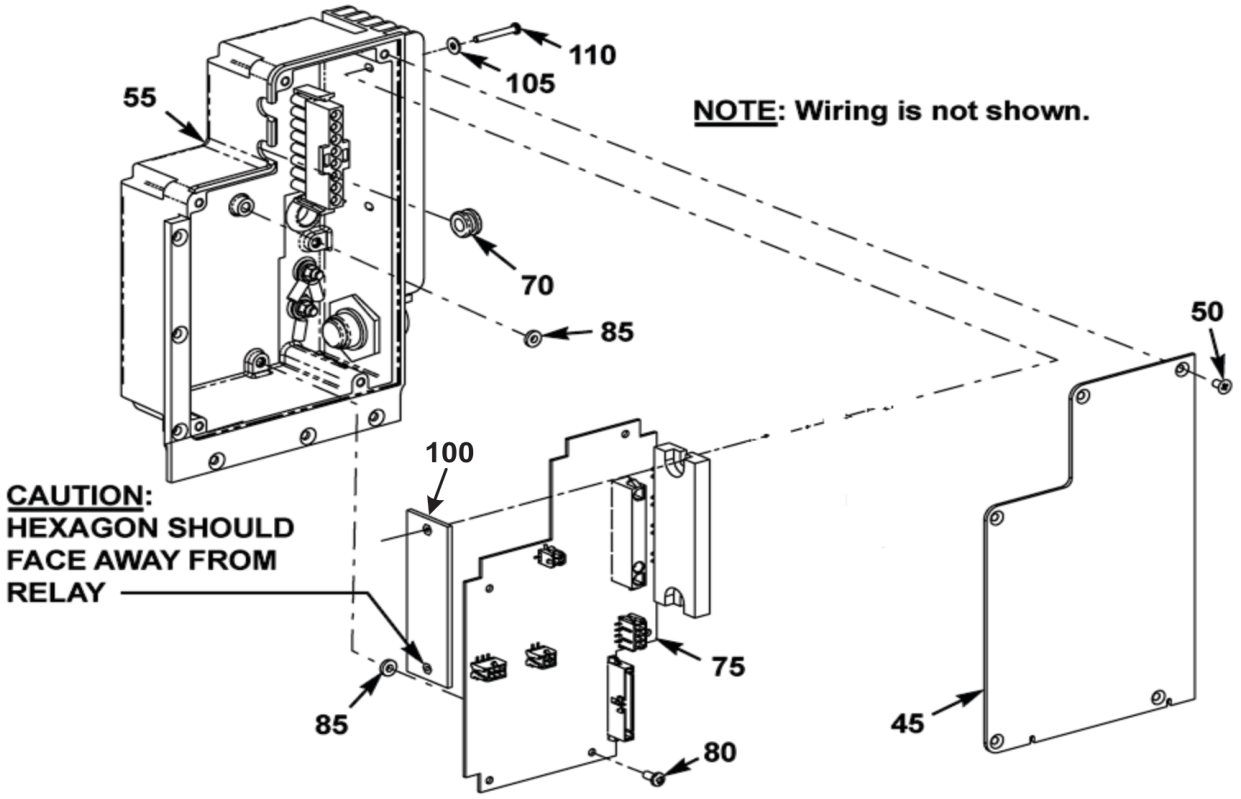

Figure 7034
Installation of Controller Board Assembly

(e) Put the screws (110), the relay on the controller board assembly (75), and the relay retainer (100).

(f) Evenly pull the relay to the side of the power module assembly (55) by tightening the screws (110) (refer to Figure 7034).

CAUTION: DO NOT TIGHTEN SCREWS TOO MUCH. THE RELAY CAN BE DESTROYED IF SCREWS ARE PUT ON TOO TIGHT.

(g) Use a torque wrench to tighten each screw (110) to one inch-pound force. Then tighten each screw (110) to two inch-pounds force.

(h) Attach the power module assembly (55) to the chassis assembly (10) using three screws (60).

(j) Attach the power module assembly (55) to the guide rail (175) using three screws (60).

NOTE: The guide rail (175) may need to be attached to the chassis assembly (10) after the platen assembly (185 or 195) has been installed.

(k) Connect all wire harness (65, IPL Figure 2 and 15, IPL Figure 4) plugs to sockets on the controller board assembly (75).

(l) Put grommets (70) on the wire harnesses (65, IPL Figures 2 and 15, IPL Figure 4) and wires from equipment options and put grommets in slots in power module assembly (55) housing. Put open side of grommet near the side of the power module assembly (55).

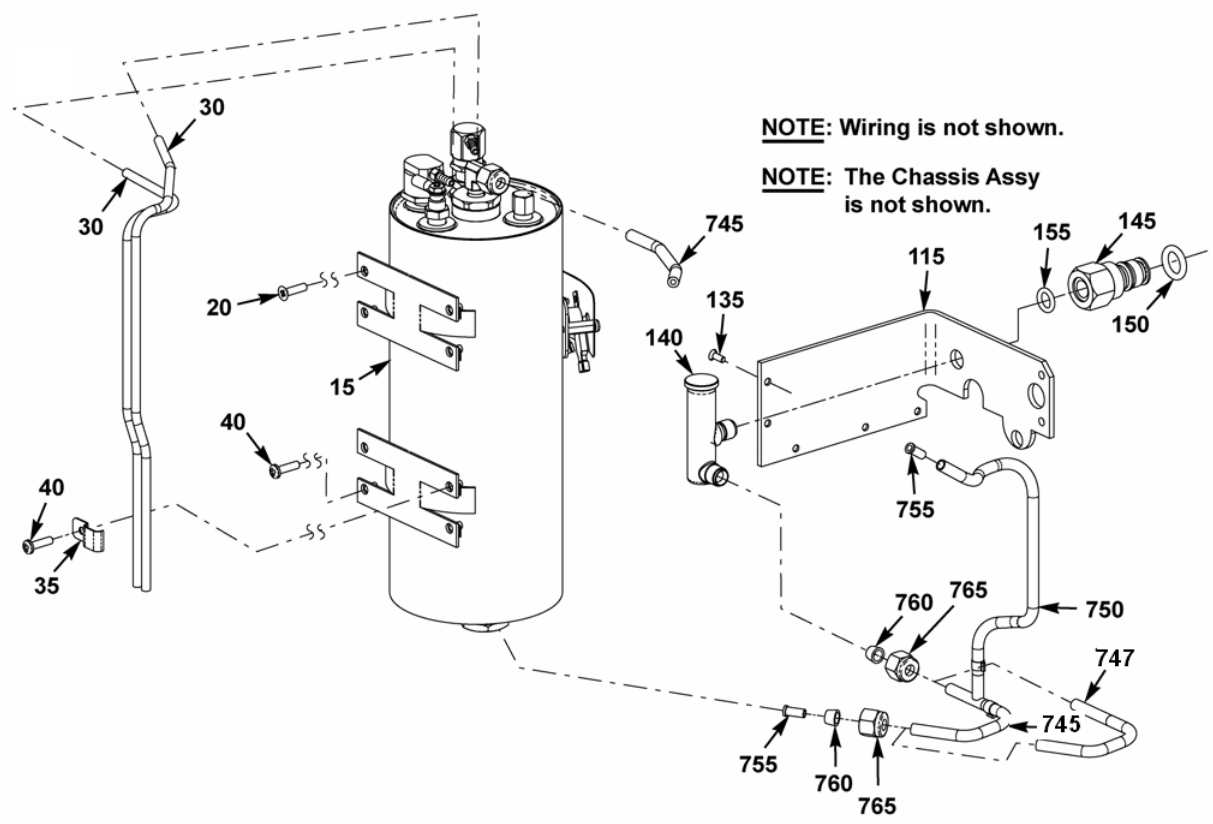

Figure 7035
Installation of Tank Assembly

(14) Attach the tank Assembly (15) to the chassis assembly (10) (refer to Figure 7035).

NOTE: The brewhead assembly (300) needs to be attached to the chassis assembly (10) after the tank assembly (15) is installed to get access to screws (20) on the upper part of the chassis assembly (10) for the tank assembly (15). Refer to ASSEMBLY, Paragraph 2.E to install the brewhead assembly (300).

CAUTION: THE IDENTIFICATION PLACARD MUST BE KEPT WITH THE BEVERAGE MAKER WHEN THE SCREWS HOLDING THE PLACARD AND TANK ASSEMBLY ON THE CHASSIS ASSEMBLY ARE REMOVED. THIS ID PLACARD IS NOT AVAILABLE AS A SPARE PART.

(a) Attach the tank assembly (15) to the chassis assembly (10) using four pan head screws (45, IPL Figure 1 and 40 and 135, IPL Figure 2) and four flat head screws (20). Refer to ASSEMBLY, Paragraph 2.L for assembly of the tank assembly (15).

NOTE: Installation of two screws (45, IPL Figures 1, 1A, 1B, 1C) low inside the chassis assembly (10) server cavity attaches the ID placard to the chassis assembly.

(b) Put wires for RTD (125, IPL Figure 3) through grommet in side of power module assembly (55) and connect plug of RTD (125, IPL Figure 3) wire to controller board (75) in the power module assembly.

(c) Attach tubes (30) to barbed fittings on the top of the tank assembly (15) if replacement or cleaning was required.

(d) Put tubes (30) through holes in chassis assembly (55) and attach the tubes inside the server cavity using a tube clamp (35) and screw (40). The tubes should be no less than one inch above the drain in the platen assembly (185 or 195).

(e) Connect tube assembly (745) to fitting (60, IPL Figure 3) of water filter on top of tank assembly (15).

(f) Use solder to attach wire to left side terminal of over temperature thermostat on tank assembly (15).

CAUTION: IDENTIFY HEATER TERMINALS AND WIRE HARNESS TERMINALS THAT GO TOGETHER. THE WIRING TERMINALS MUST BE PUSHED AND TURNED TO BE ATTACHED TO EACH OTHER. THEY ARE NOT EASY TO CONNECT BY HAND, SO TOOLS MAY BE REQUIRED. DO NOT USE TOO MUCH FORCE TO HOLD OR ATTACH THE PLASTIC TERMINALS TO EACH OTHER.

(g) Connect wire harness terminals to terminals of heater assembly (15, IPL Figure 3).

CAUTION: CAREFULLY ATTACH WIRE TERMINALS TO TERMINALS OF THE OVERTEMPERATURE THERMOSTATS SO THAT THE TERMINALS DOES NOT MOVE ON THE PLASTIC HOUSING OF THE THERMOSTATS. MOVEMENT OF A TERMINAL CAN CRACK THE THERMOSTAT HOUSING THAT CAN RESULT IN FAILURE OF THE THERMOSTAT.

(h) Attach three quick connect wiring terminals to terminals of over temperature thermostats (30, IPL Figure 3) on the tank assembly (15).

(j) Attach ring terminal to hex nut on low water sensor (120, IPL Figure 3) on the top of the tank assembly (15) using a hex nut on low water sensor.

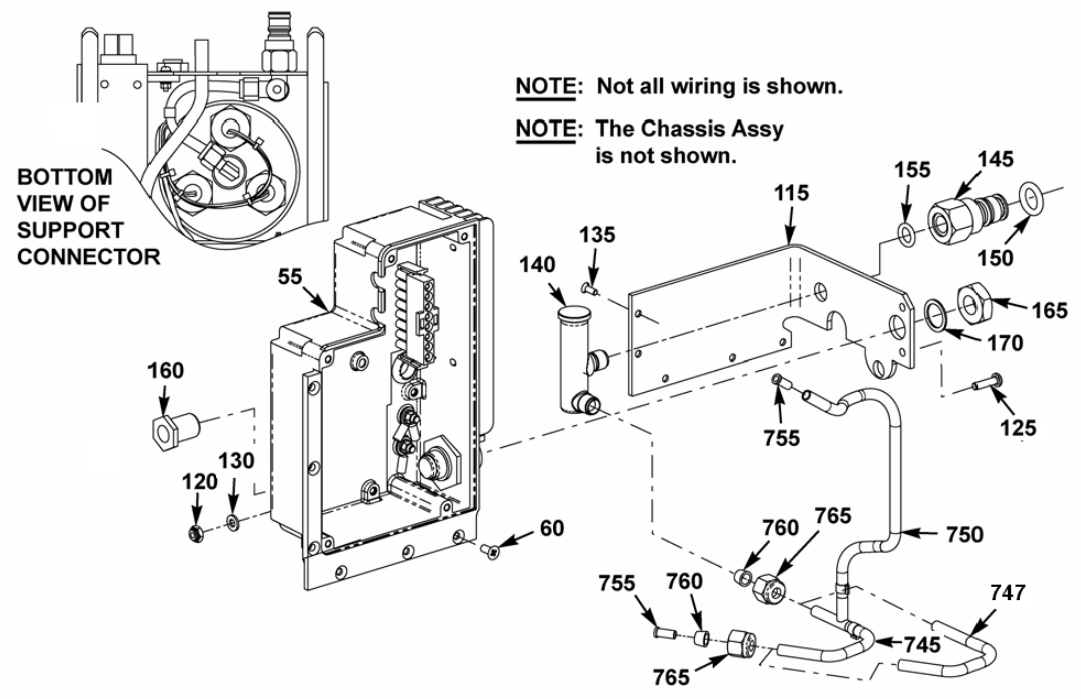

Figure 7036
Installation of Support Connector

(15) Attach support connector (115) to chassis assembly (10) (refer to Figure 7036).

(a) Refer to ASSEMBLY, Paragraph 2.H for assembly of the drain valve assembly (140).

(b) Attach the drain valve assembly (140) to the support connector (115) using an o-ring (155) o-ring (150) and water shutoff plug (145).

NOTE: Apply a thin layer of thread lubricant (refer to Table 7001) on the outer surface of o-ring (155). Remove excessive lubricant, if any.

CAUTION: CAREFULLY PUT THE SUPPORT CONNECTORON THE REAR DRAIN TUBE (IF OPTION INSTALLED) TO AVOID DAMAGE TO THE DRAIN TUBE.

(c) Attach the support connector (115) to the left side rail guide (175) using three screws (135).

(d) Attach the support connector (115) to the left side of the chassis assembly (10) using two screws (135).

(e) Attach the support connector (115) to the power module assembly (55) using a screw (130), flat washer (125), and lock nut (120).

(f) Attach the alignment pin guide (160) to the support connector(390) and power module assembly (55) using a lock washer (170) and hexagon nut (165).

(g) Put the ferrule (760), fitting (765), and insert (755) on the water supply tube assembly (745, 747 or 750).

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(h) Attach the water supply tube assembly (750, 747 or 745) to the drain valve assembly (140) and tank assembly (15).

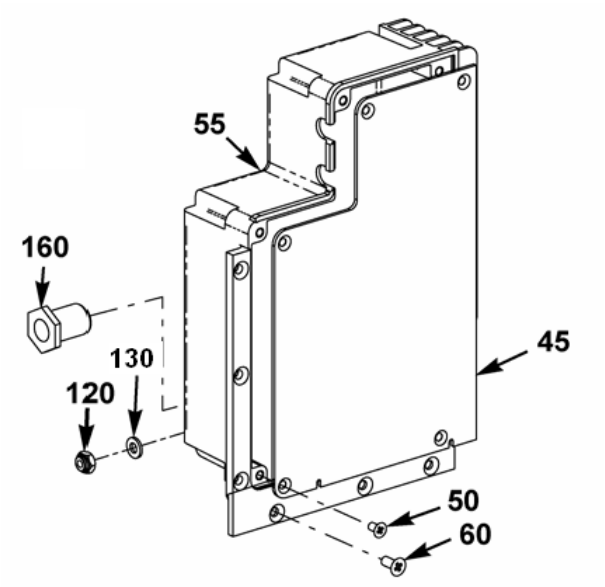

Figure 7037
Installation of Power Module Cover

(16) After necessary wire connections (refer to Figure 7013) are made to the controller board (75) in the power module assembly (55), attach the power module assembly cover (45) to the power module assembly using five screws (50) (refer to Figure 7037).

(17) Attach wire tie downs to wiring, spacing the tie downs approximately one and a half inches apart, to keep wiring from rubbing or touching heat sources (refer to FITS AND CLEARANCES, Figures 8004 and 8005 for examples of wire routing).

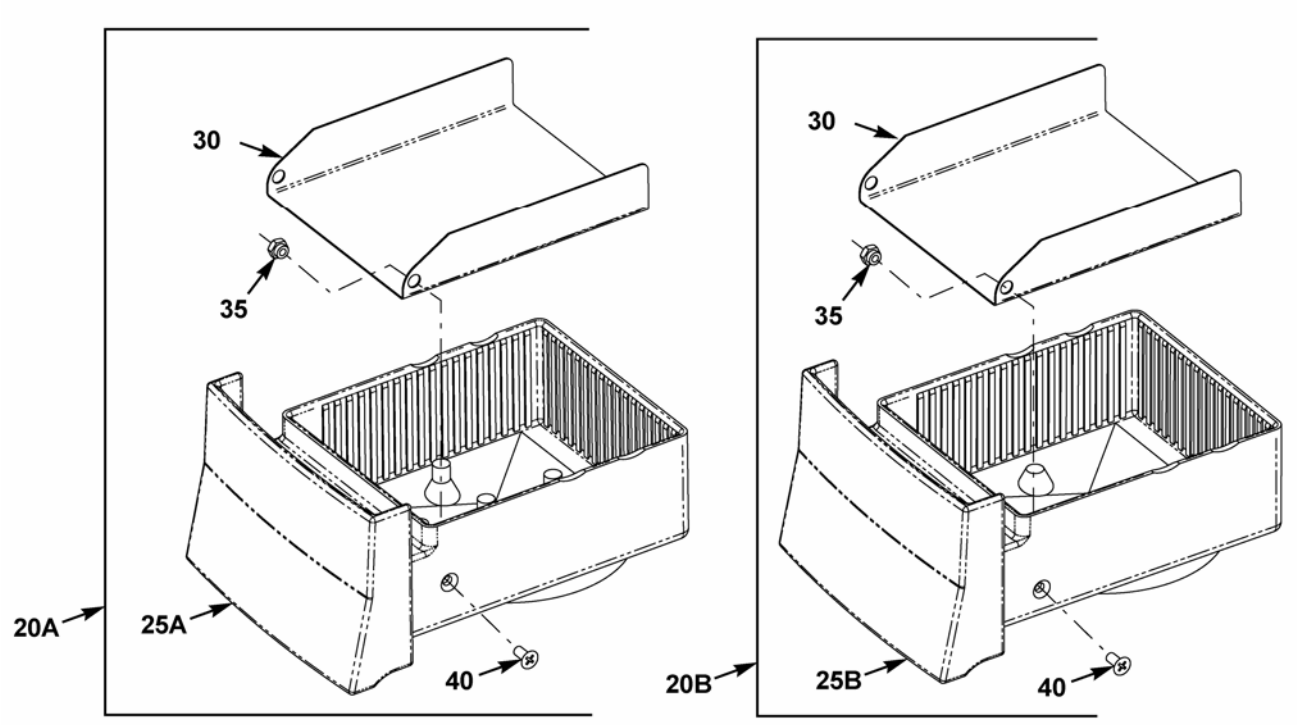

Figure 7038
Brew Cup Assembly

N. Assembly of the Beverage Maker (refer to IPL Figures 1, 1A, 1B, and 1C and Figure 7038).

NOTE: All philips screw heads inside the server cavity and brew cup cavity must be filled with translucent silicone rubber adhesive (refer to Table 7001).

Table 7006
Brew Cup Assembly PN Reference

UNIT PN	BREW CUP ASSEMBLY PN	DESCRIPTION
4510-XXUX-XX*	3520-0540-01	Brew cup assembly (upper)
4510-XXLX-XX*	3520-0540-03	Brew cup assembly (lower)

*X represents other numbers or letters in the Unit Part Number

(1) Attach the spacer plate (30) to the brew cup (25A, 25B) using two screws (40) and two nuts (35).

(2) With the brew handle in the raised position, put the brew cup (20A or 20B) in the cavity of the Beverage Maker subassembly (10).

(3) Refer to the FITS AND CLEARANCES Section for adjustment of essential assemblies.

(4) Refer to TESTING AND FAULT ISOLATION for procedures to test the Beverage Maker following ASSEMBLY.

FITS AND CLEARANCES

1. General

   A. Refer to TESTING AND FAULT ISOLATION and INSPECTION/CHECK for assistance in identifying parameters and adjustments required for full function of the Beverage Maker.

2. Procedures

   A. Plumbing

      (1) Do not tighten nut (holding solenoid coil on the solenoid guide) too much. Putting this nut on too tight can damage or hinder function of the solenoid and solenoid coil.

      (2) When putting thread seal tape on plumbing fittings:

         (a) Make sure all threads are clean.

         (b) Start tape below the first thread and wrap in the direction of the spiral of the threads.

         (c) Wrap the threads with no more than two and half layers of tape, pull tape tight so it conforms to threads, and then cut off the tape.

      (3) Trim tubes to a flat face perpendicular to the wall of the tube. An irregular face on the tube end contributes to a seal that can fail under pressure. Align tube, insert, and ferrule carefully when tightening nut and tube on fittings to make sure there is an effective seal.

      (4) When tightening plumbing fittings that use a flow washer, o-ring, or gasket (on vent valve and pressure relief valve fittings) make sure these parts are not crushed or pinched, or an ineffective seal may result.

      (5) The flow washers (310 and 538) should be installed so the concave side faces away from water flows (refer to Figure 8001).

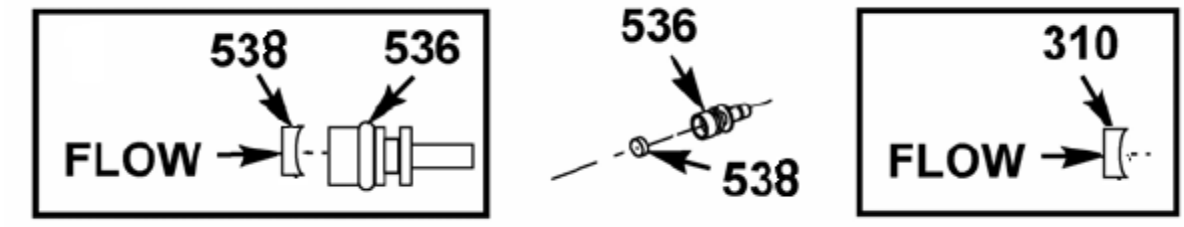

Figure 8001
Flow Washer Detail

(6) The drain tubes for the relief valve and the vent valve should be retained by the fastener and clamp on the back wall of the server cavity of the chassis assembly. The tubing should follow a smooth curve from the tank assembly valves to the back wall of the server cavity of the chassis assembly, without otherwise touching the tank assembly. The tubes should end no less than one inch (2.5 cm) above the platen assembly to avoid backup from the drain (this is a requirement that helps avoid contamination of potable water).

(7) The brew head tubing should not touch the chassis assembly but fit a smooth curve from the manifold assembly to the brew head.

(8) Tubing from the drain valve to the tank assembly inlet tube should not touch the tank assembly. It should fit a smooth curve from the drain valve fitting to the inlet tube fitting.

(9) All plumbing and tubing must fit within the envelope of the chassis assembly, power module assembly and tank assembly, to allow the Beverage Maker to smoothly and easily fit into galley installations.

B. Mechanical and Structural

(1) Align the switch assembly (585, IPL Figure 2) with the brewhead guide on the brewhead assembly (300). Tighten screws (590) so that the actuator wheel touches the brewhead guide when the brewhead assembly is lowered. When the brewhead assembly is lowered, the guide should cause the actuator to push the button on the switch (refer to Figure 8002 for example).

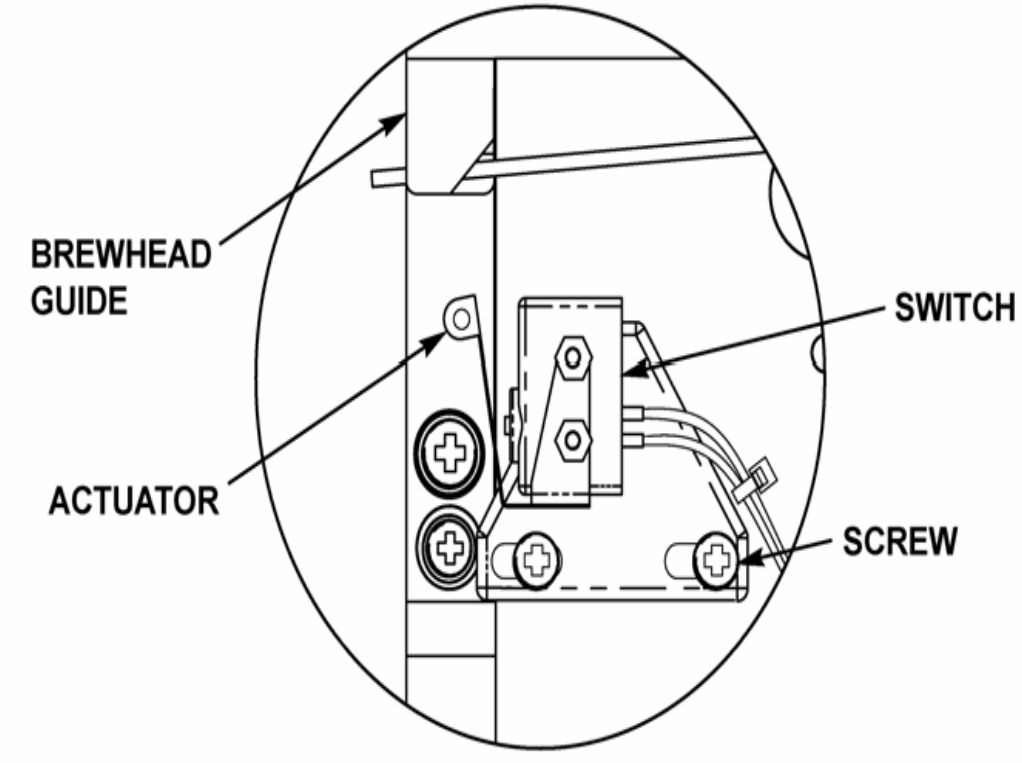

Figure 8002
Switch Assembly Detail

(2) Seal all external screw heads inside the server cavity with translucent silicone rubber adhesive, AD-50002.

NOTE: A minimum amount of sealant should be applied to fill the screw recesses and screw cavity to allow for a smooth finish. Any excess sealant should wiped clean.The sealant should be applied only to clean dry surface.

C. The lockdown lever (470 and 490, IPL Figure 2) mechanism should work smoothly and easily, without binding when being moved from either the lifted or lowered positions.

(1) The pin on the power module assembly (55, IPL Figure 2) should be at or below the surface of the outside of the support connector (115).

(2) All fasteners on the outer walls of the Beverage Maker should be at or below the surface of the wall of the beverage maker.

D. Electrical

(1) When installing the power connector (PP1) for the wire harness on the back of the power module assembly (1, IPL Figure 4), adequate wire length must be provided so no tension is put on the connector pins in the connector. Strained wiring can result in bent pins, hindering secure and complete electrical connection between the Beverage Maker and the rail assembly.

(2) All wiring must fit within the external envelope of the chassis assembly, power module assembly, and tank assembly, to allow the Beverage Maker to smoothly and easily fit into galley installations.

(3) The screws holding the relay (on the controller board assembly) to the side of the power module housing assembly should not be tightened to more than two inch-pounds torque.

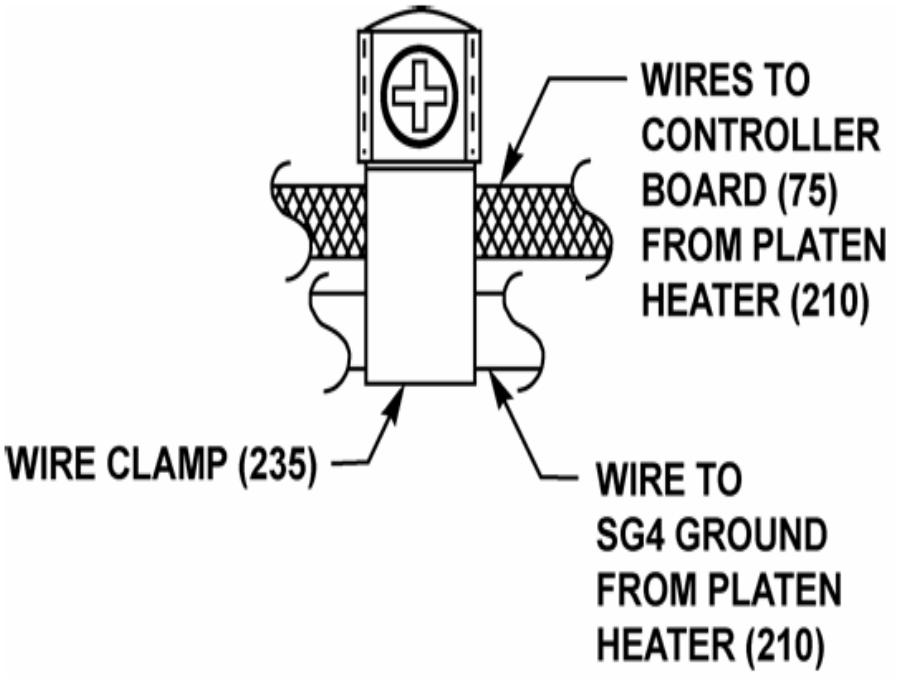

Figure 8003
Installation of Wire Clamp

(4) The platen heater wires should point toward the rear of the Beverage Maker to allow the wiring to be fastened under the clamp on the bottom of the platen assembly.

CAUTION: MAKE SURE WIRES IN WIRE CLAMP ARE SIDE BY SIDE AND NOT ON TOP OF EACH OTHER (REFER TO FIGURE 8003).

(5) Attach wire tie downs to wiring, spacing the tie downs approximately one and a half inches apart, to keep wiring from rubbing or touching heat sources (refer to Figures 8004 and 8005 for examples of wire routing). Spacing between the wire tie downs is recommended to be between 0.5 and 1.5 inches (1.27 and 3.81 cm).

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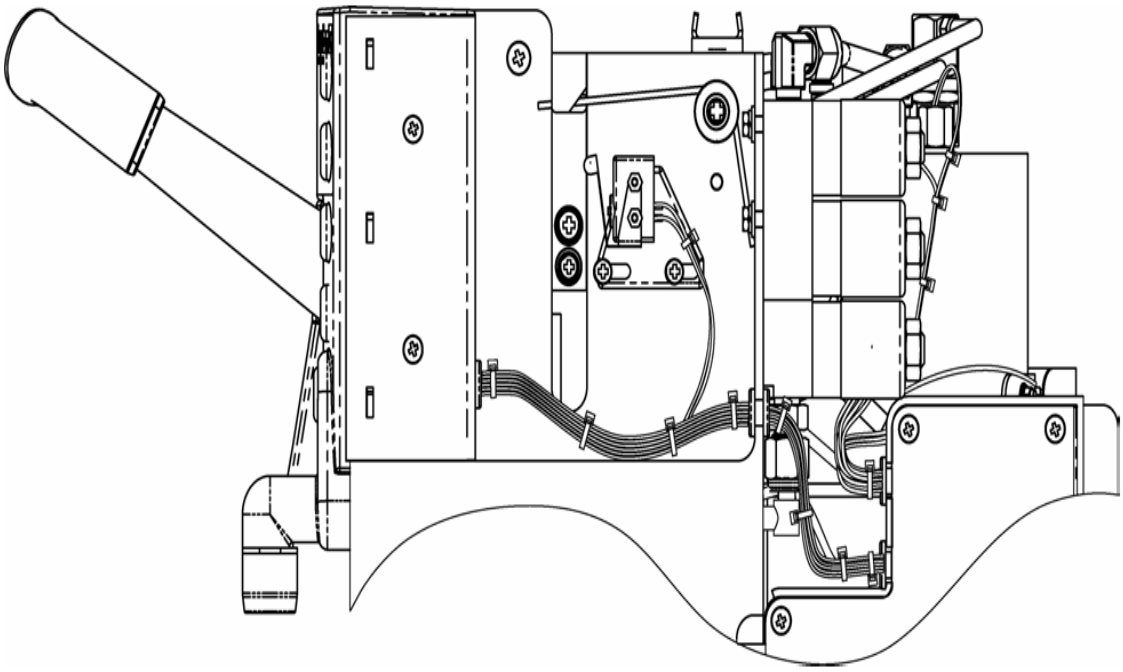

Figure 8004
Example of Wire Routing

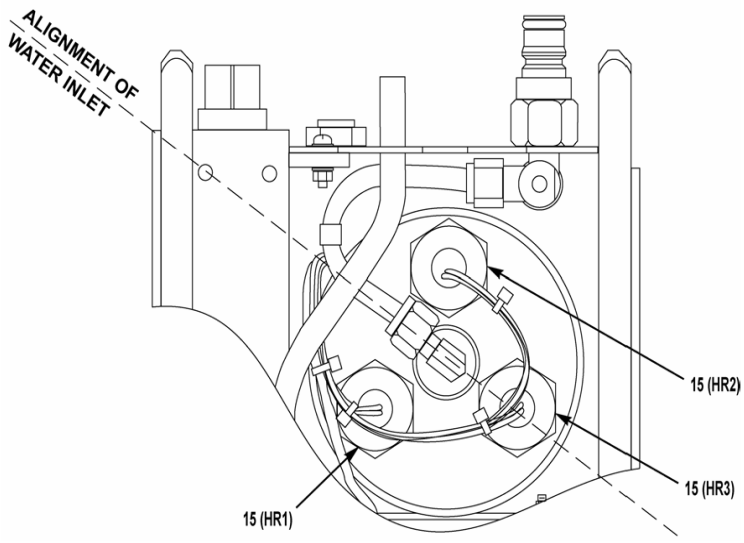

Figure 8005
Example of Heater Assembly Wire Routing

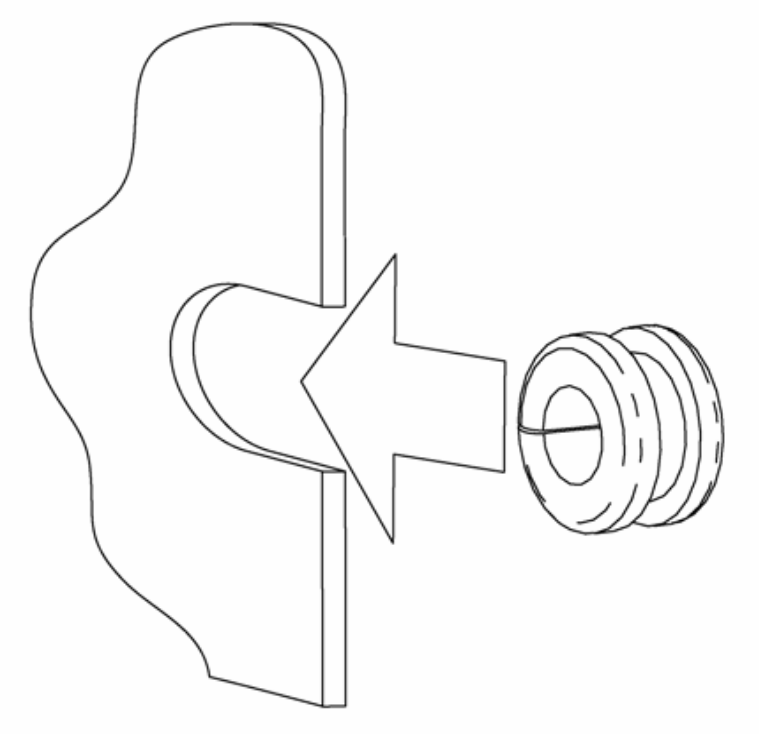

Figure 8006
Grommet Installation

(6) Put wiring in grommet, install it in the slot, then turn grommet so the side opening is near the closed end of the slot, so the grommet holds wiring correctly (refer to Figure 8006).

SPECIAL TOOLS, FIXTURES, EQUIPMENT, AND CONSUMABLES

1. General

A. All work described in this manual may be conducted in a normal work area with ambient environmental and physical conditions and whose environment has been cleaned for Beverage Maker equipment repair.

B. Special tools, fixtures, equipment, and consumables required to perform the work function listed in all paragraphs of this manual are listed in Table 9001.

NOTE: Equivalent substitutes may be used for listed items.

Table 9001
Special Tools, Fixtures, Equipment, and Consumables

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
1 in X 3 in Deep Socket, 1/2 in Drive for heater and water filter removal (refer to Figure 9001)	4409005-001	B/E Aerospace Inc. (V63367)
1/8 inch NPT plug	AN933-1D or MIT-022P	Commercially Available or FasTest, Inc. 1646 Terrace Drive Roseville, MN 55113-1317
Alodine Chemical Coating	Alodine 1201 CM-50017	Henkel Corp, (V71410) B/E Aerospace (V63367)
A food-grade aqueous acidic cleaner that removes calcium, lime, and rust deposits (scale) without damage to unit form, fit, or function	None	Commercially Available
Anti-Seize Lubricant, Food Grade	White Paste Lubricant CM-50022	Loctite Corp. (V05972) B/E Aerospace (V63367)
Black Adhesive Sealant	732 Black RTV-103 AD-052	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)
Brush, Wire, Soft-bristled	None	Commercially Available
Clean Water	None	Commercially Available
Cleaner, Mild Liquid Detergent (Dish Soap)	None	Commercially Available
Cloth, Cleaning, Low-Lint, Type 1	A-A-59323	Commercially Available
Cotton Swabs	None	Commercially Available

Table 9001 (Continued)
Special Tools, Fixtures, Equipment, and Consumables

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Electrical connector cable between Beverage Maker and power supply	998552	B/E Aerospace (V63367)
Epoxy Adhesive	E-Cast F-28 Resin F-14 Hardener AD-50029	United Resin Corp. (V08LA7) B/E Aerospace (V63367)
ESD Conductive Mat	None	Commercially Available
ESD Grounding Wrist Strap	None	Commercially Available
Flow Washer Tool (Use one flow washer tool to remove flow washer and use two tools to install flow washer) (refer to Figure 9002)	3575-1319-01	B/E Aerospace, Inc. (V63367)
Heat Sink Compound	10-8109 CM-50019	Waldom Inc. (V72653) B/E Aerospace, Inc. (V63367)
High Low Temperature RTD wiring harness with 4-position connector	997112	B/E Aerospace Inc. (V63367)
IC (Microcontroller) Extraction Tool, 20-128 pins non-ESD pull tool (refer to Figure 9003)	EX-5	OK Industries (V9X418)
Insertion Tool (for PP1 connector) (refer to Figure 9004)	MS24256A16	Commercially Available
Isopropyl Alcohol	TT-I-735	Commercially Available
Megohmmeter	None	Commercially Available
Milliohmmeter	Model 1880	QuadTech Inc (V0PK96)
Nylon Wire Tie Downs	MS3367-4-9	Commercially Available
Nitrogen Gas	A-A-59503	Commercially Available
Ohmmeter	None	Commercially Available
Petroleum Jelly	None CM-50015	Commercially Available B/E Aerospace Inc. (V63367)
Pincers (for tea tube clamps) (refer to Figure 9005)	14100082 993973	Oetiker, Inc. (V25281) B/E Aerospace Inc. (V63367)
Power Module Wrench (for nut on PP1 connector) (refer to Figure 9006)	3575-1323-03	B/E Aerospace Inc. (V63367)

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Table 9001 (Continued)
Special Tools, Fixtures, Equipment, and Consumables

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Power Supply, including: 3-phase 115 ±3 VAC, 400 Hz, 10 amps per phase; 3 ammeters (0-25 amp); and 3 voltmeters (0-250 volt) (refer to Figure 1001)	None	Commercially Available
Removal Tool (for PP1 connector) (refer to Figure 9007)	MS24256R16	Commercially Available
Server	MISC-3201	B/E Aerospace Inc. (V63367)
Red Silicone Rubber Adhesive	736 Red RTV AD-50005	Dow Corning Corp. (V71984) B/E Aerospace, Inc. (V63367)
Sleeving, Non-Fray	Varglas Type HO, Size 3/8 in TU-50039	Varflex Corp. (V79074) B/E Aerospace, Inc. (V63367)
Socket Key Tool (remove plunger guide assy from solenoid valve assy body or from solenoid manifold) (refer to Figure 9008)	3575-1321-01	B/E Aerospace Inc. (V63367)
DELETED		
Solder	Kester No. 245, No Clean, Low Residue SD-50003	Kester Solder (Div of Litton Systems, Inc.) (V75297) B/E Aerospace Inc. (V63367)
Stopwatch	None	Commercially Available
Teflon Tape - 1/4 in	MS-STR-4 TT-7454	Swagelok Companies (V02570) B/E Aerospace, Inc. (V63367)
Terminal Extractor Tool (refer to Figure 9009)	0011030043	Molex (V27264)
Thermometer, 50-250 °F (10-121 °C)	None	Commercially Available
Thread Lock Adhesive	242 Blue AD-300	Loctite Corp. (V05972) B/E Aerospace, Inc. (V63367)
Thread Lubricant	Never Seez NSWT-14 CM-50016	Bostik Inc. (V1Y0Z7) B/E Aerospace, Inc. (V63367)
Torque Tool (handle for socket key tool) (refer to Figure 9010)	3575-1320-01	B/E Aerospace Inc. (V63367)
Translucent Silicone Rubber Adhesive/Sealant	732 Clear RTV RTV 108 AD-50002	Dow Corning Corp. (V71984) Momentive Performance Materials Inc. (V01139) B/E Aerospace, Inc. (V63367)

Table 9001 (Continued)
Special Tools, Fixtures, Equipment, and Consumables

NOMENCLATURE	PART/SPEC NUMBER	SOURCE (CAGE)
Water Supply, including: pressure regulator (0-150 psig, 0-10.34 barg) and flow meter (0-0.5 GPM, 0-1.9 LPM) (refer to Figure 1001)	None	Commercially Available

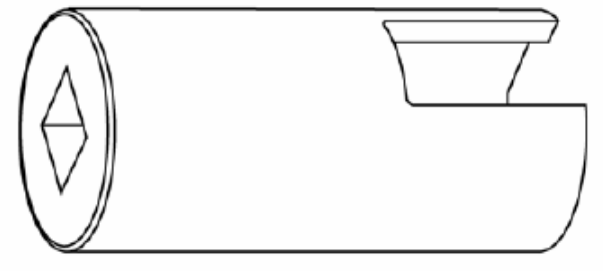

Figure 9001
1in X 3 in Deep Socket, 1/2 in Drive

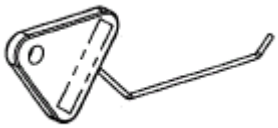

Figure 9002
Flow Washer Tool

Figure 9003
Extraction Tool

Figure 9004
Insertion Tool

Figure 9005
Pincers

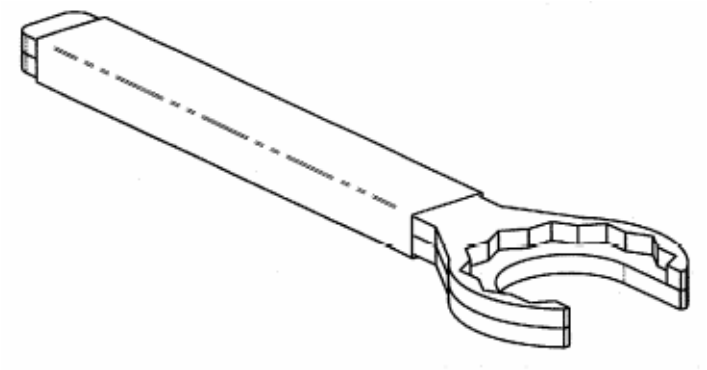

Figure 9006
Power Module Wrench

Figure 9007
Removal Tool

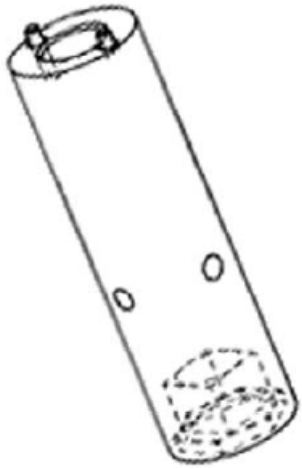

Figure 9008
Socket Key Tool

Figure 9009
Terminal Extractor Tool

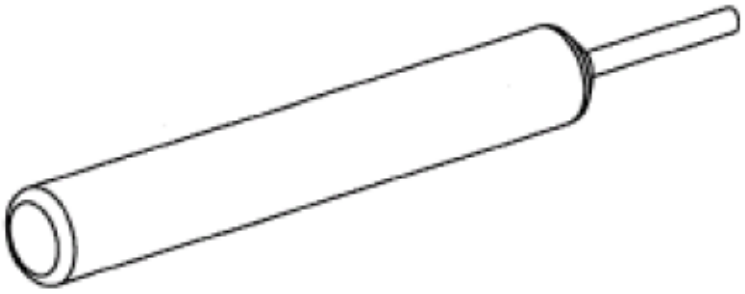

Figure 9010
Torque Tool Handle for Socket Key Tool

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STORAGE

1. General

   A. This section contains the storage procedures for the Beverage Maker.

2. Procedure

   A. Storage

      (1) After completing the Beverage Maker assembly, final adjustments, and final testing, dry out the Beverage Maker and place the Beverage Maker in a clean, dust free, moisture proof container. A sealed polyethylene bag is acceptable.

      (2) The Beverage Maker may be stored in any normal ambient temperature, pressure, and humidity environment.

      (3) Mark the Beverage Maker container with the correct part identification. Include the name of the Beverage Maker, part number, Beverage Maker condition, and date of repair.

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1. Introduction

The ILLUSTRATED PARTS LIST is intended for use in provisioning, storing, and issuing replaceable parts for the component, and in identification of new and reclaimed parts.

2. Vendor's Parts

Vendor's parts used in the unit and which are not altered by B/E Aerospace, Inc. are listed in the Detailed Parts List by the vendor's part number, vendor's description of the part, and vendor's code listed in parentheses following the description. Vendors' code symbols used in this publication are taken from the CAGE Cataloging Handbook, and consist of the applicable code symbol preceded by the Letter "V." Table 10001 shows a numerically arranged list of vendor codes used in this publication.

Table 10001
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V01139	Momentive Performance Materials Inc. 260 Hudson River Road Saratoga, NY 12188
V02570	Swagelok Companies 29500 Solon Road Solon, OH 44139-3449
V05972	Loctite Corporation 1001 Trout Brook Crossing Rocky Hill, CT 06067-3582
V08LA7	United Resin Corp 1305 E Street Gertrude Pl Suite B Orange, CA 92705
V0PK96	Quadtech Inc 734 Forest St 500 Marlborough, MA 01752
V1Y0Z7	Bostik, Inc. Construction And Distribution Division 205 Oliver Dr Marshall, MI 49068
V25281	Oetiker Inc. 6317 Euclid St Marlette, MI 48453
V27264	Molex, Inc. DBA Molex Automotive Division 2222 Wellington Court Lisle, IL 60532-1682
V63367	B/E Aerospace, Inc. 10800 Pflumm Road Lenexa, Kansas 66215

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Table 10001 (Continued)
Vendor Information

VENDOR CODE (CAGE)	ADDRESS
V71410	Henkel Corp. 32100 Stephenson Hwy Madison Heights, Mi 48071
V71984	Dow Corning Corporation 2200 W. Salzburg Road Midland, MI 48686
V72653	GC/Waldom Inc. 1801 Morgan Street Winnebago, IL 61102
V75297	Kester, Inc. 800 W Thorndale Avenue Dupage, IL 60143
V79074	Varflex Corporation 512 W Court St Rome, NY 13440
V9X418	OK Industries Inc. 134 Marbledale Road Tuckahoe, NY 10707

3. Use of the Illustrated Parts List

   A. When the part number is known:

      Turn to the Numerical Index and locate the part number. Figure and item numbers on the illustration where the part appears are listed in the column to the right of the part number. Turn to the illustration and locate the item number. The corresponding item number in the accompanying Detailed Parts List will give part number, description, assembly relationship, quantity required for that particular application, and whether or not the part appears in the illustrations.

   B. When the part number is not known:

      Look through the figures and identify the part by appearance. Note the figure number and the item number. The corresponding Fig-Item number in the Detailed Parts List shows the part number, description, assembly relationship, effectivity code, and the number of units per assembly.

4. Material Arrangement

   A. Numerical Index

      The Numerical Index is an alphanumeric listing of all the part numbers in the Detailed Parts List. Each part number is listed together with its corresponding item number, ILLUSTRATED PARTS LIST figure number, and the total required number of units per assembly. The figure-item number is composed of two parts. To the left of the dash is the figure number. It is the number of the figure in the ILLUSTRATED PARTS LIST that shows the part. To the right of the dash is the item number. It is the number that identifies a part in a figure.

B. Illustrations

The ILLUSTRATED PARTS LIST illustrations interrelate to the Numerical Index and the Detailed Parts List. The separate illustrations pictorially describe recommended disassembly/assembly sequence and support DISASSEMBLY and ASSEMBLY sections as well as the Numerical Index and the Detailed Parts List of this manual.

C. Detailed Parts List

(1) Fig-Item

The Detailed Parts List is arranged numerically by item numbers. These are the same item numbers listed in the Numerical Index. In the Detailed Parts List, the figure number portion of the figure-item number is listed only once per column. Immediately below a figure number are the item numbers that appear on that figure. A dash in front of an item number means the corresponding part is not shown in that or any other figure. Each item number is listed together with its corresponding part number, nomenclature (description, assembly relationship), effectivity code, and units per assembly.

(2) Part Number

The part number column contains the B/E Aerospace part number or if there is no B/E Aerospace Part Number the original manufacturer's part number will be shown. When standard parts are used, the standard part number is listed in this column.

(3) Airline Part Number

This column is left blank for airline internal use.

(4) Nomenclature

(a) Indenture System

The indenture system shows the relationship of parts and assembly to next higher assembly or installations, as follows:

1 2 3 4 5 6 7
Unit (Product which this manual describes)
• Detail parts for assembly into the unit
• Subassembly
• Attaching parts for subassembly
* * *
• • Detail parts for subassembly
• • Sub-subassembly
• • Attaching parts for sub-subassembly
* * *
• • • Detailed parts for sub-subassembly

(b) Attaching Parts

Attaching parts are captioned ATTACHING PARTS and are listed immediately following the parts attached. The * * * symbol follows the last item of the attaching parts group.

(c) Vendor Code

Parts manufactured by companies other than B/E Aerospace, Inc. are identified by an appropriate vendor's CAGE code following the nomenclature, and are preceded by the letter "V." Standard parts such as AN, MS, etc., are not identified by a vendor code.

(5) Effectivity Code

When two or more units (main assembly) are listed in the same ILLUSTRATED PARTS LIST, a code ("A", "B", "C", etc.) is assigned to each main unit. All subcomponents that are peculiar to a particular unit are identified by the same effectivity code letter as the unit. If parts are common to all units, the effectivity code column is left blank. The coding used in the ILLUSTRATED PARTS LIST in this manual is shown in Table 10002.

Table 10002
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
4510-31UF-00	1	A
4510-31LF-00	1	B
4510-32UF-00	1	C
4510-32LF-00	1	D
4510-33UF-00	1	E
4510-33LF-00	1	F
4510-34UF-00	1	G
4510-34LF-00	1	H
4510-35UF-00	1	J
4510-35LF-00	1	K
4510-36UF-00	1	L
4510-36LF-00	1	M
4510-37UF-00	1	N
4510-37LF-00	1	P
4510-38UF-00	1	R
4510-38LF-00	1	T
4510-41UF-00	1A	A
4510-41LF-00	1A	B
4510-42UF-00	1A	C
4510-42LF-00	1A	D
4510-43UF-00	1A	E
4510-43LF-00	1A	F

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Table 10002 (Continued)
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
4510-44UF-00	1A	G
4510-44LF-00	1A	H
4510-45UF-00	1A	J
4510-45LF-00	1A	K
4510-46UF-00	1A	L
4510-46LF-00	1A	M
4510-47UF-00	1A	N
4510-47LF-00	1A	P
4510-48UF-00	1A	R
4510-48LF-00	1A	T
4510-31UG-00	1B	A
4510-31LG-00	1B	B
4510-32UG-00	1B	C
4510-32LG-00	1B	D
4510-33UG-00	1B	E
4510-33LG-00	1B	F
4510-34UG-00	1B	G
4510-34LG-00	1B	H
4510-35UG-00	1B	J
4510-35LG-00	1B	K
4510-36UG-00	1B	L
4510-36LG-00	1B	M
4510-37UG-00	1B	N
4510-37LG-00	1B	P
4510-38UG-00	1B	R
4510-38LG-00	1B	T
4510-41UG-00	1C	A
4510-41LG-00	1C	B
4510-42UG-00	1C	C
4510-42LG-00	1C	D
4510-43UG-00	1C	E
4510-43LG-00	1C	F

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Table 10002 (Continued)
Effectivity Guide

ASSEMBLY PN	IPL FIGURE NO.	EFF CODE
4510-44UG-00	1C	G
4510-44LG-00	1C	H
4510-45UG-00	1C	J
4510-45LG-00	1C	K
4510-46UG-00	1C	L
4510-46LG-00	1C	M
4510-47UG-00	1C	N
4510-47LG-00	1C	P
4510-48UG-00	1C	R
4510-48LG-00	1C	T

(6) Units per assembly

Quantities specified in the "UNITS PER ASSEMBLY" column are the total number of each part required per assembly or subassembly and are not necessarily the total used per the complete unit.

(a) “AR” indicates “as required” and refers to variable quantity items, such as shims, spacers, or similar items.

(b) “RF” indicates “reference” and refers to items and their quantities which are listed elsewhere and are provided here as a reference only.

(c) “NP” indicates “non procurable” and refers to items which are not procurable and may not be ordered.

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5. Parts Replacement Data

Interchangeability relationship between parts is identified in the “NOMENCLATURE” column. A list of terms used and their definition is as follows.

TERM	ABBREVIATION	DEFINITION
Alternate	ALT	Part fully meets functional and structural specifications, but differs either in overall external dimensions, connection installation, and/or mounting provisions, and may require additional rework or modifications to install in a specific application.
Optional	OPT	The part is fully interchangeable in form, fit, and function with other item numbers shown.
Preferred	PRFD	Part is preferred over the other optional parts shown.
Replaced By	REPLD BY	Part is replaced by and is interchangeable with item number listed in notation.
Replaces	REPLS	Part replaces and is interchangeable with number listed in notation.
Superseded By	SUPSD BY	Part is replaced by and is not interchangeable with item number listed in notation.
Supersedes	SUPSDS	Part replaces and is not interchangeable with item number listed in notation.
Service Bulletins	PRE SB POST SB RWK SB	Identifies when a service bulletin (SB) has an effect on a part. PRE shows the condition of the part before the work in the service bulletin is done; POST shows the condition after the work is done; RWK shows that the part is reworked by the service bulletin.

6. Alpha/Numerical Index

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
AN316C9R		2	-165A	1
		2A	-165A	1
		2B	-165A	1
		2C	-165A	1
CL315		3	47	1
CL3201		2	235	1
		2A	235	1
		2B	235	1
		2C	235	1
CL50010		2	307	1
		2	539	1
		2A	307	1
		2A	539	1
		2B	307	1
		2B	539	1
		2C	307	1
		2C	539	1
CL50034		2	532	1
		2A	532	1
		2B	532	1
		2C	532	1
CO50311		4	-39C	3
CO50312		4	-39D	3
CO50313		3	-15B	1
CO50314		3	-15C	1
CO50403		4	-39A	1
CO50409		11	10	1
CO6505		4	20	1
CO6505-01		4	25	1
CO6505-03		4	30	6
CO6505-05		4	35	1
GR098		2	67	2
		2	70	2
		2	587	1
		2	-680	1
		2	-680A	DEL
		2A	67	2
		2A	70	2

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
GR098		2A	587	1
		2A	-680	1
		2A	-680A	DEL
		2B	67	2
		2B	70	2
		2B	587	1
		2B	-680	1
		2B	-680A	DEL
		2C	67	2
		2C	70	2
		2C	587	1
		2C	-680	1
		2C	-680A	DEL
GR50053		4	40	1
LW163		2	170	1
		2A	170	1
		2B	170	1
		2C	170	1
MISC013		7	25	1
		8	35	1
MISC3201		1A	60*	AR
		1	60*	AR
		1C	60*	AR
MISC50389		3	-120	1
MISC50408		4	-38	1
MISC50411		3	104A	1
		3	-104D	1
MISC50470		3	-104C	1
MISC50509		3	-104E	1
MS15795-804		2	105	2
		2	528	2
		2	-528A	2
		2A	105	2
		2A	528	2
		2A	-528A	2
		2B	105	2
		2B	528	2
		2B	-528A	2
		2C	105	2
		2C	528	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS15795-804		2C	-528A	2
MS15795-805		2	-225	3
		2A	-225	3
		2B	-225	3
		2C	-225	3
		9	40	2
		9	85	2
		10	20	3
MS15795-807		2	130	1
		2	345	2
		2	465	2
		2	740	2
		2A	130	1
		2A	345	2
		2A	465	2
		2A	740	2
		2B	130	1
		2B	345	2
		2B	465	2
		2B	740	2
		2C	130	1
		2C	345	2
		2C	465	2
		2C	740	2
		9	70	2
MS15795-808		2	705	2
		2A	705	2
		2B	705	2
		2C	705	2
MS16633-4012		2	510	1
		2A	510	1
		2B	510	1
		2C	510	1
MS16633-4031		8	15	1
MS21042L3		4	45	2
MS21043-04		3	40	6
MS21043-06		2	220	3
		2A	220	3
		2B	220	3
		2C	220	3
		9	25	4

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS21043-06		9	45	2
MS21209C0615		4	-11	10
MS21209C0810		2	12	1
		2B	12	1
		2C	12	1
		2A	12	1
MS24693C24		2	50	5
		2	280A	2
		2	690A	2
		2A	50	5
		2A	280A	2
		2A	690A	2
		2B	50	5
		2B	280A	2
		2B	690A	2
		2C	50	5
		2C	280A	2
		2C	690A	2
MS24693C26		2	-280	2
		2	-690	2
		2A	-280	2
		2A	-690	2
		2B	-280	2
		2B	-690	2
		2C	-280	2
		2C	-690	2
MS24693C27		10	14	2
		10	40	2
MS24693C272		2	370	3
		2A	370	3
		2B	370	3
		2C	370	3
MS24693C31		9	35	2
MS24693C48		2	60	6
		2	135	5
		2	180	6
I		2	-329	2
		2	425	4
		2	475	1
		2	520	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2	-555	8
MS24693C48		2	555A	4
		2A	60	6
		2A	135	5
		2A	180	6
		2A	-329	2
		2A	425	4
		2A	475	1
		2A	520	2
		2A	-555	8
		2A	555A	4
		2B	60	6
		2B	135	5
		2B	180	6
		2B	-329	2
		2B	425	4
		2B	475	1
		2B	520	2
		2B	-555	8
		2B	555A	4
		2C	60	6
		2C	135	5
		2C	180	6
		2C	-329	2
		2C	425	4
		2C	475	1
		2C	520	2
		2C	-555	8
		2C	555A	4
MS24693C49		2	557	4
		2A	557	4
		2B	557	4
		2C	557	4
MS24693C51		2	20	4
		2A	20	4
		2B	20	4
		2C	20	4
MS24693C53		2	330	2
		2A	330	2
		2B	330	2
		2C	330	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
MS35338-135		2	415	2
MS35338-135		2	527	2
		2	-527A	2
		2	542	1
MS35338-135		2A	415	2
		2A	527	2
		2A	-527A	2
		2A	542	1
		2B	415	2
		2B	527	2
		2B	-527A	2
		2B	542	1
		2C	415	2
		2C	527	2
		2C	-527A	2
		2C	542	1
MS35338-136		2	595	2
		2A	595	2
		2B	595	2
		2C	595	2
		3	85	1
		9	80	2
		10	25	3
MS35338-137		2	460	2
		2	580	4
		2	735	2
		2A	460	2
		2A	580	4
		2A	735	2
		2B	460	2
		2B	580	4
		2B	735	2
		2C	460	2
		2C	580	4
		2C	735	2
		9	65	2
MS35338-138		2	395	1
		2	700	2
		2A	395	1
		2A	700	2
		2B	395	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2B	700	2
MS35338-138		2C	395	1
		2C	700	2
MS51849-53C		2	730A	2
MS51849-53C		2A	730A	2
		2B	730A	2
		2C	730A	2
		2	-730	2
		2B	-730	2
		2A	-730	2
MS51849C53		2C	-730	2
MS51957-13		2	526	2
		2	-526A	2
		2	540	1
		2A	526	2
		2A	-526A	2
		2A	540	1
		2B	526	2
		2B	-526A	2
		2B	540	1
		2C	526	2
		2C	-526A	2
		2C	540	1
MS51957-15		2	410	2
		2A	410	2
		2B	410	2
		2C	410	2
MS51957-20		2	110	2
		2A	110	2
		2B	110	2
		2C	110	2
MS51957-26		3	90	1
MS51957-27		2	80	5
		2	190	8
		2	200	7
		2	-232	1
		2A	80	5
		2A	190	8
		2A	200	7
		2A	-232	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2A	-775	1
MS51957-27		2B	80	5
		2B	190	8
		2B	200	7
		2B	-232	1
		2C	80	5
		2C	190	8
		2C	200	7
		2C	-232	1
		2C	-775	1
		3	50	2
MS51957-29		2	230	1
		2	590	2
		2A	230	1
		2A	590	2
		2B	230	1
		2B	590	2
		2C	230	1
		2C	590	2
		9	75	2
MS51957-30		10	30	3
MS51957-45		9	60	2
MS51957-46		1A	45	2
		1	45	2
		1C	45	2
		2	40	2
		2	125	1
		2	455	2
		2A	40	2
		2A	125	1
		2A	455	2
		2B	40	2
		2B	125	1
		2B	455	2
		2C	40	2
		2C	125	1
		2C	455	2
MS51958-68		2	400	1
		2A	400	1
		2B	400	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2C	400	1
NAS1149D0316H		2	255	1
		2A	255	1
		2B	255	1
NAS1149D0316H		2C	255	1
		4	50	4
NAS1149F0332P		2	250	1
		2A	250	1
		2B	250	1
		2C	250	1
		4	55	2
NAS1189E06P5L		1A	40	2
		1	40	2
		1C	40	2
NAS1801-3-4		2	245	1
		2A	245	1
		2B	245	1
		2C	245	1
NAS1801-3-8		4	60	2
NS50113		3	55	2
OR0341		6	15	1
OR0361		2	-295	1
		2A	-295	1
		2B	-295	1
		2C	-295	1
OR0381		2	295A	1
		2A	295A	1
		2B	295A	1
		2C	295A	1
OR0571		2	306	2
		2	500	1
		2	534	1
		2A	306	2
		2A	500	1
		2A	534	1
		2B	306	2
		2B	500	1
		2B	534	1
		2C	306	2
		2C	500	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2C	534	1
OR0571		6	20	1
		7	15	1
		8	25	1
OR0662		2	155	1
		2A	155	1
		2B	155	1
		2C	155	1
OR067		2	150	1
		2A	150	1
		2B	150	1
		2C	150	1
OR0941		3	20	3
		3	95	1
OR50001		8	20	1
OR50019		3	104B	1
OR50039		2	215	1
		2A	215	1
		2B	215	1
		2C	215	1
PF102		3	60	1
		11	25A	2
		11	25B	1
PF110		11	20	2
PF143		2	765	2
		2A	765	2
		2B	765	2
		2C	765	2
PF202		2	-320	DEL
		2	755	2
		2A	-320	DEL
		2A	755	2
		2B	-320	DEL
		2B	755	2
		2C	-320	DEL
		2C	755	2
PF50002		2	760	2
		2A	760	2
		2B	760	2
		2C	760	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
PN5855		4	65	1
SC50268-30		10	30A	2
SC50270		10	32	1
SC50280-29		9	-75A	2
SC50280-45		9	-60A	2
SN431		2	-120	1
		2	350	2
		2A	-120	1
		2A	350	2
		2B	120	1
		2B	350	2
		2C	120	1
		2C	350	2
TE215		4	36	3
TE50432		3	-15A	1
TE50441		4	-37	5
		4	-39B	3
TE50442		12	-15	2
TU379		2	30	1
		2A	30	1
		2B	30	1
		2C	30	1
TU4505		2	308	1
		2	747	1
		2A	308	1
		2A	747	1
		2B	308	1
		2B	747	1
		2C	308	1
		2C	747	1
TU50020		2	745	1
		2A	745	1
		2B	745	1
		2C	745	1
TU50039		3	17	1
TU50052		11	35	1
TU50242		2	-530	1
		2A	-530	1
		2B	-530	1
		2C	-530	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
VA1003		3	160	1
VA1004		3	165	1
VA1031		3	155	1
VA50107		3	135A	1
WA120		2	240	1
		2A	240	1
		2B	240	1
		2C	240	1
WA50036		2	85	6
		2A	85	6
		2B	85	6
		2C	85	6
WI142		4	19	AR
WI156		3	35	1
WI50106		4	18	AR
352-0983-3		3	30A	3
3520-0242-01		2	-602	1
		2	-602A	DEL
		2A	-602	1
		2A	-602A	DEL
		2B	-602	1
		2B	-602A	DEL
		2C	-602	1
		2C	-602A	DEL
3520-0509-01		3	25	1
3520-0523-01		3	-100	1
3520-0525-01		2	270	1
		2A	270	1
		7	1	RF
3520-0528-01		2A	770	1
		2C	770	1
3520-0529-01		2	15	1
		2A	15	1
		2B	15	1
		2C	15	1
		3	-1	RF
3520-0530-01		2	55	1
		2A	55	1
		2B	55	1
		2C	55	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		4	-1	RF
3520-0531-01		4	10	1
3520-0533-03		2	325	1
		2A	325	1
3520-0533-03		2B	325	1
		2C	325	1
3520-0534-01		2	300	1
		2A	300	1
		2B	300	1
		2C	300	1
3520-0534-01		9	-1	RF
3520-0535-01		2	285	1
		2A	285	1
		2B	285	1
		2C	285	1
3520-0535-03		2	-290	1
		2A	-290	1
		2B	-290	1
		2C	-290	1
3520-0536-01		2	585	1
		2A	585	1
		2B	585	1
		2C	585	1
3520-0537-01		2	550	2
		2A	550	2
		2B	550	2
		2C	550	2
3520-0540-01		1A	20A	1
		1	20A	1
		1C	20A	1
3520-0540-03		1A	20B	1
		1	20B	1
		1C	20B	1
3520-0542-01		2	420	1
		2A	420	1
		2B	420	1
		2C	420	1
		4	15	1
3520-0544-03		4	-39	1
3520-0544-05		4	-17	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0548-01		2	10	1
3520-0548-01		2A	10	1
		2B	10	1
		2C	10	1
3520-0550-01		3	125	1
3520-0551-01		3	15	3
3520-0552-01		2	305	1
		2A	305	1
		2B	305	1
		2C	305	1
3520-0553-01		2	65	1
		2A	65	1
		2B	65	1
		2C	65	1
3520-0554-01		2	750	1
		2A	750	1
		2B	750	1
		2C	750	1
3520-0555-01		3	10	1
3520-0556-01		2	-605	1
		2A	-605	1
		2B	-605	1
		2C	-605	1
3520-0556-02		2	-610	1
		2A	-610	1
		2B	-610	1
		2C	-610	1
3520-0556-03		2	-615	1
		2A	-615	1
		2B	-615	1
		2C	-615	1
3520-0556-04		2	-620	1
		2A	-620	1
		2B	-620	1
		2C	-620	1
3520-0556-05		2	-625	1
		2A	-625	1
		2B	-625	1
		2C	-625	1
3520-0556-06		2	-630	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2A	-630	1
3520-0556-06		2B	-630	1
		2C	-630	1
		2	-635	1
3520-0556-07		2A	-635	1
		2B	-635	1
		2C	-635	1
3520-0556-08		2	-640	1
		2A	-640	1
		2B	-640	1
		2C	-640	1
3520-0558-01		8	30	1
3520-0559-01		2B	275	1
		2C	275	1
		8	1	RF
3520-0560-01		2	430	1
		2A	430	1
		2B	430	1
		2C	430	1
3520-0560-03		2	-435	1
		2A	-435	1
		2B	-435	1
		2C	-435	1
3520-0560-05		2	-440	1
		2A	-440	1
		2B	-440	1
		2C	-440	1
3520-0560-07		2	-445	1
		2A	-445	1
		2B	-445	1
		2C	-445	1
3520-0560-09		2	-450	1
		2A	-450	1
		2B	-450	1
		2C	-450	1
3520-0561-05		2	185	1
		2A	185	1
		2B	185	1
		2C	185	1
3520-0561-07		2	195	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2A	195	1
3520-0561-07		2B	195	1
		2C	195	1
3520-0566-01		2	100	1
3520-0566-01		2A	100	1
		2B	100	1
		2C	100	1
3520-0568-01		10	15	1
3520-0569-01		2	-545	1
		2A	-545	
		2B	-545	1
		2C	-545	1
		10	-1	RF
3520-0570-01		10	10	1
3520-0571-11		1	10A	1
		2	-1A	RF
3520-0571-12		1	10B	1
		2	-1B	RF
3520-0571-13		1	10C	1
		2	-1C	RF
3520-0571-14		1	10D	1
		2	-1D	RF
3520-0571-15		1	10E	1
		2	-1E	RF
3520-0571-16		1	10F	1
		2	-1F	RF
3520-0571-17		1	10G	1
		2	-1G	RF
3520-0571-18		1	10H	1
		2	-1H	RF
3520-0571-21		1A	10A	1
		2A	-1A	RF
3520-0571-22		1A	10B	1
		2A	-1B	RF
3520-0571-23		1A	10C	1
		2A	-1C	RF
3520-0571-24		1A	10D	1
		2A	-1D	RF
3520-0571-25		1A	10E	1
		2A	-1E	RF

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0571-26		1A	10F	1
3520-0571-26		2A	-1F	RF
3520-0571-27		1A	10G	1
		2A	-1G	RF
3520-0571-28		1A	10H	1
		2A	-1H	RF
3520-0571-31		2B	-1A	RF
3520-0571-32		2B	-1B	RF
3520-0571-33		2B	-1C	RF
3520-0571-34		2B	-1D	RF
3520-0571-35		2B	-1E	RF
3520-0571-36		2B	-1F	RF
3520-0571-37		2B	-1G	RF
3520-0571-38		2B	-1H	RF
3520-0571-41		1C	10A	1
		2C	-1A	RF
		1C	10B	1
		2C	-1B	RF
3520-0571-43		1C	10C	1
		2C	-1C	RF
3520-0571-44		1C	10D	1
		2C	-1D	RF
3520-0571-45		1C	10E	1
		2C	-1E	RF
3520-0571-46		1C	10F	1
		2C	-1F	RF
3520-0571-47		1C	10G	1
		2C	-1G	RF
3520-0571-48		1C	10H	1
		2C	-1H	RF
3520-0578-01		10	12	1
3520-0579-01		10	15A	1
3520-0580-01		2	545A	1
		2A	545A	1
		2B	545A	1
		2C	545A	1
		10	-1A	RF
3520-0581-01		3	120A	1
3520-0594-01		3	-100B	1
3520-0594-03		3	-100A	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3520-0595-01		10	10A	1
3520-0603-01		1A	-70A*	AR
		1	-70A*	AR
		1C	-70A*	AR
3520-0603-03		1A	-70B*	AR
		1	-70B*	AR
		1C	-70B*	AR
3520-0605-01		2	-710A	1
		2A	710A	1
		2B	710A	1
		2C	710A	1
		11	-1E	RF
3520-0605-03		2	-725A	1
		2A	-725A	1
		2B	-725A	1
		2C	-725A	1
		11	-1H	RF
3520-0605-05		2	-715A	1
		2A	-715A	1
		2B	-715A	1
		2C	-715A	1
		11	-1F	RF
3520-0605-07		2	-720A	1
		2A	-720A	1
		2B	-720A	1
		2C	-720A	1
		11	-1G	RF
3530-0121-01		2	75	1
		2A	75	1
		2B	75	1
		2C	75	1
		5	-1	DEL
3530-0232-01		2	710	1
		2A	-710	1
		2B	-710	1
		2C	-710	1
		11	-1A	RF
3530-0232-03		2	-725	1
		2A	-725	1
		2B	-725	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2C	-725	1
3530-0232-03		11	-1D	RF
3530-0232-05		2	-715	1
		2A	-715	1
3530-0232-05		2B	-715	1
		2C	-715	1
		11	-1B	RF
3530-0232-07		2	-720	1
		2A	-720	1
		2B	-720	1
		2C	-720	1
		11	-1C	RF
3530-0241-01		2	600	1
		2A	600	1
		2B	600	1
		2C	600	1
3530-0242-01		2	642	1
		2A	642	1
		2B	642	1
		2C	642	1
3530-0244-01		2	645	1
		2A	645	1
		2B	645	1
		2C	645	1
3530-0245-01		2	650	1
		2A	650	1
		2B	650	1
		2C	650	1
3530-0246-01		2	660	1
		2A	660	1
		2B	660	1
		2C	660	1
3530-0247-01		2	675	1
		2A	675	1
		2B	675	1
		2C	675	1
3530-0248-01		2	665	1
		2A	665	1
		2B	665	1
		2C	665	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3530-0249-01		2	670	1
3530-0249-01		2A	670	1
		2B	670	1
		2C	670	1
3530-0251-01		2	655	1
		2A	655	1
		2B	655	1
		2C	655	1
3530-0254-01		1A	-25A	1
		1	-25A	1
3530-0254-01		1C	-25A	1
3530-0254-03		1A	-25B	1
		1	-25B	1
		1C	-25B	1
3530-0261-01		2	210	1
		2A	210	1
		2B	210	1
		2C	210	1
3530-0268-01		3	-30	3
3530-0272-01		9	10A	1
3530-0274-01		2	-90	2
		2	-90A	DEL
		2A	-90	2
		2A	-90A	DEL
		2B	-90	2
		2B	-90A	DEL
		2C	-90	2
		2C	-90A	DEL
3530-0279-01		11	30A	3
		11	30C	2
		12	-1	RF
3530-0279-03		12	25	1
3530-0279-05		12	20	1
3530-0279-07		12	50	1
3530-0279-09		12	45	1
3530-0279-11		12	5	1
3530-0279-13		12	55	1
3530-0279-15		12	30	1
3530-0279-17		12	35	1
3530-0279-19		12	40	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
3530-0279-21		12	-10	1
3530-0279-23		11	30B	3
		11	30D	2
		12	-1A	RF
3530-0279-25		12	42	1
3530-0289-01		2	-525	1
		2A	-525	1
		2B	-525	1
		2C	-525	1
3530-0305-01		1A	-55*	AR
		1	-55*	AR
		1C	-55*	AR
3530-0310-01		1A	-75*	AR
		1	-75*	AR
		1C	-75*	AR
3530-0408-01		4	-13	1
3530-0879-01		2	525A	1
		2A	525A	1
		2B	525A	1
		2C	525A	1
3575-0010-01		1A	30	1
		1	30	1
		1C	30	1
3575-0050-01		2	560	1
		2A	560	1
		2B	560	1
		2C	560	1
3575-0050-02		2	565	1
		2A	565	1
		2B	565	1
		2C	565	1
3575-0189-01		9	30	1
3575-0388-01		1A	35	2
		1	35	2
		1C	35	2
3575-0435-01		2	515	1
		2A	515	1
		2B	515	1
		2C	515	1
3575-0613-01		2	360	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2A	360	1
3575-0613-01		2B	360	1
		2C	360	1
3575-0630-01		3	45	1
3575-0647-01		3	70	1
3575-0650-01		2	470	1
		2A	470	1
		2B	470	1
		2C	470	1
3575-1211-01		11	5A	1
3575-1211-03		11	5D	1
3575-1211-05		11	5B	1
3575-1211-07		11	5C	1
3575-1215-01		2	265	1
		2A	265	1
		2B	265	1
		2C	265	1
3575-1220-01		2	695	2
		2A	695	2
		2B	695	2
		2C	695	2
3575-1223-01		2	45	1
		2A	45	1
		2B	45	1
		2C	45	1
3575-1225-01		7	20	1
3575-1227-01		7	10	1
3575-1230-01		9	-10	1
3575-1236-03		10	35	2
3575-1236-05		10	13	2
3575-1244-01		2	685	1
		2A	685	1
		2B	685	1
		2C	685	1
3575-1246-01		8	10	1
3575-1250-01		4	-12	1
3575-1253-01		2	-95	1
		2	-95A	DEL
		2A	-95	1
		2A	-95A	DEL

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2B	-95	1
3575-1253-01		2B	-95A	DEL
		2C	-95	1
		2C	-95A	DEL
3575-1271-01		2	260	1
		2A	260	1
		2B	260	1
		2C	260	1
3575-1283-01		2	536	1
		2A	536	1
		2B	536	1
		2C	536	1
3575-1284-01		11	15	1
3575-1312-01		2	530A	1
		2A	530A	1
		2B	530A	1
		2C	530A	1
3575-1324-01		3	101	1
3575-1325-01		3	102	1
3575-1327-01		3	103	1
3575-1329-01		3	104	1
3575-1351-01		11	5E	1
3575-1351-03		11	5H	1
3575-1351-05		11	5F	1
3575-1351-07		11	5G	1
3575-1356-01		2	290A	1
		2A	290A	1
		2B	290A	1
		2C	290A	1
3575-1362-01		2	225A	3
		2A	225A	3
		2B	225A	3
		2C	225A	3
3575-1480-07		3	-15D	1
3575-2785-01		3	-103A	1
3575-2791-01		3	-104F	1
411-0676-3		2	385	1
		2A	385	1
		2B	385	1
		2C	385	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
411-0686-1		2	570	2
411-0686-1		2A	570	2
		2B	570	2
		2C	570	2
411-0689-1		2	575	2
		2A	575	2
		2B	575	2
		2C	575	2
411-0692-1		2	355	2
		2A	355	2
		2B	355	2
		2C	355	2
411-0693-1		9	15	1
411-0709-1		9	20	1
411-0724-1		2	335	2
		2A	335	2
		2B	335	2
		2C	335	2
411-0725-1		2	340	2
		2A	340	2
		2B	340	2
		2C	340	2
411-0748-1		9	50	1
411-0748-3		9	55	1
411-0926-1		2	-365	2
		2A	-365	2
		2B	-365	2
		2C	-365	2
416-0463-1		2	405	1
		2A	405	1
		2B	405	1
		2C	405	1
423-0078-3		2	175	2
		2A	175	2
		2B	175	2
		2C	175	2
425-0702-1		2	-328	2
		2A	-328	2
		2B	-328	2
		2C	-328	2
430531		2	-327	2

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2	-327A	DEL
430531		2A	-327	2
		2A	-327A	DEL
		2B	-327	2
		2B	-327A	DEL
		2C	-327	2
		2C	-327A	DEL
4510-31LF-00		1	-1B	RF
4510-31UF-00		1	-1A	RF
4510-32LF-00		1	-1D	RF
4510-32UF-00		1	-1C	RF
4510-33LF-00		1	-1F	RF
4510-33UF-00		1	-1E	RF
4510-34LF-00		1	-1H	RF
4510-34UF-00		1	-1G	RF
4510-35LF-00		1	-1K	RF
4510-35UF-00		1	-1J	RF
4510-36LF-00		1	-1M	RF
4510-36UF-00		1	-1L	RF
4510-37LF-00		1	-1P	RF
4510-37UF-00		1	-1N	RF
4510-38LF-00		1	-1T	RF
4510-38UF-00		1	-1R	RF
4510-41LF-00		1A	-1B	RF
4510-41LG-00		1C	-1B	RF
4510-41UF-00		1A	-1A	RF
4510-41UG-00		1C	-1A	RF
4510-42LF-00		1A	-1D	RF
4510-42LG-00		1C	-1D	RF
4510-42UF-00		1A	-1C	RF
4510-42UG-00		1C	-1C	RF
4510-43LF-00		1A	-1F	RF
4510-43LG-00		1C	-1F	RF
4510-43UF-00		1A	-1E	RF
4510-43UG-00		1C	-1E	RF
4510-44LF-00		1A	-1H	RF
4510-44LG-00		1C	-1H	RF
4510-44UF-00		1A	-1G	RF
4510-44UG-00		1C	-1G	RF
4510-45LF-00		1A	-1K	RF

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
4510-45LG-00		1C	-1K	RF
4510-45UF-00		1A	-1J	RF
4510-45UG-00		1C	-1J	RF
4510-46LF-00		1A	-1M	RF
4510-46LG-00		1C	-1M	RF
4510-46UF-00		1A	-1L	RF
4510-46UG-00		1C	-1L	RF
4510-47LF-00		1A	-1P	RF
4510-47LG-00		1C	-1P	RF
4510-47UF-00		1A	-1N	RF
4510-47UG-00		1C	-1N	RF
4510-48LF-00		1A	-1T	RF
4510-48LG-00		1C	-1T	RF
4510-48UF-00		1A	-1R	RF
4510-48UG-00		1C	-1R	RF
467-0865-1		2	309	1
		2A	309	1
		2B	309	1
		2C	309	1
467-0912-1		2	315	1
		2A	315	1
		2B	315	1
		2C	315	1
60-2001-1		2	380	1
		2A	380	1
		2B	380	1
		2C	380	1
60-2010-1		2	165	1
		2A	165	1
		2B	165	1
		2C	165	1
60-2013-1		2	485	1
		2A	485	1
		2B	485	1
		2C	485	1
62-0007-1		3	75	1
62-0008-1		6	10	1
62-0010-1		2	145	1
		2A	145	1
		2B	145	1
		2C	145	1

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PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
62-0011-1		3	80	1
62-0013-1		6	25	1
62-0014-1		2	140	1
		2A	140	1
		2B	140	1
		2C	140	1
		6	1	RF
62-0018-1		2	115	1
		2A	115	1
		2B	115	1
		2C	115	1
62-0020-1		3	65	1
62-0025-1		2	390	1
		2A	390	1
		2B	390	1
		2C	390	1
62-0037-1		2	160	1
		2A	160	1
		2B	160	1
		2C	160	1
62-0061-1		2	375	4
		2A	375	4
		2B	375	4
		2C	375	4
62-0066-7		2	310	1
		2	538	1
		2A	310	1
		2A	538	1
		2B	310	1
		2B	538	1
		2C	310	1
		2C	538	1
62-0136-11		3	115	1
		3	180	1
62-0136-7		3	105	1
		3	170	1
62-0136-9		3	110	1
		3	175	1
62-0144-1		2	35	1
		2A	35	1

- ITEM NOT ILLUSTRATED

PART NUMBER	AIRLINE PART NO.	IPL FIG.	ITEM	TTL REQ
		2B	35	1
		2C	35	1
62-0210-1		2	490	1
		2A	490	1
		2B	490	1
		2C	490	1
62-0210-2		2	505	1
62-0210-2		2A	505	1
		2B	505	1
		2C	505	1
62-0210-3		2	495	1
		2A	495	1
		2B	495	1
		2C	495	1
62-0211-1		2	480	1
		2A	480	1
		2B	480	1
		2C	480	1
		3	130	RF
62-0223-11		3	150	1
62-0223-3		3	135	1
62-0223-5		3	145	1
62-0223-9		3	140	1

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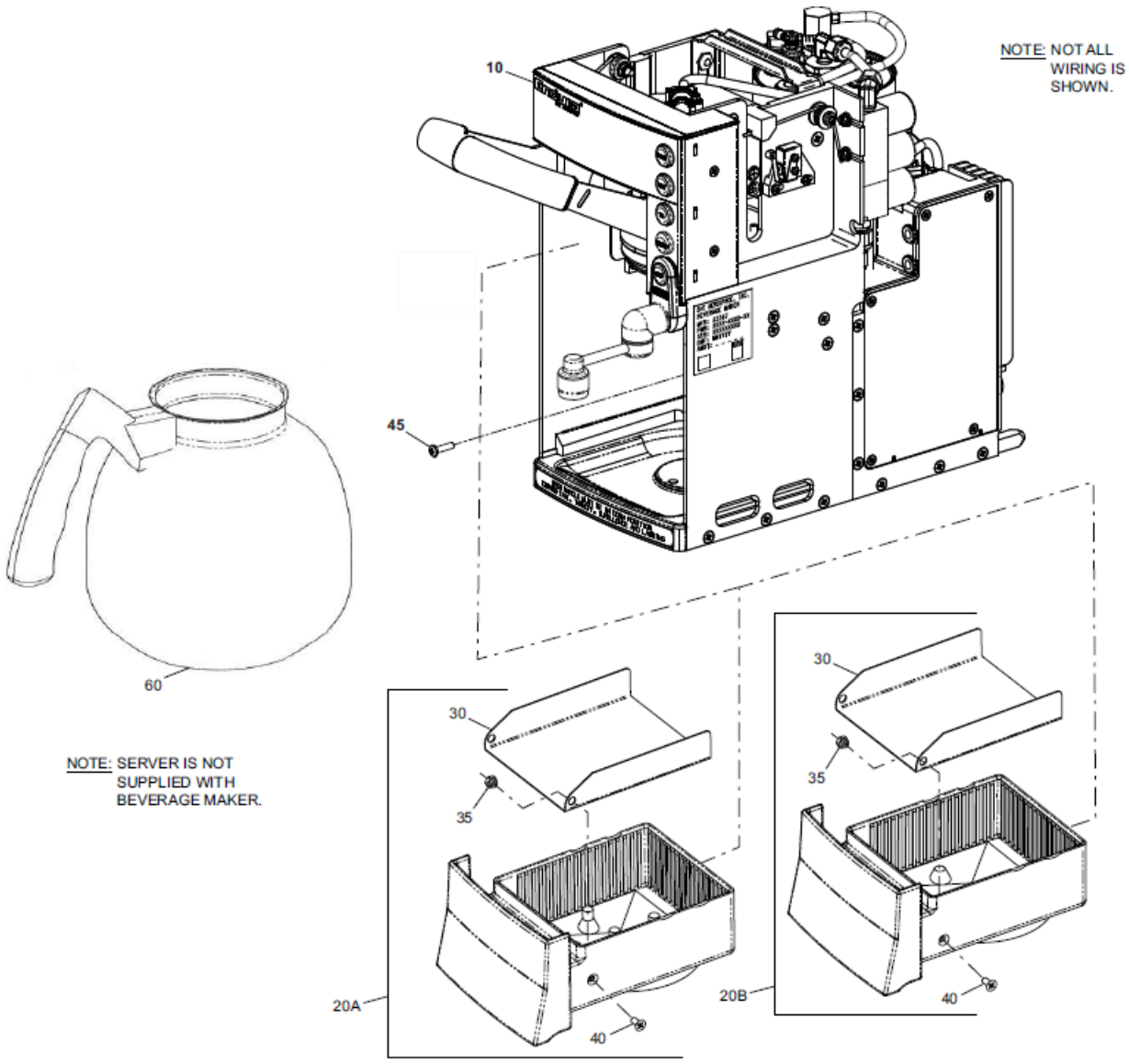

IPL Figure 1
Beverage Maker

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
-1A	4510-31UF-00		BEVERAGE MAKER	A	RF
-1B	4510-31LF-00		BEVERAGE MAKER	B	RF
-1C	4510-32UF-00		BEVERAGE MAKER	C	RF
-1D	4510-32LF-00		BEVERAGE MAKER	D	RF
-1E	4510-33UF-00		BEVERAGE MAKER	E	RF
-1F	4510-33LF-00		BEVERAGE MAKER	F	RF
-1G	4510-34UF-00		BEVERAGE MAKER	G	RF
-1H	4510-34LF-00		BEVERAGE MAKER	H	RF
-1J	4510-35UF-00		BEVERAGE MAKER	J	RF
-1K	4510-35LF-00		BEVERAGE MAKER	K	RF
-1L	4510-36UF-00		BEVERAGE MAKER	L	RF
-1M	4510-36LF-00		BEVERAGE MAKER	M	RF
-1N	4510-37UF-00		BEVERAGE MAKER	N	RF
-1P	4510-37LF-00		BEVERAGE MAKER	P	RF
-1R	4510-38UF-00		BEVERAGE MAKER	R	RF
-1T	4510-38LF-00		BEVERAGE MAKER	T	RF
10A	3520-0571-11		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	A,B	1
10B	3520-0571-12		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	C,D	1
10C	3520-0571-13		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	E,F	1
10D	3520-0571-14		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	G,H	1
10E	3520-0571-15		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	J,K	1
10F	3520-0571-16		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	L,M	1
10G	3520-0571-17		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	N,P	1
10H	3520-0571-18		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2 FOR DET BKDN)	R,T	1
20A	3520-0540-01		• CUP ASSY, BREW (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1					
20B	3520-0540-03		• CUP ASSY, BREW (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • CUP, BREW (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	1
-25B	3530-0254-03		• • CUP, BREW (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	1
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 6-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER		AR
60*	MISC3201		• SERVER (TPN MISC-3201) (OPT TO ITEM 55)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			*NOTE: NOT SUPPLIED WITH BEVERAGE MAKER		

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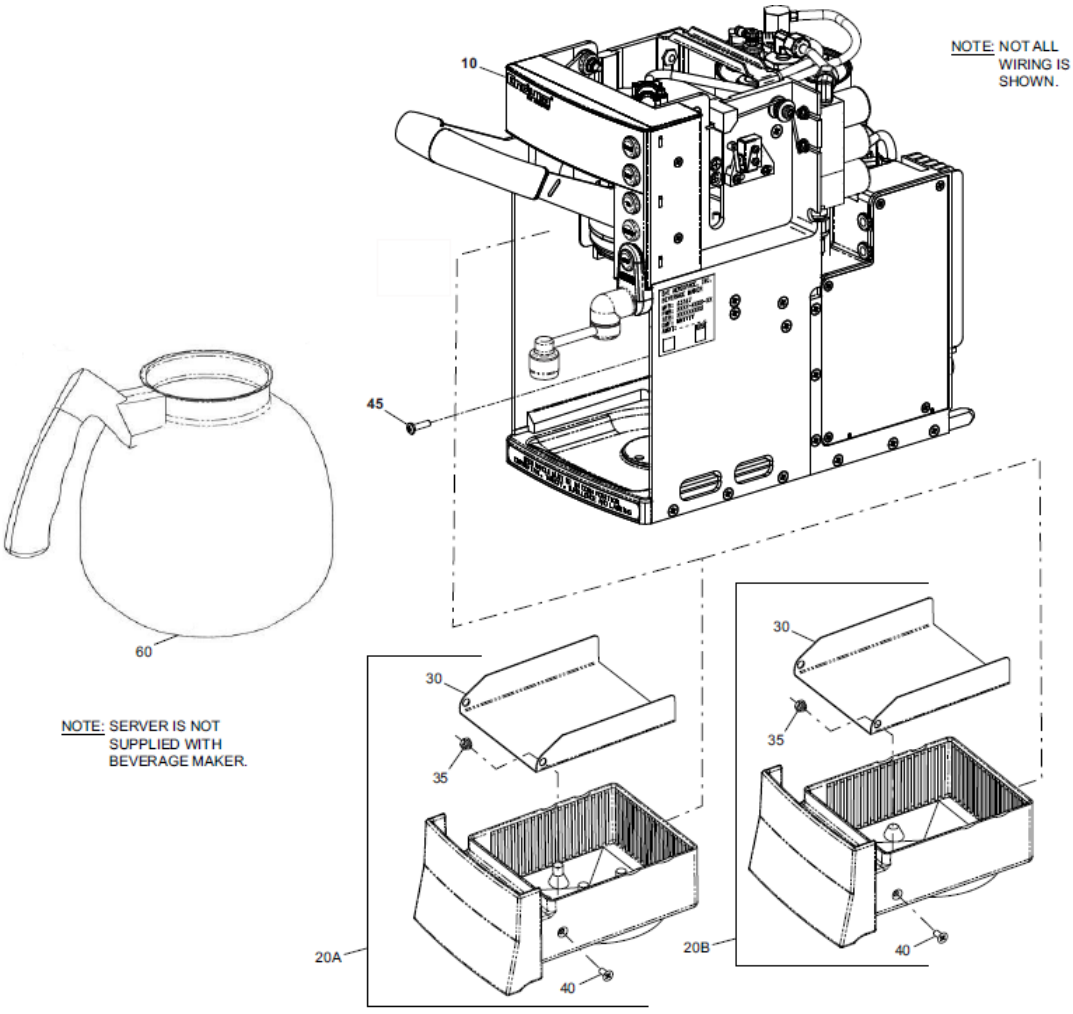

IPL Figure 1A
Beverage Maker

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1A					
-1A	4510-41UF-00		BEVERAGE MAKER	A	RF
-1B	4510-41LF-00		BEVERAGE MAKER	B	RF
-1C	4510-42UF-00		BEVERAGE MAKER	C	RF
-1D	4510-42LF-00		BEVERAGE MAKER	D	RF
-1E	4510-43UF-00		BEVERAGE MAKER	E	RF
-1F	4510-43LF-00		BEVERAGE MAKER	F	RF
-1G	4510-44UF-00		BEVERAGE MAKER	G	RF
-1H	4510-44LF-00		BEVERAGE MAKER	H	RF
-1J	4510-45UF-00		BEVERAGE MAKER	J	RF
-1K	4510-45LF-00		BEVERAGE MAKER	K	RF
-1L	4510-46UF-00		BEVERAGE MAKER	L	RF
-1M	4510-46LF-00		BEVERAGE MAKER	M	RF
-1N	4510-47UF-00		BEVERAGE MAKER	N	RF
-1P	4510-47LF-00		BEVERAGE MAKER	P	RF
-1R	4510-48UF-00		BEVERAGE MAKER	R	RF
-1T	4510-48LF-00		BEVERAGE MAKER	T	RF
10A	3520-0571-21		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	A,B	1
10B	3520-0571-22		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	C,D	1
10C	3520-0571-23		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	E,F	1
10D	3520-0571-24		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	G,H	1
10E	3520-0571-25		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	J,K	1
10F	3520-0571-26		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	L,M	1
10G	3520-0571-27		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	N,P	1
10H	3520-0571-28		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2A FOR DET BKDN)	R,T	1
20A	3520-0540-01		• CUP ASSY, BREW (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1A					
20B	3520-0540-03		• CUP ASSY, BREW (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • CUP, BREW (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	1
-25B	3530-0254-03		• • CUP, BREW (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	1
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 6-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER (NOT SUPPLIED WITH THE UNIT)		AR
60*	MISC3201		• SERVER (TPN MISC-3201) (OPT TO ITEM 55)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			*NOTE: NOT SUPPLIED WITH BEVERAGE MAKER		

- ITEM NOT ILLUSTRATED

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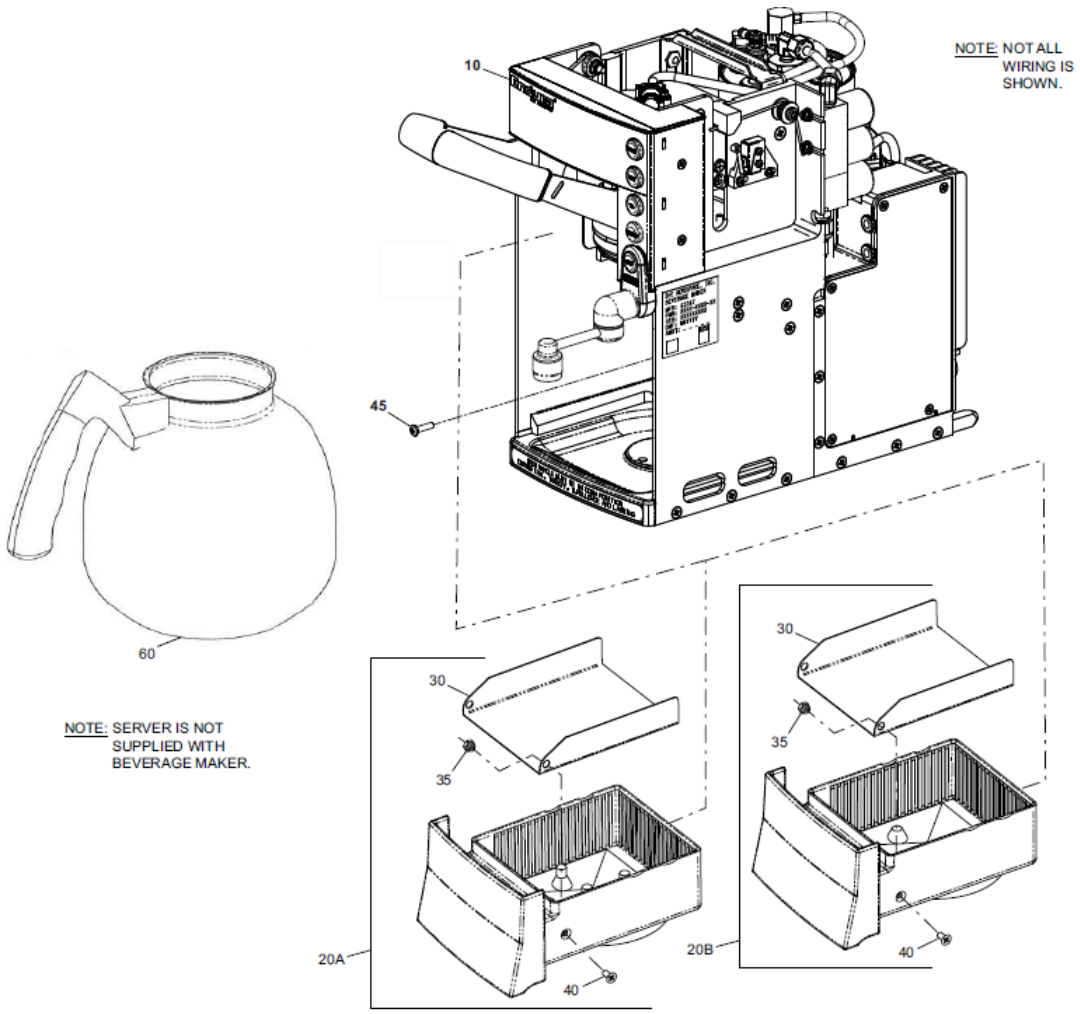

IPL Figure 1B
Beverage Maker

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1B					
-1A	4510-31UG-00		• BEVERAGE MAKER	A	RF
-1B	4510-31LG-00		• BEVERAGE MAKER	B	RF
-1C	4510-32UG-00		• BEVERAGE MAKER	C	RF
-1D	4510-32LG-00		• BEVERAGE MAKER	D	RF
-1E	4510-33UG-00		• BEVERAGE MAKER	E	RF
-1F	4510-33LG-00		• BEVERAGE MAKER	F	RF
-1G	4510-34UG-00		• BEVERAGE MAKER	G	RF
-1H	4510-34LG-00		• BEVERAGE MAKER	H	RF
-1J	4510-35UG-00		• BEVERAGE MAKER	J	RF
-1K	4510-35LG-00		• BEVERAGE MAKER	K	RF
-1L	4510-36UG-00		• BEVERAGE MAKER	L	RF
-1M	4510-36LG-00		• BEVERAGE MAKER	M	RF
-1N	4510-37UG-00		• BEVERAGE MAKER	N	RF
-1P	4510-37LG-00		• BEVERAGE MAKER	P	RF
-1R	4510-38UG-00		• BEVERAGE MAKER	R	RF
-1T	4510-38LG-00		• BEVERAGE MAKER	T	RF
10A	3520-0571-31		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	A,B	1
10B	3520-0571-32		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	C,D	1
10C	3520-0571-33		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	E,F	1
10D	3520-0571-34		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	G,H	1
10E	3520-0571-35		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	J,K	1
10F	3520-0571-36		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	L,M	1
10G	3520-0571-37		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	N,P	1
10H	3520-0571-38		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2B FOR DET BKDN)	R,T	1
20A	3520-0540-01		• CUP ASSY, BREW (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1B					
20B	3520-0540-03		• CUP ASSY, BREW (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • CUP, BREW (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	1
-25B	3530-0254-03		• • CUP, BREW (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	1
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 6-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER (NOT SUPPLIED WITH THE UNIT)		AR
60*	MISC3201		• SERVER (TPN MISC-3201) (OPT TO ITEM 55)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			*NOTE: NOT SUPPLIED WITH BEVERAGE MAKER		

- ITEM NOT ILLUSTRATED

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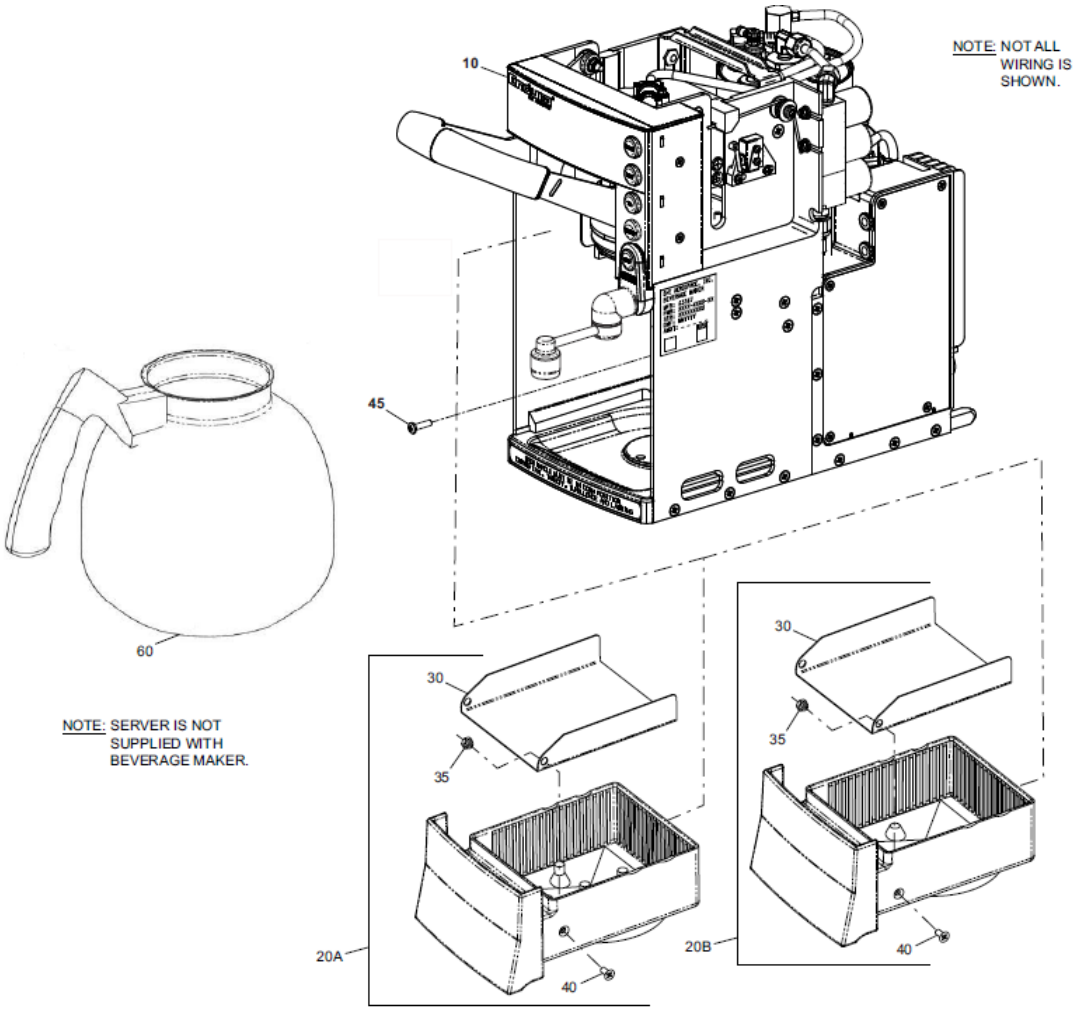

IPL Figure 1C
Beverage Maker

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1C					
-1A	4510-41UG-00		BEVERAGE MAKER	A	RF
-1B	4510-41LG-00		BEVERAGE MAKER	B	RF
-1C	4510-42UG-00		BEVERAGE MAKER	C	RF
-1D	4510-42LG-00		BEVERAGE MAKER	D	RF
-1E	4510-43UG-00		BEVERAGE MAKER	E	RF
-1F	4510-43LG-00		BEVERAGE MAKER	F	RF
-1G	4510-44UG-00		BEVERAGE MAKER	G	RF
-1H	4510-44LG-00		BEVERAGE MAKER	H	RF
-1J	4510-45UG-00		BEVERAGE MAKER	J	RF
-1K	4510-45LG-00		BEVERAGE MAKER	K	RF
-1L	4510-46UG-00		BEVERAGE MAKER	L	RF
-1M	4510-46LG-00		BEVERAGE MAKER	M	RF
-1N	4510-47UG-00		BEVERAGE MAKER	N	RF
-1P	4510-47LG-00		BEVERAGE MAKER	P	RF
-1R	4510-48UG-00		BEVERAGE MAKER	R	RF
-1T	4510-48LG-00		BEVERAGE MAKER	T	RF
10A	3520-0571-41		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	A,B	1
10B	3520-0571-42		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	C,D	1
10C	3520-0571-43		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	E,F	1
10D	3520-0571-44		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	G,H	1
10E	3520-0571-45		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	J,K	1
10F	3520-0571-46		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	L,M	1
10G	3520-0571-47		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	N,P	1
10H	3520-0571-48		• BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 2C FOR DET BKDN)	R,T	1
20A	3520-0540-01		• CUP ASSY, BREW (UPPER)	A,C,E,G, J,L,N,R	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
1C					
20B	3520-0540-03		• CUP ASSY, BREW (LOWER)	B,D,F,H, K,M,P,T	1
-25A	3530-0254-01		• • CUP, BREW (UPPER) (NOT AVAILABLE AS A SPARE)	A,C,E,G, J,L,N,R	1
-25B	3530-0254-03		• • CUP, BREW (LOWER) (NOT AVAILABLE AS A SPARE)	B,D,F,H, K,M,P,T	1
30	3575-0010-01		• • PLATE, SPACER		1
			ATTACHING PARTS		
35	3575-0388-01		• • NUT, HEX, SHOULDER		2
40	NAS1189E06P5L		• • SCREW, FH 6-32 X 31 SELF LKG		2
			* * *		
45	MS51957-46		• SCREW, PH, 6-32 X 63		2
-55*	3530-0305-01		• INSULATED SERVER (NOT SUPPLIED WITH THE UNIT)		AR
60*	MISC3201		• SERVER (TPN MISC-3201) (OPT TO ITEM 55)		AR
-70A*	3520-0603-01		• SERVER COVER ASSEMBLY (BLACK)		AR
-70B*	3520-0603-03		• SERVER COVER ASSEMBLY (ORANGE)		AR
-75*	3530-0310-01		• LANYARD SERVER COVER		AR
			*NOTE: NOT SUPPLIED WITH BEVERAGE MAKER		

- ITEM NOT ILLUSTRATED

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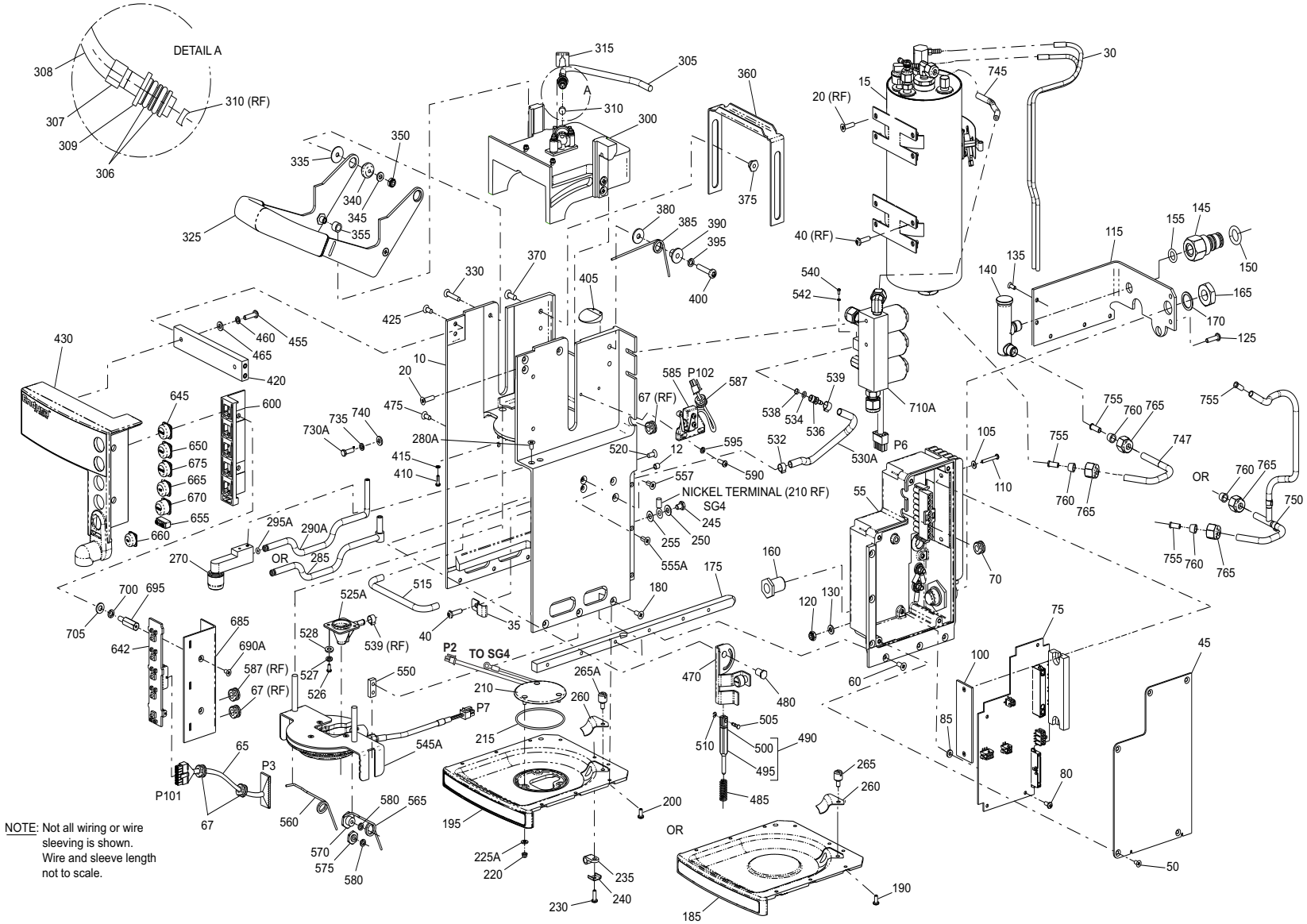

IPL Figure 2
Beverage Maker Main Subassembly

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-1A	3520-0571-11		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	A	RF
-1B	3520-0571-12		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	B	RF
-1C	3520-0571-13		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	C	RF
-1D	3520-0571-14		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	D	RF
-1E	3520-0571-15		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	E	RF
-1F	3520-0571-16		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	F	RF
-1G	3520-0571-17		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	G	RF
-1H	3520-0571-18		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1 FOR NHA)	H	RF
10	3520-0548-01		• ASSY, CHASSIS		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• ASSY, TANK (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		1
35	62-0144-1		• CLIP, HOSE		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		2
			* * *		
45	3575-1223-01		• COVER, POWER MODULE		1
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
55	3520-0530-01		• ASSY, POWER MODULE (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
65	3520-0553-01		• ASSY, HARNESS		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
75	3530-0121-01		• PC BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1
			ATTACHING PARTS		
-120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
140	62-0014-1		• ASSY, DRAIN VALVE (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• ASSY, WATER SHUTOFF PLUG		1
150	OR067		• • O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• GUIDE, ALIGNMENT PIN		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		• NUT, PLAIN HEXAGON (SUPSDS ITEM 165)		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-3		• GUIDE, RAIL		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-05		• ASSY, PLATEN	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	8
			* * *		
195	3520-0561-07		• ASSY, PLATEN (USED WITH PLATEN HEATER)	D,E,F,H	1
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	7
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
210	3530-0261-01		• HEATER, PLATEN	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X 44 (WITH WARMER)	D,E,F,H	1
-232	MS51957-27		• SCREW, PH, 6-32 X .31 (WITHOUT WARMER)	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,F,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1271-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
265	3575-1215-01		• FASTENER, SCREEN, DRAIN		1
			* * *		
270	3520-0525-01		• ASSY, FAUCET (SEE IPL FIGURE 7 FOR DET BKDN)		1
			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 280A)		2
280A	MS24693C24		• SCREW, PH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 280)		2
			* * *		
285	3520-0535-01		• ASSY, FAUCET TUBE	B,C,E, F,G,H	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-290	3520-0535-03		• ASSY, FAUCET TUBE (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• ASSY, BREWHEAD (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• ASSY, BREW TUBE		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±.2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• WASHER, FLOW		1
315	467-0912-1		• CLIP		1
-320	PF202		DELETED INSERT, TUBE (TPN PF-202)		DEL
			* * *		
325	3520-0533-03		• ASSY, BREW HANDLE		1
-327	430531		• • DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
-328	425-0702-1		• • ASSEMBLY, GUIDE BUSHING		2
-329	MS24693C48		• • SCREW, FH		2
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
355	411-0692-1		• BUSHING, HANDLE		2

- ITEM NOT ILLUSTRATED

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 375)		2
370	MS24693C272		• SCREW, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		
420	3520-0542-01		• ASSEMBLY, BRACE, CHASSIS		1
			ATTACHING PARTS		
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		4
			* * *		
430	3520-0560-01		• ASSY, PANEL, FRONT	F,H	1
-435	3520-0560-03		• ASSY, PANEL, FRONT	B,D,G	1
-440	3520-0560-05		• ASSY, PANEL, FRONT	A	1
-445	3520-0560-07		• ASSY, PANEL, FRONT	E	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-450	3520-0560-09		• ASSY, PANEL, FRONT	C	1
			ATTACHING PARTS		
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8		2
			* * *		
470	3575-0650-01		• LEVER, PIN, LOCATING		1
			ATTACHING PARTS		
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• ASSY, PIN, PIVOT		1
			* * *		
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• ASSY, PIN, LOCATING		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING (TPN OR-0571)		1
			ATTACHING PARTS		
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING		1
			* * *		
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3530-0289-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A) (PRE SB 4510-25-14)	B,F,G,H	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
525A	3530-0879-01		• FUNNEL, TEA TUBE (SUPSDS ITEM 525) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
526	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSD BY ITEM 526A)	B,F,G,H	2
-526A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 526)		2
527	MS35338-135		• WASHER, LK, (SUPSD BY ITEM 527A)	B,F,G,H	2
-527A	MS35338-135		• WASHER, LK, (SUPSDS ITEM 527)		2
528	MS15795-804		• WASHER, FL, #4 (SUPSD BY ITEM 528A)	B,F,G,H	2
-528A	MS15795-804		• WASHER, FL, #4 (SUPSDS ITEM 528)		2
-530	TU50242		• TUBE, 80 IN LONG (TPN TU-50242) (PRE SB 4510-25-02) (SUPSD BY ITEM 530A)	B,F,G,H	1
530A	3575-1312-01		• TUBE, TEA, HEAT FORMED (POST SB 4510-25-02) (SUPSDS ITEM 530)	B,F,G,H	1
532	CL50034		• CLAMP, TUBE (TPN CL-50034)	B,F,G,H	1
534	OR0571		• O-RING (TPN OR-0571)	B,F,G,H	1
536	3575-1283-01		• CARTRIDGE, REGULATOR	B,F,G,H	1
538	62-0066-7		• WASHER, FLOW	B,F,G,H	1
539	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010)	B,F,G,H	1
540	MS51957-13		• SCREW, PNH, 4-40 X .25	B,F,G,H	1
542	MS35338-135		• WASHER, LK, #4	B,F,G,H	1
			* * *		
-545	3520-0569-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (PRE SB 4510-25-03) (SUPSD BY ITEM 545A)		1
545A	3520-0580-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (POST SB 4510-25-03) (SUPSDS ITEM 545)		1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
			ATTACHING PARTS		
550	3520-0537-01		• ASSY, GUIDE, LLS		2
-555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSD BY ITEM 555A)		8
555A	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS ITEM 555)		4
557	MS24693C49		• SCREW, FH, (TPN MS24693-C49)		4
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1
565	3575-0050-02		• SPRING, RETAINER, SERVER, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• ASSY, SWITCH		1
587	GR098		• • GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2
			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A	3520-0242-01		DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)		1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C,D E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 690)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-710A	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0232-05		• ASSY, MANIFOLD (SUPSD BY ITEM 715A) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715A	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0232-07		• ASSY, MANIFOLD (SUPSD BY ITEM 720A) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2					
-720A	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-725	3530-0232-03		• ASSY, MANIFOLD (SUPSD BY ITEM 725A) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725A	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		• SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
745	TU50020		• TUBING, (1/4" OD X .063" WALL) 8.5 IN (TPN TU-50020)		1
747	TU4505		• TUBING-TEFLON (1/4" OD X .030" WALL), 4.0 IN (TPN TU-4505)	A,B,D,H	1
750	3520-0554-01		• ASSY, TUBE, INLET, WATER	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		

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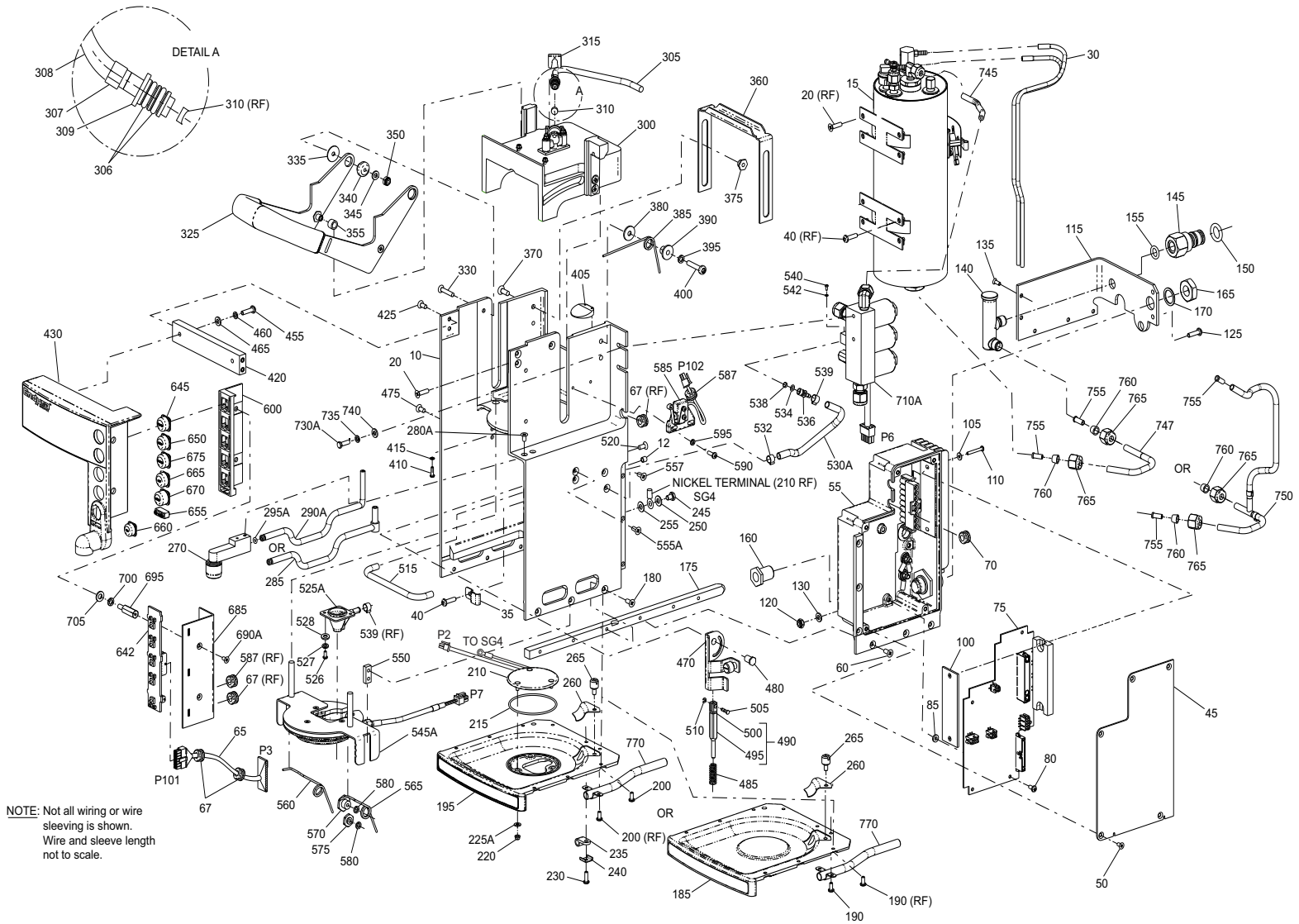

IPL Figure 2A
Beverage Maker Main Subassembly

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
-1A	3520-0571-21		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	A	RF
-1B	3520-0571-22		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	B	RF
-1C	3520-0571-23		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	C	RF
-1D	3520-0571-24		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	D	RF
-1E	3520-0571-25		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	E	RF
-1F	3520-0571-26		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	F	RF
-1G	3520-0571-27		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	G	RF
-1H	3520-0571-28		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1A FOR NHA)	H	RF
10	3520-0548-01		• ASSY, CHASSIS		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• ASSY, TANK (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		1
35	62-0144-1		• CLIP, HOSE		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		2
			* * *		
45	3575-1223-01		• COVER, POWER MODULE		1
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
55	3520-0530-01		• ASSY, POWER MODULE (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (MS24693-C48)		6
			* * *		
65	3520-0553-01		• ASSY, HARNESS		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
75	3530-0121-01		• PC BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1
			ATTACHING PARTS		
-120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
140	62-0014-1		• ASSY, DRAIN VALVE (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• ASSY, WATER SHUTOFF PLUG		1
150	OR067		• • O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• GUIDE, ALIGNMENT PIN		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		• NUT, PLAIN HEXAGON (SUPSDS ITEM 165)		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-3		• GUIDE, RAIL		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-05		• ASSY, PLATEN	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	8
			* * *		
195	3520-0561-07		• ASSY, PLATEN	D,E,F,H	1
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	7
			* * *		

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
210	3530-0261-01		• HEATER, PLATEN	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44 (WITH WARMER)	D,E,F,H	1
-232	MS51957-27		• SCREW, PH, 6-32 X .31 (WITHOUT WARMER)	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,F,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1271-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
265	3575-1215-01		• FASTENER, SCREEN, DRAIN (ALUMINUM)		1
			* * *		
270	3520-0525-01		• ASSY, FAUCET (SEE IPL FIGURE 7 FOR DET BKDN)		1
			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 280A)		2
280A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 280)		2
			* * *		

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
285	3520-0535-01		• ASSY, FAUCET TUBE	B,C,E, F,G,H	1
-290	3520-0535-03		• ASSY, FAUCET TUBE (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• ASSY, BREWHEAD (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• ASSY, BREW TUBE		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• WASHER, FLOW		1
315	467-0912-1		• CLIP		1
-320	PF202		DELETED INSERT, TUBE (TPN PF-202)		DEL
			* * *		
325	3520-0533-03		• ASSY, BREW HANDLE		1
-327	430531		• • DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
-328	425-0702-1		• • ASSEMBLY, GUIDE BUSHING		2
-329	MS24693C48		• • SCREW, FH		2
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
355	411-0692-1		• BUSHING, HANDLE		2
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 375)		2
			ATTACHING PARTS		
370	MS24693C272		• SCREW, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		
420	3520-0542-01		• ASSEMBLY, BRACE, CHASSIS		1
			ATTACHING PARTS		
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		4
			* * *		
430	3520-0560-01		• ASSY, PANEL, FRONT	F,H	1
-435	3520-0560-03		• ASSY, PANEL, FRONT	B,D,G	1
-440	3520-0560-05		• ASSY, PANEL, FRONT	A	1
-445	3520-0560-07		• ASSY, PANEL, FRONT	E	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
-450	3520-0560-09		• ASSY, PANEL, FRONT	C	1
			ATTACHING PARTS		
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8		2
			* * *		
470	3575-0650-01		• LEVER, PIN, LOCATING		1
			ATTACHING PARTS		
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• ASSY, PIN, PIVOT		1
			* * *		
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• ASSY, PIN, LOCATING		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING (TPN OR-0571)		1
			ATTACHING PARTS		
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING		1
			* * *		
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3530-0289-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A) (PRE SB 4510-25-14)	B,F,G,H	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
525A	3530-0879-01		• FUNNEL, TEA TUBE (SUPSDS ITEM 525) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
526	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSD BY ITEM 526A)	B,F,G,H	2
-526A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 526)		2
527	MS35338-135		• WASHER, LK, (SUPSD BY ITEM 527A)	B,F,G,H	2
-527A	MS35338-135		• WASHER, LK, (SUPSDS ITEM 527)		2
528	MS15795-804		• WASHER, FL, #4 (SUPSD BY ITEM 528A)	B,F,G,H	2
-528A	MS15795-804		• WASHER, FL, #4 (SUPSDS ITEM 528)		2
-530	TU50242		• TUBE, 80 IN LONG (TPN TU-50242) (PRE SB 4510-25-02) (SUPSD BY ITEM 530A)	B,F,G,H	1
530A	3575-1312-01		• TUBE, TEA, HEAT FORMED (POST SB 4510-25-02) (SUPSDS ITEM 530)	B,F,G,H	1
532	CL50034		• CLAMP, TUBE (TPN CL-50034)	B,F,G,H	1
534	OR0571		• O-RING (TPN OR-0571)	B,F,G,H	1
536	3575-1283-01		• CARTRIDGE, REGULATOR	B,F,G,H	1
538	62-0066-7		• WASHER, FLOW	B,F,G,H	1
539	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010)	B,F,G,H	1
540	MS51957-13		• SCREW, PNH, 4-40 X .25	B,F,G,H	1
542	MS35338-135		• WASHER, LK, #4	B,F,G,H	1
			* * *		
-545	3520-0569-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (PRE SB 4510-25-03) (SUPSD BY ITEM 545A)		
545A	3520-0580-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (POST SB 4510-25-03) (SUPSDS ITEM 545)		1
			ATTACHING PARTS		

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
550	3520-0537-01		• ASSY, GUIDE, LLS		2
-555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS BY ITEM 555A)		8
555A	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS ITEM 555)		4
557	MS24693C49		• SCREW, FH, (TPN MS24693-C49)		4
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1
565	3575-0050-02		• SPRING, RETAINER, SERVER, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• ASSY, SWITCH		1
587	GR098		• GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2
			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A	3520-0242-01		DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D, E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C,D E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 690)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
710A	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0232-05		• ASSY, MANIFOLD (SUPSD BY ITEM 715A) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715A	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0232-07		• ASSY, MANIFOLD (SUPSD BY ITEM 720A) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-720A	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2A					
-725	3530-0232-03		• ASSY, MANIFOLD (SUPSD BY ITEM 725A) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725A	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		• SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
745	TU50020		• TUBING, (1/4" OD X .063" WALL) 8.5 IN (TPN TU-50020)		1
747	TU4505		• TUBING-TEFLON (1/4" OD X .030" WALL), 4.0 IN (TPN TU-4505)	A,B,D,H	1
750	3520-0554-01		• ASSY, TUBE, INLET, WATER	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		
770	3520-0528-01		• ASSY, TUBE, DRAIN		1
			ATTACHING PARTS		
-775	MS51957-27		• SCREW, PH, 6-32 X .31		1
			* * *		

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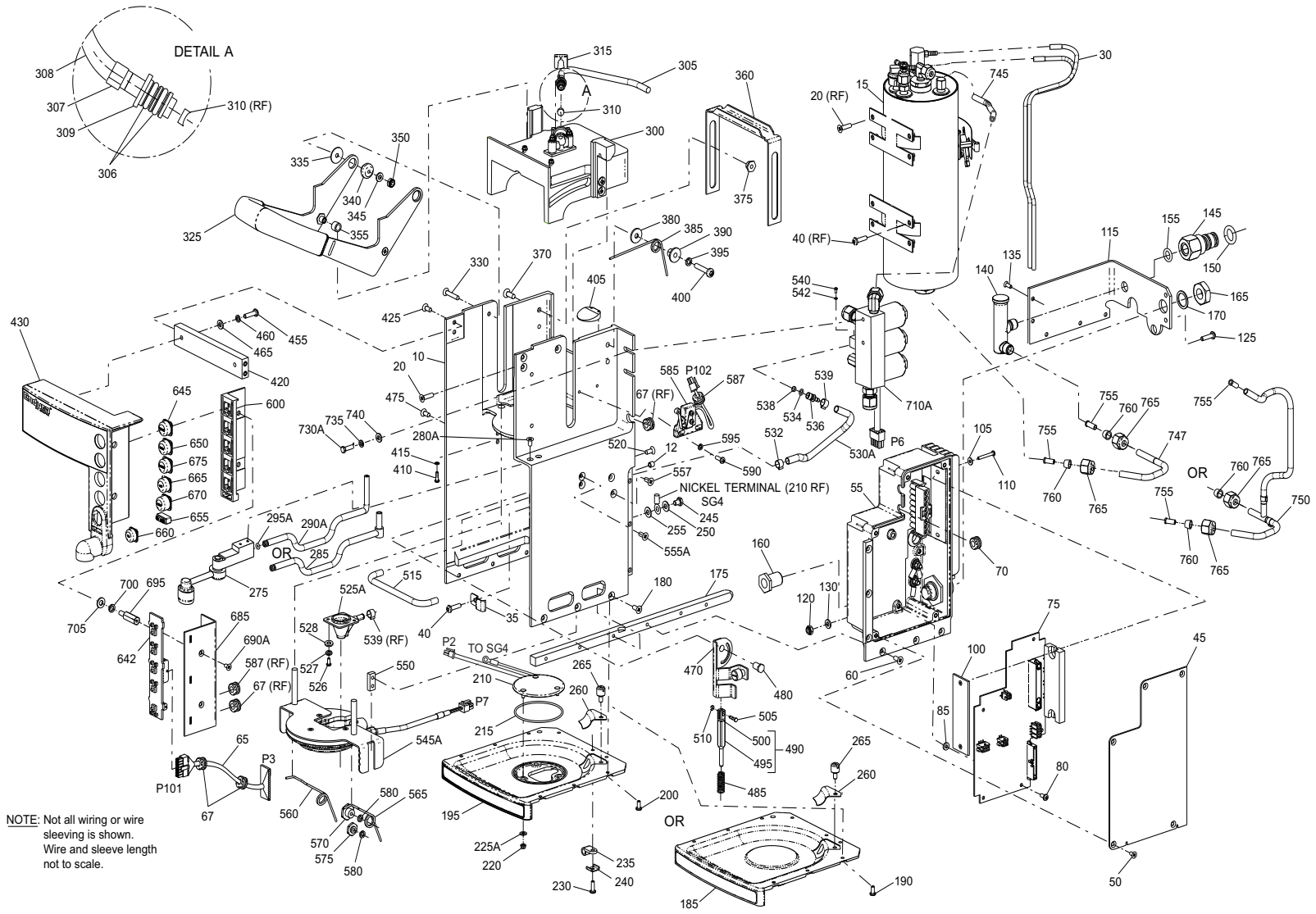

IPL Figure 2B
Beverage Maker Subassembly

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
-1A	3520-0571-31		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	A	RF
-1B	3520-0571-32		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	B	RF
-1C	3520-0571-33		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	C	RF
-1D	3520-0571-34		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	D	RF
-1E	3520-0571-35		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	E	RF
-1F	3520-0571-36		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	F	RF
-1G	3520-0571-37		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	G	RF
-1H	3520-0571-38		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1B FOR NHA)	H	RF
10	3520-0548-01		• ASSY, CHASSIS		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• ASSY, TANK (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		1
35	62-0144-1		• CLIP, HOSE		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		2
			* * *		
45	3575-1223-01		• COVER, POWER MODULE		1
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
55	3520-0530-01		• ASSY, POWER MODULE (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
65	3520-0553-01		• ASSY, HARNESS		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
75	3530-0121-01		• PC BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1
			ATTACHING PARTS		
120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
140	62-0014-1		• ASSY, DRAIN VALVE (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• ASSY, WATER SHUTOFF PLUG		1
150	OR067		• • O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• GUIDE, ALIGNMENT PIN		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		• NUT, PLAIN HEXAGON (SUPSDS ITEM 165)		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-3		• GUIDE, RAIL		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-05		• ASSY, PLATEN	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	8
			* * *		
195	3520-0561-07		• ASSY, PLATEN	D,E,F,H	1
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	7
			* * *		

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
210	3530-0261-01		• HEATER, PLATEN	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X 44 (WITH WARMER)	D,E,F,H	1
-232	MS51957-27		• SCREW, PH, 6-32 X .31 (WITHOUT WARMER)	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,F,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1271-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
265	3575-1215-01		• FASTENER, SCREEN, DRAIN		1
			* * *		
275	3520-0559-01		• ASSY, SWIVEL FAUCET (SEE IPL FIGURE 8 FOR DET BKDN)		1
			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 280A)		2
280A	MS24693C24		• SCREW, PH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 280)		2
			* * *		
285	3520-0535-01		• ASSY, FAUCET TUBE	B,C,E, F,G,H	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
-290	3520-0535-03		• ASSY, FAUCET TUBE (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• ASSY, BREWHEAD (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• ASSY, BREW TUBE		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• WASHER, FLOW		1
315	467-0912-1		• CLIP		1
-320	PF202		DELETED INSERT, TUBE (TPN PF-202)		DEL
			* * *		
325	3520-0533-03		• ASSY, BREW HANDLE		1
-327	430531		• • DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
-328	425-0702-1		• • ASSEMBLY, GUIDE BUSHING		2
-329	MS24693C48		• • SCREW, FH		2
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
355	411-0692-1		• BUSHING, HANDLE		2

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 375)		2
370	MS24693C272		• SCREW, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		
420	3520-0542-01		• ASSEMBLY, BRACE, CHASSIS		1
			ATTACHING PARTS		
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		4
			* * *		
430	3520-0560-01		• ASSY, PANEL, FRONT	F,H	1
-435	3520-0560-03		• ASSY, PANEL, FRONT	B,D,G	1
-440	3520-0560-05		• ASSY, PANEL, FRONT	A	1
-445	3520-0560-07		• ASSY, PANEL, FRONT	E	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
-450	3520-0560-09		• ASSY, PANEL, FRONT	C	1
			ATTACHING PARTS		
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8		2
			* * *		
470	3575-0650-01		• LEVER, PIN, LOCATING		1
			ATTACHING PARTS		
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• ASSY, PIN, PIVOT		1
			* * *		
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• ASSY, PIN, LOCATING		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING (TPN OR-0571)		1
			ATTACHING PARTS		
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING		1
			* * *		
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3530-0289-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A) (PRE SB 4510-25-14)	B,F,G,H	1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
525A	3530-0879-01		• FUNNEL, TEA TUBE (SUPSDS ITEM 525) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
526	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSD BY ITEM 526A)	B,F,G,H	2
-526A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 526)		2
527	MS35338-135		• WASHER, LK, (SUPSD BY ITEM 527A)	B,F,G,H	2
-527A	MS35338-135		• WASHER, LK, (SUPSDS ITEM 527)		2
528	MS15795-804		• WASHER, FL, #4 (SUPSD BY ITEM 528A)	B,F,G,H	2
-528A	MS15795-804		• WASHER, FL, #4 (SUPSDS ITEM 528)		2
-530	TU50242		• TUBE, 80 IN LONG (TPN TU-50242) (PRE SB 4510-25-02) (SUPSD BY ITEM 530A)	B,F,G,H	1
530A	3575-1312-01		• TUBE, TEA, HEAT FORMED (POST SB 4510-25-02) (SUPSDS ITEM 530)	B,F,G,H	1
532	CL50034		• CLAMP, TUBE (TPN CL-50034)	B,F,G,H	1
534	OR0571		• O-RING (TPN OR-0571)	B,F,G,H	1
536	3575-1283-01		• CARTRIDGE, REGULATOR	B,F,G,H	1
538	62-0066-7		• WASHER, FLOW	B,F,G,H	1
539	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010)	B,F,G,H	1
540	MS51957-13		• SCREW, PNH, 4-40 X .25	B,F,G,H	1
542	MS35338-135		• WASHER, LK, #4	B,F,G,H	1
			* * *		
-545	3520-0569-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (PRE SB 4510-25-03) (SUPSD BY ITEM 545A)		1
545A	3520-0580-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (POST SB 4510-25-03) (SUPSDS ITEM 545)		1

- ITEM NOT ILLUSTRATED

FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
			ATTACHING PARTS		
550	3520-0537-01		• ASSY, GUIDE, LLS		2
-555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS BY ITEM 555A)		8
555A	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS ITEM 555)		4
557	MS24693C49		• SCREW, FH, (TPN MS24693-C49)		4
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1
565	3575-0050-02		• SPRING, RETAINER, SERVER, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• ASSY, SWITCH		1
587	GR098		• GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2
			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A	3520-0242-01		DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C,D E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 690)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
710A	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0232-05		• ASSY, MANIFOLD (SUPSD BY ITEM 715A) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715A	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0232-07		• ASSY, MANIFOLD (SUPSD BY ITEM 720A) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1

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FIG ITEM	PART NUMBER	AIRLINE PART NO	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2B					
-720A	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-725	3530-0232-03		• ASSY, MANIFOLD (SUPSD BY ITEM 725A) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725A	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		• SCREW, HEX, 8-32 X .38 (TPN MS51849-C53) (SUPSD BY ITEM 730A)		2
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
745	TU50020		• TUBING, (1/4" OD X .063" WALL) 8.5 IN (TPN TU-50020)		1
747	TU4505		• TUBING-TEFLON (1/4" OD X .030" WALL), 4.0 IN (TPN TU-4505)	A,B,D,H	1
750	3520-0554-01		• ASSY, TUBE, INLET, WATER	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		

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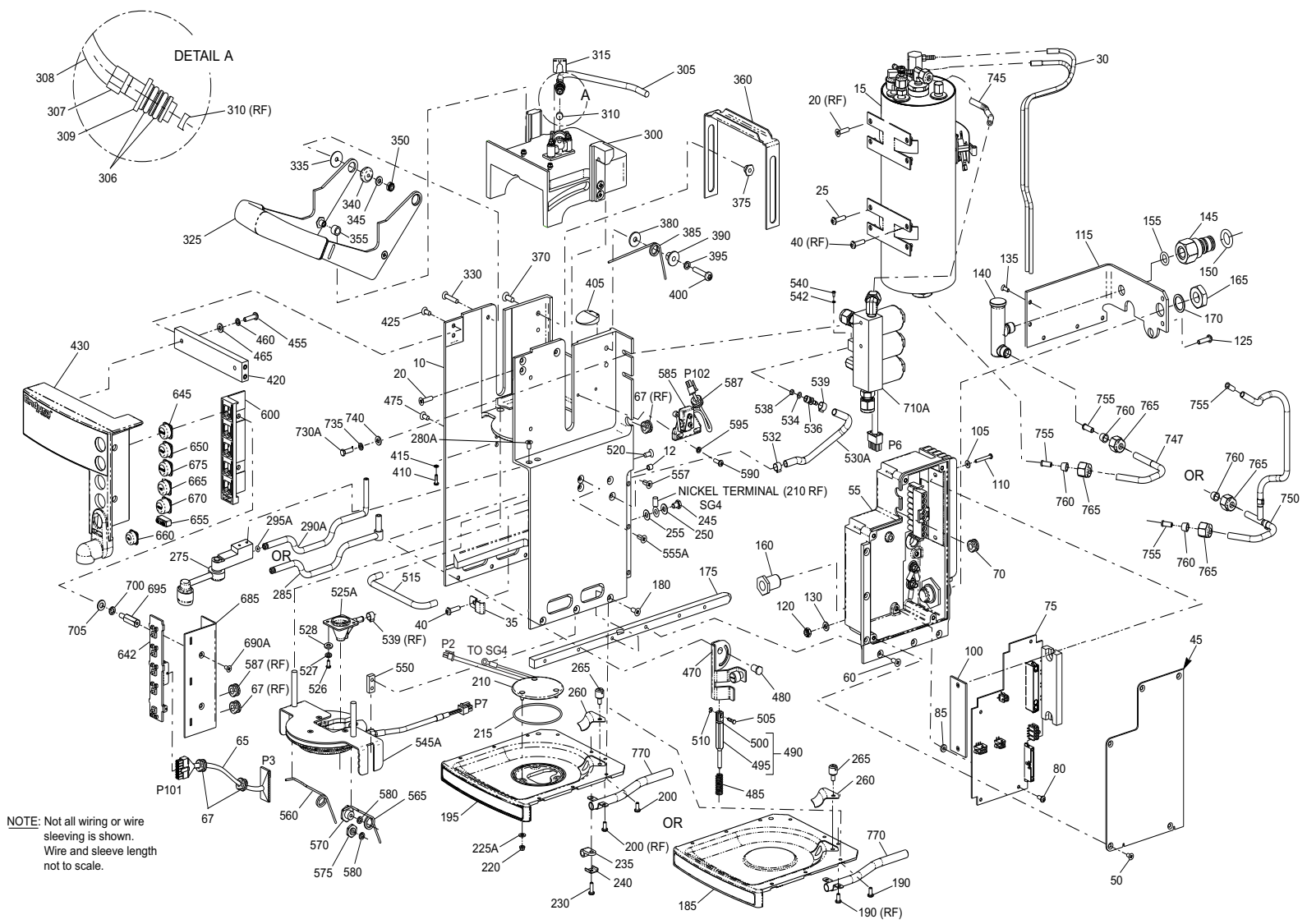

IPL Figure 2C
Beverage Maker Main Subassembly

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
-1A	3520-0571-41		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	A	RF
-1B	3520-0571-42		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	B	RF
-1C	3520-0571-43		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	C	RF
-1D	3520-0571-44		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	D	RF
-1E	3520-0571-45		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	E	RF
-1F	3520-0571-46		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	F	RF
-1G	3520-0571-47		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	G	RF
-1H	3520-0571-48		BEVERAGE MAKER SUBASSEMBLY (SEE IPL FIGURE 1C FOR NHA)	H	RF
10	3520-0548-01		• ASSY, CHASSIS		1
12	MS21209C0810		• • HELICAL COIL INSERT, 8-32 X .164 LG		1
15	3520-0529-01		• ASSY, TANK (SEE IPL FIGURE 3 FOR DET BKDN)		1
			ATTACHING PARTS		
20	MS24693C51		• SCREW, FH, 8-32 X .63 (TPN MS24693-C51)		4
			* * *		
30	TU379		• TUBING, 13.5 IN (TPN TU-379)		1
35	62-0144-1		• CLIP, HOSE		1
			ATTACHING PARTS		
40	MS51957-46		• SCREW, PH, 8-32 X .63		2
			* * *		
45	3575-1223-01		• COVER, POWER MODULE		1
			ATTACHING PARTS		
50	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24)		5
			* * *		

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
55	3520-0530-01		• ASSY, POWER MODULE (SEE IPL FIGURE 4 FOR DET BKDN)		1
			ATTACHING PARTS		
60	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
65	3520-0553-01		• ASSY, HARNESS		1
67	GR098		• • GROMMET (TPN GR-098)		2
70	GR098		• GROMMET (TPN GR-098)		2
75	3530-0121-01		• PC BOARD ASSY (REFER TO CMM 25-33-32)		1
			ATTACHING PARTS		
80	MS51957-27		• SCREW, PH, 6-32 X .31		5
85	WA50036		• WASHER, SHOULDER, #6, FIBRE (TPN WA-50036)		6
-90	3530-0274-01		• PAD, INSULATOR (SUPSD BY ITEM 90A)		2
-90A	3530-0274-01		DELETED (SUPSDS ITEM 90)		DEL
-95	3575-1253-01		• SPACER, RELAY (SUPSD BY ITEM 95A)		1
-95A	3575-1253-01		DELETED (SUPSDS ITEM 95)		DEL
100	3520-0566-01		• RETAINER, RELAY		1
105	MS15795-804		• WASHER, FL, #4		2
110	MS51957-20		• SCREW, PH, 4-40 X .875		2
			* * *		
115	62-0018-1		• CONNECTOR, SUPPORT		1
			ATTACHING PARTS		
120	SN431		• NUT, LK, #8 (TPN SN-431)		1
125	MS51957-46		• SCREW, PH, 6-32 X .63		1
130	MS15795-807		• WASHER, FL, #8		1
135	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		5
			* * *		

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
140	62-0014-1		• ASSY, DRAIN VALVE (SEE IPL FIGURE 6 FOR DET BKDN)		1
			ATTACHING PARTS		
145	62-0010-1		• ASSY, WATER SHUTOFF PLUG		1
150	OR067		• • O-RING (SUPPLIED WITH PLUG) (TPN OR-067)		1
155	OR0662		• O-RING (TPN OR-0662)		1
			* * *		
160	62-0037-1		• GUIDE, ALIGNMENT PIN		1
			ATTACHING PARTS		
165	60-2010-1		• NUT, PLAIN HEXAGON (SUPSD BY ITEM 165A)		1
-165A	AN316C9R		• NUT, PLAIN HEXAGON (SUPSDS ITEM 165)		1
170	LW163		• WASHER, LK, 9/16 (TPN LW-163)		1
			* * *		
175	423-0078-3		• GUIDE, RAIL		2
			ATTACHING PARTS		
180	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		6
			* * *		
185	3520-0561-05		• ASSY, PLATEN	A,B,C,G	1
			ATTACHING PARTS		
190	MS51957-27		• SCREW, PH, 6-32 X .31	A,B,C,G	8
			* * *		
195	3520-0561-07		• ASSY, PLATEN	D,E,F,H	1
			ATTACHING PARTS		
200	MS51957-27		• SCREW, PH, 6-32 X .31	D,E,F,H	7
			* * *		

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
210	3530-0261-01		• HEATER, PLATEN	D,E,F,H	1
215	OR50039		• O-RING (TPN OR-50039)	D,E,F,H	1
			ATTACHING PARTS		
220	MS21043-06		• NUT, LK, #6	D,E,F,H	3
-225	MS15795-805		• WASHER, FL, #6 (SUPSD BY ITEM 225A)	D,E,F,H	3
225A	3575-1362-01		• WASHER, SHOULDER (SUPSDS ITEM 225)	D,E,F,H	3
230	MS51957-29		• SCREW, PH, 6-32 X .44 (WITH WARMER)	D,E,F,H	1
-232	MS51957-27		• SCREW, PH, 6-32 X .31 (WITHOUT WARMER)	D,E,F,H	1
235	CL3201		• CLAMP, NYLON, BLACK, .250 DIAMETER (TPN CL-3201)	D,E,F,H	1
240	WA120		• WASHER, CLAMP (TPN WA-120)	D,E,F,H	1
245	NAS1801-3-4		• SCREW, HEX, 10-32 X .25 (SG4 CONNECTION)		1
250	NAS1149F0332P		• WASHER, FL, #10		1
255	NAS1149D0316H		• WASHER, FL, #10		1
			* * *		
260	3575-1271-01		• SCREEN, DRAIN		1
			ATTACHING PARTS		
265	3575-1215-01		• FASTENER, SCREEN, DRAIN		1
			* * *		
275	3520-0559-01		• ASSY, SWIVEL FAUCET (SEE IPL FIGURE 8 FOR DET BKDN)		1
			ATTACHING PARTS		
-280	MS24693C26		• SCREW, PH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 280A)		2
280A	MS24693C24		• SCREW, PH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 280)		2
			* * *		
285	3520-0535-01		• ASSY, FAUCET TUBE	B,C,E, F,G,H	1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
-290	3520-0535-03		• ASSY, FAUCET TUBE (SUPSD BY ITEM 290A)	A,D	1
290A	3575-1356-01		• FAUCET TUBE (SUPSDS ITEM 290)	A,D	1
-295	OR0361		• O-RING (TPN OR-0361) (SUPSD BY ITEM 295A)		1
295A	OR0381		• O-RING (TPN OR-0381) (SUPSDS ITEM 295)		1
300	3520-0534-01		• ASSY, BREWHEAD (SEE IPL FIGURE 9 FOR DET BKDN)		1
305	3520-0552-01		• ASSY, BREW TUBE		1
306	OR0571		• • O-RING (TPN OR-0571)		2
307	CL50010		• • CLAMP (TPN CL-50010)		1
308	TU4505		• • TUBING, 5.6 ±2 IN (TPN TU-4505)		1
309	467-0865-1		• • BREW FITTING		1
			ATTACHING PARTS		
310	62-0066-7		• WASHER, FLOW		1
315	467-0912-1		• CLIP		1
-320	PF202		DELETED INSERT, TUBE (TPN PF-202)		DEL
			* * *		
325	3520-0533-03		• ASSY, BREW HANDLE		1
-327	430531		• • DECAL, DIE CUT (SUPSD BY ITEM 327A)		2
-327A	430531		DELETED (SUPSDS ITEM 327)		DEL
-328	425-0702-1		• • ASSEMBLY, GUIDE BUSHING		2
-329	MS24693C48		• • SCREW, FH		2
			ATTACHING PARTS		
330	MS24693C53		• SCREW, FH, 8-32 X .88 (TPN MS24693-C53)		2
335	411-0724-1		• WASHER, HANDLE		2
340	411-0725-1		• BUSHING		2
345	MS15795-807		• WASHER, FL, #8		2
350	SN431		• NUT, LK, #8 (TPN SN-431)		2
355	411-0692-1		• BUSHING, HANDLE		2

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2C					
			* * *		
360	3575-0613-01		• HANDLE, CARRYING		1
			ATTACHING PARTS		
-365	411-0926-1		• SCREW, GUIDE (SUPSD BY ITEM 375)		2
370	MS24693C272		• SCREW, 10-32 X .50 (TPN MS24693-C272)		3
375	62-0061-1		• BUSHING, HANDLE (SUPSDS ITEM 365 AFTER 2006/01/20, REQUIRING COUNTERSUNK HOLES IN CHASSIS ASSEMBLY ITEM 10A)		4
380	60-2001-1		• WASHER, METALLIC		1
385	411-0676-3		• SPRING, TENSION		1
390	62-0025-1		• RETAINER, SPRING		1
395	MS35338-138		• WASHER, LK, #10		1
400	MS51958-68		• SCREW, PH, 10-32 X 1.25		1
			* * *		
405	416-0463-1		• RETAINER, CUP		1
			ATTACHING PARTS		
410	MS51957-15		• SCREW, PH, 4-40 X .38		2
415	MS35338-135		• WASHER, LK, #4		2
			* * *		
420	3520-0542-01		• ASSEMBLY, BRACE, CHASSIS		1
			ATTACHING PARTS		
425	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		4
			* * *		
430	3520-0560-01		• ASSY, PANEL, FRONT	F,H	1
-435	3520-0560-03		• ASSY, PANEL, FRONT	B,D,G	1
-440	3520-0560-05		• ASSY, PANEL, FRONT	A	1
-445	3520-0560-07		• ASSY, PANEL, FRONT	E	1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
-450	3520-0560-09		• ASSY, PANEL, FRONT	C	1
			ATTACHING PARTS		
455	MS51957-46		• SCREW, PH, 8-32 X .63		2
460	MS35338-137		• WASHER, LK, #8		2
465	MS15795-807		• WASHER, FL, #8		2
			* * *		
470	3575-0650-01		• LEVER, PIN, LOCATING		1
			ATTACHING PARTS		
475	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		1
480	62-0211-1		• ASSY, PIN, PIVOT		1
			* * *		
485	60-2013-1		• SPRING, HELICAL		1
490	62-0210-1		• ASSY, PIN, LOCATING		1
495	62-0210-3		• • PIN, MACHINED		1
500	OR0571		• • O-RING (TPN OR-0571)		1
			ATTACHING PARTS		
505	62-0210-2		• PIN		1
510	MS16633-4012		• RING, RETAINING		1
			* * *		
515	3575-0435-01		• HANDLE		1
			ATTACHING PARTS		
520	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48)		2
			* * *		
-525	3530-0289-01		• TUBE, FUNNEL (SUPSD BY ITEM 525A) (PRE SB 4510-25-14)	B,F,G,H	1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
525A	3530-0879-01		• FUNNEL, TEA TUBE (SUPSDS ITEM 525) (POST SB 4510-25-14)		1
			ATTACHING PARTS		
526	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSD BY ITEM 526A)	B,F,G,H	2
-526A	MS51957-13		• SCREW, PH, 4-40 X .25 (SUPSDS ITEM 526)		2
527	MS35338-135		• WASHER, LK, (SUPSD BY ITEM 527A)	B,F,G,H	2
-527A	MS35338-135		• WASHER, LK, (SUPSDS ITEM 527)		2
528	MS15795-804		• WASHER, FL, #4 (SUPSD BY ITEM 528A)	B,F,G,H	2
-528A	MS15795-804		• WASHER, FL, #4 (SUPSDS ITEM 528)		2
-530	TU50242		• TUBE, 80 IN LONG (TPN TU-50242) (PRE SB 4510-25-02) (SUPSD BY ITEM 530A)	B,F,G,H	1
530A	3575-1312-01		• TUBE, TEA, HEAT FORMED (POST SB 4510-25-02) (SUPSDS ITEM 530)	B,F,G,H	1
532	CL50034		• CLAMP, TUBE (TPN CL-50034)	B,F,G,H	1
534	OR0571		• O-RING (TPN OR-0571)	B,F,G,H	1
536	3575-1283-01		• CARTRIDGE, REGULATOR	B,F,G,H	1
538	62-0066-7		• WASHER, FLOW	B,F,G,H	1
539	CL50010		• CLAMP, TUBE, 9/32 INCH (TPN CL-50010)	B,F,G,H	1
540	MS51957-13		• SCREW, PNH, 4-40 X .25	B,F,G,H	1
542	MS35338-135		• WASHER, LK, #4	B,F,G,H	1
			* * *		
-545	3520-0569-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (PRE SB 4510-25-03) (SUPSD BY ITEM 545A)		1
545A	3520-0580-01		• ASSY, RETAINER, SERVER (SEE IPL FIGURE 10 FOR DET BKDN) (POST SB 4510-25-03) (SUPSDS ITEM 545)		1
			ATTACHING PARTS		

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
550	3520-0537-01		• ASSY, GUIDE, LLS		2
-555	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS BY ITEM 555A)		8
555A	MS24693C48		• SCREW, FH, 8-32 X .38 (TPN MS24693-C48) (SUPSDS ITEM 555)		4
557	MS24693C49		• SCREW, FH, (TPN MS24693-C49)		4
560	3575-0050-01		• SPRING, RETAINER, SERVER, LEFT HAND		1
565	3575-0050-02		• SPRING, RETAINER, SERVER, RIGHT HAND		1
570	411-0686-1		• BUSHING, SPRING		2
575	411-0689-1		• BUSHING, GUIDE, SPRING		2
580	MS35338-137		• WASHER, LK, #8		4
			* * *		
585	3520-0536-01		• ASSY, SWITCH		1
587	GR098		• GROMMET (TPN GR-098)		1
			ATTACHING PARTS		
590	MS51957-29		• SCREW, PH, 6-32 X .44		2
595	MS35338-136		• WASHER, LK, #6		2
			* * *		
600	3530-0241-01		• FRAME, BUTTON		1
-602	3520-0242-01		• ASSY, SWITCH (UNCONFIGURED) (SUPSD BY ITEM 602A)		1
-602A	3520-0242-01		DELETED (SEE ITEM 642) (SUPSDS ITEM 602)		DEL
-605	3520-0556-01		• ASSY, SWITCH (CONFIGURED)	A	1
-610	3520-0556-02		• ASSY, SWITCH (CONFIGURED)	B	1
-615	3520-0556-03		• ASSY, SWITCH (CONFIGURED)	C	1
-620	3520-0556-04		• ASSY, SWITCH (CONFIGURED)	D	1
-625	3520-0556-05		• ASSY, SWITCH (CONFIGURED)	E	1
-630	3520-0556-06		• ASSY, SWITCH (CONFIGURED)	F	1
-635	3520-0556-07		• ASSY, SWITCH (CONFIGURED)	G	1
-640	3520-0556-08		• ASSY, SWITCH (CONFIGURED)	H	1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
642	3530-0242-01		• • PC BOARD SWITCH ASSY (UNCONFIGURED)	A,B,C,D E,F,G,H	1
645	3530-0244-01		• BUTTON, POWER		1
650	3530-0245-01		• BUTTON, BREW		1
655	3530-0251-01		• INDICATOR		1
660	3530-0246-01		• BUTTON, TEA	B,F,G,H	1
665	3530-0248-01		• BUTTON, HOT	A,B,C, D,E,H	1
670	3530-0249-01		• BUTTON, COLD	C,E,F,G	1
675	3530-0247-01		• BUTTON, PLATEN	D,E,F,H	1
-680	GR098		• GROMMET, WIRE (TPN GR-098) (SUPSD BY ITEM 680A)		1
-680A	GR098		DELETED (SEE ITEM 67) (SUPSDS ITEM 680)		DEL
685	3575-1244-01		• COVER, SWITCH		1
			ATTACHING PARTS		
-690	MS24693C26		• SCREW, FH, 6-32 X .38 (TPN MS24693-C26) (SUPSD BY ITEM 690A)		2
690A	MS24693C24		• SCREW, FH, 6-32 X .25 (TPN MS24693-C24) (SUPSDS ITEM 690)		2
695	3575-1220-01		• STANDOFF, COVER, SWITCH		2
700	MS35338-138		• WASHER, LK, #10		2
705	MS15795-808		• WASHER, FL, #10		2
			* * *		
-710	3530-0232-01		• ASSY, MANIFOLD (SUPSD BY ITEM 710A) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
710A	3520-0605-01		• MANIFOLD ASSY (SUPSDS ITEM 710) (SEE IPL FIGURE 11 FOR DET BKDN)	F,G	1
-715	3530-0232-05		• ASSY, MANIFOLD (SUPSD BY ITEM 715A) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-715A	3520-0605-05		• MANIFOLD ASSY (SUPSDS ITEM 715) (SEE IPL FIGURE 11 FOR DET BKDN)	C,E	1
-720	3530-0232-07		• ASSY, MANIFOLD (SUPSD BY ITEM 720A) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1
-720A	3520-0605-07		• MANIFOLD ASSY (SUPSDS ITEM 720) (SEE IPL FIGURE 11 FOR DET BKDN)	A,D	1

- ITEM NOT ILLUSTRATED

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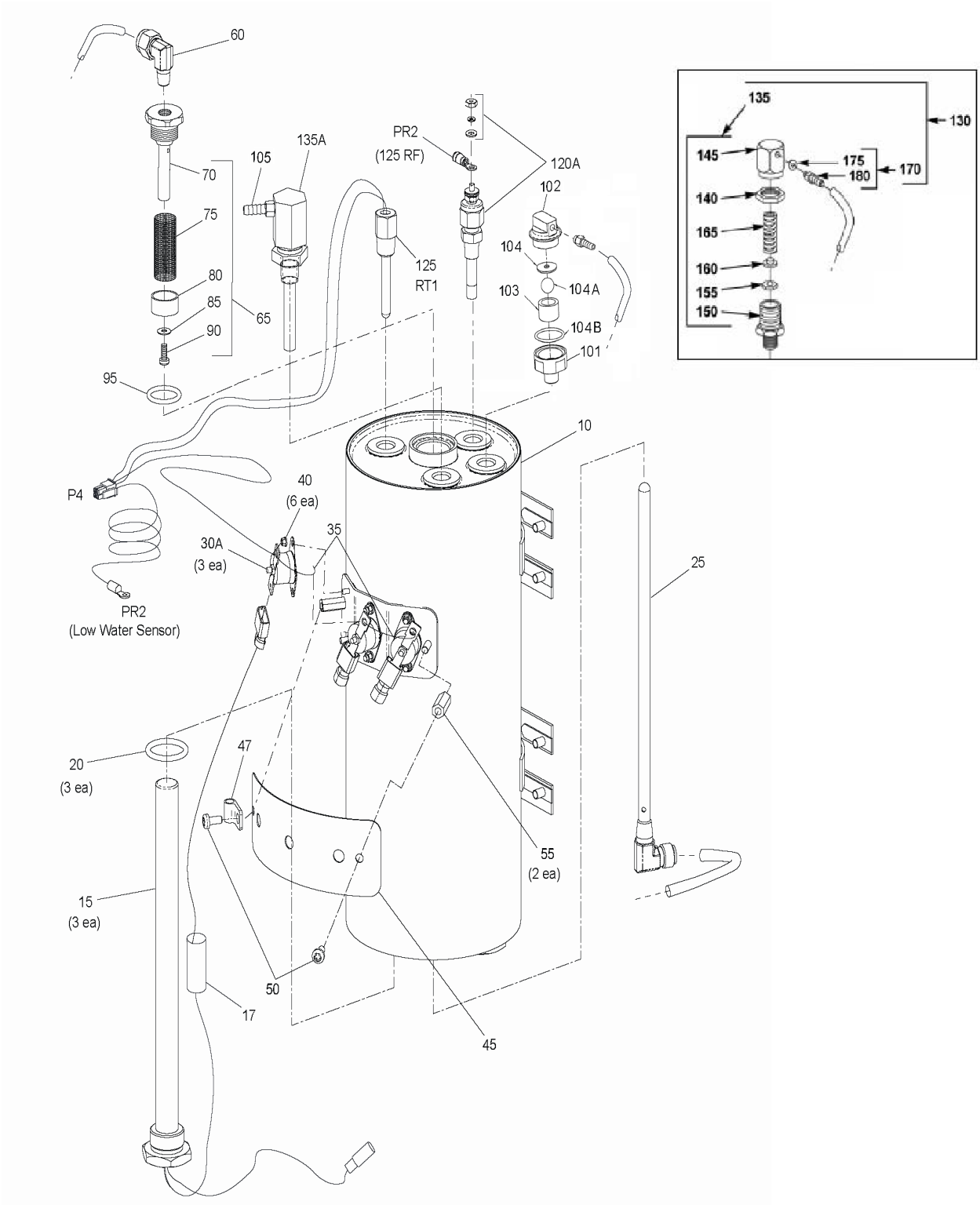
FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
2C					
-725	3530-0232-03		• ASSY, MANIFOLD (SUPSD BY ITEM 725A) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
-725A	3520-0605-03		• MANIFOLD ASSY (SUPSDS ITEM 725) (SEE IPL FIGURE 11 FOR DET BKDN)	B,H	1
			ATTACHING PARTS		
-730	MS51849C53		• SCREW, HEX, 8-32 X .38(TPN MS51849-C53) (SUPSD BY ITEM 730A)		2
730A	MS51849-53C		• SCREW, HEX, 8-32 X .38 (SUPSDS ITEM 730)		2
735	MS35338-137		• WASHER, LK, #8		2
740	MS15795-807		• WASHER, FL, #8		2
			* * *		
745	TU50020		• TUBING, (1/4" OD X .063" WALL) 8.5 IN (TPN TU-50020)		1
747	TU4505		• TUBING-TEFLON (1/4" OD X .030" WALL), 4.0 IN (TPN TU-4505)	A,B,D,H	1
750	3520-0554-01		• ASSY, TUBE, INLET, WATER	C,E,F,G	1
			ATTACHING PARTS		
755	PF202		• INSERT, TUBE (TPN PF-202)		2
760	PF50002		• FERRULE, TUBING (TPN PF-50002)		2
765	PF143		• NUT, FITTING (TPN PF-143)		2
			* * *		
770	3520-0528-01		• ASSY, TUBE, DRAIN		1
			ATTACHING PARTS		
-775	MS51957-27		• SCREW, PH, 6-32 X .31		1
			* * *		

- ITEM NOT ILLUSTRATED



IPL Figure 3
Tank Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
-1	3520-0529-01		TANK ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)		RF
10	3520-0555-01		• TANK WELDED ASSY		1
15	3520-0551-01		• TANK HEATER ASSY (HR1, HR2, AND HR3)		3
-15A	TE50432		• • TERMINAL, FAST-ON (TPN TE-50432)		1
-15B	CO50313		• • CONNECTOR (TPN CO-50313)		1
-15C	CO50314		• • CONNECTOR PIN (TPN CO-50314)		1
-15D	3575-1480-07		• • HEATER ASSEMBLY TANK		1
17	TU50039		• SLEEVING, VARGLAS, BLACK, 8 IN (TPN TU-50039)		1
20	OR0941		• O-RING (TPN OR-0941)		3
25	3520-0509-01		• ASSY, TUBE		1
-30	3530-0268-01		• THERMOSTAT, OVERTEMPERATURE (TS1, TS2, TS3) (SUPSD BY ITEM 30A) (PRE SIL 4510-SIL-02)		3
30A	352-0983-3		• THERMOSTAT, OVERTEMPERATURE (TS1, TS2, TS3) (SUPSDS ITEM 30) (POST SIL 4510-SIL-02)		3
			ATTACHING PARTS		
35	WI156		• WIRE, BUSS, SOLID, 3 IN, 16 GA (TPN WI-156)		1
40	MS21043-04		• NUT, LK, 4-40		6
			* * *		
45	3575-0630-01		• COVER, SWITCH, THERMAL		1
47	CL315		• CLAMP (TPN CL-315)		1
			ATTACHING PARTS		
50	MS51957-27		• SCREW, PH, 6-32 X .31		2
55	NS50113		• SPACER, HEX, 1/4 IN DIA, 6-32, .625 IN (TPN NS-50113)		2
			* * *		
60	PF102		• FITTING, ELBOW (TPN PF-102)		1

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
65	62-0020-1		• ASSEMBLY, FILTER		1
70	3575-0647-01		• • JET, WATER		1
75	62-0007-1		• • SCREEN, FILTER		1
80	62-0011-1		• • RETAINER		1
85	MS35338-136		• • WASHER, LK, #6		1
90	MS51957-26		• • SCREW, PH, 6-32 X .25		1
95	OR0941		• O-RING (TPN OR-0941)		1
-100	3520-0523-01		• VENT VALVE ASSY (PRE SB 4510-25-07) (SUPSD BY ITEM 100A)		1
-100A	3520-0594-03		• VENT VALVE ASSY (POST SB 4510-25-07) (SUPSDS ITEM 100)		1
-100B	3520-0594-01		• VENT VALVE ASSY (POST SB 4510-25-07)		1
101	3575-1324-01		• • ADAPTER, VENT VALVE		1
102	3575-1325-01		• • BODY, VENT VALVE		1
103	3575-1327-01		• • LINER, VENT VALVE (SUPSD BY ITEM 103A) (PRE SB 3520-0594-25-003)		1
-103A	3575-2785-01		• • LINER, VENT VALVE (SUPSDS ITEM 103) (POST SB 3520-0594-25-003)		1
104	3575-1329-01		• • WASHER, VENT VALVE (SUPSD BY ITEM 104F) (PRE SB 3520-0594-25-003)		1
-104F	3575-2791-01		• • WASHER, VENT SEAL (SUPSDS ITEM 104) (POST SB 3520-0594-25-003)		1
104A	MISC50411		• • BALL, VENT VALVE (TPN MISC-50411) (PRE SB 3520-0594-25-001) (SUPSD BY ITEM 104C)		1
104B	OR50019		• • O-RING (TPN OR-50019)		1
-104C	MISC50470		• • BALL, VENT VALVE (TPN MISC-50470) (POST SB 3520-0594-25-001) (PRE SB 3520-0594-25-002) (PRE SB 3520-0594-25-003) (SUPSDS ITEM 104A) (SUPSD BY ITEM 104E)		1
-104D	MISC50411		• • BALL, VENT VALVE (TPN MISC-50411) (POST SB 3520-0594-25-002) (OPT TO ITEM 104C) (SUPSD BY ITEM 104E)		1

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
3					
-104E	MISC50509		• • BALL, VENT VALVE (TPN MISC-50509) (SUPSDS ITEM 104C and 104D) (POST SB 3520-0594-25-003)		1
105	62-0136-7		• FITTING, BARBED		1
110	62-0136-9		• • GASKET, OD .24 X ID .15 X .022 THICK		1
115	62-0136-11		• • FITTING, NICKEL PLATE		1
-120	MISC50389		• SENSOR, WATER, LOW (TPN MISC-50389) (SUPSD BY ITEM 120A) (PRE SB 4510-25-08)		1
120A	3520-0581-01		• LOW WATER SENSOR ASSY (SUPSDS ITEM 120) (POST SB 4510-25-08)		1
125	3520-0550-01		• ASSY, RTD		1
130	62-0223-1		• PRESSURE RELIEF VALVE ASSY (SUPSD BY ITEM 135A) (PRE SB 4510-25-009)		RF
135	62-0223-3		• • RELIEF VALVE SUBASSY (SUPSD BY ITEM 135A) (PRE SB 4510-25-009)		1
135A	VA50107		• VALVE, PRESSURE RELIEF (TPN VA-50107) (SUPSDS ITEMS 130 AND 135) (POST SB 4510-25-009)		1
140	62-0223-9		• • • NUT, LOCK		1
145	62-0223-5		• • • CAP, ADJUSTING		1
150	62-0223-11		• • • BODY, VALVE		1
155	VA1031		• • • ASSY, VALVE		1
160	VA1003		• • • REST, SPRING		1
165	VA1004		• • • SPRING		1
170	62-0136-7		• • FITTING, BARBED		1
175	62-0136-9		• • • GASKET, OD .24 X ID .15 X .022 THICK		1
180	62-0136-11		• • • FITTING, NICKEL PLATE		1

- ITEM NOT ILLUSTRATED

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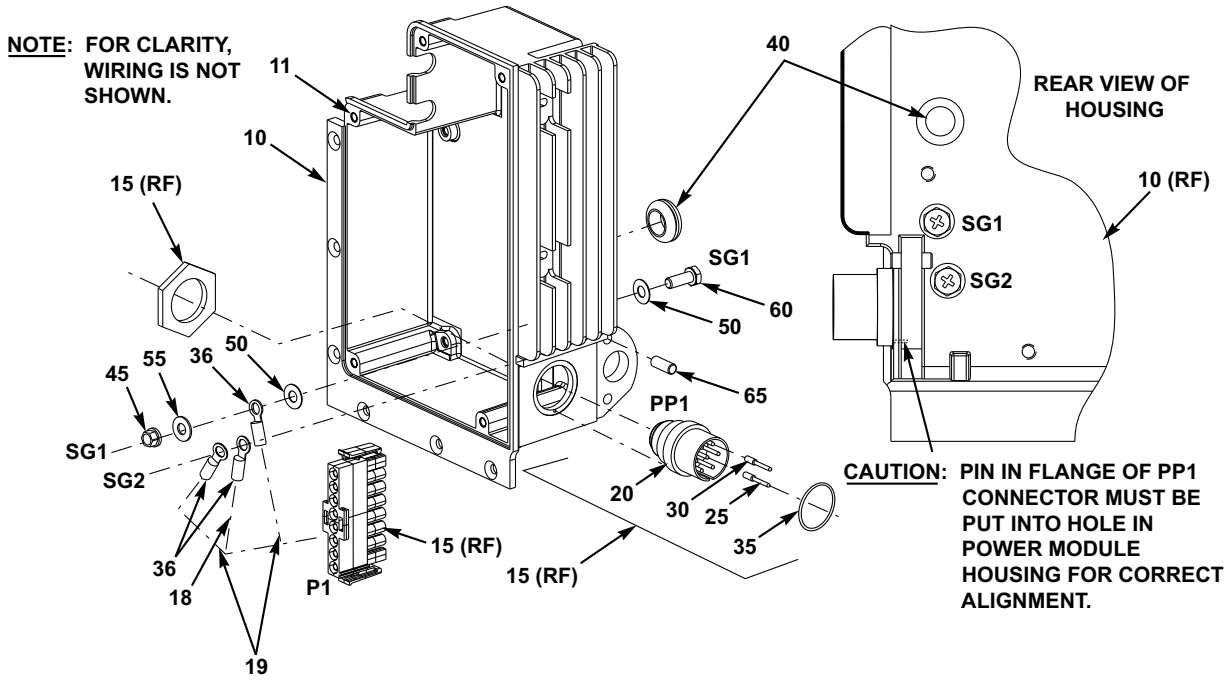

IPL Figure 4
Power Module Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
4					
-1	3520-0530-01		POWER MODULE ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)		RF
10	3520-0531-01		• POWER MODULE HOUSING ASSY		1
-11	MS21209C0615		• • INSERT, HELICAL COIL		10
-12	3575-1250-01		• • INSULATOR, PC BOARD		1
-13	3530-0408-01		• • HOUSING, POWER MODULE		1
15	3520-0544-01		• HARNESS ASSEMBLY		1
-17	3520-0544-05		• • ASSEMBLY, CONNECTOR HARNESS		1
18	WI50106		• • • WIRE, WHITE, 14 GA (1.75 LG.) (TPN WI-50106)		AR
19	WI142		• • • WIRE, WHITE, 16 GA (69.5 LG.) (TPN WI-142)		AR
20	CO6505		• • • CONNECTOR, 7-PIN (TPN CO-6505)		1
25	CO6505-01		• • • • PIN (TPN CO-6505-01)		1
30	CO6505-03		• • • • PIN (TPN CO-6505-03)		6
35	CO6505-05		• • • • O-RING (TPN CO-6505-05)		1
36	TE215		• • • TERMINAL, RING (TPN TE-215)		3
-37	TE50441		• • • SOCKET (TPN TE-50441)		5
-38	MISC50408		• • • RING CORE FERRITE (TPN MISC-50408)		1
-39	3520-0544-03		• • CONNECTOR HARNESS ASSEMBLY		1
-39A	CO50403		• • • PLUG, 8 CIRCUIT (TPN CO-50403)		1
-39B	TE50441		• • • SOCKET (TPN TE-50441)		3
-39C	CO50311		• • • RECEPTACLE, CONNECTOR (TPN CO-50311)		3
-39D	CO50312		• • • CONTACT SOCKET (TPN CO-50312)		3
40	GR50053		• GROMMET (TPN GR-50053)		1
45	MS21042L3		• NUT, LK, 10-32		2
50	NAS1149D0316H		• WASHER, FL, #10		4
55	NAS1149F0332P		• WASHER, FL, #10		2
60	NAS1801-3-8		• SCREW, HEX, 10-32 X .50 (SG1 AND SG2)		2
65	PN5855		• PIN (TPN PN-5855)		1

- ITEM NOT ILLUSTRATED

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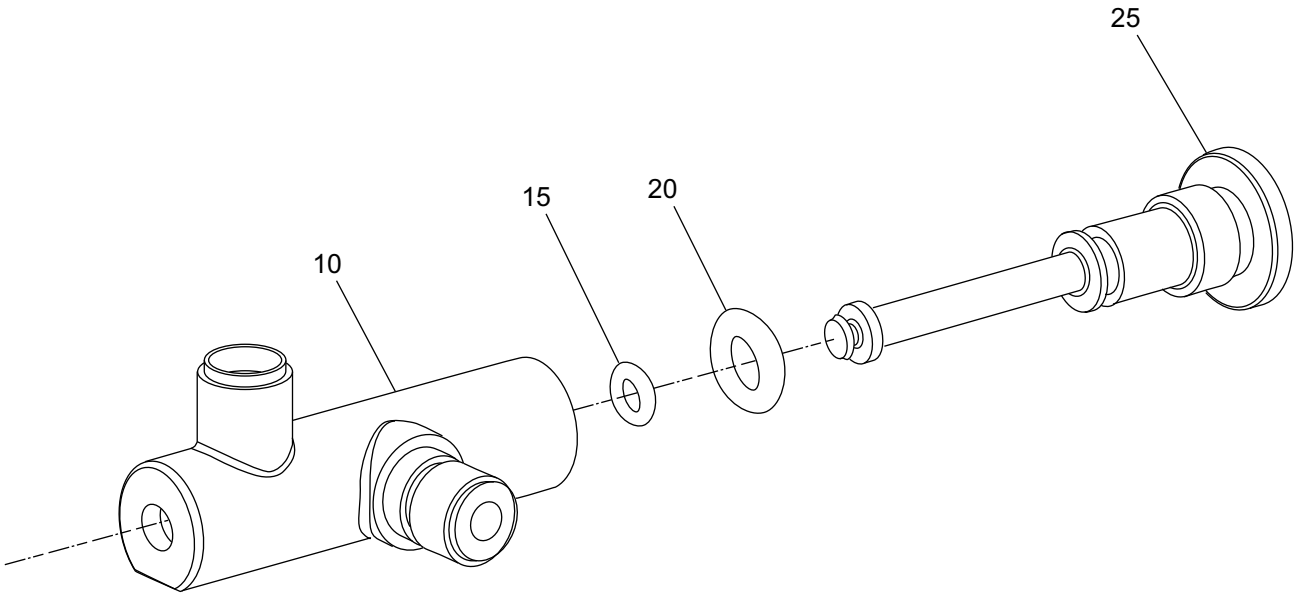
IPL Figure 5 (Deleted)
Controller Board Assembly (Refer to CMM 25-33-32)

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
5					
-1	3530-0121-01		DELETED (REFER TO CMM 25-33-32)		DEL

- ITEM NOT ILLUSTRATED



IPL Figure 6
Drain Valve Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
6					
1	62-0014-1		DRAIN VALVE ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA)		RF
10	62-0008-1		• BODY		1
15	OR0341		• O-RING (TPN OR-0341)		1
20	OR0571		• O-RING (TPN OR-0571)		1
25	62-0013-1		• STEM ASSY		1

- ITEM NOT ILLUSTRATED

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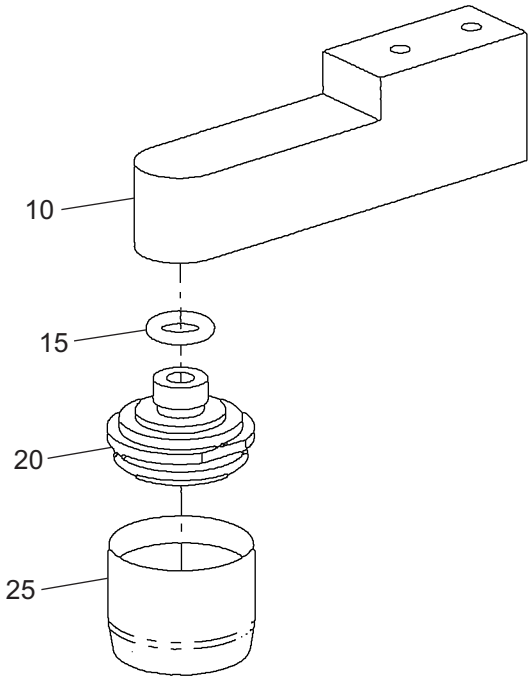
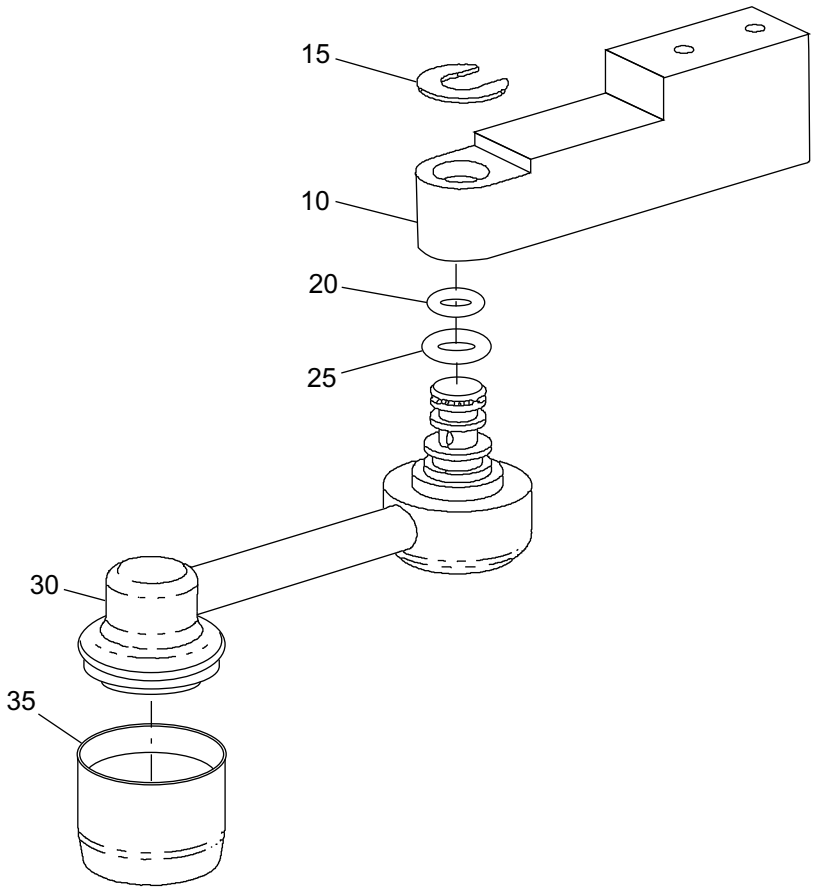


IPL Figure 7
Faucet Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
7					
1	3520-0525-01		FAUCET ASSY (SEE IPL FIGURES 2, AND 2A FOR NHA)		RF
10	3575-1227-01		• BAR, FAUCET		1
15	OR0571		• O-RING (TPN OR-0571)		1
20	3575-1225-01		• ADAPTER, AERATOR		1
25	MISC013		• AERATOR, FAUCET (TPN MISC-013)		1

- ITEM NOT ILLUSTRATED



IPL Figure 8
Swivel Faucet Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
8					
1	3520-0559-01		ASSY, SWIVEL FAUCET (SEE IPL FIGURES 2B AND 2C FOR NHA)		RF
10	3575-1246-01		• BAR, FAUCET, SWIVEL		1
15	MS16633-4031		• RING, RETAINING		1
20	OR50001		• O-RING (TPN OR-50001)		1
25	OR0571		• O-RING (TPN OR-0571)		1
30	3520-0558-01		• WELDED FAUCET ASSY		1
35	MISC013		• AERATOR (TPN MISC-013)		1

- ITEM NOT ILLUSTRATED

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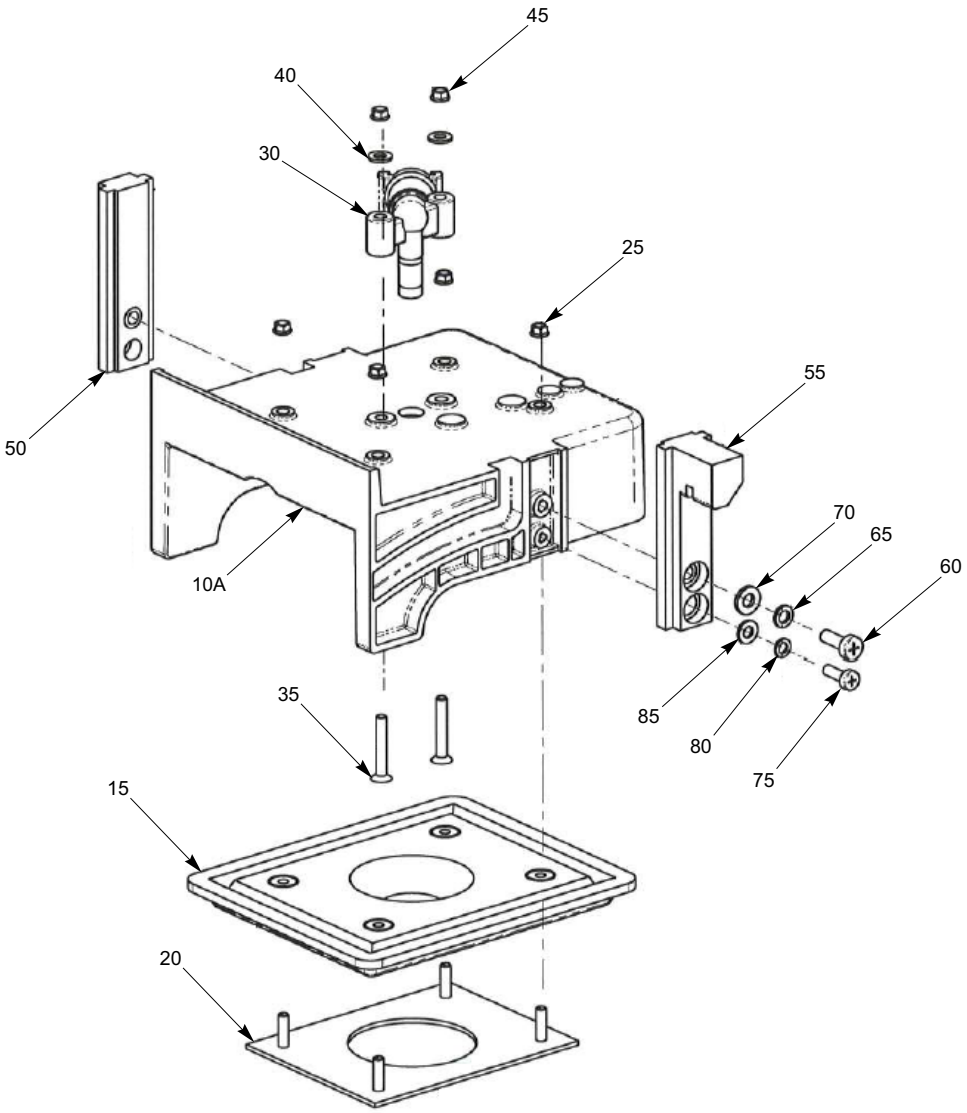

IPL Figure 9
Brew Head Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
9					
-1	3520-0534-01		ASSY, BREWHEAD (SEE IPL FIGURES 2, 2A, 2B AND 2C FOR NHA)		RF
-10	3575-1230-01		• BREW HEAD		1
10A	3530-0272-01		• BREW HEAD (OPT FOR ITEM 10)		1
15	411-0693-1		• GASKET		1
			ATTACHING PARTS		
20	411-0709-1		• RETAINER, GASKET		1
25	MS21043-06		• NUT, LK, 6-32		4
			* * *		
30	3575-0189-01		• FITTING, BREW		1
			ATTACHING PARTS		
35	MS24693C31		• SCREW, FH, 6-32 X .87 (TPN MS24693-C31)		2
40	MS15795-805		• WASHER, FL, #6		2
45	MS21043-06		• NUT, LK, 6-32		2
			* * *		
50	411-0748-1		• GUIDE ASSY (LEFT HAND)		1
55	411-0748-3		• GUIDE ASSY (RIGHT HAND)		1
			ATTACHING PARTS		
60	MS51957-45		• SCREW, PH, 8-32 X .50 (SUPSD BY ITEM 60A) (PRE SB 3520-0534-25-001)		2
-60A	SC50280-45		• SCREW, PH, 8-32 X .50 (TPN SC-50280-45) (SUPSDS ITEM 60) (POST SB 3520-0534-25-001)		2
65	MS35338-137		• WASHER, LK, #8		2
70	MS15795-807		• WASHER, FL, #8		2
75	MS51957-29		• SCREW, PH, 6-32 X .44 (SUPSD BY ITEM 75A) (PRE SB 3520-0534-25-001)		2
-75A	SC50280-29		• SCREW, PH, 6-32 X .44 (TPN SC-50280-29) (SUPSDS ITEM 75) (POST SB 3520-0534-25-001)		2

- ITEM NOT ILLUSTRATED

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FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
9					
80	MS35338-136		• WASHER, LK, #6		2
85	MS15795-805		• WASHER, FL, #6		2
			* * *		

- ITEM NOT ILLUSTRATED

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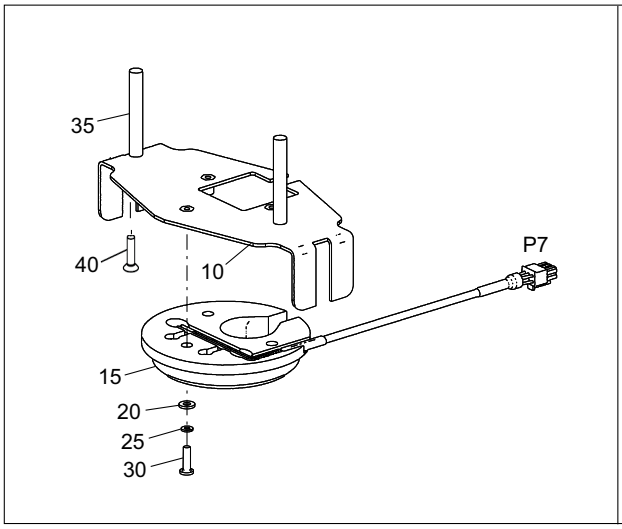
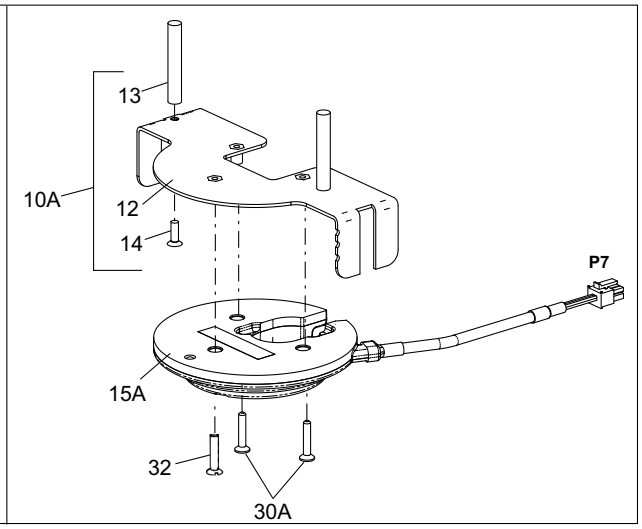


3520-0569-01



3520-0580-01

IPL Figure 10
Retainer Plate Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
10					
-1	3520-0569-01		SERVER RETAINER ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (PRE SB 4510-25-03) (SUPSD BY ITEM 1A)	A	RF
-1A	3520-0580-01		SERVER RETAINER ASSY (SEE IPL FIGURES 2, 2A, 2B, AND 2C FOR NHA) (POST SB 4510-25-03) (SUPSDS ITEM 1)	B	RF
10	3520-0570-01		• ASSY, RETAINER	A	1
10A	3520-0595-01		• RETAINER PLATE AND ROD ASSY	B	1
12	3520-0578-01		• • RETAINER PLATE ASSY	B	1
13	3575-1236-05		• • RETAINER ROD, LLS	B	2
			ATTACHING PARTS		
14	MS24693C27		• • SCREW, FH, 6-32 X .44 (TPN MS24693-C27)	B	2
			* * *		
15	3520-0568-01		• ASSY, LIQUID LEVEL SENSOR (LLS)	A	1
15A	3520-0579-01		• INFRARED SENSOR ASSY	B	1
20	MS15795-805		• WASHER, FL, #6	A	3
25	MS35338-136		• WASHER, LK, #6	A	3
30	MS51957-30		• SCREW, PH, 6-32 X .50 (SUPSD BY ITEM 30A)	A	3
30A	SC50268-30		• SCREW, FH, NYLOC THREADS, 6-32 X .75 (TPN SC-50268-30) (SUPSDS ITEM 30)	B	2
32	SC50270		• SCREW, FH, NYLON, 6-32 X .625 (TPN SC-50270)	B	1
			* * *		
35	3575-1236-03		• ROD, RETAINING, LLS	A	2
			ATTACHING PARTS		
40	MS24693C27		• SCREW, FH, 6-32 X .44 (TPN MS24693-C27)	A	2
			* * *		

- ITEM NOT ILLUSTRATED

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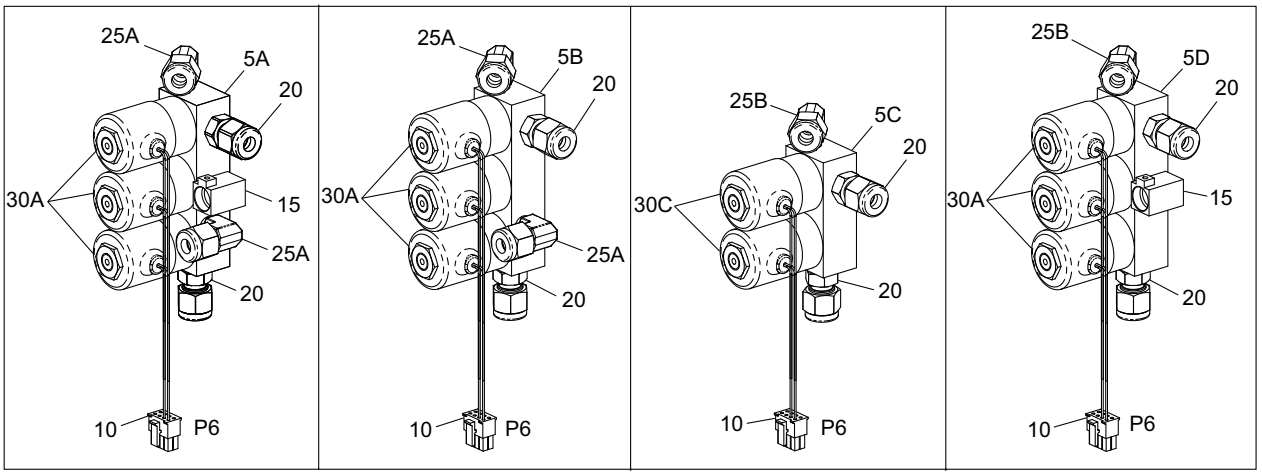
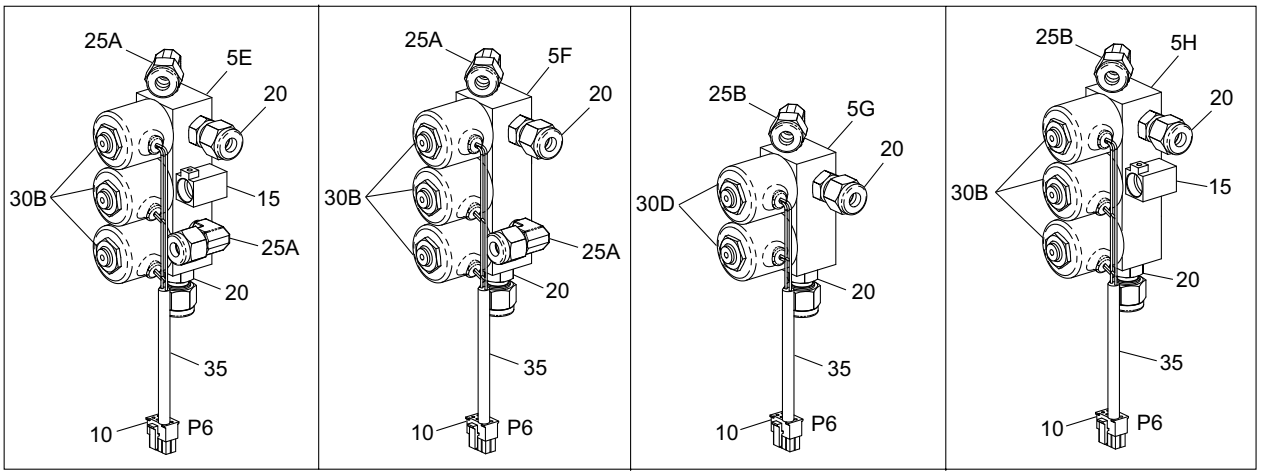


3530-0232-01 3530-0232-05 3530-0232-07 3530-0232-03



3520-0605-01 3520-0605-05 3520-0605-07 3520-0605-03

IPL Figure 11
Manifold Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
11					
-1A	3530-0232-01		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1E)	A	RF
-1B	3530-0232-05		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AD 2C FOR NHA) (SUPSD BY ITEM 1F)	B	RF
-1C	3530-0232-07		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1G)	C	RF
-1D	3530-0232-03		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSD BY ITEM 1H)	D	RF
-1E	3520-0605-01		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1A)	E	RF
-1F	3520-0605-05		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1B)	F	RF
-1G	3520-0605-07		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1C)	G	RF
-1H	3520-0605-03		ASSY, MANIFOLD (SEE IPL FIGURE 2, 2A, 2B, AND 2C FOR NHA) (SUPSDS ITEM 1D)	H	RF
5A	3575-1211-01		• BODY, MANIFOLD	A	1
5B	3575-1211-05		• BODY, MANIFOLD	B	1
5C	3575-1211-07		• BODY, MANIFOLD	C	1
5D	3575-1211-03		• BODY, MANIFOLD	D	1
5E	3575-1351-01		• MANIFOLD, SOLENOID	E	1
5F	3575-1351-05		• MANIFOLD, SOLENOID	F	1
5G	3575-1351-07		• MANIFOLD, SOLENOID	G	1
5H	3575-1351-03		• MANIFOLD, SOLENOID	H	1
10	CO50409		• CONNECTOR, EIGHT PIN, DUAL ROW (P6) (TPN CO-50409)		1
15	3575-1284-01		• FITTING, REGULATOR	A,D,E, H	1
20	PF110		• FITTING, STRAIGHT (SUPLD W/BODY, NUT, AND FERRULE) (TPN PF-110)		2
25A	PF102		• FITTING, 90 DEGREE, ELBOW (SUPLD W/BODY, NUT, AND FERRULE) (TPN PF-102)	A,B,E,F	2
25B	PF102		• FITTING, 90 DEGREE, ELBOW (SUPLD W/BODY, NUT, AND FERRULE) (TPN PF-102)	C,D,G, H	1
30A	3530-0279-01		• ASSY, SOLENOID VALVE (SEE IPL FIGURE 12 FOR DET BKDN)	A,B,D	3

- ITEM NOT ILLUSTRATED

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
11					
30B	3530-0279-23		• ASSY, SOLENOID VALVE (SEE IPL FIGURE 12 FOR DET BKDN)	E,F,H	3
30C	3530-0279-01		• ASSY, SOLENOID VALVE (SEE IPL FIGURE 12 FOR DET BKDN)	C	2
30D	3530-0279-23		• ASSY, SOLENOID VALVE (SEE IPL FIGURE 12 FOR DET BKDN)	G	2
35	TU50052		• VARGLAS, NON-FRAY SLEEVING (TPN TU-50052)	E,F,G,H	1

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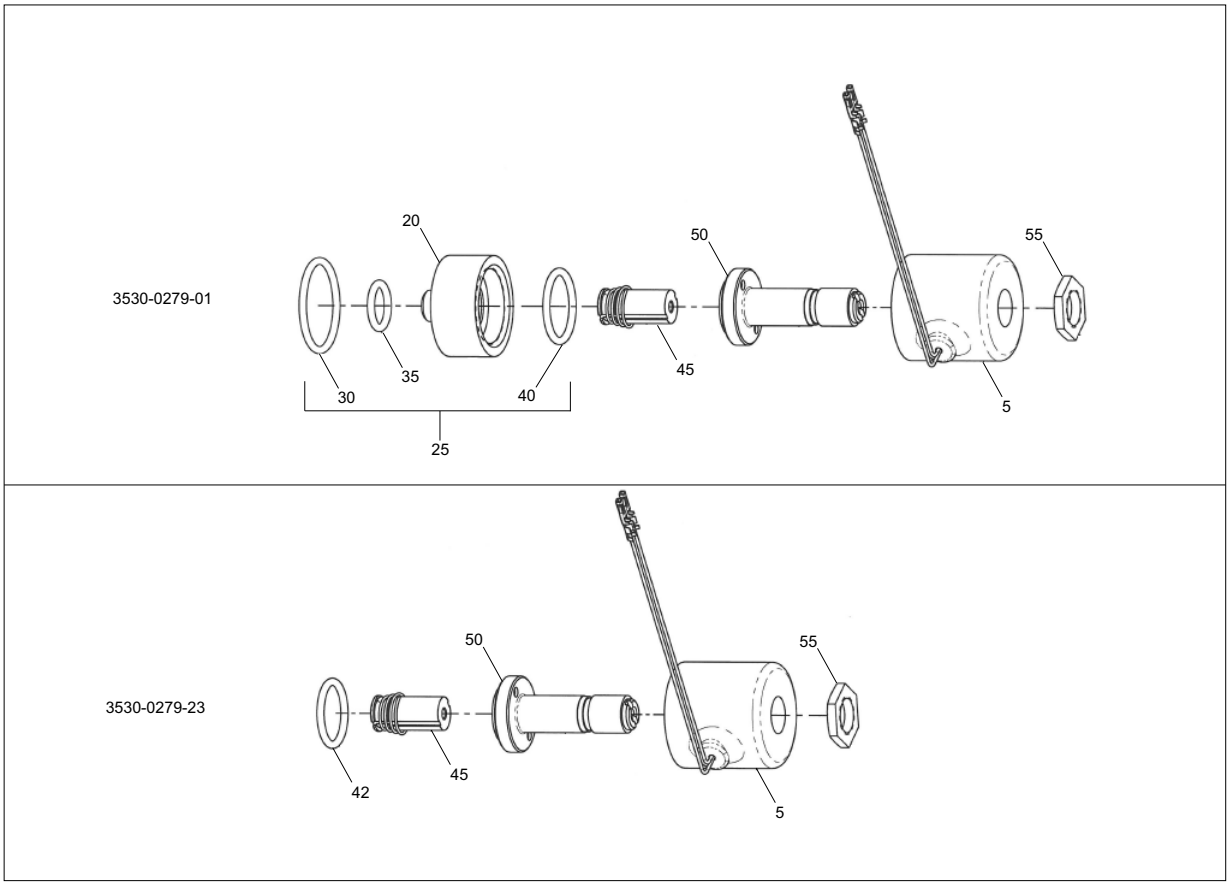

IPL Figure 12
Solenoid Valve Assembly

FIG. ITEM	PART NUMBER	AIRLINE PART NO.	NOMENCLATURE 1 2 3 4 5 6 7	EFF CODE	UNITS/ ASSY
12					
-1	3530-0279-01		SOLENOID VALVE ASSY (SEE IPL FIGURE 11 FOR NHA) (SUPSD BY ITEM 1A)	A	RF
-1A	3530-0279-23		SOLENOID VALVE ASSY (SEE IPL FIGURE 11 FOR NHA) (SUPSDS ITEM 1)	B	RF
5	3530-0279-11		• COIL ASSY		1
-10	3530-0279-21		• • COIL ASSY		1
-15	TE50442		• • CRIMP TERMINAL (TPN TE-50442)		2
20	3530-0279-05		• BODY	A	1
25	3530-0279-03		• KIT, O-RING	A	1
30	3530-0279-15		• • O-RING	A	1
35	3530-0279-17		• • O-RING	A	1
40	3530-0279-19		• • O-RING	A	1
42	3530-0279-25		• O-RING	B	1
45	3530-0279-09		• PLUNGER ASSY		1
			ATTACHING PARTS		
50	3530-0279-07		• GUIDE ASSY		1
55	3530-0279-13		• NUT		1
			* * *		

- ITEM NOT ILLUSTRATED

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